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Abstract

Cyclic codes are an important family of linear codes that can be efficiently encoded using simple shift registers. They were introduced in 1950 and have been expanded upon in various forms over different alphabets. Cyclic codes and their generalizations are useful because they can be examined using polynomial rings. New linear codes with better parameters have been obtained from cyclic codes and their generalizations. A new method for studying cyclic codes and constacyclic codes has been introduced by Radkova et al. This method involves treating them as invariant subspaces using an operator called the λ-cyclic shift. Similar investigations have been conducted for right polycyclic codes and monomial codes using a special operator. These studies have revealed that the operators used are cyclic and the codes under examination can be viewed as ideals in the polynomial ring $\mathbb{F}_q[x]/\langle f(x)\rangle$, where $f(x)$ is the minimal polynomial of the operator. Therefore, an interesting question arises: can the results from these previous works be extended to linear codes that are invariant under both a linear transformation and a pseudo-linear transformation T?

In this dissertation we are interested to answer this question, where we investigate the structure of linear codes that are invariant under linear transformations and pseudo-linear transformations T. In particular, the first part considers T-codes, which are linear codes invariant under linear transformations, while the second part considers (T, σ, δ)-codes, which are linear codes invariant under (σ, δ)-pseudo-linear transformations. The operator T is decomposed into a direct sum of cyclic and pseudo-cyclic operators T_i, leading to $(T_1, T_2, \ldots, T_r)$-multi-cyclic codes and $(T_1, T_2, \ldots, T_r, \sigma, \delta)$-multi-cyclic codes. Special cases of these codes include quasi T_1-cyclic codes and quasi (T_1, σ, δ)-skew cyclic codes. The dissertation also investigates M-codes, which are linear codes invariant under the right multiplication by a square matrix M, and (M, σ, δ)-skew codes, which are linear codes invariant under the right (σ, δ)-multiplication by a given matrix M. Similarity relations between matrices are used to find connections between M-codes, and various types of linear codes with cyclical property in the literature. In order to use polynomials rings in the study of M-codes, we show a one-to-one correspondence between M-codes and $\mathbb{F}_q[x]$-submodules of $\prod_{i=1}^{r} R_{f_i}$, where $R_{f_i} := \mathbb{F}_q[x]/\langle f_i(x)\rangle$ and $(f_1(x), f_2(x), \ldots, f_r(x))$ is the sequence of invariant factors of M. A similar correspondence was proven between (M, σ, δ)-skew codes and $\mathbb{F}_q[x, \sigma, \delta]$-submodules of $\prod_{i=1}^{r} R_{f_i}$, where $R_{f_i} := \mathbb{F}_q[x, \sigma, \delta]/\mathbb{F}_q[x, \sigma, \delta]f_i(x)$ and $(f_1(x), f_2(x), \ldots, f_r(x))$ is the sequence of "skew invariant factors of M". We also prove some bounds for the different classes discussed above, such as BCH-like and Hartmann-Tzeng-like bounds, and, as an application of the theory developed here, we construct some optimal linear codes. We characterize also the structure of **1**-generator quasi M-cyclic codes, as well as their additive structure under a basis of $\mathbb{F}_{q^r}/\langle f(x)\rangle$. This leads to a characterization of quasi-cyclic codes of index r as submodules of $\mathbb{F}_{q^r}/\mathbb{F}_q[x]/\langle f(x)\rangle$.

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Publications

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   (c) H. Ou-azzou, B. Ialou, E. Mouloua, A class of **1**-generator right quasi-polycyclic codes, 2nd International Conference on Innovative Research in Applied Science, Engineering and Technology (IRASET), (2022).

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Introduction

Coding theory's main goal is to provide reliable communication over a noisy channel, where a sent message can be received with possible errors. The challenge of coding theory is, therefore, to detect and correct these errors. Even more challenging is decoding: the message is supposed to be encoded with as little redundancy as possible while remaining capable of a certain level of error correction through an adequate decoding process. Information theory and coding theory are generally considered to have been born in 1948 when Claude Shannon published his famous paper "A mathematical theory of communication", [82], where the existence of good codes was proved, but without any explicit method of their construction. Explicit construction of good codes turned out to be a major challenge for coding theorists since then.

Cyclic codes are one of the most important families of linear codes because of their algebraic structure, which allows for easy implementation. Cyclic codes have been introduced in 1950 by Prange in [72] as linear codes, in which the cyclic shift of any codeword is another codeword. Since then, they have been generalized in various ways, over different alphabets (fields and rings). Some of these generalizations include constacyclic codes [19], monomial codes [30], polycyclic codes [80], quasi-cyclic codes (QC), and others. The advantage of the class of cyclic codes and their generalization is in the use of the polynomial ring to investigate the structure of these codes. Over the finite field $\mathbb{F}_q$, cyclic codes are characterized by the ideals of $\mathbb{F}_q[x]/\langle x^n-1\rangle$, constacyclic codes by the ideals of $\mathbb{F}_q[x]/\langle x^n-\lambda\rangle$ where λ is a non-zero element in $\mathbb{F}_q$, and polycyclic codes by the ideals of $\mathbb{F}_q[x]/\langle f(x)\rangle$, for some non-zero polynomial $f(x)$. A remarkable generalization of cyclic codes is the family of quasi-cyclic codes, which were first introduced in [89] as linear codes of length $n = mr$ such that the cyclic shift of any codeword by r positions is another codeword. Such a code is called a quasi-cyclic code of index r, and when $r = 1$ we obtain cyclic code. The structure of quasi-cyclic codes has been studied from multiple perspectives. For example, in[88], the authors studied quasi-cyclic codes via a polynomial approach, where quasi-cyclic codes of index r are viewed as linear codes of length r over the polynomial ring $\mathbb{F}_q[x]/\langle x^m-1\rangle$. In [42], the structure of quasi-cyclic codes was investigated via Gröbner basis. Another generalization in this context is the family of generalized quasi-cyclic codes (GQC), introduced first by Siap and others in [78], and are further studied [28, 33]. Similarly, constacyclic codes are generalized to quasi-twisted (QT) codes of index r (equivalent name, quasi-constacyclic codes of index r), see [92, 8, 54]. Still more general is the class of multi-twisted codes (generalized quasi-constacyclic codes) in [10, 83].

Using the theory of non-commutative polynomials, cyclic codes are generalized to skew-cyclic codes introduced in [12] and are viewed as left $\mathbb{F}_q[x,\sigma]$-submodules of $R/R\cdot(x^n-1)$, where $R := \mathbb{F}_q[x,\sigma]$ is a non-commutative ring, namely the skew polynomial ring, and σ is an automorphism on $\mathbb{F}_q$. Similarly to classical cyclic codes, skew cyclic codes are generalized in various ways, such as skew constacyclic codes, skew-negacyclic codes, and others. In [14] the structure of skew codes with derivation δ was investigated, which involves the ring $\mathbb{F}_q[x,\sigma,\delta]$. Several papers present new linear codes with better parameters that are obtained from skew cyclic codes (e.g., [13], [87], [73], [7]).

In [74] the authors have introduced a new approach to study constacyclic codes based on the theory of invariant subspaces by a given operator T_λ, called the λ-quasi-twisted codes, a

similar study for right polycyclic codes is given in [85], and for monomial codes in [30], with a special operator in each case. Based on these works, we observe that the operators used are cyclic, and the code considered is seen as an ideal of the polynomial ring $\mathbb{F}_q[x]/\langle f(x)\rangle$, where $f(x)$ is the minimal polynomial of the operator. Therefore, a natural question arises: *can the results from these previous works be extended to linear codes that are invariant under both a linear transformation and a pseudo-linear transformation T?*

The answer to this question is yes, and we address it in two parts. In the first part, we consider linear codes invariant under a linear transformation T, called T-code. In the second part, we consider linear codes invariant under a (σ,δ)-pseudo-linear transformation T, called (T,σ,δ)-codes, where σ is an automorphism of $\mathbb{F}_q$ and δ a σ-derivation of $\mathbb{F}_q$. For more details on (σ,δ)-pseudo-linear transformations, see [49, 50, 40]. According to the cyclic decomposition theorem, we decompose T as a direct sum of cyclic and pseudo-cyclic operators T_i i.e. $T = T_1 \oplus T_2 \oplus \ldots \oplus T_r$. Hence we talk about $(T_1, T_2, \ldots, T_r)$**-multi-cyclic codes** and $(T_1, T_2, \ldots, T_r, \sigma, \delta)$**-multi-cyclic codes**. In the particular case when $T_1 = T_2 = \ldots = T_r$ we talk about **quasi** T_1**-cyclic codes** and **quasi** (T_1, σ, δ)**-skew cyclic codes**. Since every operator can be represented by a matrix with respect to a given basis, and all of these matrices are similar, for simplicity we restrict ourselves to linear codes invariant under the right multiplication by a square matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$, which we call M-codes. Similarly, linear codes invariant under the right (σ,δ)-multiplication by a given matrix M are called (M,σ,δ)-skew codes.

The plan of this dissertation is organized as follows:

In Chapter 1, we recall the necessary background on cyclic codes, based on their correspondence to the ideals of $\mathbb{F}_q[x]/\langle x^n-1\rangle$. We also describe a method for factoring x^n-1 from cyclotomic cosets. Next, we recall the definition of polycyclic codes and prove some of their algebraic properties. Finally, we conclude with a brief discussion of generalized quasi-cyclic and quasi-cyclic codes.

In Chapter 2, we study linear codes invariant under right multiplication by a cyclic matrix M, that we call M-cyclic codes, and when M is the companion matrix C_f of a given non-zero polynomial $f(x)$ we call them f-cyclic codes. The similarity relation between matrices helps us find connections between M-cyclic codes and f-cyclic codes, where $f(x)$ is the minimal polynomial of M. The class of M-cyclic codes contains cyclic codes and their various generalizations such as constacyclic codes, right and left polycyclic codes, monomial codes, and others as special cases. As common in the study of cyclic codes and their generalizations, we make use of the one-to-one correspondence between M-cyclic codes and ideals of the polynomial ring $R_f := \mathbb{F}_q[x]/\langle f(x)\rangle$, where $f(x)$ is the minimal polynomial of M. This correspondence leads to some basic characterizations of these codes such as generator and parity check polynomials among others. Next, we study the duality of these codes, where we prove that the b-dual of an M-cyclic code is an M^*-cyclic code, where M^* is the adjoint matrix of M with respect to b, and we explore some important results on the duality of these codes. Finally, we give examples as applications of some of the results and we construct some optimal codes.

In Chapter 3, we are interested in linear codes of length $n = mr$ that are invariant under right multiplication by a matrix $D = \operatorname{diag}(\overbrace{M, M, \ldots, M}^{r \text{ times}})$, where M is an $m \times m$ cyclic matrix. Then, we talk about quasi M-cyclic codes of index r and quasi f-cyclic codes when M is the companion matrix of $f(x)$. This class covers quasi-cyclic codes and all their generalizations, such as quasi-twisted codes and quasi-negacyclic codes, and also naturally permits us to define left and right quasi-polycyclic and quasi-monomial codes. As is common in coding theory, we prove the one-to-one correspondence between quasi f-cyclic codes of index r and R_f-submodules of R_f^r, with $R_f := \mathbb{F}_q[x]/\langle f(x)\rangle$ and $f(x)$ is the minimal polynomial of M, via an isomorphism $\varphi_{f,r}$. The same correspondence was proved for quasi M-cyclic codes by considering another

isomorphism $\psi_{f,r} := \varphi_{f,r} \circ P^{-1}$, where $D = \operatorname{diag}(\overbrace{M, M, \ldots, M}^{r \text{ times}}) = P^{-1} \operatorname{diag}(\overbrace{C_f, C_f, \ldots, C_f}^{r \text{ times}})P$. Based on this correspondence, we explore some important results on the structure of $\mathbf{1}$-generator quasi f-cyclic codes, and we prove the BCH-like and the Hartmann-Tzeng-like bounds. Next, we are interested in the additive structure of quasi f-cyclic codes of index r, where we map them from R_f^r to $\mathbb{F}_{q^r}/\langle f(x)\rangle$ via an $\mathbb{F}_{q^r}$-module morphism. We characterize their $\mathbb{F}_{q^r}$-linear images with respect to a basis of the extension $\mathbb{F}_{q^r}/\mathbb{F}_q$, and show that such codes must be invariant under certain matrices related to the basis of the extension $\mathbb{F}_{q^r}/\mathbb{F}_q$. Moreover, when the basis of the extension is normal the characterization is particularly simple.

In Chapter 4, we are interested in linear codes invariant under right multiplication by a square matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$ and we call them M-codes. In particular we distinguish three fundamental types: M-cyclic codes, quasi M-cyclic codes and $(M_1, M_2, \ldots, M_r)$-multi-cyclic codes. The approach developed here allows us to realize cyclic codes and many of their generalizations as special cases. It also helps us better understand the structure of linear codes invariant under some special matrices such as a permutation matrix or a monomial matrix. Next, we study the duality of M-codes, where we show that the b-dual of an M-code is an M^*-code, where M^* is the adjoint matrix of M with respect to a non-degenerate bilinear form b, and we explore some important results on the duality of these codes. In order to use polynomial rings in the study of M-codes, we show a one-to-one correspondence between M-codes and $\mathbb{F}_q[x]$-submodules of $\prod_{i=1}^r R_{f_i}$, where $R_{f_i} := \mathbb{F}_q[x]/\langle f_i(x)\rangle$ and $(f_1(x), f_2(x), \ldots, f_r(x))$ is the sequence of invariant factors of M. Finally, we investigate the structure of $\mathbf{1}$-generator M-codes and we prove BCH-like and Hartmann-Tzeng-like bounds for them. As an application of the theory developed here, we construct some optimal linear codes.

In Chapter 5, we are interested in linear codes invariant under the (σ, δ)-action of a given matrix M, that we call (M, σ, δ)-skew codes. Various subclasses of these codes, depending on the (σ, δ)-cyclical property of M, are derived. In particular, we distinguish between three fundamental types: (M, σ, δ)-cyclic, quasi (M, σ, δ)-cyclic and $(M_1, M_2, \ldots, M_r, \sigma, \delta)$-multi-cyclic. These different subclasses allow us to obtain some of the well-known types of linear codes as special cases depending on the choice of M. We observe that the (σ, δ)-similarity in $\mathbb{M}_n(\mathbb{F}_q)$ gives rise to various relations between these codes. As common in the study of cyclic codes and their generalizations, we use the skew polynomial ring $\mathbb{F}_q[x, \sigma, \delta]$ to study (M, σ, δ)-skew codes and we investigate their algebraic structure as in the commutative case.

Finally, we mention a brief conclusion and discuss some open problems and interesting perspectives.

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Chapter 1

Preliminaries

In this chapter, we recall the necessary background that we need throughout this document, about fundamental notions on linear codes, cyclic codes, and polycyclic codes. For more details, we invite the reader to see [37, 66, 53].

1.1 Introduction to coding theory

Coding theory originated as the mathematical foundation for the transmission of messages over noisy communication channels and deals with the problem of detecting and correcting transmission errors caused by the channel noise. Suppose that we want to send a message m of a certain length via a noisy channel. The problem is to ensure that the intended message can be obtained from what is received. A naive solution to this problem is to repeat each symbol of the message many times, or, more generally, by adding some redundancy symbols to the original message. We define the **information rate** R as the ratio of the number of bits in the original message (information vector) to the number of bits in the vector after adding redundancy.

For example, suppose we want to send the message 0 or 1 via a noisy channel where the probability of bit flipping is p. We repeat each symbol of the message 3 times and we obtain $c = 000$ or $c = 111$. We send c via a noisy channel. So, if for example, we receive $c^{'} = 001$, then by applying the majority decoding model, we decode $c^{'}$ to 000 and by deleting the redundancy symbols we obtain our initial message $m = 0$. In this example, the information rate R is $\frac{1}{3}$. As the probability of one error is p, the probability that we send 000 and receive 001 is $(1-p)^2p$, and the probability that we send 111 and receive 001 is $p^2(1-p)$. So, if $p < \frac{1}{2}$ then it is more likely that we sent 000 than that we sent 111. Hence, we decode this message as 000.

In the above situation, we supposed that at most one error can be occurring. Next, we ask for the probability that more than one error occurs. We do this exactly in the case where there are two or more errors. The probability of this occurring is:

$$\left(\begin{array}{c} 3 \\ 2 \end{array} \right) (1-p)p^2 + \left(\begin{array}{c} 3 \\ 3 \end{array} \right) p^3 \approx 3p^2 \ll p,$$

since when p is small, the p^3 term is much smaller than the p^2 term. If we do no encoding and just send a single bit then the probability that it is received incorrectly is p. Therefore, we see that when p is small, this repetition helps us decode correctly.

More generally, suppose we repeat each symbol n times, where for convenience we assume that n is odd. With the same analysis as above, we see that the probability of decoding incorrectly is

$$\left(\begin{array}{c} n \\ \frac{n+1}{2} \end{array} \right) p^{\frac{n+1}{2}} (1-p)^{\frac{n-1}{2}} + \ldots + \left(\begin{array}{c} n \\ n \end{array} \right) p^n \approx \left(\begin{array}{c} n \\ \frac{n-1}{2} \end{array} \right) p^{\frac{n+1}{2}}$$

when p is small. So for $p < \frac{1}{2}$ the probability of incorrect decoding approaches 0 as n goes to infinity.
In general, the probability of making an error decreases as the number of repetitions increases, which is not desirable or even possible in some situations. However, by using other coding schemes, it is possible to obtain a higher probability of decoding to the correct message. Thus, one of the main problems of coding theory is to construct coding schemes (or codes). In 1948, Claude Shannon proved that there are "good" codes for long lengths, but did not give a way to construct these codes. More precisely, Shannon's theorem says that it is possible to transmit information through a noisy channel at some rate R, depending on the codes used, less than the channel capacity C, with an arbitrarily small probability of decoding error. In the above example, the rate of the codes is $\frac{1}{3}$.

To find these good codes that are relatively easy to implement, coding theorists have been led to look for codes with algebraic or geometric structures. We limit ourselves to block codes, i.e., the words in the message have a fixed length of symbols. Let us start with the following definition.

Definition 1.1 (Definition 2.1.1 in [53])
Let F be a finite set of size q, which we refer to as a code alphabet and whose elements are called code symbols.

1. A q-ary word of length n over F is a sequence $c = c_0c_1 \dots c_{n-1}$ with each $c_i \in F$ for all i. Equivalently, c may also be regarded as the vector $(c_0, c_1 \dots, c_{n-1})$
2. A q-ary block code of length n over F is a non-empty set $C \subseteq F^n$.
3. An element of C is called a codeword in C.
4. The number of codewords in C, denoted by $|C|$, is called the size of C.
5. The information rate of a code C of length n is defined to be $\frac{\log_q(|C|)}{n}$.
6. A code of length n and size M is called a (n, M)-code.

An important notion in coding theory is the Hamming distance, which allows us to measure the number of positions in which two words of a code differ.

Definition 1.2
1. The **Hamming distance** on F^n between two elements $x = \left(x_0, x_1, \dots, x_{n-1}\right)$ and $y = (y_0, y_1, \dots, y_{n-1})$ is defined by:
$$d(x, y) := |\{0 \leq i \leq n-1 : x_i \neq y_i\}|$$
and it is a metric on F^n.
2. The **minimum Hamming distance** of a code C is defined by
$$d(C) = \min\{d(x, y) \ : \ x, y \in C, x \neq y\}$$
3. If C is a code of length n over F and $|C| = M$ with minimum Hamming distance d, then we say that C is a q-ary (n, M, d)-code or $(n, M, d)_q$-code.

In the following remark, we give some particularity of the above definition when the alphabet F has an algebraic structure.

Remark 1.1
Let F be an alphabet and $C \subseteq F^n$ a code of length n over F.

1. If F has the zero element, then the **Hamming weight** of an element $x = \left(x_0, x_1, \ldots, x_{n-1}\right) \in F^n$ is defined as its distance from the zero vector $\mathbf{0} = (0, 0, \ldots, 0)$ (i.e.)

$$wt_H(x) := d(x, 0) = |\{i : x_i \neq 0\}|$$

and the minimum Hamming weight of C is

$$wt_H(C) := \min\{wt_H(x) \ : \ x \neq 0\}.$$

2. If F has the structure of an additive group, then

$$d(x, y) = wt_H(x - y) \quad \text{for any} \quad x, y \in F^n$$

and the Hamming weight of C is

$$d(C) = \min\{d(x, y) \ : \ x, y \in C \text{ and } x \neq y\}$$

It turns out that the distance of a code is closely related to the error detection and correction capabilities of the code.

Definition 1.3 (Definition 2.5.4 and Definition 2.5.8 in [53])
Let C be a code of length n over an alphabet F, and let t be a positive integer.

1. C is t-error-detecting if whenever at most t, but at least one, error is made in a codeword, the resulting word is not a codeword. A code C is exactly t-error-detecting if it is t-error-detecting but not $(t + 1)$-error-detecting.

2. C is t-error-correcting if minimum distance decoding is able to correct t or less errors that may occur in any codeword. A code C is exactly t-error-correcting if it is t-error-correcting but not $(t + 1)$-error-correcting.

Theorem 1.1 (Theorem 2.5.6 and Theorem 2.5.10 in [53])
Let C be a block code of length n over F having minimum distance d.

1. C is t-error-detecting if and only if $d \geq t + 1$, i.e., a code with distance d is an exactly $(d - 1)$-error-detecting code.

2. C is t-error-correcting if and only if $d \geq 2t + 1$, i.e., a code with distance d is an exactly $\lfloor (d - 1)/2 \rfloor$-error-correcting code, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x.

1.2 Linear codes

In order to simplify the encoding and decoding methods, if we impose an additional structure on a code, then we may have many practical advantages. The most popular block codes that are studied are codes over a finite field $\mathbb{F}_q$. So a $[n, k, d]_q$ linear code C is a $\mathbb{F}_q$-subspace of $\mathbb{F}_q^n$ of dimension k and minimum Hamming distance d. Let us now give the following definition.

Definition 1.4
Let $C \subseteq \mathbb{F}_q^n$ be a $[n, k, d]_q$ linear code.

1. A generator matrix for C is a $k \times n$ matrix G whose rows form a basis for C.

2. If G is a generator matrix of C, then $C = \left\{ {}^tGm \ : \ m \in \mathbb{F}_q^k \right\}$ or $C = \left\{ mG \ : \ m \in \mathbb{F}_q^k \right\}$.

3. A generator matrix of the form $G = (I_k \mid A)$, where I_k denotes the $k \times k$ identity matrix, is said to be in standard form.

Remark 1.2
Observe that if C is a $[n, k, d]_q$ linear code then its size is $M = q^k$, and its rate is $R = \frac{k}{n}$.

Let us recall that the Euclidean inner product in $\mathbb{F}_q^n$ of two vectors $x = (x_0, x_1, \ldots, x_{n-1})$ and $y = (y_0, y_1, \ldots, y_{n-1})$ is defined by

$$\langle x, y \rangle = \sum_{i=0}^{n-1} x_i y_i$$

and the dual code of a (linear) code C is defined by

$$C^{\perp} := \left\{ x \in \mathbb{F}_q^n \quad : \quad \langle x, y \rangle = 0, \quad \forall y \in C \right\}.$$

We have the following characterization of the Euclidean dual of a linear code over $\mathbb{F}_q$.

Proposition 1.1 ([53])
Let C be a $[n, k, d]_q$ linear code, with a generator matrix G.

1. $C^{\perp}$ is a $[n, n-k]_q$ code
2. a generator matrix of $C^{\perp}$ is a $(n-k) \times n$ square matrix H such that $H\,{}^t c = 0$ for all $c \in C$. We call H a parity check matrix of C.
3. If $G = (I_k \mid A)$ is a generator matrix of C, then $H = (-\,{}^t A \mid I_{n-k})$ is parity check matrix of C.

Proposition 1.2 ([53])
Let C be a linear code and let H be a parity-check matrix for C. Then

1. C has distance $\geq d$ if and only if any $d-1$ columns of H are linearly independent.
2. C has distance $\leq d$ if and only if H has d columns that are linearly dependent.

Now we give the definition of the equivalence concept between two codes, i.e., when two codes are "essentially the same."

Definition 1.5 (Section 1.6, and 1.7 in [37])
Let C_1 and C_2 be two linear codes of length n over $\mathbb{F}_q$. Then C_1 and C_2 are called **permutation equivalent** if there exists an $n \times n$ permutation matrix P such that $C_1 = C_2 P$. If P is a monomial matrix we say that C_1 and C_2 are **monomially equivalent**.

Remark 1.3
Two equivalent codes have the same minimum Hamming distance and in general same parameters.

We close this section by talking about a challenging problem in coding theory. Consider a linear code with three parameters: the length n, the size $M = q^k$ and the minimum distance d. We would like to have good codes, that is, codes of **large size** (to encode a large number of source messages) and **large minimum distance**(to correct a large number of errors). However, these are two **conflicting goals.** So, finding an optimal solution to this problem is one of the main problems of coding theory. For example, given the **length** n and the **dimension** k, we look for the **largest possible value** $d_q(n, k)$ **of the minimum distance**. There are some important theorems that give upper bounds for minimum Hamming distances of codes (not necessarily linear). For more details, we refer the reader to [37, Chapter 2]. There is also an online database which gives information about best known linear codes (BKLC) over small finite fields $\mathbb{F}_q$, $q \leq 9$. http://www.codetables.de/.

In the following, we recall the Singleton bound theorem and the definition of an MDS code.

Theorem 1.2 (Singleton bound, [37])
For any integer $q > 1$, any positive integer n and any integer d such that $1 \leq d \leq n$, we have

$$A_q(n, d) \leq q^{n-d+1}.$$

where $A_q(n, d)$ denotes the largest possible size M of a q-ary linear code of length n and minimum distance d. In particular, when q is a prime power, the parameters $[n, k, d]_q$ of any linear code over $\mathbb{F}_q$ satisfy

$$k + d \leq n + 1.$$

Definition 1.6 (MDS code)
An $[n, k, d]_q$ linear code with parameters such that $k + d = n + 1$ is called a maximum distance separable (MDS) code.

1.3 Cyclic codes and their generalizations

Cyclic codes are one of the most studied subclasses of linear codes, due to their simple algebraic structure that allows easy implementation. Moreover, many linear codes with good parameters are cyclic codes or one of their generalizations. In this section, we recall the basic background of cyclic codes, quasi-cyclic codes and generalized quasi-cyclic codes.

1.3.1 Structure of cyclic codes

Let us start with the following definition of a cyclic code.

Definition 1.7
A linear code $C \subseteq \mathbb{F}_q^n$ is called a cyclic code if for each codeword $c = (c_0, c_1, \ldots, c_{n-1})$ we have $(c_{n-1}, c_0, \ldots, c_{n-2}) \in C$.

Define the cyclic shift operator T on $\mathbb{F}_q^n$ by:

$$T(v_0, v_1, \ldots, v_{n-1}) = (v_{n-1}, v_0, \ldots, v_{n-2}) \tag{1.1}$$

Proposition 1.3
A linear code $C \subseteq \mathbb{F}_q^n$ is cyclic if it is invariant under T.

Proposition 1.4
The Euclidean dual code $C^\perp$ of a cyclic code C is also cyclic.

Proof
If $v = (v_0, v_1, \ldots, v_{n-1}) \in C^\perp$ then $\langle v \ , \ c\rangle = 0$ for all $c = (c_0, \ldots, c_{n-1}) \in C$. Thus, we have

$$\begin{aligned} \langle\ T(v)\ ,\ c\rangle &= \langle (v_{n-1}, v_0, \ldots, v_{n-2}), (c_0, c_1, \ldots, c_{n-1})\rangle \\ &= v_{n-1}c_0 + v_0c_1 + \ldots + v_{n-2}c_{n-1} \\ &= \langle v \ , \ T^{n-1}(c)\rangle = 0 \ \text{ since } T^{n-1}(c) \in C \end{aligned}$$

It follows that $T(v) \in C^\perp$ and so $C^\perp$ is a cyclic code.

Define φ_n the $\mathbb{F}_q$-vector space isomorphism between $\mathbb{F}_q^n$ and $\mathbb{F}_q[x]/\langle x^n - 1\rangle$ by:

$$\begin{array}{ccccc} \varphi_n & : & \mathbb{F}_q^n & \longrightarrow & \mathbb{F}_q[x]/\langle x^n - 1\rangle \\ & & v = (v_0, v_1, \ldots, v_{n-1}) & \longmapsto & \varphi_n(v) = v_0 + v_1x + \ldots + v_{n-1}x^{n-1} + \langle x^n - 1\rangle . \end{array} \tag{1.2}$$

With $X = x + \langle x^n - 1\rangle$ is the coset of x in $\mathbb{F}_q[x]/\langle x^n - 1\rangle$. This leads to the following characterization of cyclic codes by ideals of the polynomial ring $\mathbb{F}_q[x]/\langle x^n - 1\rangle$.

Theorem 1.3 ([53])
A linear code $C \subseteq \mathbb{F}_q^n$ is cyclic if and only if $\varphi_n(C)$ is an ideal of $\mathbb{F}_q[x]/\langle x^n-1\rangle$.

Proof
Suppose that C is cyclic and let $c=(c_0,\ldots,c_{n-1}) \in C$, then $\varphi_n(c) = \sum_{i=0}^{n-1} c_i x^i + \langle x^n-1\rangle \in \varphi_n(C)$. It follows that

$$\begin{aligned} X\cdot\varphi_n(c) =& c_0x + c_1x^2 + \ldots + c_{n-2}x^{n-1} + c_{n-1}x^n \\ =& c_{n-1} + c_0x + c_1x^2 + \ldots + c_{n-2}x^{n-1} \ \ (\text{ since } x^n = 1 \mod (x^n-1)) \end{aligned}$$

Hence, $\varphi_n^{-1}(x\cdot\varphi_n(c)) = (c_{n-1},c_0,\ldots,c_{n-2}) \in C$, since C is cyclic. This means that $X\varphi_n(c) \in \varphi_n(C)$. By linearity of C and φ_n, it follows that $\varphi_n(C)$ is an ideal of $\mathbb{F}_q[x]/\langle x^n-1\rangle$.

Remark 1.4
From now to the end of this document we identify the elements of the quotient ring $\mathbb{F}_q[x]/\langle x^n-1\rangle$ with polynomials of degree strictly less than n, which means that φ_n defined above will be expressed as follows

$$\varphi_n(v) = v_0 + v_1x + \ldots + v_{n-1}x^{n-1} \in \mathbb{F}_q[x]/\langle x^n-1\rangle.$$

Consequently, we focus on the ideals of $\mathbb{F}_q[x]/\langle x^n-1\rangle$ containing $J := \langle x^n-1\rangle$ and identify I/J with $\{g(x) : g(x) \in I \text{ and } \deg(g(x)) < n$ and, for all $g(x)$ such that $\deg(g(x)) < \deg(f(x))$, we identify the equivalence class $g(x)+J$ with $g(x)$. A similar identification will be used when we study constacyclic codes, polycyclic codes and other types of codes with cyclical properties.

Definition 1.8
Let $C \subseteq \mathbb{F}_q^n$ be a cyclic code of length n over $\mathbb{F}_q$. The **standard generator polynomial** of C is the monic polynomial $g(x)$ of least degree in $\mathbb{F}_q[x]/\langle x^n-1\rangle$ such that $\varphi_n(C) = \langle g(x)\rangle$.

The following are well-known facts about cyclic codes.

Theorem 1.4 ([53])
Let C be a cyclic code of length n over $\mathbb{F}_q$ such that $\varphi_n(C) = \langle g(x)\rangle$, where $g(x)$ is the standard generator polynomial. Then the following holds

1. $g(x)$ is a divisor of x^n-1 over $\mathbb{F}_q$. Hence, $x^n-1 = g(x)h(x)$ for some $h(x) \in \mathbb{F}_q[x]$.
2. The polynomial $h(x)$ is called the (parity) check polynomial and it has the property that $v(x)$ is in $\varphi_n(C)$ if and only if $h(x)v(x) = 0$ in $\mathbb{F}_q[x]/\langle x^n-1\rangle$.
3. The dimension of C is $k = n - \deg(g(x)) = \deg(h(x))$ and a basis for C is $\left\{g(x), xg(x), \ldots, x^{k-1}g(x)\right\}$
4. If $g(x) = g_0 + g_1x + \ldots + g_{n-k}x^{n-k}$ then $g_0 \neq 0$ then the following circulant matrix G is a generator matrix for C, where each row is a cyclic shift of the previous row.

$$G = \begin{pmatrix} \varphi_n^{-1}(g(x)) \\ \varphi_n^{-1}(xg(x)) \\ \vdots \\ \varphi_n^{-1}\left(x^{k-1}g(x)\right) \end{pmatrix} = \begin{pmatrix} g_0 & g_1 & \ldots & g_{n-k} & 0 & \ldots & \ldots & 0 \\ 0 & g_0 & g_1 & \ldots & g_{n-k} & 0 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \ddots & \vdots \\ 0 & \ldots & & 0 & g_0 & g_1 & \ldots & g_{n-k} \end{pmatrix} \tag{1.3}$$

5. There is a one-to-one correspondence between divisors of x^n-1 and cyclic codes of length n over $\mathbb{F}_q$.

Now we will give a characterization of the Euclidean dual of a cyclic code, but let us recall the following definition of the reciprocal polynomial.

Definition 1.9
Let $h(x) = \sum_{i=0}^{k} h_i x^i$ be a polynomial of degree k over $\mathbb{F}_q$. Then the reciprocal polynomial $h(x)$ is the polynomial

$$h^*(x) := x^k \cdot h(1/x) = \sum_{i=0}^{k} a_{k-i} x^i.$$

Theorem 1.5 ([53], Theorem 7.3.7)
Let $C \subseteq \mathbb{F}_q^n$ be a cyclic code generated by $g(x)$, and having $h(x)$ as a parity check polynomial. Then

1. $h^*(x)$ is a generator polynomial of the Euclidean dual $C^\perp$.
2. $h^*(x)$ divides $x^n - 1$.
3. The set $\{h^*(x), xh^*(x), \ldots, x^{k-1}h^*(x)\}$ forms a basis of $C^\perp$ and the dimension of C is $k = n - \deg(h^*(x)) = n - \deg(h(x)) = \deg(g(x))$.
4. A parity check matrix H of C is given by:

$$H = \begin{pmatrix} \varphi_n^{-1}(h^*) \\ \varphi_n^{-1}(x \cdot h^*) \\ \vdots \\ \varphi_n^{-1}\left(x^{n-k-1} \cdot h^*\right) \end{pmatrix} = \begin{pmatrix} h_k & h_{k-1} & \ldots & \ldots & h_0 & 0 & 0 & 0 & \ldots & 0 \\ 0 & h_k & h_{k-1} & \ldots & \ldots & h_0 & 0 & 0 & \ldots & 0 \\ \vdots & & & & & & & & \vdots & \\ 0 & 0 & \ldots & \ldots & h_k & h_{k-1} & \ldots & \ldots & \ldots & h_0 \end{pmatrix} \tag{1.4}$$

where $k = n - \deg(h(x))$ and $h(x) = \sum_{i=0}^{n-k} h_i x^i$.

1.3.2 Factorization of $x^n - 1$.

As seen in the above section, the polynomial $x^n - 1$ over $\mathbb{F}_q$ is very important for the study of cyclic codes. In this section we study this polynomial, its roots and its factorization. Let n and q be such that $n = n'p^k$ where $\gcd(n', q) = 1$ and p the characteristic of $\mathbb{F}_q$. Then

$$x^n - 1 = x^{n'p^k} - 1 = \left(x^{n'} - 1\right)^{p^k}.$$

Therefore it suffices to consider the case $\gcd(n, q) = 1$. Let $\mathbb{L}/\mathbb{F}_q$ be a field extension of $\mathbb{F}_q$ which contains all roots of $x^n - 1$. Such an extension exists and $\mathbb{L} := \mathbb{F}_{q^m}$ where m is the multiplicative order of q modulo n, and it is the smallest such extension, see [9, Corollary 1.3.3.]. Note that the set of roots of $x^n - 1$ forms a subgroup of $\mathbb{F}_{q^m}^*$ and any one of its generators is called a **primitive n^{th} root of unity**. Let α be a generator of the multiplicative group $\mathbb{F}_{q^m}^*$, called a **primitive element**, then $\beta = \alpha^{\frac{q^m-1}{n}}$ is a primitive n^{th} root of unity. Hence, all roots of $x^n - 1$ can be expressed as powers of β,

$$\beta, \beta^2, \ldots, \beta^{n-1}, \beta^n = 1 = \beta^0.$$

Let us recall that for each $a \in \mathbb{F}_{q^m}$, the minimal polynomial of a is the polynomial $m_a(x) \in \mathbb{F}_q[x]$ of least degree such that $m_a(x) = 0$. Note that $m_a(x)$ is irreducible and has $d = \deg(m_a(x))$ roots in a field extension of $\mathbb{F}_q$. These roots are called the conjugates of a.

For $0 \leq i \leq n-1$, the conjugates of β^i are

$$\beta^i, \beta^{iq}, \beta^{iq^2}, \ldots, \beta^{iq^{d-1}}.$$

where d is the smallest positive integer for which $\beta^{iq^d} = \beta^i$, and so the minimal polynomial of β^i is given by

$$m_{\beta^i}(x) = \prod_{k=0}^{d-1}(x - \beta^{iq^k})$$

Note that the set of exponents of β in the last product, with d as specified, is called i^{th} q-cyclotomic coset modulo n. More precisely, we have the following definition.

Definition 1.10 ([68], Chapter 4)
For any integer t, the q-cyclotomic coset C_t of t modulo n is defined by

$$C_t = \left\{ t \cdot q^j (\mathrm{mod} n) \mid j = 0, 1, \ldots \right\}.$$

Let $\{i_0 = 0, i_1, \ldots, i_s\}$ be a complete set of representatives of all q-cyclotomic cosets modulo n. This means that $\{0, i_1, \ldots, i_s\}$ is a subset of $\{0, 1, \ldots, n-1\}$, $C_{i_0} = \{0\}, C_{i_1}, C_{i_2}, \ldots, C_{i_s}$ are distinct and $\bigcup_{k=0}^{s} C_{i_k} = \{0, 1, \ldots, n-1\}$.

It follows that if $C_0, C_1, \ldots, C_s$ is a complete family then

$$x^n - 1 = \prod_{j=0}^{s} m_{\beta^i}(x)$$

1.3.3 BCH and Hartmann-Tzeng bounds for cyclic codes

An important subclass of cyclic codes having the property that their minimum distance is lower bounded, is the class of BCH codes, and their lower bound is called the BCH bound. Binary BCH codes were first discovered by A. Hocquenghem [36] in 1959 and independently by R. C. Bose and D. K. Ray-Chaudhuri [16] in 1960. Generalizations of the binary BCH codes to q-ary codes were obtained by D. Gorenstein and N. Zierler [35] in 1961.

Theorem 1.6 (BCH bound, Theorem 8.1.9, [53])
Let $2 \leq \delta \leq n$ be an integer and $\beta \in \mathbb{F}_{q^m}/\mathbb{F}_q$ a primitive n^{th} root of unity, where m is the multiplicative order of q modulo n. Let C be a cyclic code of length n over $\mathbb{F}_q$ generated by $g(x)$, such that

$$g(\beta^{a+ib}) = 0 \quad \text{for} \quad i = 0, \ldots, \delta - 2,$$

where a and b are non-zero integers with $\gcd(n, b) = 1$. Then the minimum Hamming distance of C is at least δ, we refer to this lower bound δ as the BCH bound.

Definition 1.11 (Definition 8.1.6, [53])
A code C that is defined as in Theorem 1.6 is called a **BCH code** with **designed distance** δ. If $a = 1$ we say that C is a **narrow-sense** BCH code. If $n = q^m - 1$, the BCH code is called **primitive.**

Proposition 1.5 (Chapter 8, [53])
Let C be a **BCH code** of designed distance δ as in the above definition. Then the following conditions are equivalent

1. $g(x) = \mathrm{lcm}(m_{\beta^b}(x), m_{\beta^{a+b}}(x), \ldots, m_{\beta^{a+(\delta-2)b}}(x))$, is a generator of least degree of C, with $m_{\beta^i}(x)$ is the minimal polynomials of β^i.

2. a parity check matrix of C is

$$H = \begin{pmatrix} 1 & \beta^{b} & \dots & \beta^{b(n-1)} \\ 1 & \beta^{b+1} & \dots & \beta^{(b+1)(n-1)} \\ & & \cdots & \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \beta^{b+\delta-2} & \dots & \beta^{(b+\delta-2)(n-1)} \end{pmatrix}$$

3. $c(x) \in \varphi_{n}(C)$ if and only if $c(\beta^{a+ib}) = 0$ for $i = 0, \dots, \delta - 2$.

Theorem 1.7 (Hartmann-Tzeng bound)
Let $2 \leq \delta \leq n$, r be some integers and $\beta \in \mathbb{F}_{q^m}/\mathbb{F}_{q}$ a primitive n^{th} root of unity, where m is the multiplicative order of q modulo n. Let C be a cyclic code of length n over $\mathbb{F}_{q}$ generated by $g(x)$, such that

$$g(\beta^{a+ib+jc}) = 0 \quad \text{for} \quad i = 0, \dots, \delta - 2, \quad j = 0 \dots, r.$$

for some positive integers a, b, c such that $\gcd(n, b) = 1$ and $\gcd(n, c) = 1$. Then the minimum Hamming distance of C is at least $\delta + r$. We refer to this lower bound $\delta + r$ as the Hartmann-Tzeng bound.

1.3.4 Quasi-cyclic codes

Quasi-cyclic codes form an important class of linear codes which include cyclic codes. Much research has been done on quasi-cyclic codes and they have been shown to be useful in solving one of the main problems of coding theory. A large number of codes with optimal or best-known parameters are quasi-cyclic. We begin with the following definition of a quasi-cyclic code.

Definition 1.12 ([54])
A linear code C of length $n = mr$ over $\mathbb{F}_{q}$ is called quasi-cyclic code of index r if r is the smallest integer such that for each codeword

$$c = \left(\overbrace{c_{1,0}, c_{1,1}, \dots, c_{1,m-1}}^{c_1}, \overbrace{c_{2,0}, c_{2,1}, \dots, c_{2,m-1}}^{c_2}, \dots, \overbrace{c_{r,0}, c_{r,1}, \dots, c_{r,m-1}}^{c_r} \right)$$

we have

$$c^{'} = \left(\overbrace{c_{r,0}, c_{r,1}, \dots, c_{r,m-1}}^{c_r}, \overbrace{c_{1,0}, c_{1,1}, \dots, c_{1,m-1}}^{c_1}, \dots, \overbrace{c_{r-1,0}, c_{r-1,1}, \dots, c_{r-1,m-1}}^{c_{r-1}} \right),$$

is again a codeword.

Remark 1.5
Note that quasi-cyclic codes are cyclic codes when $r = 1$.

From the above definition, the following result is immediate.

Proposition 1.6
A linear code C is quasi-cyclic of index r if it is invariant under T^{r}, where T is cyclic shift on $\mathbb{F}_{q}^{n}$ as in (1.1).

Let $\langle x^m-1\rangle$ be the principal ideal of $\mathbb{F}_q[x]$ generated by x^m-1, and $R_m := \mathbb{F}_q[x]/\langle x^m-1\rangle$ be the quotient ring. The operator T^r, defined above, permits to equip $\mathbb{F}_q^{mr}$ with an $\mathbb{F}_q[x]$-module structure where the multiplication is given by:

$$\begin{array}{ccc} \mathbb{F}_q[x]\times\mathbb{F}_q^{mr} & \longrightarrow & \mathbb{F}_q^{mr} \\ (P,v) & \longmapsto & P.v := P(T)(v) \end{array} \tag{1.5}$$

Since x^m-1 is the minimal polynomial of T, $\langle x^m-1\rangle.\mathbb{F}_q^{mr}=\{0\}$, and so $\mathbb{F}_q^{mr}$ is an R_m-module. It follows that each quasi-cyclic code can be seen as an R_m-submodule of $\mathbb{F}_q^{mr}$.
Let us now define the following R_m-module isomorphism $\varphi_{m,r}$ by:

$$\begin{array}{ccccc} \varphi_{m,r}: & \left(\mathbb{F}_q^m\right)^r & \longrightarrow & R_m^r \\ & (c_1,c_2,\ldots,c_r) & \longmapsto & (c_1(x),c_2(x),\ldots,c_r(x)) \end{array} \tag{1.6}$$

where $c_i(x)=\sum_{j=0}^{m-1} c_{i,j}x^j \in R,\ 1\le i\le r$. It follows that.

Theorem 1.8
The map $\varphi_{m,r}$ induces a one-to-one correspondence between quasi-cyclic codes and linear codes of length r over R_m.

Now, we give the following definition of an **s**-generator quasi-cyclic code.

Definition 1.13
Let $C\subseteq\mathbb{F}_q^{mr}$ be a quasi-cyclic code of index r. We say that C is an **s**-generator quasi-cyclic code if $\varphi_{m,r}(C)$ is generated by s elements as an R_m-submodule of R_m^r, i.e., there exist $\underline{\mathbf{a}_1}(x),\underline{\mathbf{a}_2}(x),\ldots,\underline{\mathbf{a}_s}(x)$ in R_m^r such that $\varphi_{m,r}(C)$ is generated by $\underline{\mathbf{a}_1}(x),\underline{\mathbf{a}_2}(x),\ldots,\underline{\mathbf{a}_s}(x)$ where each $\underline{\mathbf{a}_i}(x)=(a_{i,1}(x),a_{i,2}(x),\ldots,a_{i,r}(x))$ in R_m^r.

In the following result, we show that every quasi-cyclic code can be decomposed as a direct sum of cyclic codes of length m.

Corollary 1.1
If C is a quasi-cyclic code of length $n=mr$ over $\mathbb{F}_q$, then C can be decomposed as a direct sum

$$C=C_1\oplus C_2\oplus\ldots\oplus C_r$$

where each $C_j,\ j=1,\ldots,r$, is a cyclic code of length m over $\mathbb{F}_q$.

Another decomposition of quasi-cyclic codes over some field extension of $\mathbb{F}_q$, called **CRT decomposition** is possible. For details we invite the reader to [34, 54]. Assume that $\gcd(m,q)=1$ and factor the polynomial x^m-1 into pairwise distinct irreducible polynomials in $\mathbb{F}_q[x]$ as

$$x^m-1=f_1(x)f_2(x)\ldots f_s(x)$$

Hence, by Chinese Remainder Theorem, we have the following ring isomorphism:

$$R_m\cong\bigoplus_{i=1}^{s}\mathbb{F}_q[x]/\langle f_i(x)\rangle$$

Since each $f_i(x)$ divides x^m-1, its roots are powers of some fixed primitive m^{th} root of unity ξ. For each $i=1,\ldots,s$, let u_i be the smallest non-negative integer such that $f_i(\xi^{u_i})=0$. Since each $f_i(x)$ is irreducible, $\mathbb{E}_i:=\mathbb{F}_q[x]/\langle f_i(x)\rangle$ is a field extension of $\mathbb{F}_q$. It follows that

$$R_m^r\cong\mathbb{E}_1^r\oplus\ldots\oplus\mathbb{E}_s^r$$

So we deduce the following decomposition of quasi-cyclic codes.

Theorem 1.9 ([34])
Let C be a quasi-cyclic code of length $n = mr$ over $\mathbb{F}_q$. Then $\varphi_{m,r}(C)$ is an $(\mathbb{E}_1 \oplus \ldots \oplus \mathbb{E}_s)$-submodule of $\mathbb{E}_1^r \oplus \ldots \oplus \mathbb{E}_s^r$ and

$$\varphi_{m,r}(C) = C_1 \oplus \ldots \oplus C_s \tag{1.7}$$

where each C_i is a linear code of length r over $\mathbb{E}_i$, for each i, and is called **the constituent of** C**.**

Proposition 1.7 (Lemma 2.1 in [34])
Let C be a quasi-cyclic code such that $\varphi_{m,r}(C)$ is generated by $\underline{\mathbf{a}}(x) = (a_1(x), a_2(x), \ldots, a_r(x)) \in R_m^r$ as an R_m-submodule of R_m^r. Then the constituents of C in the above theorem are given by

$$C_i := \mathrm{Span}_{\mathbb{E}_i}\left\{(a_1(\xi^{u_j}), a_2(\xi^{u_j}) \ldots, a_r(\xi^{u_j}))\right\}, \quad i = 1, \ldots, s.$$

1.3.5 Generalized quasi-cyclic codes

In this section, we are interested in an even more general class of linear codes, called generalized quasi-cyclic codes. They were introduced first in [78].

Definition 1.14
Let $(n_1, n_2, \ldots, n_r)$ be positive integers and $n = n_1 + n_2 + \ldots + n_r$. A linear code $C \subseteq \mathbb{F}_q^n$ is called a generalized quasi-cyclic code of length n over $\mathbb{F}_q$, if for each

$$c = \left(\overbrace{c_{1,0}, c_{1,1}, \ldots, c_{1,n_1-1}}^{c_1}, \overbrace{c_{2,0}, c_{2,1}, \ldots, c_{2,n_2-1}}^{c_2}, \ldots, \overbrace{c_{r,0}, c_{r,1}, \ldots, c_{r,n_r-1}}^{c_r}\right) \in C.$$

we have

$$c' = \left(\overbrace{c_{1,n_1-1}, c_{1,0}, c_{1,1}, \ldots, c_{1,n_1-2}}, \overbrace{c_{2,n_2-1}, c_{2,0}, c_{2,1}, \ldots, c_{2,n_2-2}}, \ldots, \overbrace{c_{r,n_r-1}, c_{r,0}, c_{r,1}, \ldots, c_{r,n_r-2}}\right),$$

is again a codeword.

Remark 1.6
Let C be a generalized quasi-cyclic code of length $n = n_1 + n_2 + \ldots + n_r$ over $\mathbb{F}_q$. Then

1. for $n_1 = n_2 = \ldots = n_r$, C is a quasi-cyclic code.
2. for $r = 1$, C is a cyclic code.

Proposition 1.8
Let C be a linear code of length $n = n_1 + n_2 + \ldots + n_r$ over $\mathbb{F}_q$. Then C is a generalized quasi-cyclic code if it is invariant under the operator $\widetilde{T} = T_1 \oplus T_2 \oplus \ldots \oplus T_r$, with each T_i is the (right) cyclic shift on $\mathbb{F}_q^{n_i}$.

As in the case of quasi-cyclic codes, the operator $\widetilde{T}$ permits us to endow $\mathbb{F}_q{}^n$ with an $\mathbb{F}_q[x]$-module structure. Let $\varphi := (\varphi_{n_1}, \varphi_{n_2}, \ldots, \varphi_{n_r})$ where each φ_{n_i} is the $\mathbb{F}_q$-vector space isomorphism between $\mathbb{F}_q^{n_i}$ and $R_{n_i} := \mathbb{F}_q[x]/\langle x^{n_i} - 1\rangle$, defined by

$$\varphi_{n_i}((v_{i,0}, v_{i,1}, \ldots, v_{i,n_i-1})) = \sum_{j=0}^{n_i-1} v_{i,j} x^j.$$

It follows that φ is an $\mathbb{F}_q[x]$-module isomorphism between $\mathbb{F}_q^n$ and $\prod_{i=1}^{r} R_{n_i}$. Thus we deduce the following result.

Proposition 1.9 ([78])
A linear code $C \subseteq \mathbb{F}_q^n$ is a generalized quasi-cyclic code of length n if and only if $\varphi(C)$ is an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{n_i}$

1.4 Constacyclic codes and their generalizations

Constacyclic codes form a remarkable generalization of cyclic codes and they were first introduced by Berlekamp in [19]. In this section, we recall the basic background of constacyclic codes and their generalizations: quasi-twisted codes and multi-twisted codes.

Definition 1.15 (Constacyclic codes)
Let C be a linear code of length n over $\mathbb{F}_q$ and $\lambda \in \mathbb{F}_q^*$. We say that C is a λ-constacyclic code if for each codeword $c = (c_0, c_1, \ldots, c_{n-1})$ in C, the vector $(\lambda c_{n-1}, c_0, \ldots, c_{n-2}) \in C$, is again a codeword.

The following characterization of constacyclic codes is immediate.

Proposition 1.10
A linear code $C \subseteq \mathbb{F}_q^n$ is λ-constacyclic if and only if $\varphi_\lambda(C)$ is an ideal of $\mathbb{F}_q[x]/\langle x^n - \lambda\rangle$, where φ_λ is the $\mathbb{F}_q$-vector space isomorphism between $\mathbb{F}_q^n$ and $\mathbb{F}_q[x]/\langle x^n - \lambda\rangle$ which is defined by $\varphi_\lambda(c_0, c_1, \ldots, c_{n-1}) = c_0 + c_1 x + \ldots + c_{n-1}x^{n-1}$.

As in the case of cyclic codes the irreducible factorization of $x^n - \lambda$ permits us to construct all possible λ-constacyclic codes. So, we recall factorization of the polynomial $x^n - \lambda$ and its relation to cyclotomic cosets, for more details we invite the reader to [8, 91]. Assume that $\gcd(n, q) = 1$ and let $\beta \in \mathbb{F}_{q^m}$ be an n-th root of unity, where m is the order of q modulo n. The roots of $x^n - \lambda$ are

$$\xi, \xi\beta, \xi\beta^2, \ldots, \xi\beta^{n-2}, \xi\beta^{n-1},$$

where $\xi^n = \lambda$ (ξ belongs to an extension of $\mathbb{F}_q$). Since $\xi^n = \lambda, \xi^{nr} = \lambda^r = 1$, where r is the order of λ in the multiplicative group $\mathbb{F}_q^*$. Hence $\xi \in \mathbb{F}_{q^s}$ where s is the order of q modulo nr. Now, $q^s - 1 \equiv 0 \bmod nr$ so $q^s - 1 \equiv 0 \bmod n$. This implies that $m \mid s$. Consequently, $\mathbb{F}_{q^m} \subseteq \mathbb{F}_{q^s}$. Hence, the field $\mathbb{F}_{q^s}$ contains both β and ξ and we may take $\xi = \alpha^t$ and $\beta = \alpha^{rt}$ where α is a primitive element of $\mathbb{F}_{q^s}$ and $q^s - 1 = ntr$, for some integer t. So $\beta = \xi^r$, and $x^n - \lambda$ factors as follows:

$$x^n - \lambda = \prod_{i=0}^{n-1} \left(x - \xi\beta^i\right) = \prod_{i=0}^{n-1} \left(x - \alpha^{t(1+ir)}\right) = \prod_{i=0}^{n-1} \left(x - \xi^{1+ir}\right).$$

Hence the roots of $x^n - \lambda$ are precisely the elements of the form $\alpha^{1+ri} = \xi\beta^i$ for $0 \leqslant i \leqslant n-1$. They correspond to the set $\theta_{r,n} = \{j : 0 \leqslant j < rn, j \equiv 1(\mod r)\}$, and each irreducible factor of $x^n - \lambda$ corresponds to a cyclotomic coset modulo nr (not necessarily modulo n).

Definition 1.16 (Quasi-twisted codes, [8])
Let C be a linear code of length n over $\mathbb{F}_q$, $\lambda \in \mathbb{F}_q^*$ and r be a positive integer. We say that C is a λ-quasi-twisted code of index r if r is the smallest integer such that for each codeword

$$c = \left(\overbrace{c_{1,0}, c_{1,1}, \ldots, c_{1,m-1}}^{c_1}, \overbrace{c_{2,0}, c_{2,1}, \ldots, c_{2,m-1}}^{c_2}, \ldots, \overbrace{c_{r,0}, c_{r,1}, \ldots, c_{r,m-1}}^{c_r}\right) \in C$$

we have

$$c' = \left(\overbrace{\lambda c_{r,0}, \lambda c_{r,1}, \ldots, \lambda c_{r,m-1}}^{\lambda c_r}, \overbrace{c_{1,0}, c_{1,1}, \ldots, c_{1,m-1}}^{c_1}, \ldots, \overbrace{c_{r-1,0}, c_{r-1,1}, \ldots, c_{r-1,m-1}}^{c_{r-1}}\right) \in C,$$

is again a codeword.

As in the case of quasi-cyclic codes, the following characterization of quasi-twisted codes is immediate.

Proposition 1.11
Let $C \subseteq \mathbb{F}_q^n$ be a λ-quasi-twisted code of index r over $\mathbb{F}_q$. Then

1. for $r = 1$, C is a λ-constacyclic code.
2. C is invariant under the operator $T_{\lambda,r}$ defined by
$$T_{\lambda,r}(x_0, x_1, \ldots, x_{n-1}) = (\lambda x_{n-r}, \lambda x_{n-r+1}, \ldots, \lambda x_{n-1}, x_0, \ldots, x_{n-r-1})$$
3. $\varphi_\lambda(C)$ is an R_λ-submodule of R_λ^r, where $R_\lambda := \mathbb{F}_q[x]/\langle x^n - \lambda\rangle$.

We close this section by another generalization of quasi-twisted codes, called multi-twisted codes. They were introduced first in [10].

Definition 1.17 (Multi-twisted codes)
Let $(n_1, n_2, \ldots, n_r)$ be positive integers and $n = n_1 + n_2 + \ldots + n_r$, and $\lambda_1, \lambda_2, \ldots, \lambda_r$ be non-zero elements of $\mathbb{F}_q$. We say that a linear code $C \subseteq \mathbb{F}_q^n$ is a $(\lambda_1, \lambda_2, \ldots, \lambda_r)$-multi-twisted code of length n over $\mathbb{F}_q$, if for each
$$c = \left(\overbrace{c_{1,0}, c_{1,1}, \ldots, c_{1,n_1-1}}^{c_1}, \overbrace{c_{2,0}, c_{2,1}, \ldots, c_{2,n_2-1}}^{c_2}, \ldots, \overbrace{c_{r,0}, c_{r,1}, \ldots, c_{r,n_r-1}}^{c_r}\right) \in C.$$
we have
$$\left(\overbrace{\lambda_1 c_{1,n_1-1}, c_{1,0}, c_{1,1}, \ldots, c_{1,n_1-2}}, \overbrace{\lambda_2 c_{2,n_2-1}, c_{2,0}, c_{2,1}, \ldots, c_{2,n_2-2}}, \ldots, \overbrace{\lambda_r c_{r,n_r-1}, c_{r,0}, c_{r,1}, \ldots, c_{r,n_r-2}}\right),$$
is again a codeword.

Proposition 1.12
Let $C \subseteq \mathbb{F}_q^n$ be a $(\lambda_1, \lambda_2, \ldots, \lambda_r)$-multi-twisted code of length n over $\mathbb{F}_q$. Then

1. for $n_1 = n_2 = \ldots = n_r$ and $\lambda_1 = \lambda_2 = \ldots = \lambda_r$ then C is a quasi-twisted code. Moreover, if $r = 1$, then C is a constacyclic code.
2. C is invariant under the operator $\widetilde{T}_\lambda = T_{\lambda_1} \oplus T_{\lambda_2} \oplus \ldots \oplus T_{\lambda_r}$, where each T_{λ_i} is the operator of $\mathbb{F}_q^{n_i}$, defined by $T_{\lambda_i}(x_0, x_1, \ldots, x_{n_i-1}) = (\lambda_i x_{n_i-1}, x_0, \ldots, x_{n_i-2})$
3. $\varphi(C)$ is an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{\lambda_i}$, where $R_{\lambda_i} := \mathbb{F}_q[x]/\langle x^{n_i} - \lambda_i\rangle$ and $\varphi := (\varphi_{\lambda_1}, \varphi_{\lambda_2}, \ldots, \varphi_{\lambda_r})$ with each φ_{λ_i} is the $\mathbb{F}_q$-vector space isomorphism between $\mathbb{F}_q^{n_i}$ and R_{λ_i} defined by
$$\varphi_{\lambda_i}((v_{i,0}, v_{i,1}, \ldots, v_{i,n_i-1})) = \sum_{j=0}^{n_i-1} v_{i,j} x^j.$$

Remark 1.7
Note that we can decompose quasi-twisted and multi-twisted codes into direct sums of constacyclic codes (and a CRT decomposition), similar to the generalized quasi-cyclic and quasi-cyclic codes presented in the previous section, if we replace $x^n - 1$ by $x^n - \lambda$.

1.5 Polycyclic codes

Polycyclic codes (known under the name pseudo-cyclic [71]) of length n over a finite field $\mathbb{F}_q$ with q elements, are an important subclass of linear codes, which generalize cyclic codes. These can be described by the ideals of the polynomial ring $\mathbb{F}_q[x]/\langle f(x)\rangle$, where $f(x)$ is a non-zero polynomial in $\mathbb{F}_q[x]$. When $f(x) = x^n - \lambda$ they yield constacyclic codes. They are better studied under that name in [80] over $\mathbb{F}_q$.

Definition 1.18
Let C be a linear code of length n over $\mathbb{F}_q$ and $a = \left(a_0, a_1, \ldots, a_{n-1}\right) \in \mathbb{F}_q^n$. We say that C is

1. a right polycyclic code induced by the vector a if for each codeword $c = \left(c_0, c_1, \ldots, c_{n-1}\right) \in C$ we have $\left(0, c_0, \ldots, c_{n-2}\right) + c_{n-1}\left(a_0, a_1, \ldots, a_{n-1}\right) \in C$.
2. a left polycyclic code induced by the vector a if for each codeword $c = \left(c_0, c_1, \ldots, c_{n-1}\right) \in C$ we have $\left(c_1, \ldots, c_{n-1}, 0\right) + c_0\left(a_0, a_1, \ldots, a_{n-1}\right) \in C$.

Remark 1.8
λ-constacyclic codes are right polycyclic codes associated with the vector $(\lambda, 0, \ldots, 0)$, and left polycyclic codes associated with the vector $(0, 0, \ldots, \lambda)$. It follows that cyclic codes ($\lambda = 1$) and negacyclic codes ($\lambda = -1$) are special cases of polycyclic codes.

Let $a = \left(a_0, a_1, \ldots, a_{n-1}\right) \in \mathbb{F}_q^n$. We define the right polyshift operator $T_{\overrightarrow{a}}$ on $\mathbb{F}_q^n$ by

$$T_{\overrightarrow{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(0, v_0, \ldots, v_{n-2}\right) + v_{n-1}\left(a_0, a_1, \ldots, a_{n-1}\right). \tag{1.8}$$

and the left polyshift operator $T_{\overleftarrow{a}}$ of $\mathbb{F}_q^n$ is defined by

$$T_{\overleftarrow{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(v_1, \ldots, v_{n-1}, 0\right) + v_0\left(a_0, a_1, \ldots, a_{n-1}\right). \tag{1.9}$$

It follows that

Proposition 1.13
Let $C \subseteq \mathbb{F}_q^n$ be a linear code of length n over $\mathbb{F}_q$. The following assertions are equivalent:

1. C is a right (resp. left) polycyclic code induced by the vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right)$.
2. C is an invariant subspace of $\mathbb{F}_q^n$ under $T_{\overrightarrow{a}}$ (resp. $T_{\overleftarrow{a}}$).

As is common in coding theory, let us associate to each vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right) \in \mathbb{F}_q^n$, the right polynomial $\overrightarrow{a}(x) = \sum_{i=0}^{n-1} a_i x^i$ and the left polynomial $\overleftarrow{a}(x) = \sum_{i=0}^{n-1} a_i x^{n-1-i}$. It is shown in [80] that each right (resp., left) polycyclic code induced by the vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right)$ corresponds to an ideal in the polynomial ring $R_{\overrightarrow{a}} := \mathbb{F}_q[x]/\langle x^n - \overrightarrow{a}(x)\rangle$ (resp., $R_{\overleftarrow{a}} := \mathbb{F}_q[x]/\langle x^n - \overleftarrow{a}(x)\rangle$). In the following proposition, we collect some basic results on right polycyclic codes that are presented in [80].

Proposition 1.14
Let $C \subseteq \mathbb{F}_q^n$ be a right polycyclic code induced by the vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right)$, and let $\varphi_{\overrightarrow{a}}$ be the isomorphism between $\mathbb{F}_q^n$ and $R_{\overrightarrow{a}}$ defined by $\varphi_{\overrightarrow{a}}(c_0, c_1, \ldots, c_{n-1}) = \sum_{i=0}^{n-1} c_i x^i$. Then

1. $\varphi_{\overrightarrow{a}}(C)$ is an ideal of the polynomial ring $R_{\overrightarrow{a}}$.

2. There is a monic polynomial $g(x) \in \mathbb{F}_q[x]$ of least degree which divides $x^n - \overrightarrow{a}(x)$ and $\varphi_{\overrightarrow{a}}(C) = \langle g(x) \rangle$.

3. The set $\{g(x), xg(x), \ldots, x^{n-\deg(g)-1}g(x)\}$ forms a basis of $\varphi_{\overrightarrow{a}}(C)$ and the dimension of C is $n - \deg(g(x))$.

4. A generator matrix G of C is given by :

$$G = \begin{pmatrix} \varphi_{\overrightarrow{a}}^{-1}(g(x)) \\ \varphi_{\overrightarrow{a}}^{-1}(xg(x)) \\ \vdots \\ \vdots \\ \varphi_{\overrightarrow{a}}^{-1}\left(x^{k-1}g(x)\right) \end{pmatrix} = \begin{pmatrix} g_0 & g_1 & \ldots & g_{n-k} & 0 & \ldots & \ldots & 0 \\ 0 & g_0 & g_1 & \ldots & g_{n-k} & 0 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \ddots & \vdots \\ 0 & \ldots & & 0 & g_0 & g_1 & \ldots & g_{n-k} \end{pmatrix}$$

where $k = n - \deg(g(x))$ and $g(x) = \sum_{i=0}^{n-k} g_i x^i$.

Remark 1.9
As is the norm in the literature, we identify C with $\varphi_{\overrightarrow{a}}(C)$, and often we will simply write C instead of $\varphi_{\overrightarrow{a}}(C)$.

We now give the definition of a generator polynomial for a right (resp. left) polycyclic code.

Definition 1.19
Let $C \subseteq \mathbb{F}_q^n$ be a right polycyclic code of length n over $\mathbb{F}_q$. Then the **standard generator polynomial** of C is the monic polynomial $g(x)$ of least degree in $R_{\overrightarrow{a}}$ such that $\varphi_{\overrightarrow{a}}(C) = \langle g(x) \rangle$.

We close this chapter with the following example.

Example 1.1
In this example, we construct right polycyclic codes with associate vector $a = (1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 0)$ of length 11 over $\mathbb{F}_2$. Let $\overrightarrow{a}(x) = x^8 + x^7 + x^5 + x^3 + x^2 + x + 1$, then $f(x) = x^{11} - \overrightarrow{a}(x)$. The irreducible factorization of $f(x)$ is

$$f(x) = (x+1)\left(x^2 + x + 1\right)\left(x^4 + x^3 + 1\right)\left(x^4 + x^3 + x^2 + x + 1\right)$$

As $\gcd(2, 15) = 1$, the linear code C generated by

$$g(x) = \left(x^2 + x + 1\right)\left(x^4 + x^3 + x^2 + x + 1\right) = x^6 + x^4 + x^3 + x^2 + 1$$

is a right polycyclic code of length 11 with a generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \end{pmatrix}$$

and the minimum Hamming distance $d_H = 4$.

More generally, in the following table, we list all the non-trivial binary right polycyclic codes with associate vector R and all the binary cyclic codes of length 11.

Type of code	Generator polynomial of the code	Dimension	d_H
Right polycyclic codes	$x^7 + x^6 + x^4 + 1$	4	4
	$x^9 + x^8 + x^5 + x^4 + x^3 + 1$	2	6
	$x^{10} + x^9 + x^8 + x^6 + x^5 + x^2 + 1$	1	7
	$x^8 + x^4 + x^2 + x + 1$	3	5
	$x^5 + x^3 + x + 1$	6	4
	$x^6 + x^4 + x^3 + x^2 + 1$	5	4
	$x^5 + 1$	6	2
	$x^3 + 1$	8	2
	$x + 1$	10	2
	$x^6 + x^3 + x^2 + x + 1$	5	3
	$x^4 + x^3 + x^2 + x + 1$	7	2
	$x^7 + x^6 + x^5 + x^2 + x + 1$	4	4
	$x^4 + x^3 + 1$	7	3
	$x^2 + x + 1$	9	2
Cyclic codes	$x + 1$	10	2
	$x^{10} + x^9 + x^8 + x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$	1	11

Table 1.1: Binary polycyclic codes of length 11 induced by $a = (1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 0)$

Chapter 2

Linear codes invariant under a cyclic operator

In this chapter, we study linear codes invariant under a cyclic operator T, called T-cyclic codes. Since every cyclic operator can be represented by a cyclic matrix with respect to a given basis, all of these matrices are similar. For simplicity we restrict ourselves to linear codes invariant under the right multiplication by a cyclic matrix M, that we call M-cyclic codes, and when M is the companion matrix C_f of a given non-zero polynomial $f(x)$ we call them f-cyclic codes. The similarity relation between matrices helps us find connections between M-cyclic codes and f-cyclic codes, where $f(x)$ is the minimal polynomial of M. The class of M-cyclic codes contains cyclic codes and their various generalizations such as constacyclic codes, right and left polycyclic codes, monomial codes, and others. As common in the study of cyclic codes and their generalizations, we make use of the one-to-one correspondence between M-cyclic codes and ideals of the polynomial ring $R_f := \mathbb{F}_q[x]/\langle f(x)\rangle$, where $f(x)$ is the minimal polynomial of M. This correspondence leads to some basic characterizations of these codes such as generator and parity check polynomials among others. Next, we study the duality of these codes, where we show that the b-dual of an M-cyclic code is an M^*-cyclic code, where M^* is the adjoint matrix of M with respect to b, and we explore some important results on the duality of these codes. Finally, we give examples as applications of some of the results and we construct some optimal codes.

2.1 Cyclic operators and cyclic matrices

Let $\mathbb{F}_q$ be a finite field with q elements, where $q = p^s$ with p a prime number and s a positive integer. Let $\mathbb{F}_q^n$ be the $\mathbb{F}_q$-vector space of dimension $n \geq 1$.

Definition 2.1 (Cyclic operator)
An operator $T : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q^n$ is called a cyclic operator if there is a vector $v \in \mathbb{F}_q^n$ such that the set $B_v := \left\{v, T(v), \ldots, T^{n-1}(v)\right\}$ is a basis of $\mathbb{F}_q^n$, in this case v is called a T-cyclic vector of $\mathbb{F}_q^n$ and B_v a T-cyclic basis of $\mathbb{F}_q^n$.

Definition 2.2 (Cyclic matrix)
A square matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$ is called a cyclic matrix if there is a vector $v \in \mathbb{F}_q^n$ such that $B_v := \{v, Mv, \ldots, M^{n-1}v\}$ forms a basis of $\mathbb{F}_q^n$.

Note that any operator T can be represented by a matrix with respect to any given basis of $\mathbb{F}_q^n$ and all of these matrices are similar to each other by the transition matrices between the

bases. For simplicity, if M is the representation matrix of T with respect to a given basis B, we will denote T by $T_{_M}$, the associated operator of M, and we write

$$T_{_M}(v) = Mv^t \quad \text{for} \quad v = (v_{_0}, v_{_1}, \ldots, v_{_{n-1}}) \in \mathbb{F}_{_q}^{^n}.$$

Let us recall that two matrices M and $M^{'}$ are called similar if there is an invertible matrix P such $M^{'} = P^{-1}MP$. This allows us to define an equivalence relation on $\mathbb{M}_n(\mathbb{F}_{_q})$, and we denote the similarity class of M by $\Delta(M)$. Two operators are similar if their representation matrices, with respect to a given basis of $\mathbb{F}_{_q}^{^n}$, are similar.

Remark 2.1 (Notation)
From now to the end of this document we avoid the use of transpose, i.e., the use of Mv will be understood as Mv^t and also more generally we identify the vector $(Mv^t)^t$ by Mv.

Remark 2.2

1. If M is a cyclic matrix then all elements of $\Delta(M)$ are cyclic.
2. A cyclic operator T can be represented by different cyclic matrices, which are elements of $\Delta(C_f)$, where C_f is the companion matrix of $f(x)$, the minimal polynomial of T.
3. Two similar cyclic matrices represent the same cyclic operator T with respect to different bases.

In the following theorem, we collect some basic properties of cyclic operators that one can find easily in the literature. For example in [38].

Theorem 2.1 ([38])
Let T be a cyclic operator of $\mathbb{F}_{_q}^{^n}$ with minimal polynomial $f(x)$, and let $B_v := \{v, T(v), ..., T^{n-1}(v)\}$ be a T-cyclic basis of $\mathbb{F}_{_q}^{^n}$ and $b : \mathbb{F}_{_q}^{^n} \times \mathbb{F}_{_q}^{^n} \longrightarrow \mathbb{F}_{_q}$ be a non-degenerate bilinear form of $\mathbb{F}_{_q}^{^n}$. Then,

1. $f(x) = x^n - \sum_{i=0}^{n-1} a_i x^i$, where $T^{^n}(v) = \sum_{i=0}^{n-1} a_i T^{^i}(v)$.
2. The representation matrix M of T with respect to B_v is the companion matrix of $f(x)$, i.e.,
$$M = \begin{pmatrix} 0 & 0 & \ldots & 0 & a_0 \\ 1 & 0 & \ldots & 0 & a_1 \\ 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \ldots & 0 & 1 & a_{n-1} \end{pmatrix} \tag{2.1}$$
3. The minimal polynomial of T coincides with its characteristic polynomial.
4. The adjoint operator T^* of T with respect to b is also cyclic with minimal polynomial $f(x)$.

Now we give some useful examples of cyclic operators in coding theory.

Example 2.1

1. The operator T_1 defined by
$$T_1\left(v_{_0}, v_{_1}, \ldots, v_{_{n-1}}\right) = \left(v_{_{n-1}}, v_{_0}, \ldots, v_{_{n-2}}\right),$$
is a cyclic operator with minimal polynomial $f_{_{T_1}}(x) = x^n - 1$, called the **cyclic shift.**

2. The operator T_{-1} defined by

$$T_{-1}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(-v_{n-1}, v_0, \ldots, v_{n-2}\right),$$

is a cyclic operator with minimal polynomial $f_{T_{-1}}(x) = x^n + 1$, called the **negacyclic shift.**

3. Let λ be a non-zero element of $\mathbb{F}_q$. The operator T_λ defined by

$$T_\lambda\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(\lambda v_{n-1}, v_0, \ldots, v_{n-2}\right),$$

is a cyclic operator with minimal polynomial $f_\lambda(x) = x^n - \lambda$, called the **$\lambda$-cyclic shift.**

4. The operator $T_{\overline{a}}$ defined by

$$T_{\overline{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(a_{n-1}v_{n-1}, a_0 v_0, \ldots, a_{n-2}v_{n-2}\right),$$

is a cyclic operator with minimal polynomial $f_{\overline{a}}(x) = x^n - \prod_{i=0}^{n-1} a_i$, called the **monomial shift induced by the vector** $a = (a_0, a_1, \ldots, a_{n-1})$.

5. The operator $T_{\overrightarrow{a}}$ defined by

$$T_{\overrightarrow{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(0, v_0, \ldots, v_{n-2}\right) + v_{n-1}\left(a_0, a_1, \ldots, a_{n-1}\right),$$

is a cyclic operator with minimal polynomial $f(x) = x^n - \sum_{i=0}^{n-1} a_i x^i$, called the **right polyshift induced by the vector** $a = (a_0, a_1, \ldots, a_{n-1})$.

6. The operator $T_{\overleftarrow{a}}$ defined by

$$T_{\overleftarrow{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(v_1, \ldots, v_{n-1}, 0\right) + v_0\left(a_0, a_1, \ldots, a_{n-1}\right),$$

is a cyclic operator with minimal polynomial $f_{\overleftarrow{a}}(x) = x^n - \sum_{i=0}^{n-1} a_i x^{n-1-i}$, called the **left polyshift induced by the vector** $a = (a_0, a_1, \ldots, a_{n-1})$.

7. Let $\sigma \in S_n$ be an n-cycle, then T_σ defined by

$$T_\sigma\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(v_{\sigma(0)}, v_{\sigma(1)}, \ldots, v_{\sigma(n-1)}\right),$$

is a cyclic operator with minimal polynomial $f_\sigma(x) = x^n - 1$, called the **permutation operator.**

8. Let $M_{\sigma,\overline{a}} = \mathrm{diag}(a_0, a_1, \ldots, a_{n-1})P_\sigma$ be a monomial matrix where $\sigma \in S_n$ is an n-cycle, and P_σ its associated permutation matrix. The operator $T_{\sigma,\overline{a}}$ associated to $M_{\sigma,\overline{a}}$ defined by

$$T_{\sigma,\overline{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = (a_{\sigma(0)}v_{\sigma(0)}, a_{\sigma(1)}v_{\sigma(1)}, \ldots, a_{\sigma(n-1)}v_{\sigma(n-1)}),$$

is a cyclic operator with minimal polynomial $f_{\sigma,\overline{a}}(x) = x^n - \prod_{i=0}^{n-1} a_i$, called the **$\sigma$-monomial shift induced by the vector** $a = (a_0, a_1, \ldots, a_{n-1})$.

Remark 2.3
Recall that a square matrix M is called a monomial matrix of order n if it contains exactly one non-zero element in each row and each column. Note also that if M is a monomial matrix of order n with non-zero components $a_0, a_1, \ldots, a_{n-1} \in \mathbb{F}_q$, then there is a permutation $\sigma \in S_n$ (not necessarily an n-cycle) such that $M = \mathrm{diag}(a_0, a_1, \ldots, a_{n-1})P_\sigma$, where P_σ is the associated permutation matrix to σ. In the above definition we assume that σ is an n-cycle to define σ-monomial codes.

Below is a list of certain types of linear codes with cyclical properties.

Definition 2.3 ([19, 30, 80, 81])
A linear code $C \subseteq \mathbb{F}_q^n$ of length n over $\mathbb{F}_q$ is called a

1. **λ-constacyclic code** if for every $c = \left(c_0, c_1, \ldots, c_{n-1}\right) \in C$ we have $c' = (\lambda c_{n-1}, c_0, \ldots, c_{n-2}) \in C$, where λ is a non-zero element of $\mathbb{F}_q$. Hence it is invariant under the cyclic operator T_λ. For $\lambda = \pm 1$ we obtain cyclic codes and negacyclic codes as special cases. Hence they are invariant under the cyclic operators T_1 and T_{-1}.

2. **monomial code** induced by the vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right) \in (\mathbb{F}_q^*)^n$, if for each $c = \left(c_0, c_1, \ldots, c_{n-1}\right) \in C$ we have $c' = \left(a_{n-1}c_{n-1}, a_0c_0, \ldots, a_{n-2}c_{n-2}\right) \in C$. Hence it is invariant under the cyclic operator $T_{\overline{a}}$.

3. **right polycyclic code** induced by the vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right) \in \mathbb{F}_q^n$ if for each $c = \left(c_0, c_1, \ldots, c_{n-1}\right) \in C$ we have $c' = \left(c_{n-1}a_0, c_0 + c_{n-1}a_1, \ldots, c_{n-2} + c_{n-1}a_{n-1}\right) \in C$. Hence it is invariant under the cyclic operator $T_{\overrightarrow{a}}$.

4. **left polycyclic code** induced by the vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right) \in \mathbb{F}_q^n$ if for each $c = \left(c_0, c_1, \ldots, c_{n-1}\right) \in C$ we have $c' = \left(c_1 + c_0a_0, \ldots, c_{n-2} + c_0a_{n-2}, c_0a_{n-1}\right) \in C$. Hence it is invariant under the cyclic operator $T_{\overleftarrow{a}}$.

5. **σ-cyclic code** if for each $c = \left(c_0, c_1, \ldots, c_{n-1}\right) \in C$ we have $c' = (c_{\sigma(0)}, c_{\sigma(1)}, \ldots, c_{\sigma(n-1)}) \in C$. Hence it is invariant under the cyclic operator T_σ.

6. **σ-monomial code** induced by the vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right) \in \mathbb{F}_q^n$ if for each $c = \left(c_0, c_1, \ldots, c_{n-1}\right) \in C$ we have $c' = (a_{\sigma(0)}c_{\sigma(0)}, a_{\sigma(1)}c_{\sigma(1)}, \ldots, a_{\sigma(n-1)}c_{\sigma(n-1)}) \in C$. Hence it is invariant under the cyclic operator $T_{\sigma,\overline{a}}$, when σ is an n-cycle.

2.2 Order of polynomials

In the following Lemma, we give the definition of the order of polynomials in $\mathbb{F}_q[x]$ which will be used later.

Lemma 2.1 ([52], Lemma 3.1.)
Let $f(x) \in \mathbb{F}_q[x]$ be a polynomial of degree n with $f(0) \neq 0$. Then there is a positive integer $e \leq q^n - 1$ such that $f(x)$ divides $x^e - 1$. The least positive integer e with this property is called the order of $f(x)$, which will be denoted by $\mathrm{ord}(f(x)) = e$.

Remark 2.4
If $f(x) \in \mathbb{F}_q[x]$ is a polynomial of order e such that $\gcd(e, q) = 1$, then all roots of $f(x)$ are simple and there is a primitive e^{th} root of unity α in some extension field $\mathbb{L}/\mathbb{F}_q$ such that all roots of $f(x)$ can be expressed as powers of α.

In the following proposition, we prove some results on the order of polynomials over $\mathbb{F}_q$, using technical facts from linear algebra.

Proposition 2.1
Let $f(x) \in \mathbb{F}_q[x]$ be a polynomial of degree n and of order e such that $f(0) \neq 0$. Let C_f be its companion matrix. Then,

1. The companion matrix, C_f, of f is invertible.
2. The order e of $f(x)$ is the least positive integer such that $C_f^e = \mathrm{I}_n$.
3. The order of C_f, as an element of the finite linear group $\mathrm{GL_n}(\mathbb{F}_q)$, is e.
4. The order of any matrix $M \in \Delta(C_f)$ is e.
5. If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of C_f, then $e = \mathrm{lcm}(\mathrm{ord}(\lambda_1), \mathrm{ord}(\lambda_2), \ldots, \mathrm{ord}(\lambda_n))$ where $\mathrm{ord}(\lambda_i)$ is the order of λ_i in $\mathbb{F}_q^*$.

Proof
1. Since $f(x)$ is the minimal polynomial of C_f, $\det(C_f) = f(0) \neq 0$.
2. Since e is the order of $f(x)$,
$$\begin{array}{rcl} f(x) \text{ divides } x^e - 1 & \iff & (x^e - 1)\,(C_f) = 0 \\ & \iff & C_f^e - \mathrm{I}_n = 0 \\ & \iff & C_f^e = \mathrm{I}_n\,. \end{array}$$
3. Follows from the statement 2 and the definition of the polynomial order e.
4. Follows from the statement 3.
5. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of C_f, and $\mathrm{ord}(\lambda_i)$ be the order of each λ_i in $\mathbb{F}_q^*$. Then there is an invertible matrix $A \in \mathrm{GL_n}(\mathbb{F}_q)$ such that $C_f = A^{-1}\,\mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)A$. It follows that
$$C_f^{\mathrm{lcm}(\mathrm{ord}(\lambda_1),\mathrm{ord}(\lambda_2),\ldots,\mathrm{ord}(\lambda_n))} = \mathrm{I}_n\,.$$
This means that e divides $\mathrm{lcm}(\mathrm{ord}(\lambda_1), \mathrm{ord}(\lambda_2), \ldots, \mathrm{ord}(\lambda_n))$.
On the other hand, we have
$$C_f^e = A^{-1}\,\mathrm{diag}(\lambda_1^e, \lambda_2^e, \ldots, \lambda_n^e)A = \mathrm{I}_n\,.$$
It follows that $\lambda_i^e = 1$, for every $i = 1, \ldots, r$. Hence $\mathrm{ord}(\lambda_i)$ divides e and $\mathrm{lcm}(\mathrm{ord}(\lambda_1), \mathrm{ord}(\lambda_2), \ldots, \mathrm{ord}(\lambda_r))$ divides e.

2.3 Linear codes invariant under cyclic operators

Let M be an $n \times n$ square matrix and let T_M be its associate operator on $\mathbb{F}_q^n$.

Definition 2.4
A linear code $C \subseteq \mathbb{F}_q^n$ of length n over $\mathbb{F}_q$ is called an

1. M-code if it is invariant under $T_{_M}$, i.e., for every $c = (c_0, c_1, \ldots, c_{n-1}) \in C$, we have

$$T_{_M}(c_0, c_1, \ldots, c_{n-1}) = M(c_0, c_1, \ldots, c_{n-1})^t \in C.$$

2. M-cyclic code if it is invariant under $T_{_M}$, where M is a cyclic matrix.

3. f-cyclic code if it is invariant under T_{C_f}, where C_f is the companion matrix of a non-zero polynomial $f(x) \in \mathbb{F}_{_q}[x]$ of degree n.

Remark 2.5
Note that f-cyclic codes were already known under the name of pseudo-cyclic codes in [71] or right polycyclic codes [80], where in [71] they are defined as ideals of the polynomial ring $\mathbb{F}_{_q}[x]/\langle f(x)\rangle$. However, in [80] they are defined as subspaces of $\mathbb{F}_{_q}^n$ invariant under the right polycyclic shift.

Proposition 2.2
Let M be a cyclic matrix with similarity class $\Delta(M)$ and let $C \subseteq \mathbb{F}_{_q}^n$ be a linear code. Then the following assertions are equivalent.

1. C is an M-cyclic code.

2. $P^{-1}C$ is an $M^{'}$-cyclic code, for every $M^{'} \in \Delta(M)$ such that $M^{'} = P^{-1}MP$.

Proof
Assume that C is an M-cyclic code of length n over $\mathbb{F}_{_q}$ then $MC \subseteq C$. Hence

$$M^{'}P^{-1}C = P^{-1}MPP^{-1}C = P^{-1}MC \subseteq P^{-1}C.$$

Conversely, if $P^{-1}C$ is an $M^{'}$-cyclic code for each $M^{'} \in \Delta(M)$ such that $M^{'} = P^{-1}MP$, then $M \in \Delta(M^{'})$ by P^{-1}, i.e., $M = PM^{'}P^{-1}$. From the first implication we obtain that $P(P^{-1}C) = C$ is an M-cyclic code.

Since each cyclic matrix is similar to the companion matrix of its minimal polynomial, we deduce the following result.

Corollary 2.1
Let M be a cyclic matrix with minimal polynomial $f(x) = x^n - \sum_{i=0}^{n-1} a_i x^i$, v be an M-cyclic vector of $\mathbb{F}_{_q}^n$, and $C \subseteq \mathbb{F}_{_q}^n$ be a linear code of length n. Then the following assertions are equivalent.

1. C is an M-cyclic code.

2. $P^{-1}C$ is an f-cyclic code, with $P = \left[v, T_M(v), \ldots, T_M^{n-1}(v)\right] \in \mathrm{GL_n}\left(\mathbb{F}_{_q}\right)$ and $C_f = P^{-1}MP$.

3. $P^{-1}C$ is a right polycyclic code induced by the vector $a = (a_0, a_1, \ldots, a_{n-1})$.

Remark 2.6
The result proved in the above theorem is equivalent to changing the basis of the code C, where P represents the transition matrix between two bases of $\mathbb{F}_{_q}^n$. Then C and $P^{-1}C$ have the same minimum Hamming distance if P is a monomial matrix. In general, C and $P^{-1}C$ may have different Hamming weights in two different bases of $\mathbb{F}_{_q}^n$.

Let $f(x) \in \mathbb{F}_q[x]$, and consider the $\mathbb{F}_q$-vector space isomorphism φ_f between $\mathbb{F}_q^n$ and $\mathbb{F}_q[x]/\langle f(x)\rangle$ defined by:

$$\begin{array}{rccc} \varphi_f : & \mathbb{F}_q^n & \longrightarrow & \mathbb{F}_q[x]/\langle f(x)\rangle \\ & v := (v_0, v_1, \ldots, v_{n-1}) & \longmapsto & v(x) = \sum_{i=0}^{n-1} v_i x^i \end{array} \tag{2.2}$$

Theorem 2.2
Let $C \subseteq \mathbb{F}_q^n$ be a linear code, M be a cyclic matrix with minimal polynomial $f(x)$, and v be an M-cyclic vector. Then

1. C is an f-cyclic code if and only if $\varphi_f(C)$ is an ideal of R_f.
2. C is an M-cyclic code if and only if $\psi_f(C)$ is an ideal of R_f, where $\psi = \varphi_f \circ P_v^{-1}$.

Proof

1. Suppose that C is an f-cyclic code, and let $c = (c_0, c_1, \ldots, c_{n-1}) \in C$. Then

$$\begin{aligned} x\varphi_f(c) &= \sum_{i=0}^{n-1} c_i x^{i+1} \\ &= c_{n-1}a_0 + \sum_{i=1}^{n-1}(c_{i-1} + c_{n-1}a_i)x^i \quad (\mod f(x)) \\ &= \varphi_f\left(c_{n-1}a_0, c_0 + c_{n-1}a_0, ..., c_{n-2} + c_{n-1}a_{n-1}\right) \\ &= \varphi_f(C_f c^t) \end{aligned} \tag{2.3}$$

   Thus by induction we show that for $k \in \{0, ..., n-1\}$

$$x^k\varphi_f(c) = \varphi_f(C_f^k c^t).$$

   It follows that $\varphi_f(C)$ is an ideal of R_f.

2. If C is an M-cyclic code, then by Corollary 2.1, $P_v^{-1}C$ is an f-cyclic code. Hence by the first statement, we deduce that $\psi_f(C) = \varphi_f(P_v^{-1}C)$ is an ideal of R_f.

Now we deduce the following characterization of f-cyclic codes.

Theorem 2.3
Let $C \subseteq \mathbb{F}_q^n$ be an f-cyclic code.

1. There is a monic polynomial of least degree $g(x) \in \mathbb{F}_q[x]$ such that $g(x)$ divides $f(x)$ and $\varphi_f(C) = \langle g(x)\rangle$.
2. The set $\{g(x), xg(x), \ldots, x^{k-1}g(x)\}$ forms a basis of $\varphi_f(C)$ and the dimension of C is $k = n - \deg(g(x))$.
3. A generator matrix G of C is given by:

$$G = \begin{pmatrix} \varphi_f^{-1}(g(x)) \\ \varphi_f^{-1}(xg(x)) \\ \vdots \\ \varphi_f^{-1}\left(x^{k-1}g(x)\right) \end{pmatrix} \begin{pmatrix} g_0 & g_1 & \cdots & g_{n-k} & 0 & \cdots & \cdots & 0 \\ 0 & g_0 & g_1 & \cdots & g_{n-k} & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \ddots & \vdots \\ 0 & \ldots & & 0 & g_0 & g_1 & \cdots & g_{n-k} \end{pmatrix} \tag{2.4}$$

   where $k = n - \deg(g)$ and $g(x) = \sum_{i=0}^{n-k} g_i x^i$.

4. There is a one-to-one correspondence between divisors of $f(x)$ and f-cyclic codes of length n over $\mathbb{F}_q$.

Proof
According to Theorem 2.2, $\varphi_f(C)$ is an ideal of the principal ideal ring $\mathbb{F}_q[x]/\langle f(x)\rangle$, hence there is a monic polynomial $g(x)$ such that $\varphi_f(C) = \langle g(x)\rangle$. Since

$$x^i.g(x) = \varphi_f\left(C_f^i(\varphi_f^{-1}(g(x)))\right) \quad \text{for} \quad i \in \{0, \ldots, n-1\},$$

the set $\{g(x), xg(x), \ldots, x^{n-\deg(g)-1}g(x)\}$ forms a basis of $\varphi_f(C)$ and

$$\dim_{\mathbb{F}_q}(C) = \dim_{\mathbb{F}_q}(\varphi_f(C)) = n - \deg(g).$$

Finally, as $\{\varphi_f^{-1}(g(x)), \varphi_f^{-1}(xg(x)), \ldots, \varphi_f^{-1}(x^{n-\deg(g)-1}g(x))\}$ is a basis of C, it follows that its generator matrix has the form desired.

Remark 2.7
Note that the result of the above theorem is still valid for M-cyclic codes except for generator matrices which may be in a different form. For example if C is a left polycyclic code induced by the vector $a = (a_0, a_1, \ldots, a_{n-1})$, then its generator matrix has the form

$$G = \begin{pmatrix} 0 & \cdots & \cdots & 0 & g_{n-k} & \cdots & g_1 & g_0 \\ 0 & \cdots & 0 & g_{n-k} & \cdots & g_1 & g_0 & 0 \\ \vdots & \iddots & \iddots & & \iddots & \iddots & \iddots & \vdots \\ 0 & \iddots & & \iddots & \iddots & \iddots & \iddots & \vdots \\ g_{n-k} & \cdots & g_1 & g_0 & 0 & \cdots & \iddots & 0 \end{pmatrix}$$

where $k = n - \deg(g)$ and $g(x) = \sum_{i=0}^{n-k} g_i x^i$ is divisor of $x^n - \sum_{i=0}^{n-1} a_i x^{n-i-1}$.

We now give the definition of a generator polynomial of an f-cyclic code. Similar definitions can be given for M-cyclic codes as well using ψ_f. Like an ordinary cyclic code, an f-cyclic code C has many generators but among all of its generators, there is a special unique one, called the standard generator of C. Any other generator of C is a multiple of the standard generator. When we say "the generator" of such a code, we mean this unique standard generator.

Definition 2.5
Let $C \subseteq \mathbb{F}_q^n$ be an f-cyclic (resp. M-cyclic) code of length n over $\mathbb{F}_q$. Then the **standard generator polynomial** of C is the monic polynomial $g(x)$ of least degree in R_f such that $\varphi_f(C) = \langle g(x)\rangle$ (resp. $\psi_f(C) = \langle g(x)\rangle$).

Note that each f-cyclic (resp. M-cyclic) code C can be viewed as an R_f-submodule of R_f. Hence there is an element $a(x) \in R_f$ such that $\varphi_f(C) = R_f \cdot a(x)$. In the following proposition we give a characterization of their generators.

Proposition 2.3
Let $C \subseteq \mathbb{F}_q^n$ be an f-cyclic code of length n over $\mathbb{F}_q$ with the standard generator polynomial $g(x)$, and $f(x) = h(x)g(x)$.

1. If $\varphi_f(C) = R_f \cdot a(x)$, for some $a(x) \in \mathbb{F}_q[x]$ then $g(x) = \gcd(a(x), f(x))$.
2. If $\gcd(p(x), h(x)) = 1$, then $p(x)g(x)$ is another generator of C, i.e., $\varphi_f(C) = \langle p(x)g(x)\rangle$.

Proof

1. Let $\alpha(x) = \gcd(a(x), f(x))$. Since $a(x) \in \varphi_f(C)$, $a(x) = q(x)g(x)$, $g(x)$ is a divisor of $\alpha(x)$. On the other hand, let $g(x) = q_1(x)\alpha(x) + r(x)$, with $\deg(r(x)) < \deg(\alpha(x)) \leq \deg(g(x))$ or $r(x) = 0$. As $g(x)$ divides $\alpha(x)$, one can write $g(x) = q_1(x)q_2g(x) + r(x) = q'(x)g(x) + r(x)$. It follows that $r(x) = g(x) - q'(x)g(x) = (1 - q'(x))g(x) \in \varphi_f(C)$. Hence by minimality of the degree of $g(x)$ we deduce that $r(x) = 0$, which means that $\alpha(x)$ is a divisor of $g(x)$. Therefore, $g(x) = \alpha(x) = \gcd(a(x), h(x))$.

2. Let C be an f-cyclic code with a generator $g(x)$. Since $\gcd(p(x), h(x)) = 1$, there exist polynomials $a(x), b(x) \in R_f$ such that $a(x)p(x) + b(x)h(x) = 1$. It follows that $a(x)p(x)g(x) + b(x)h(x)g(x) = g(x)$, i.e., $a(x)p(x)g(x) + b(x)f(x) = g(x)$. Hence $a(x)p(x)g(x) = g(x) \mod f(x)$, which implies that $g(x) \in \varphi_f(C) = \langle p(x)g(x) \rangle$, and so $p(x)g(x)$ generates C.

Example 2.2 (f-cyclic codes)
Let $q = 4$ and $\mathbb{F}_4 = \mathbb{F}_2(\alpha)$ where $\alpha^2 + \alpha + 1 = 0$ and $f(x) = x^7 + (\alpha + 1)\, x + \alpha + 1$. The irreducible factorization of $f(x)$ over $\mathbb{F}_4$ is given by:

$$f(x) = (x + \alpha) \cdot (x^2 + \alpha x + \alpha) \cdot (x^4 + x^2 + (\alpha + 1)\, x + 1).$$

So one can construct $2^3 - 2$ non-trivial f-cyclic codes with generators

$$\begin{aligned}
g_1(x) &= x^2 + \alpha x + \alpha \\
g_2(x) &= x^4 + x^2 + (\alpha + 1)\, x + 1 \\
g_3(x) &= x^5 + \alpha x^4 + x^3 + x^2 + \alpha \\
g_4(x) &= x^6 + \alpha x^5 + (\alpha + 1)\, x^4 + x^3 + \alpha x^2 + (\alpha + 1)\, x + \alpha \\
g_5(x) &= x^3 + x + \alpha + 1 \\
g_6(x) &= x + \alpha
\end{aligned}$$

For example $g_3(x) = x^5 + \alpha x^4 + x^3 + x^2 + \alpha$ is a divisor of $f(x)$, and generates a $[7, 2, 5]_4$ f-cyclic code. Note that this code is optimal according to the online database [27]. Moreover, all of these f-cyclic codes are optimal and their generators and parameters are listed in the following table. Note also that these codes are right polycyclic codes induced by the vector $a = (\alpha + 1, \alpha + 1, 0, 0, 0, 0, 0) \in \mathbb{F}_4^7$.

$g(x)$	$h(x)$	$[n, k, d]_q$
$x^2 + \alpha x + \alpha$	$x^5 + \alpha x^4 + x^3 + x^2 + \alpha$	$[7, 5, 2]_4$
$x^4 + x^2 + (\alpha + 1)\, x + 1$	$x^3 + x + \alpha + 1$	$[7, 3, 4]_4$
$x^5 + \alpha x^4 + x^3 + x^2 + \alpha$	$x^2 + \alpha x + \alpha$	$[7, 2, 5]_4$
$x^6 + \alpha x^5 + (\alpha + 1)\, x^4 + x^3 + \alpha x^2 + (\alpha + 1)\, x + \alpha$	$x + \alpha$	$[7, 1, 7]_4$
$x^3 + x + \alpha + 1$	$x^4 + x^2 + (\alpha + 1)\, x + 1$	$[7, 4, 3]_4$
$x + \alpha$	$x^6 + \alpha x^5 + (\alpha + 1)\, x^4 + x^3 + \alpha x^2 + (\alpha + 1)\, x + \alpha$	$[7, 6, 2]_4$

Remark 2.8 (Construction of M-cyclic codes)
To construct M-cyclic codes we need to determine the irreducible factorization of $f(x)$, the minimal polynomial of M, and an invertible matrix P such that $C_f = P^{-1}MP$, where v is an M-cyclic vector. More precisely we follow these steps:

1. Compute the irreducible factorization of $f(x) = \prod_{i=1}^{r} f_i^{\alpha_i}(x)$

2. Take $g(x) = \prod_{i=1}^{r} f_i^{k_i}(x)$ where $0 \leq k_i \leq \alpha_i$, as the generator polynomial of an f-cyclic code C.

3. Compute P such that $C_f = P^{-1}MP$.

4. Finally the linear code $C^{'} = PC$ is an M-cyclic code.

Example 2.3 (M-cyclic codes)
Let

$$M := \begin{pmatrix} 1 & 0 & 1 & 0 & 2 \\ 1 & 2 & 0 & 0 & 1 \\ 2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 0 & 1 & 0 & 1 \end{pmatrix}$$

be a 5×5 cyclic matrix over $\mathbb{F}_3$ and T_M its associated linear operator. The minimal polynomial of M is $f(x) = x^5 - 2x^4 - x - 1$. It follows that M is similar to C_f, the companion matrix of $f(x)$ by the invertible matrix

$$P = \begin{pmatrix} 0 & 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

Let φ_f be the usual isomorphism between $\mathbb{F}_3^5$ and R_f. The irreducible factorization of $f(x)$ is

$$f(x) = (x+2) \cdot (x+1)^2 \cdot (x^2+1).$$

So, for example the linear code C with $\varphi_f(C) = \langle x^3 + x^2 + 2x + 2 \rangle$ is a $[5, 2, 3] - f$-cyclic code, with a generator matrix

$$G = \begin{pmatrix} 2 & 2 & 1 & 1 & 0 \\ 0 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

It follows that PC is a $[5, 2, d] - M$-cyclic code with a generator matrix $G\,{}^tP$. By using Magma Software, we found $d = 3$. In general, $d = 3$ if P is a monomial matrix. On the other hand, let C be an f-cyclic code with $\varphi_f(C) = \langle x^2 + 1 \rangle$. Then C is a $[5, 3, 2]_3 - f$-cyclic code with a generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

In this case PC is a $[5, 3, 1]_3 - M$-cyclic code.

2.4 Connection between M-cyclic codes and other types of codes

This section is devoted to connections between M-codes and other well-known types of codes in the literature. In line with Example 2.1 and Definition 2.3, we give the following proposition.

Proposition 2.4
The following codes are f-cyclic codes

1. Constacyclic codes are f_λ-cyclic codes with $f_\lambda(x) = x^n - \lambda$, since they are invariant under M_λ, the companion matrix of f_λ.

2. Cyclic codes are f_1-cyclic codes with $f_1(x) = x^n - 1$, since they are invariant under M_1, the companion matrix of f_1.

3. Negacyclic codes are f_{-1}-cyclic codes with $f_{-1}(x) = x^n + 1$, since they are invariant under M_{-1}, the companion matrix of f_{-1}.

4. Right polycyclic codes induced by a vector $a = (a_0, a_1, \ldots, a_{n-1})$ are $f_{\overrightarrow{a}}$-cyclic code, since they are invariant under $M_{\overrightarrow{a}}$, the companion matrix of $f_{\overrightarrow{a}}(x) = x^n - \sum_{i=0}^{n-1} a_i x^i$.

On the other hand, the following codes are M-cyclic codes.

1. Left polycyclic codes induced by a vector $a = (a_0, a_1, \ldots, a_{n-1})$ are $M_{\overleftarrow{a}}$-cyclic codes, where
$$M_{\overleftarrow{a}} = \begin{pmatrix} a_0 & 1 & 0 & \ldots & 0 \\ a_1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & & \ddots & 1 \\ a_{n-1} & 0 & \ldots & \ldots & 0 \end{pmatrix}$$
is a cyclic matrix with minimal polynomial $f_{\overleftarrow{a}} = x^n - \sum_{i=0}^{n-1} a_i x^{n-1-i}$.

2. Monomial codes induced by $a = (a_0, a_1, \ldots, a_{n-1})$ are $M_{\overline{a}}$-cyclic codes, where
$$M_{\overline{a}} = \begin{pmatrix} 0 & 0 & \ldots & 0 & a_{n-1} \\ a_0 & 0 & \ldots & 0 & 0 \\ 0 & a_1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \ldots & 0 & a_{n-2} & 0 \end{pmatrix}$$
is a cyclic matrix with minimal polynomial $f_{\overline{a}} = x^n - \prod_{i=0}^{n-1} a_i$.

3. σ-cyclic codes are P_σ-cyclic codes, where σ is an n-cycle with associated matrix P_σ.

4. σ-monomial codes induced by a vector $a = (a_0, a_1, \ldots, a_{n-1})$ are $M_{\sigma,\overline{a}}$-cyclic codes, where σ is an n-cycle and
$$M_{\sigma,\overline{a}} = \mathrm{diag}(a_0, a_1, \ldots, a_{n-1}) P_\sigma$$
is a cyclic matrix with minimal polynomial $f_{\sigma,\overline{a}} = x^n - \prod_{i=0}^{n-1} a_i x^i$.

In the following proposition we give some connections between different types of codes of Definition 2.3. First let σ be the n-cycle defined by $\sigma(1, 2, \ldots, n) = (n, 1, \ldots, n-1)$. We refer to this n-cycle as a **standard n-cycle** or as an n-cycle of **standard form**.

Proposition 2.5
Let C be a linear code of length n over $\mathbb{F}_q$.

1. C is a **right** polycyclic code induced by the vector $a = \left(a_0, a_1, \ldots, a_{n-1}\right)$ if and only if $P^{-1}C$ is a **left** polycyclic code induced by the vector $b = \left(a_{n-1}, a_{n-2}, \ldots, a_0\right)$, where
$$P^{-1} = P = \begin{pmatrix} 0 & \ldots & \ldots & 0 & 1 \\ 0 & \ldots & 0 & 1 & 0 \\ \vdots & \iddots & \iddots & \iddots & \vdots \\ 0 & \iddots & \iddots & & \vdots \\ 1 & 0 & \ldots & \ldots & 0 \end{pmatrix}$$

2. C is a monomial code induced by the vector $a = (a_0, a_1, \ldots, a_{n-1})$ if and only if $Q^{-1}C$ is a $\prod_{i=0}^{n-1} a_i$-constacyclic code, where
$$Q = \begin{pmatrix} 0 & 0 & \ldots & 0 & \prod_{i=0}^{n-1} a_i \\ a_0 & 0 & \ldots & 0 & 0 \\ 0 & a_0 a_1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \ldots & 0 & \prod_{i=0}^{n-2} a_i & 0 \end{pmatrix}, \quad Q^{-1} = \begin{pmatrix} 0 & a_0^{-1} & 0 & \ldots & 0 \\ 0 & 0 & (a_0 a_1)^{-1} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & \vdots & & \ddots & \prod_{i=0}^{n-2} a_i^{-1} \\ \prod_{i=0}^{n-1} a_i^{-1} & 0 & \ldots & \ldots & 0 \end{pmatrix}$$

3. C is a σ-cyclic code if and only if $P_\rho^{-1}C$ is a cyclic code, where $\rho \in S_n$ is such that $\sigma = \rho\sigma^{'}\rho^{-1}$ and $\sigma^{'}$ is of standard form.

4. C is a σ-cyclic monomial code with associated vector $a = (a_0, a_1, \ldots, a_{n-1})$ if and only if $Q_\rho^{-1}C$ is a $\prod_{i=0}^{n-1} a_i$-constacyclic code, where $Q_\rho = P_\rho Q$ and $\rho, \sigma^{'} \in S_n$ is such that $\sigma = \rho\sigma^{'}\rho^{-1}$ and $\sigma^{'}$ is a standard n-cycle.

Proof

1. Since $M_{\overrightarrow{b}} = P^{-1}M_{\overleftarrow{a}}P$.

2. Since $M_\lambda = Q^{-1}M_{\overline{a}}Q$, with $\lambda = \prod_{i=0}^{n-1} a_i$.

3. Since $C_{f_1} = M_{\sigma'} = P_\rho^{-1}P_\sigma P_\rho$, where $f_1 = x^n - 1$.

4. Since $M_\lambda = Q_\rho^{-1}M_{\sigma,\overline{a}}Q_\rho$, with $\lambda = \prod_{i=0}^{n-1} a_i$.

Now we give some examples of isomorphisms that we use to relate various types of codes, cited in Definition 2.3, with ideals of the polynomial ring R_f for some special polynomials $f(x) \in \mathbb{F}_q[x]$.

Remark 2.9

1. In [80], right polycyclic codes induced by a vector a are characterized by the ideals of $R_{\overrightarrow{a}} := \mathbb{F}_q[x]/\langle f_{\overrightarrow{a}}(x)\rangle$, with $f_{\overrightarrow{a}} = x^n - \sum_{i=0}^{n-1} a_i x^i$, by using the isomorphism $\varphi_{\overrightarrow{a}}$, defined by

$$\varphi_{\overrightarrow{a}}(v_0, v_1, \ldots, v_{n-1}) = v_0 + v_1 x + \ldots + v_{n-1}x^{n-1}$$

Since right polycyclic codes induced by a are exactly $f_{\overrightarrow{a}}$-cyclic codes, we obtain Theorem 2.2.

2. In [80], left polycyclic codes induced by a vector a are characterized by the ideals of $R_{\overleftarrow{a}} := \mathbb{F}_q[x]/\langle f_{\overleftarrow{a}}(x)\rangle$, with $f_{\overleftarrow{a}}(x) = x^n - \sum_{i=0}^{n-1} a_i x^{n-1-i}$, by using the **left isomorphism** $\varphi_{\overleftarrow{a}}$, defined by

$$\varphi_{\overleftarrow{a}}(v_0, v_1, \ldots, v_{n-1}) = v_{n-1} + v_{n-2}x + \ldots + v_1 x^{n-2} + v_0 x^{n-1}$$

From Proposition 2.5, we know that left polycyclic codes induced by a vector a are $M_{\overleftarrow{a}}$-cyclic. We verify that

$$M_{\overrightarrow{b}} = P^{-1}M_{\overleftarrow{a}}P,$$

where $b = (a_{n-1}, a_{n-2}, \ldots, a_0) \in \mathbb{F}_q^n$ and $M_{\overrightarrow{b}}$ is the companion matrix of $f_{\overleftarrow{a}}(x) = x^n - \sum_{i=0}^{n-1} a_i x^{n-1-i}$. By Theorem 2.2, we have $\varphi_{\overrightarrow{a}} = \varphi_f \circ P^{-1}$.

3. From Proposition 2.5, we know that monomial codes are $M_{\overline{a}}$-cyclic codes. Since $M_\lambda = Q^{-1}M_{\overline{a}}Q, \quad \lambda = \prod_{i=0}^{n-1} a_i$, the **monomial isomorphism** $\psi_{\overline{a}} := \varphi_{f_{\overline{a}}} \circ P^{-1}$ given by

$$\psi_{\overline{a}}(v_0, v_1, \ldots, v_{n-1}) = v_1 a_0^{-1} + (a_0 a_1)^{-1} v_2 x + \ldots + v_{n-1}\prod_{i=0}^{n-2} a_i^{-1} x^{n-2} + v_0 \prod_{i=0}^{n-1} a_i^{-1} x^{n-1}$$

induces the one-to-one correspondence between monomial codes associated with the vector $a = (a_0, a_1, \ldots, a_{n-1})$ and the ideals of the polynomial ring $R_{\overline{a}} := \mathbb{F}_q[x]/\langle f_{\overline{a}}(x)\rangle$.

4. To connect σ-cyclic codes with the ideals of $\mathbb{F}_q[x]/\langle x^n-1\rangle$ we will use the isomorphism φ_ρ defined by
$$\varphi_\rho(v_0, v_1, \ldots, v_{n-1}) = \sum_{i=0}^{n-1} v_{\rho^{-1}(i)} x^i$$
where $\rho \in S_n$ is such that $\sigma = \rho\sigma'\rho^{-1}$ and σ' is of standard form.

5. From Proposition 2.5, we know that σ-cyclic monomial codes induced by the vector $a = (a_0, a_1, \ldots, a_{n-1})$ are $M_{\sigma,\overline{a}}$-cyclic codes. Since $M_\lambda = Q_\rho^{-1} M_{\overline{a}} Q_\rho$, $\lambda = \prod_{i=0}^{n-1} a_i$, the isomorphism $\psi_{\sigma,\overline{a}} := \varphi_{f_{\overline{a}}} \circ Q_\rho^{-1}$, gives a one-to-one correspondence between σ-cyclic monomial codes and the ideals of the polynomial ring $R_{\overline{a}} := \mathbb{F}_q[x]/\langle f_{\overline{a}}(x)\rangle$.

Example 2.4 (Monomial codes)
Let $q = 9$ and $\mathbb{F}_9 = \mathbb{F}_3(\alpha)$, with $\alpha^2 + 2\alpha + 2 = 0$ and $a = (1, \alpha, 2, \alpha^2, 4) \in \mathbb{F}_9^5$. Our goal in this example is to construct monomial codes induced by the vector a, i.e., linear codes that are invariant under the monomial matrix $M_{\overline{a}}$. Let
$$Q = \begin{pmatrix} 0 & 0 & 0 & 0 & 2\alpha+1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & \alpha & 0 & 0 & 0 \\ 0 & 0 & 2\alpha & 0 & 0 \\ 0 & 0 & 0 & 2\alpha^3 & 0 \end{pmatrix}$$
be an invertible matrix such that $C_{f_{\overline{a}}} = Q^{-1} M_{\overline{a}} Q$ as in the above Proposition. Hence, the monomial isomorphism $\psi_{\overline{a}} = \varphi_{f_{\overline{a}}} \circ Q^{-1}$ permits us to give a one-to-one correspondence between monomial codes and the ideals of $R_{\overline{a}} := \mathbb{F}_q[x]/\langle f_{\overline{a}}(x)\rangle$, where
$$f_{\overline{a}} = x^5 - (1 \times \alpha \times 2 \times \alpha^2 \times 4) = x^5 - (2\alpha + 1)$$
and its irreducible factorization is given by
$$f_{\overline{a}}(x) = x^5 - (2\alpha+1) = (x + 2\alpha + 1) \cdot (x^2 + x + 2\alpha + 2) \cdot (x^2 + (\alpha+1)\, x + 2\alpha + 2).$$
Hence, one can construct $2^3 - 2 = 6$ non-trivial monomial codes of length 5 over $\mathbb{F}_9$ induced by the vector $a = (1, \alpha, 2, \alpha^2, 4)$, that we list in the following table.

Generator polynomial	Parity check polynomial	Parameters
$x^4 + (\alpha+2)\, x^3 + (2\alpha+2)\, x^2 + 2\alpha x + 2$	$x + 2\alpha + 1$	$[5, 1, 5]_9$
$x^2 + x + 2\alpha + 2$	$x^3 + 2x^2 + (\alpha+2)\, x + \alpha$	$[5, 3, 3]_9$
$x^3 + (2\alpha+2)\, x^2 + \alpha x + \alpha$	$x^2 + (\alpha+1)\, x + 2\alpha + 2$	$[5, 2, 4]_9$
$x^2 + (\alpha+1)\, x + 2\alpha + 2$	$x^3 + (2\alpha+2)\, x^2 + \alpha x + \alpha$	$[5, 3, 3]_9$
$x + 2\alpha + 1$	$x^4 + (\alpha+2)\, x^3 + (2\alpha+2)\, x^2 + 2\alpha x + 2$	$[5, 4, 2]_9$
$x^3 + 2x^2 + (\alpha+2)\, x + \alpha$	$x^2 + x + 2\alpha + 2$	$[5, 2, 4]_9$

2.5 M-cyclic codes as invariant subspaces

In this section, we are interested in M-cyclic codes as invariant subspaces, where we give their characterization by the theory of invariant subspaces. In [58], we investigate the structure of polycyclic codes as invariant subspaces.
Let $f(x)$ be the minimal polynomial of M, with its irreducible factorization given by
$$f(x) = \prod_{i=1}^{r} f_i^{\alpha_i}(x) \tag{2.5}$$
where $f_i(x)$ is an irreducible polynomial over $\mathbb{F}_q$. By the Cayley-Hamilton theorem [38, Theorem 4, P. 194], the matrix M satisfies
$$f(M) = 0$$

For $i = 1, \ldots, r$, let U_i be the null space of each $f_i^{\alpha_i}(x)$, i.e.,

$$U_i := \ker\left(f_i^{\alpha_i}(M)\right) = \left\{v \in \mathbb{F}_q^n \ : \ f^{\alpha_i}(M)\, v = 0\right\} \tag{2.6}$$

Theorem 2.4
The subspaces U_i of $\mathbb{F}_q^n$ satisfy the following conditions :

1. $\mathbb{F}_q^n = U_1 \oplus U_2 \oplus \ldots \oplus U_r$.
2. U_i is an invariant subspace of $\mathbb{F}_q^n$ by T_M.
3. If W is an invariant subspace of $\mathbb{F}_q^n$ by M and $W_i = W \cap U_i$, for $i = 1, \ldots, r$, then W_i is also invariant under M and $W = W_1 \oplus W_2 \oplus \ldots \oplus W_r$.
4. $\psi_f(U_i)$ is the ideal of $\mathbb{F}_q[x]/\langle f(x)\rangle$ generated by $\widehat{f_i^{\alpha_i}}(x)$, where $\widehat{f_i^{\alpha_i}}(x) = \dfrac{f(x)}{f_i^{\alpha_i}(x)}$.
5. The dimension of U_i is given by $\dim_{\mathbb{F}_q}(U_i) = \deg\left(f_i^{\alpha_i}\right) = \alpha_i \deg(f_i)$.
6. If $P(x)$ is a divisor of $f(x)$ in $\mathbb{F}_q[x]$, then $\ker(P(M))$ is a direct sum of invariant subspaces of $\mathbb{F}_q^n$.
7. If $\gcd(e, p) = 1$, then U_i is a minimal invariant subspace of $\mathbb{F}_q^n$ by T_M, where p is the characteristic of $\mathbb{F}_q$.
8. If $\gcd(e, p) = 1$ and $P(x)$ is a divisor of $f(x)$ in $\mathbb{F}_q[x]$, then $\ker(P(M))$ is a direct sum of minimal invariant subspaces of $\mathbb{F}_q^n$.

Proof

1. Since $f_1, f_2, \ldots, f_r$ are distinct monic irreducible polynomials, according to the primary decomposition theorem [38, Theorem 12, P. 220] we have that
$$\mathbb{F}_q^n = U_1 \oplus U_2 \oplus \ldots \oplus U_r.$$
2. Let $a \in U_i$. Then
$$f_i^{\alpha_i}(T_M(a)) = [f_i^{\alpha_i}(M)M]a = [Mf_i^{\alpha_i}(M)]a = 0.$$
It follows that $T_M(a) \in U_i$.
3. Since U_i is an invariant subspace, then $W_i = W \cap U_i$ is also an invariant subspace, for all $i = 1, \ldots, r$. From the statement 1, we have
$$W_1 \oplus W_2 \oplus \ldots \oplus W_r \subseteq \mathbb{F}_q^n \cap W = W \tag{2.7}$$
On the other hand $\gcd\left(\widehat{f_1^{\alpha_1}}(x), \widehat{f_2^{\alpha_2}}(x), \ldots, \widehat{f_r^{\alpha}}(x)\right) = 1$. By the Euclidean algorithm there are polynomials $(a_1(x), a_2(x), \ldots, a_r(x))$ such that
$$a_1(x)\widehat{f_1^{\alpha_1}}(x) + a_2(x)\widehat{f_2^{\alpha_2}}(x) + \ldots + a_r(x)\widehat{f_r^{\alpha_r}}(x) = 1$$
Then for every vector $w \in W$,
$$[a_1(M)\widehat{f_1^{\alpha_1}}(M)]w + [a_2(M)\widehat{f_2^{\alpha_2}}(M)]w + \ldots + [a_r(M)\widehat{f_r^{\alpha_r}}(M)]w = w$$
Let $w_i = \widehat{f_i^{\alpha_i}}(M)w$. Then $[f_i^{\alpha_i}(M)]w_i = [f_i^{\alpha_i}(M)\widehat{f_i^{\alpha_i}}(M)]w = [f(M)]w = 0$. Hence $w_i \in U_i \cap W = W_i$. Finally we show that
$$W_1 \oplus W_2 \oplus \ldots . \oplus W_r = W$$

4. Since U_i is an invariant subspace, the result follows by Theorem 2.2.

5. As $\psi(U_i) = \langle \widehat{f_i^{\alpha_i}}(x) \rangle$, by a similar argument as in Theorem 2.3

$$\dim_{\mathbb{F}_q}(U_i) = n - \deg(\widehat{f_i^{\alpha_i}}(x)) = \alpha_i \deg(f_i).$$

6. In statement 3, take $W = \ker(P(M)))$.

7. If $\gcd(e, n) = 1$, then $\alpha_i = 1$ in (2.5) for each $i = 1, \ldots, r$. Let $U \subseteq U_i$. The polynomial $f_i^{\alpha_i}(x) = f_i(x)$ is irreducible, therefore $\dim_{\mathbb{F}_q}(U) = \dim_{\mathbb{F}_q}(U_i)$ and $U = U_i$.

8. It is immediate from the above statement.

According to the above theorem, we have the following characterization of M-cyclic codes.

Theorem 2.5
Let C be an M-cyclic code of length n over $\mathbb{F}_q$ generated by $g(x) = \prod_{i=1}^{r} f_i^{k_i}(x)$, $0 \leq k_i \leq \alpha_i$ and $h(x) \in \mathbb{F}_q[x]$ be such that $f(x) = h(x)g(x)$. Then

1. $\psi_f(C) = \psi_f(\ker(h(M)))$
2. $C = \ker(h(M))$ and $\dim_{\mathbb{F}_q}(C) = \operatorname{rank}(h(M)) = \deg(g)$.
3. $C = C_1 \oplus C_2 \oplus \ldots \oplus C_r$ where C_i is an M-cyclic code generated by $g_i(x) = \operatorname{lcm}\left(g(x), \widehat{f_i^{\alpha_i}}(x)\right) = f_i^{k_i}(x)\widehat{f_i^{\alpha_i}}(x)$, for all $i = 1, \ldots, r$.
4. If $\gcd(\operatorname{ord}(f), p) = 1$, then $C = C_1 \oplus C_2 \oplus \ldots \oplus C_r$, where C_i is a minimal M-cyclic code.

Proof
1. Let $c(x) = \sum_{i=0}^{n-1} c_i x^i \in \psi_f(C)$. Since $\psi_f(C) = \langle g(x) \rangle$, there is $a(x) \in \mathbb{F}_q[x]$ such that $c(x) = a(x)g(x)$. It follows that

$$h(x)c(x) = h(x)a(x)g(x) = 0 \bmod f(x).$$

Then

$$h(x)c(x) = \sum_{i=0}^{\deg(h)} h_i x^i c(x) = \sum_{i=0}^{\deg(h)} h_i \psi_f\left(T_M^i(c)\right) = \psi_f\left(\sum_{i=0}^{\deg(h)} h_i T_M^i(c)\right) = \psi_f\left([h(M)]c\right)$$

Hence, $h(M)c = 0$, which means that $c \in \ker(h(M))$ and $\psi_f(C) \subseteq \psi_f(\ker(h(M)))$. Conversely, let $a(x) \in \psi_f(\ker(h(M)))$. Then there exist $q(x), r(x) \in \mathbb{F}_q[x]$, such that

$$a(x) = q(x)g(x) + r(x), \text{ where } \deg(r(x)) < \deg(g(x)).$$

Then $h(x)r(x) = 0 \bmod f(x)$. Assume that $r(x) \neq 0$. Let $K(x) \in \mathbb{F}_q[x]$ be such that $h(x)r(x) = K(x)f(x)$. Since $f(x)$ is monic, then

$$\deg(h(x)r(x)) = \deg(K(x)f(x)) = \deg(K(x)) + \deg(f(x)) \geq n.$$

Otherwise, as $h(x)$ is monic we have

$$\deg(h(x)r(x)) = \deg(h) + \deg(r(x)) < n,$$

a contradiction. Hence $r(x)) = 0$ and $a(x) \in \psi_f(C)$.

2. The result follows from the statement above.

3. Let $C_i = C \cap U_i$, for all $i = 1, \ldots, r$. Then the M-cyclic code C_i is generated by

$$g_i(x) = \text{lcm}\left(g(x), \widehat{f_i^{\alpha_i}}(x)\right) = f_i^{k_i}(x)\widehat{f_i^{\alpha_i}}(x)$$

   By statement 3 of Theorem 2.4, we have that $C = C_1 \oplus C_2 \oplus \ldots \oplus C_r$.

4. It is immediate from the statement above.

2.6 Dual of linear codes invariant under cyclic operators

In this section, we are interested in the duals of T-cyclic codes with respect to a non-degenerate bilinear form b on $\mathbb{F}_q^n$. Let $b : \mathbb{F}_q^n \times \mathbb{F}_q^n \longrightarrow \mathbb{F}_q$ be a non-degenerate bilinear form on $\mathbb{F}_q^n$. Then the b-dual code of a linear code $C \subseteq \mathbb{F}_q^n$ is defined by

$$C^{\perp b} := \left\{ x \in \mathbb{F}_q^n \quad : \quad b(x, y) = 0, \quad \forall y \in C \right\}.$$

When b is the Euclidean inner product on $\mathbb{F}_q^n$ we will denote $C^{\perp b}$ by $C^{\perp}$. Recall that if b is a non-degenerate bilinear form on $\mathbb{F}_q^n$, then the adjoint operator of T with respect to b is the unique operator T^* satisfying

$$b(x, T^*(y)) = b(T(x), y).$$

Note that T^* can be represented by different matrices with respect to different bases of $\mathbb{F}_q^n$ and all of these matrices are similar to each other by the transition matrices between these bases. For simplicity, if M is the representation matrix of T with respect to a given basis B, we denote by M^* the representation matrix of T^* in the same basis B. Now, the following theorem is easy to prove.

Theorem 2.6
If $C \subseteq \mathbb{F}_q^n$ is an M-code, then $C^{\perp b}$ is an M^*-code.

Proof
Let C be an M-code, $c \in C$ and $v \in C^{\perp b}$ then

$$b(c, T_{M^*}(v)) = b(T_M(c), v) = 0, \quad \text{since} \quad T_M(c) \in C. \tag{2.8}$$

Thus, $T_{M^*}(v) \in C^{\perp b}$, which means that $C^{\perp b}$ is a M^*-code.

From the above theorem we easily deduce the following result.

Corollary 2.2
Let $C \subseteq \mathbb{F}_q^n$ be a linear code and M a cyclic matrix with minimal polynomial $f(x)$. The following assertions are equivalent.

1. C is an M-cyclic code.
2. $C^{\perp b}$ is an M^*-cyclic code.
3. $P^{-1}C^{\perp b}$ is an f-cyclic code where $C_f = P^{-1}M^*P$.

Let us recall that if M and M' commute that is, $MM' = M'M$, then each invariant subspace of $\mathbb{F}_q^n$ under M is also invariant under $T_{M'}$. Denote by $\mathrm{Z}(M)$ be the set of all matrices which commute with M, i.e.,

$$\mathrm{Z}(M) = \left\{ M' \in \mathbb{M}_n(\mathbb{F}_q) \ : \ MM' = M'M \right\}.$$

We deduce the following result.

Theorem 2.7
Let $M \in \mathbb{M}_n(\mathbb{F}_q)$ be a cyclic matrix, b be a non-degenerate bilinear form of $\mathbb{F}_q^n$ and C be an M-cyclic code of length n over $\mathbb{F}_q$. Then for each $M' \in \mathrm{Z}(M^*)$ we have that $C^{\perp_b}$ is an M'-code and an M'-cyclic code if M' is a cyclic matrix. Moreover,

1. If M is symmetric (i.e., $M^* = M$), then the b-dual code $C^{\perp_b}$ is also an M-cyclic code.
2. If M is orthogonal (i.e., $MM^* = M^*M = \mathrm{I}_n$), then $C^{\perp_b}$ is both an M^{-1}-cyclic code and an M-cyclic code.

Proof
If $M' \in \mathrm{Z}(M^*)$ then M' and M^* have the same invariant subspaces. Since $C^{\perp_b}$ is invariant under M^* we deduce that $C^{\perp_b}$ is invariant under M', and so it is an M'-code.

1. As M is normal, $M = M^*$, hence $C^{\perp_b}$ is invariant under M which means that $C^{\perp_b}$ is an M-cyclic code.
2. If M is orthogonal, then $M^* = M^{-1}$ and $C^{\perp_b}$ is an M^{-1}-cyclic code. On the other hand, as $M \in \mathrm{Z}(M^{-1})$, we obtain that $C^{\perp_b}$ is also an M-cyclic code.

As an application of the above theorem, we obtain some well-known results on the duality of cyclic codes and their generalization of constacyclic codes and polycyclic codes.

Corollary 2.3
1. The Euclidean dual of a λ-constacyclic code is a λ^{-1}-constacyclic code.
2. The Euclidean dual of a cyclic (resp. negacyclic) code is also a cyclic (resp. negacyclic) code.
3. The Euclidean dual of a monomial code induced by a vector $a = (a_0, a_1, \ldots, a_{n-1})$ is a monomial code induced by the vector $b = (a_0^{-1}, a_1^{-1}, \ldots, a_{n-1}^{-1})$.

Proof
1. Follows from Theorem 2.7 and the fact $M_\lambda^t M_{\lambda^{-1}} = M_{\lambda^{-1}} M_\lambda^t$, where M_λ the companion matrix of $x^n - \lambda$.
2. Follows from the above statement when $\lambda = \pm 1$.
3. Note that the Euclidean adjoint matrix of $M_{\overline{a}}$ is $M_{\overline{b}}$, with $b = (a_0^{-1}, a_1^{-1}, \ldots, a_{n-1}^{-1})$. We verify by computation that $M_{\overline{b}} \in Z(M_{\overline{a}}^t)$, where $M_{\overline{a}}$ is the monomial matrix as in Proposition 2.4. Therefore, by applying Theorem 2.7, the result holds.

2.7 Bounds on the Hamming distance of f-cyclic codes

Let $f(x) \in \mathbb{F}_q[x]$ with $f(0) \neq 0$, and let e be its order such that $\gcd(e, q) = 1$. Then all roots of $f(x)$ are simple, and can be expressed as powers of a primitive e^{th} root of unity α in a field extension $\mathbb{F}_{q^m}$, where m is the multiplicative order of q modulo e.

Let C be an f-cyclic code generated by $g(x)$ i.e., $\varphi_f(C) = \langle g(x) \rangle$, and define $V(C)$ to be the set containing all roots of $g(x)$ i.e.,

$$V(C) = \left\{ \alpha^i \in \mathbb{F}_{q^m} \ : \ g(\alpha^i) = 0 \right\}.$$

Following [39], in the following theorems we prove the BCH-like and Hartmann-Tzeng-like bounds for f-cyclic codes. Note that similar bounds were proven in [85, 51, 58], for right polycyclic codes. Here we prove them differently for f-cyclic codes.

Theorem 2.8 (BCH-like bound for f-cyclic codes)
Let $C \subseteq \mathbb{F}_q^n$ be an f-cyclic code generated by $g(x)$ such that $V(C)$ contains the set

$$T_g = \left\{ \alpha^{a+ib} \quad : \quad i = 0, \ldots, \delta - 2 \right\}$$

where, $\gcd(e, b) = 1$. Then C has minimum Hamming distance $d \geq \delta$.

Proof
Suppose that $c \in C$ is a codeword of weight $w < \delta$. Since C is f-cyclic, we can assume $c(x) := \varphi_f(c) \in \varphi_f(C)$ is of the form

$$c(x) = c_0 + \sum_{i=1}^{w-1} c_i x^{k_i} \quad \text{with} \quad c_0 \neq 0 \text{ and } \{k_0, \ldots, k_{w-1}\} \subseteq \{1, \ldots, n-1\}.$$

Following [39], we set for each integer j

$$S_j = \sum_{i=1}^{w-1} c_i \alpha^{k_i j} = c(\alpha^j) - c_0$$

Define the polynomial $p(x)$ by:

$$P(x) = \prod_{i=1}^{w-1} (x - \alpha^{k_i b}) = \sum_{i=0}^{w-1} P_i x^i \in \mathbb{F}_q[x].$$

Note that $P(1) \neq 0$, since α is a primitive e^{th} root of unity and $\gcd(e, b) = 1$.
On the other hand, as $\alpha^{k_i b}, i = 1, \ldots, w-1$ are roots of $p(x)$, we have

$$\begin{aligned}
0 &= \sum_{i=1}^{w-1} c_i \alpha^{k_i a} P\left(\alpha^{k_i b}\right) \\
&= \sum_{i=1}^{w-1} c_i \alpha^{k_i a} \sum_{j=0}^{w-1} P_j \alpha^{b k_i j} \\
&= \sum_{j=0}^{w-1} P_j \sum_{i=1}^{w-1} c_i \alpha^{k_i (a+jb)} \\
&= \sum_{j=0}^{w-1} P_j S_{a+jb} \\
&= -\sum_{j=0}^{w-1} P_j c_0 \quad \text{since} \quad S_j = -c_0 \text{ for } j \in V(C) \\
&= -c_0 \sum_{j=0}^{w-1} P_j \\
&= -c_0 P(1)
\end{aligned} \tag{2.9}$$

Since $P(1) \neq 0$, we obtain a contradiction. It follows that C does not contain any nonzero codeword of weight less than δ. Hence, $d \geq \delta$.

Now, by a similar argument as in the proof of the above theorem, we prove the Hartmann-Tzeng-like bound for f-cyclic codes.

Theorem 2.9 (Hartmann-Tzeng-like bound for f-cyclic codes)
Let C be an f-cyclic code of length n over $\mathbb{F}_q$ such that $V(C)$ contains the set

$$T_g = \left\{ \alpha^{a+ib+jc} \quad : \quad i = 0, \ldots, \delta - 2, j = 0, \ldots, s \right\}$$

where $\gcd(e, b) = 1$, and $\gcd(e, c) = 1$. Then C has minimum Hamming distance $d \geq \delta + s$.

Proof
Let $c \in C$ be a codeword of weight w. Then by the above theorem $\delta \leq w$. Assume that $\delta \leq w < \delta + s$, and $c(x) := \varphi_f(c) \in \varphi_f(C)$ is of the form

$$c(x) := c_0 + \sum_{i=1}^{w-1} c_i x^{k_i} \in \varphi_f(C), \quad \text{with} \quad c_0 \neq 0 \text{ and } \{k_0, \ldots, k_{w-1}\} \subseteq \{1, \ldots, n-1\}.$$

For each positive integer j, let

$$S_j = \sum_{i=1}^{w-1} c_i \alpha^{k_i j} = c(\alpha^j) - c_0$$

and

$$P(x) = P^{(1)}(x) P^{(2)}(x) \in \mathbb{F}_q[x]$$

where

$$P^{(1)}(x) = \prod_{i=1}^{\delta-2} (x - \alpha^{k_i b}) = \sum_{i=0}^{\delta-1} P_i^{(2)} x^i \quad \text{and} \quad P^{(2)}(x) = \prod_{j=\delta-1}^{w-1} (x - \alpha^{k_j c}) = \sum_{j=0}^{w-\delta+1} P_j^{(2)} x^j$$

It follows that $P^{(1)}(1) \neq 0$ and $P^{(2)}(1) \neq 0$, and so $p(1) \neq 0$, since $\gcd(e, b) = \gcd(e, c) = 1$. Again as in the proof of the above theorem we have:

$$\begin{aligned}
0 &= \sum_{i=1}^{w-1} c_i \alpha^{k_i a} P^{(1)}(\alpha^{k_j b}) P^{(2)}(\alpha^{k_i c}) \\
&= \sum_{i=1}^{w-1} c_i \alpha^{k_i a} \left(\sum_{l=0}^{\delta-2} P_l^{(1)} \alpha^{k_i bl} \right) \left(\sum_{h=0}^{w-\delta+1} P_h^{(2)} \alpha^{k_i ch} \right) \\
&= \sum_{i=1}^{w-1} c_i \alpha^{k_i a} \left[\sum_{l=0}^{\delta-2} P_l^{(1)} \left(\sum_{h=0}^{w-\delta+1} P_h^{(2)} \alpha^{k_i (bl+ch)} \right) \right] \\
&= \sum_{l=0}^{\delta-2} P_l^{(1)} \sum_{h=0}^{w-\delta+1} P_h^{(2)} \left(\sum_{i=1}^{w-1} c_i \alpha^{k_i (bl+ch)} \right) \\
&= \sum_{l=0}^{\delta-2} P_l^{(1)} \sum_{h=0}^{w-\delta+1} P_h^{(2)} S_{bl+ch} \\
&= \sum_{l=0}^{\delta-2} P_l^{(1)} \sum_{h=0}^{w-\delta+1} P_h^{(2)} c_0, \quad \text{since} \quad S_j = -c_0 \text{ for } j \in V(C) \\
&= -c_0 p(1)
\end{aligned} \tag{2.10}$$

Then as $p(1) \neq 0$, we obtain a contradiction. It follows that C does not contain any nonzero codeword of weight less than $\delta + s$. Hence, $d \geq \delta + s$.

By an inductive argument, we can easily generalize the previous theorem as follows.

Corollary 2.4
Let C be an f-cyclic code of length n over $\mathbb{F}_q$ such that $V(C)$ contains the set

$$T_g = \left\{ \alpha^{a+ib+\sum_{k=1}^{r} j_k c_k} \quad \text{for} \quad i = 0, \ldots, \delta-2, \; j_k = 1, \ldots, s_k \right\}$$

where $\gcd(e, b) = 1$ and $\gcd(e, c_k) = 1$ for each $j_k = 1, \ldots, s_k$. Then, $d \geq \delta + \sum_{k=1}^{r} s_k$.

Now we give a definition of a BCH f-cyclic code with designed distance δ.

Definition 2.6
Let C be an f-cyclic code of length n over $\mathbb{F}_q$ generated by $g(x)$ and let δ be a positive integer. We say that C is a BCH f-cyclic code with designed distance δ, if $V(C)$ contains the set $\left\{\alpha^{a+ib} \mid i = 0, \ldots, \delta - 2\right\}$ where $\gcd(e, b) = 1$, i.e., C satisfies the BCH-like bound.

Example 2.5 (BCH-like bound)
Let $q = 9$ and $\mathbb{F}_9 = \mathbb{F}_3(\alpha)$, with $\alpha^2 + 2\alpha + 2 = 0$ and $f(x) = x^{11} + 2x^9 + 2x^3 + 2$. The order of $f(x)$ over $\mathbb{F}_9$ is $e = 242$, and the multiplicative order of 242 modulo 9 equals 5. It follows that the field $\mathbb{F}_{9^5}$ contains a primitive e^{th} root of unity β. The irreducible factorization of $f(x)$ is given by:

$$\begin{aligned} f(x) = &\ (x+1)\cdot(x^5 + x^3 + x + 2)\cdot(x^5 + 2x^4 + 2x^3 + x^2 + 1) \\ = &\ f_1(x)\cdot f_2(x)\cdot f_3(x) \end{aligned}$$

and the 9-cyclotomic cosets modulo 242 that correspond to the irreducible factors of $f(x)$ are:

$$C_{121} = \{121\}\,,\ C_2 = \{2, 6, 18, 54, 162\}\,,\ C_{17} = \{17, 51, 153, 167, 217\}\,.$$

It follows that the set S, which correspond to all roots of $f(x)$ is

$$\mathrm{S} = \{2, 6, 17, 18, 51, 54, 121, 153, 162, 167, 217\}\,.$$

For example the f-cyclic code $C \subseteq \mathbb{F}_9^{11}$ generated by $g(x) = \langle x^6 + x^5 + x^4 + x^3 + x^2 + 2\rangle$ is a BCH f-cyclic code with designed distance $\delta = 3$, since β^{18} and $\beta^{121} = \beta^{18+103}$ are among the roots of $g(x)$. Its actual distance is $d = 4$ and it is a $[11, 5, 4]_9$ linear code.

Example 2.6
Let $q = 9$ and $\mathbb{F}_9 = \mathbb{F}_3(\alpha)$, and $f(x) = x^{11} + x^5 + x^4 + 2$. The order of $f(x)$ over $\mathbb{F}_9$ is $e = 364$, and the multiplicative order of 364 modulo 9 equals 3. It follows that the field $\mathbb{F}_{9^3}$ contains a primitive e^{th} root of unity β. The irreducible factorization of $f(x)$ is given by:

$$\begin{aligned} f(x) = &\ (x+\alpha+1)\cdot(x+2\alpha+2)\cdot(x^3 + x^2 + x + 2)\cdot(x^3 + x^2 + \alpha x + 1)\cdot(x^3 + x^2 + (2\alpha+1)\,x + 1) \\ = &\ f_1(x)\cdot f_2(x)\cdot f_3(x)\cdot f_4(x)\cdot f_5(x) \end{aligned}$$

and the 9-cyclotomic cosets modulo 364 that correspond to the irreducible factors of $f(x)$ are:

$$C_{91} = \{91\}\ ;\ C_{273} = \{273\}\ ;\ C_{28} = \{28, 84, 252\}\ ;\ C_{30} = \{30, 246, 270\}\ ;\ C_{10} = \{10, 82, 90\}\,.$$

It follows that the set R, which corresponds to all roots of $f(x)$ is

$$R = \{10, 28, 30, 82, 84, 90, 91, 246, 252, 270, 273\}\,.$$

In the following table, we list all possible subsets of R containing consecutive integers in Theorem 2.8 with the condition that its associated polynomial $g(x) = \mathrm{lcm}(m_{\beta^i}(x))$, for $i \in K$ and $M_{\beta^i}(x)$ is the minimal polynomial of β^i, is a divisor of $f(x)$. Hence we construct all possible BCH f-cyclic codes.

δ	a	b	The set T_g	Generator polynomial	Parameters
2	10	–	$\{10\}$	$x^3 + x^2 + (2\alpha + 1)\, x + 1$	$[11, 8, 3]_9$
	28	–	$\{28\}$	$x^3 + x^2 + x + 2$	$[11, 8, 3]_9$
	30	–	$\{30\}$	$x^3 + x^2 + \alpha x + 1$	$[11, 8, 3]_9$
	91	–	$\{91\}$	$x + \alpha + 1$	$[11, 10, 2]_9$
	273	–	$\{273\}$	$x + 2\alpha + 2$	$[11, 10, 2]_9$
3	10	81	$\{10, 91\}$	$x^4 + (\alpha + 2)x^3 + 2x^2 + (2\alpha + 1)x + \alpha + 1$	$[11, 7, 3]_9$
	10	263	$\{10, 273\}$	$x^4 + 2\alpha x^3 + \alpha x^2 + (\alpha + 1)x + 2\alpha + 2$	$[11, 7, 3]_9$
	30	61	$\{30, 91\}$	$x^4 + (\alpha + 2)x^3 + (2\alpha + 1)x^2 + (2\alpha + 2)x + \alpha + 1$	$[11, 7, 4]_9$
	30	243	$\{30, 273\}$	$x^4 + 2\alpha x^3 + 2x^2 + \alpha x + 2\alpha + 2$	$[11, 7, 3]_9$

Table 2.1: BCH $f-$cyclic codes of length 11 over $\mathbb{F}_9$

Example 2.7 (BCH-like bound of right polycyclic codes)
Let $R = (1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 0) \in \mathbb{F}_8^{11}$, then $R(x) = x^8 + x^7 + x^5 + x^3 + x^2 + x + 1$. The order of $f(x) = x^{11} - R(x)$ over $\mathbb{F}_8$ is $e = 15$, the multiplicative order of 15 modulo 8 is $m = 4$ and the field extension $\mathbb{F}_{8^4} = \mathbb{F}_8(\alpha)$ contains the eth primitive root of unity $\beta = \alpha^{\frac{8^4 - 1}{15}} = \alpha^{273}$.
It follows that the 8−cyclotomic cosets modulo 15 are:

$$C_0 = \{0\}\,; C_1 = \{1, 2, 4, 8\}\,; C_3 = \{3, 6, 9, 12\}\,; C_5 = \{5, 10\}\,; C_7 = \{7, 11, 13, 14\}\,.$$

In the following table, we list all the BCH-like right polycyclic codes with associate vector $R \in \mathbb{F}_8^{11}$ of length 11 and all the BCH codes of length 11 over $\mathbb{F}_8$.

2.8 Examples

In this section, we present a few examples of f-cyclic codes with good parameters. These examples of codes are optimal and most of them are presented in the online database of best-known codes [27]. Moreover, some of them cannot be obtained as constacyclic, cyclic, and negacyclic codes. Considering the class of f-cyclic codes in a computer search to discover new linear codes is both more promising and more challenging than in the constacyclic case, because the search space is larger.

Example 2.8
In Table 2.3, we list some optimal or best-known linear codes of different lengths over $\mathbb{F}_3$ that are f-cyclic. However, the parameters of these codes are not new. They may now be regarded as f-cyclic codes.

	δ	a	b	Defining consecutive set T_g	Generator polynomial	k	d_H
Right polycyclic codes	2	0	$--$	$\{0\}$	$x+1$	10	2
		11	$--$	$\{11\}$	x^4+x^3+1	7	3
		9	$--$	$\{9\}$	$x^4+x^3+x^2+x+1$	7	2
		10	$--$	$\{10\}$	x^2+x+1	9	2
	3	0	7	$\{0,7\}$	x^5+x^3+x+1	6	4
		3	4	$\{3,7\}$	$x^8+x^4+x^2+x+1$	3	5
		3	7	$\{3,10\}$	$x^6+x^4+x^3+x^2+1$	5	4
		5	2	$\{5,7\}$	$x^6+x^3+x^2+x+1$	5	3
		13	1	$\{13,14\}$	x^4+x^3+1	7	3
	4	0	7	$\{0,7,14\}$	x^5+x^3+x+1	6	4
		3	2	$\{3,5,7\}$	$x^{10}+x^9+x^8+x^6+x^5+x^2+1$	1	7
		7	2	$\{7,9,11\}$	$x^8+x^4+x^2+x+1$	3	5
	5	9	1	$\{9,10,11,12\}$	$x^{10}+x^9+x^8+x^6+x^5+x^2+1$	1	7
		11	1	$\{11,12,13,14\}$	$x^8+x^4+x^2+x+1$	3	5
	6	5	2	$\{5,7,9,11,13\}$	$x^{10}+x^9+x^8+x^6+x^5+x^2+1$	1	7
	7	3	2	$\{3,5,7,9,11,13\}$	$x^{10}+x^9+x^8+x^6+x^5+x^2+1$	1	7
Cyclic codes	2	0	$--$	$\{0\}$	$x+1$	10	2
	11	1	1	$\{1,2,3,4,5,6,7,8,9,10\}$	$x^{10}+x^9+x^8+x^7+x^6+x^5+$ $x^4+x^3+x^2+x+1$	1	11

Table 2.2: BCH-like bound of right polycyclic codes

Example 2.9 (f-cyclic codes)
Let $q=2,\ n=50$ and $f(x)=x^{50}+x^{49}+x^{38}+x^{36}+x^{34}+x^{32}+x^{29}+x^{27}+x^{26}+x^{25}+x^{23}+x^{22}+x^{18}+x^{17}+x^{15}+x^9+x^6+1$. The irreducible factorization of $f(x)$ is given by :

$$\begin{aligned} f(x) = \ & (x+1)^3\cdot(x^3+x^2+1)\cdot(x^4+x+1)\cdot(x^9+x^8+x^6+x^4+x^3+x+1)\ldots. \\ & \ldots(x^{12}+x^{11}+x^7+x^5+1)\cdot(x^{19}+x^{18}+x^{17}+x^{13}+x^9+x^6+x^4+x+1). \end{aligned}$$

In Table 2.4, we list some optimal or best-known linear codes of length 50 over $\mathbb{F}_2$ that are f-cyclic

Length	The polynomials $f(x)$	Generator polynomial of the code	Parameters
$n = 60$	$f(x) = x^{60} + x^{29} + 2x^{28} + x^{27} + x^{26} +$ $x^{20} + x^{19} + 2x^{17} + x^{15} + 2x^{14} + 2x^{11} +$ $2x^{10} + 2x^9 + 2x^6 + 2x^5 + 2x^3 + 2$	$x^8 + 2x^6 + x^5 + x + 1$	$[60, 52, 4]_3$
		$x^7 + x^6 + x^4 + x^3 + x^2 + x + 2$	$[60, 53, 4]_3$
		$x^6 + 2x^2 + 2x + 1$	$[60, 54, 3]_3$
$n = 30$	$f(x) = x^{30} + x^{25} + x^{24} + 2x^{22} + x^{21} + 2x^{17} + 2x^{16} + 2x^{12}$ $+x^{11} + 2x^{10} + 2x^9 + x^7 + x^6 + x^5 + x^3 + 2$	$x^{11} + 2x^9 + x^8 + 2x^7 + x^6 + x^3 + x^2 + 1$	$[30, 19, 6]_3$
		$x^9 + x^8 + x^4 + 2x^2 + 2x + 1$	$[30, 21, 5]_3$
		$x^5 + x^3 + 2$	$[30, 25, 3]_3$
		$x^4 + 2x^3 + 2x^2 + x + 2$	$[30, 26, 3]_3$
	$f(x) = x^{30} + x^{23} + x^{15} + x^{12} + 2x^8 + 1$	$x^6 + 2x^5 + 2x^4 + 2x + 1$	$[30, 24, 4]_3$
		$x^3 + 2x^2 + 2$	$[30, 27, 2]_3$
		$x^2 + x + 2$	$[30, 28, 2]_3$
	$f(x) = x^{30} + x^{28} + 2x^{22} + 2x^{18} + 2x^{16} + 2x^{12} +$ $2x^{11} + 2x^9 + 2x^6 + x^4 + x^2 + x + 2$	$x^7 + 2x^6 + x^5 + x^4 + x^3 + x + 2$	$[30, 23, 4]_3$
$n = 20$	$f(x) = 2x^{20} + 2x^{19} + x^{18} + 2x^{17} + 2x^{16} + 2x^{15} + 2x^{11}$ $+2x^{10} + x^9 + x^8 + 2x^7 + x^6 + x^5 + 2x^4 + 2x^3 + x^2 + 2$	$x^6 + x^5 + x^4 + 2x + 2$	$[20, 14, 4]_3$
		$x^4 + x^3 + x^2 + 2$	$[20, 16, 3]_3$
		$x^3 + x + 2$	$[20, 17, 2]_3$
		$x^2 + 2x + 2$	$[20, 18, 2]_3$

Table 2.3: Some optimal or best known f-cyclic codes over $\mathbb{F}_3$.

Length	Generator polynomial	Parameters
$n = 50$	$x^{47} + x^{44} + x^{43} + x^{42} + x^{40} + x^{37} + x^{36} + x^{34} + x^{33} + x^{32} + x^{31} + x^{28} + x^{27} + \ldots$ $\ldots + x^{25} + x^{24}x^{21} + x^{19} + x^{16} + x^{14} + x^{12} + x^9 + x^8 + x^7 + x^6 + x^4 + x^3 + x^2 + 1$	$[50, 3, 28]_2$
	$x^{22} + x^{21} + x^{20} + x^{17} + x^{16} + x^{15} + x^{14} + x^{13} + x^{12} + x^{10} + x^9 + x^7 + x^6 + x^3 + x^2 + 1$	$[50, 28, 8]_2$
	$x^{16} + x^{12} + x^9 + x^8 + x^6 + x^5 + x^4 + x^3 + x + 1$	$[50, 34, 6]_2$
	$x^{15} + x^{14} + x^{13} + x^{12} + x^8 + x^5 + x^3 + 1$	$[50, 35, 6]_2$
	$x^{14} + x^{13} + x^{12} + x^{11} + x^9 + x^5 + x^2 + 1$	$[50, 36, 6]_2$
	$x^{11} + x^{10} + x^9 + x^5 + x^4 + x^2 + x + 1$	$[50, 39, 4]_2$
	$x^{10} + x^6 + x^5 + x^3 + x^2 + 1$	$[50, 40, 4]_2$
	$x^9 + x^8 + x^7 + x^6 + x^4 + x^3 + x + 1$	$[50, 41, 4]_2$
	$x^8 + x^6 + x^3 + 1$	$[50, 42, 4]_2$
	$x^7 + x^6 + x^5 + 1$	$[50, 43, 4]_2$
	$x^5 + x^4 + x^2 + 1$	$[50, 45, 2]_2$
	$x^4 + x + 1$	$[50, 46, 2]_2$
	$x^3 + x^2 + x + 1$	$[50, 47, 2]_2$
	$x + 1$	$[50, 49, 2]_2$

Table 2.4: Some optimal or best known f-cyclic codes over $\mathbb{F}_2$.

Example 2.10
In Table 2.5, we list some MDS codes of length $n = 3, 4, \ldots, 10$ over $\mathbb{F}_{27}$ of different dimensions, except four cases that we mark with $[n, k, d]^*_{27}$ and which satisfy $n - k = d$. The polynomial used to generate $\mathbb{F}_{27} = \mathbb{F}_3(\alpha)$ is $x^3 + 2x + 1$.

Length	The polynomial $f(x)$	Generator polynomial of the $f-$cyclic code	Parameters
$n=3$	$f(x)=x^3+2$	x^2+x+1	$[3,1,3]_{27}$
		$x+2$	$[3,2,2]_{27}$
$n=4$	$f(x)=x^4+\alpha x^3+\alpha$	$x^3+\alpha^2x^2+\left(\alpha^2+\alpha+1\right)x+\alpha^2+\alpha$	$[4,1,4]_{27}$
		$x^2+\left(2\alpha^2+\alpha+2\right)x+\alpha^2+2$	$[4,2,3]_{27}$
		$x+2\alpha^2+\alpha$	$[4,3,2]_{27}$
$n=5$	$f(x)=x^5+\alpha x^3+1$	$x^4+\alpha x^3+\alpha^{10}x^2+\alpha^{11}x+\alpha^{12}$	$[5,1,5]_{27}$
	$f(x)=x^5+\alpha x^2+\alpha$	$x^3+\alpha^{16}x^2+\alpha x+\alpha^6$	$[5,2,4]_{27}$
		$x^2+\alpha^3x+\alpha^{21}$	$[5,3,3]_{27}$
	$f(x)=x^5+\alpha x^3+1$	$x+\alpha^{14}$	$[5,4,2]_{27}$
$n=6$	$f(x)=x^6+\alpha x^2+\alpha$	$x^5+\alpha^{20}x^4+\alpha^{14}x^3+\alpha^5x^2+\alpha^{25}x+\alpha^{19}$	$[6,1,6]_{27}$
		$x^4+\alpha^{22}x^3+\alpha^4x^2+\alpha^{22}x+1$	$[6,2,5]_{27}$
		$x^3+\alpha^2x^2+\alpha^{22}x+\alpha^7$	$[6,3,4]_{27}$
		$x^2+\alpha^9x+1$	$[6,4,3]_{27}$
		$x+\alpha^{19}$	$[6,5,2]_{27}$
$n=7$	$f(x)=x^7+\alpha x^6+\alpha x^3+2$	$x^6+\alpha^{24}x^5+\alpha^{18}x^4+\alpha^{12}x^3+\alpha^{18}x^2+\alpha^{12}x+\alpha^6$	$[7,1,7]_{27}$
		$x^5+\alpha^{25}x^4+\alpha^7x^3+x^2+\alpha^{21}x+\alpha^{18}$	$[7,2,6]_{27}$
		$x^4+\alpha^{10}x^3+\alpha^8x^2+\alpha^{23}x+\alpha^{17}$	$[7,3,5]_{27}$
		$x^3+\alpha^{16}x^2+\alpha^3x+\alpha^{10}$	$[7,4,4]_{27}$
		$x^2+\alpha^{11}x+\alpha^{21}$	$[7,5,3]_{27}$
		$x+\alpha^{14}$	$[7,6,2]_{27}$
$n=8$	$f(x)=x^8+\alpha^3x^3+1$	$x^7+\alpha^2x^6+\alpha^4x^5+\alpha^6x^4+\alpha^8x^3+\alpha^7x^2+\alpha^9x+\alpha^{11}$	$[8,1,8]_{27}$
		$x^6+\alpha^{11}x^5+\alpha^{17}x^4+\alpha^6x^3+\alpha^{10}x^2+\alpha^{15}x+\alpha^{15}$	$[8,2,7]_{27}$
		$x^5+\alpha^{11}x^4+\alpha^8x^3+\alpha^9x^2+\alpha^{25}x+\alpha^{17}$	$[8,3,6]_{27}$
		$x^4+\alpha^{18}x^3+\alpha^2x^2+\alpha^{24}x+\alpha^{21}$	$[8,4,5]_{27}$
		$x^3+\alpha^{18}x^2+\alpha^6x+1$	$[8,5,4]_{27}$
		$x^2+\alpha^{24}x+\alpha^{11}$	$[8,6,3]_{27}$
		$x+\alpha^{15}$	$[8,7,2]_{27}$
$n=9$	$f(x)=x^9+x^4+\alpha x+1$	$x^8+\alpha^{19}x^7+\alpha^{12}x^6+\alpha^5x^5+\alpha^{24}x^4+\alpha^{20}x^3+2x^2+\alpha^6x+\alpha^{20}$	$[9,1,9]_{27}$
		$x^7+\alpha^{25}x^6+\alpha^{12}x^5+x^4+\alpha^3x^3+\alpha^3x^2+\alpha^{14}x+\alpha^{19}$	$[9,2,8]_{27}$
		$x^6+x^5+\alpha^8x^4+\alpha^6x^3+\alpha^{20}x^2+x+\alpha^4$	$[9,3,7]_{27}$
		$x^4+\alpha^9x^3+\alpha^7x^2+\alpha^6x+1$	$[9,4,5]^*_{27}$
		$x^4+\alpha^{10}x^3+\alpha^{16}x^2+\alpha^{10}x+\alpha^2$	$[9,5,5]_{27}$
		$x^3+\alpha^{21}x^2+\alpha^2x+\alpha^{25}$	$[9,6,4]_{27}$
		$x^2+\alpha^{14}x+\alpha^{19}$	$[9,7,3]_{27}$
		$x+\alpha^3$	$[9,8,2]_{27}$
$n=10$	$f(x)=x^{10}+x^8+x^7+\alpha x^3+1$	$x^9+\alpha^{20}x^8+\alpha^{16}x^7+\alpha^6x^6+x^5+\alpha^{20}x^4+\alpha^{14}x^3+\alpha^5x^2+\alpha^{25}x+\alpha^{19}$	$[10,1,10]_{27}$
		$x^8+\alpha^3x^7+\alpha^2x^6+\alpha^{12}x^5+\alpha^{16}x^4+\alpha^{22}x^3+\alpha^3x^2+\alpha^{23}x+\alpha^{23}$	$[10,2,9]_{27}$
		$x^7+\alpha^{23}x^6+\alpha x^5+\alpha^{15}x^4+2x^3+\alpha^6x^2+\alpha x+\alpha^{16}$	$[10,3,8]_{27}$
	$f(x)=x^{10}+x^9+x^6+\alpha x^4+\alpha$	$x^6+\alpha^{14}x^5+\alpha^7x^4+\alpha^{10}x^3+\alpha^9x^2+\alpha^{23}x+1$	$[10,4,6]^*_{27}$
		$x^5+\alpha^{24}x^4+\alpha^4x^3+\alpha^9x^2+\alpha^6x+\alpha^4$	$[10,5,5]^*_{27}$
		$x^4+\alpha^{alpha}x^3+2x^2+\alpha^{11}x+\alpha^{16}$	$[10,6,4]^*_{27}$
	$f(x)=x^{10}+x^9+\alpha x^3+x+1$	$x^3+\alpha^9x^2+\alpha^2x+\alpha^5$	$[10,7,4]_{27}$
		$x^2+x+\alpha^4$	$[10,8,3]_{27}$
		$x+\alpha^{11}$	$[10,9,2]_{27}$

Table 2.5: MDS Codes over $\mathbb{F}_{27}$.

Chapter 3

Linear codes invariant under a quasi-cyclic operator

In this chapter, we study linear codes of length $n = mr$ that are invariant under an operator $T = T_1 \oplus T_1 \oplus \ldots \oplus T_1$ (where there are r copies of T_1), where T_1 is a cyclic operator on $\mathbb{F}_q^m$. In this case, we say that T is a quasi-cyclic operator of index r. We simplify the discussion by focusing on linear codes that are right-multiplied by a matrix $D = \mathrm{diag}(M, M, \ldots, M)$, where M is an $m \times m$ cyclic matrix. These codes are called quasi-M-cyclic codes of index r, and they are called quasi-f-cyclic codes when M is the companion matrix of a polynomial $f(x)$. This class contains quasi-cyclic codes and its generalizations, such as quasi-twisted codes, quasi-negacyclic codes, and others as special cases.

We prove a one-to-one correspondence between quasi f-cyclic codes of index r and R_f-submodules of R_f^r, where $R_f = \mathbb{F}_q[x]/\langle f(x)\rangle$ and $f(x)$ is the minimal polynomial of M, via an isomorphism $\varphi_{f,r}$. We also show the corresponding result for quasi-M-cyclic codes by using the isomorphism $\psi_{f,r} = \varphi_{f,r} \circ P^{-1}$, where $D = \mathrm{diag}(M, M, \ldots, M) = P^{-1}\,\mathrm{diag}(C_f, C_f, \ldots, C_f)P$.

We then focus on the structure of **1**-generator quasi f-cyclic codes and prove the BCH-like and Hartmann-Tzeng-like bounds. Afterward, we study the additive structure of quasi f-cyclic codes by mapping them to $\mathbb{F}_{q^r}/\langle f(x)\rangle$ via an $\mathbb{F}_{q^r}$-module morphism. We characterize those codes that have $\mathbb{F}_{q^r}$-linear images with respect to a basis of the extension $\mathbb{F}_{q^r}/\mathbb{F}_q$ and show that such codes must be invariant under certain matrices related to the basis of the extension $\mathbb{F}_{q^r}/\mathbb{F}_q$. If the basis of the extension is normal, the characterization is particularly simple.

3.1 Cyclic decomposition theorem

Let T be an operator on $\mathbb{F}_q^n$. For each vector $v \in \mathbb{F}_q^n$, the **T-cyclic subspace** of $\mathbb{F}_q^n$ generated by v is the smallest $\mathbb{F}_q$-vector subspace given by

$$F_v = \left\{ P(T)(v) \in \mathbb{F}_q^n \ : \ P(x) \in \mathbb{F}_q[x] \right\}.$$

Note that if $F_v = \mathbb{F}_q^n$ then T is called a cyclic operator. We now recall the following theorem that states that each operator T can be decomposed as a direct sum of cyclic operators T_i. Equivalently each square matrix M is similar to a block diagonal matrix of cyclic matrices.

Theorem 3.1 (Theorem 2 [41])
Let T be an operator of $\mathbb{F}_q^n$ with minimal polynomial $f(x)$. Then there exist $w_1, w_2, \ldots, w_r$ in $\mathbb{F}_q^n$ such that

1. $\mathbb{F}_q^n = F_{w_1} \oplus F_{w_2} \oplus \cdots \oplus F_{w_r}$, where $\oplus$ is the direct sum of subspaces in $\mathbb{F}_q^n$.

2. For $i \in \{1, \ldots, r\}$ the restriction operator $T_i := T_{|_{F_{w_i}}}$ of T on F_{w_i} is a cyclic operator of F_{w_i}.

3. If f_i is the minimal polynomial of each T_i, then $f_r | f_{r-1} | \ldots | f_1 = f$, where $|$ means "divides in $\mathbb{F}_q[x]$." The sequence $(f_1(x), f_2(x), \ldots, f_r(x))$ is called the sequence of **invariant factors of** T.

4. T is similar to the operator $T^{\oplus} := T_1 \oplus T_2 \oplus \ldots \oplus T_r$, where each T_i, $i = 1, \ldots, r$, is a cyclic operator of $\mathbb{F}_q^{n_i}$ with minimal polynomial $f_i(x)$.

Definition 3.1 (Rational canonical form)
If M is a matrix with sequence of invariant factors $f_1, f_2, \ldots, f_r$ then there is an invertible matrix P such that

$$D = \operatorname{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r}) = P^{-1}MP$$

where each C_{f_i} is the companion matrix of $f_i(x)$. The matrix $\operatorname{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$ is called the **rational canonical form** of M.

From now to the end of this chapter, we will be interested in linear codes invariant under a special operator T having the same invariant factors, that is $f_1 = f_2 = \ldots = f_r$. The general case will be studied in the next chapter.

3.2 Quasi M-cyclic codes

We start with the following definition.

Definition 3.2
Let C be a linear code of length $n = mr$ over $\mathbb{F}_q$. Then we say that C is a/an

1. **M-code** if it is invariant under T_M, where M is an $n \times n$ square matrix.

2. **Quasi M-cyclic code** of **index** r if it is invariant under $\operatorname{diag}(\overbrace{M, M, \ldots, M}^{r \text{ times}})$, where M is an $m \times m$ cyclic matrix.

3. **Quasi f-cyclic code** of **index** r if it is invariant under $\operatorname{diag}(\overbrace{C_f, C_f, \ldots, C_f}^{r \text{ times}})$ where $f(x)$ is a non-zero polynomial of degree $m = \frac{n}{r}$.

Proposition 3.1
Let C be a linear code of length $n = mr$ over $\mathbb{F}_q$ and M be a non-cyclic matrix with sequence of invariant factors $f_1(x), f_2(x), \ldots, f_r(x)$, such that $f_1(x) = f_2(x) = \ldots = f_r(x)$. Then C is an M-code if and only if $P^{-1}C$ is a quasi f-cyclic code of index r, where $D = \operatorname{diag}(\overbrace{C_f, C_f, \ldots, C_f}^{r \text{ times}}) = P^{-1}MP$ and $f(x) = f_1(x)$ is the minimal polynomial of M.

Proof
Suppose that C is an M-code, then $T_M(C) \subseteq C$. By Theorem 3.1 there is an invertible matrix P such $D = P^{-1}MP$. Hence $P^{-1}C$ is invariant under D and so it is a quasi f-cyclic code of index r.

The proof of the following Proposition is immediate.

Proposition 3.2
Let C be a linear code of length $n = mr$ and M an $m \times m$ cyclic matrix with minimal polynomial $f(x) = x^m - \sum_{i=0}^{m-1} a_i x^i$. The following assertions are equivalent.

1. C is a quasi M-cyclic code.
2. $P^{-1}C$ is a quasi f-cyclic code of index r with $D = P^{-1}MP$.

Theorem 3.2
Let $C \subseteq \mathbb{F}_q^{mr}$ be a linear code. Then,

1. C is a quasi f-cyclic code of index r if and only if $C = C_1 \oplus C_2 \oplus \ldots \oplus C_r$, where each C_i is an f-cyclic code of length m.
2. C is a quasi M-cyclic code of index r if and only if $C = C_1 \oplus C_2 \oplus \ldots \oplus C_r$, where each C_i is an M-cyclic code of length m.

Proof

1. Let C be a quasi f-cyclic code of index r. Then $T_D(C) \subseteq C$, where $D = \mathrm{diag}(\overbrace{C_f, C_f, \ldots, C_f}^{r \text{ times}})$. From Theorem 3.1, one can write $C = C_1 \oplus C_2 \oplus \ldots \oplus C_r \subseteq \mathbb{F}_q^n$, with each $C_i = C \cap F_{w_i}$. It follows that

$$T_D(C) = T_{C_f}(C_1) \oplus T_{C_f}(C_2) \oplus \ldots \oplus T_{C_f}(C_r) \subseteq C_1 \oplus C_2 \oplus \ldots \oplus C_r.$$

   Hence $T_{C_f}(C_i) \subseteq C_i$ for $i = 1, \ldots, r$, and each C_i is an f-cyclic code.

2. Similar argument as in the above statement.

Corollary 3.1
Let C be an M-code of length $n = mr$, where M is similar to $D = \mathrm{diag}(C_f, C_f, \ldots, C_f)$ by an invertible matrix P. Then $P^{-1}C$ can be decomposed as follows:

$$P^{-1}C = C_1 \oplus C_2 \oplus \ldots \oplus C_r,$$

where each C_i, $i = 1, \ldots, r$, is an f-cyclic code of length m over $\mathbb{F}_q$.

Theorem 3.3
Let $f(x) = x^m - \sum_{i=0}^{m-1} a_i x^i \in \mathbb{F}_q[x]$ be a non-zero polynomial of degree m, and C be a linear code of length $n = mr$ over $\mathbb{F}_q$. Then the following assertions are equivalent.

1. C is a quasi f-cyclic code of index r and length $n = mr$ over $\mathbb{F}_q$.
2. There is an $n \times n$ cyclic matrix $\widetilde{M}$ of minimal polynomial $\tilde{f}(x) = x^n - \sum_{i=0}^{m-1} a_i x^{ri}$ and an invertible matrix P such that PC is an $\widetilde{M}^r$-code.

Proof
If C is a quasi f-cyclic code of index r and of length $n = mr$ over $\mathbb{F}_q$. Then C is invariant under T_D, where $D = \mathrm{diag}(\overbrace{C_f, C_f, \ldots, C_f}^{r \text{ times}})$. Let $\widetilde{M}$ be the companion matrix of $\tilde{f}(x) = x^n - \sum_{i=0}^{m-1} a_i x^{ir}$, then $\widetilde{M}$ is an $n \times n$ cyclic matrix. Let the isomorphism σ of $\mathbb{F}_q^{mr}$ be defined by

$$\sigma(v) = \left(\overbrace{v_{1,0}, v_{2,0}, \ldots, v_{r,0}}, \overbrace{v_{1,1}, v_{2,1}, \ldots, v_{r,1}}, \ldots, \overbrace{v_{1,m-1}, v_{2,m-1}, \ldots, v_{r,m-1}}\right) \in (\mathbb{F}_q^m)^r \qquad (3.1)$$

where $v = \left(\overbrace{v_{1,0}, v_{1,1}, \ldots, v_{1,m-1}}^{x_1}, \overbrace{v_{2,0}, v_{2,1}, \ldots, v_{2,m-1}}^{v_2}, \ldots, \overbrace{v_{r,0}, v_{r,1}, \ldots, v_{r,m-1}}^{v_r}\right) \in (\mathbb{F}_q^r)^m$.

Set P the representation matrix of σ on the standard basis of $\mathbb{F}_q^n$. We verify that $D = P^{-1}\widetilde{M}^r P$, then finally by Proposition 3.1 PC is an $\widetilde{M}^r$-code.
Conversely, if PC is an $\widetilde{M}^r$-code, then as $D = P^{-1}\widetilde{M}^r P$, we obtain that $C = P^{-1}PC$ is a quasi f-cyclic code of index r.

Corollary 3.2
Let $\widetilde{M}$ be an $n \times n$ cyclic matrix of minimal polynomial $\widetilde{f}(x) = x^n - \sum_{i=0}^{m-1} a_i x^{ir}$, and C be a linear code of length $n = mr$ over $\mathbb{F}_q$. Then the following assertions are equivalent.

1. C is an $\widetilde{M}^r$-code
2. There is an invertible matrix P such that $P^{-1}C$ is a quasi f-cyclic code of index r, where $f(x) = x^m - \sum_{i=0}^{m-1} a_i x^i$.
3. For each $m \times m$ cyclic matrix M of minimal polynomial $f(x) = x^m - \sum_{i=0}^{m-1} a_i x^i$, there is an invertible matrix Q such that $Q^{-1}C$ is a quasi M-cyclic code.

In order to use the polynomial ring in the study of quasi f-cyclic codes, let $\langle f(x)\rangle$ be the principal ideal of $\mathbb{F}_q[x]$ generated by $f(x)$. Since $f(x)$ is the minimal polynomial of C_f, $\langle f(x)\rangle \cdot \mathbb{F}_q^{mr} = \{0\}$. Therefore, the operator T_D, where $D = \operatorname{diag}(\overbrace{C_f, C_f, \ldots, C_f}^{r \text{ times}})$, permits us to endow $\mathbb{F}_q^{mr}$ with an R_f-module structure where the multiplication is given by

$$\begin{array}{ccc} R_f \times \mathbb{F}_q^{mr} & \longrightarrow & \mathbb{F}_q^{mr} \\ (P, v) & \longmapsto & P.v := P(D)(v) \end{array} \tag{3.2}$$

Define the following R_f-module isomorphism $\varphi_{f,r}$ by

$$\begin{array}{ccccc} \varphi_{f,r} : & \mathbb{F}_q^{mr} & \longrightarrow & R_f^r \\ & (v_1, v_2, \ldots, v_r) & \longmapsto & (v_1(x), v_2(x), \ldots, v_r(x)) \end{array} \tag{3.3}$$

where for $1 \leq i \leq r$, $v_i(x) = \varphi_f(v_{i,0}, v_{i,1}, \ldots, v_{i,m-1}) = \sum_{j=0}^{m-1} v_{i,j} x^j$. Thus we deduce the following result.

Theorem 3.4
Let $C \subseteq \mathbb{F}_q^{mr}$ be a linear code and M be an $m \times m$ cyclic matrix with minimal polynomial $f(x)$. Then

1. C is a quasi f-cyclic code of index r if and only if $\varphi_{f,r}(C)$ is an R_f-submodule of R_f^r.
2. C is a quasi M-cyclic code of index r if and only if $\psi_{f,r}(C)$ is an R_f-submodule of R_f^r, where $\psi_{f,r} := \varphi_{f,r} \circ P^{-1}$ and $\operatorname{diag}(\overbrace{C_f, C_f, \ldots, C_f}^{r \text{ times}}) = P^{-1} \operatorname{diag}(\overbrace{M, M, \ldots, M}^{r \text{ times}})P$.

Proof

1. If C is a quasi f-cyclic code of index r, then by Theorem 3.2 one can decompose C as

$$C = C_1 \oplus C_2 \oplus \ldots \oplus C_r,$$

where each C_i is an f-cyclic code. It follows that

$$\varphi_{f,r}(C) = \varphi_f(C_1) \oplus \varphi_f(C_2) \oplus \ldots \oplus \varphi_f(C_r).$$

Now if $c = (c_1, c_2, \ldots, c_r) \in C$ with each $c_i \in C_i,\ i = 1, \ldots, r$, then according to Theorem 2.2, Chapter 2, we deduce that

$$x \cdot \varphi_{f,r}(c) = (xc_1(x), xc_2(x), \ldots, xc_r(x)) \in \varphi_{f,r}(C).$$

Finally, $\varphi_{f,r}(C)$ is an R_f-submodule of R_f^r.

2. If C is a quasi M-cyclic code, then by Proposition 3.2, $P^{-1}C$ is a quasi f-cyclic code. So, by the above statement $\psi_{f,r}(C)$ is an R_f-submodule of R_f^r.

Now, we give the following definition of an **s**-generator quasi f-cyclic code. Analogously, a similar definition of an **s**-generator quasi M-cyclic code can be obtained by considering the R_f-module isomorphism $\psi_{f,r}$.

Definition 3.3
A quasi f-cyclic code is called **s**-generator if $\varphi_{f,r}(C)$ is generated by s elements as an R_f-module of R_f^r, i.e., there exist $\underline{\mathbf{a}_1}(x), \underline{\mathbf{a}_2}(x), \ldots, \underline{\mathbf{a}_s}(x)$ in R_f^r such that

$$\varphi_{f,r}(C) = R_f\underline{\mathbf{a}_1}(x) + R_f\underline{\mathbf{a}_2}(x) + \ldots + R_f\underline{\mathbf{a}_s}(x),$$

where each $\underline{\mathbf{a}_i}(x) = (a_{i,1}(x), a_{i,2}(x), \ldots, a_{i,r}(x)) \in R_f^r$.

We close this section with the following property of an **s**-generator quasi f-cyclic code.

Proposition 3.3
Let $C \subseteq \mathbb{F}_q^{mr}$ be an **s**-generator quasi f-cyclic code generated by $\underline{\mathbf{a}_1}(x), \underline{\mathbf{a}_2}(x), \ldots, \underline{\mathbf{a}_s}(x)$ where $\underline{\mathbf{a}_i}(x) = (a_{i,1}(x), a_{i,2}(x), \ldots, a_{i,r}(x))$ for $1 \leq i \leq s$. Then, there exist $b_{i,j}(x),\ 1 \leq j \leq s,\ 1 \leq j \leq r$, such that $a_{i,j}(x) = b_{i,j}(x)g_{i,j}(x)$ where $g_{i,j}(x)$ divides $f(x)$.

Proof
As C is an **s**-generator quasi f-cyclic code, we have

$$\varphi_{f,r}(C) = R_f\underline{\mathbf{a}_1}(x) + R_f\underline{\mathbf{a}_2}(x) + \ldots + R_f\underline{\mathbf{a}_r}(x)$$

for some $\underline{\mathbf{a}_i}(x) = (a_{i,1}(x), a_{i,2}(x), \ldots, a_{i,r}(x)) \in R_f^r,\ 1 \leq i \leq s$. As each $R_f\underline{\mathbf{a}_i}(x),\ i = 1, \ldots, s$, is a 1-generator quasi f-cyclic code, by Theorem 3.2 one can decompose each $R_f\underline{\mathbf{a}_i}(x),\ i = 1, \ldots, s$ as

$$R_f\underline{\mathbf{a}_i}(x) = R_f a_{i,1}(x) \oplus R_f a_{i,2}(x) \oplus \ldots \oplus R_f a_{i,r}(x),$$

where $R_f a_{i,j}(x),\ j = 1, \ldots, r$ is a f-cyclic code of length m. Let $g_{i,j}(x)$ be its generator polynomial. Then there exists $b_{i,j}(x) \in R_f$ such that $a_{i,j}(x) = b_{i,j}g_{i,j}(x)$.

3.3 Connection between quasi M-cyclic codes and other types of codes

In relation to some well-known types of codes in the literature, we give the following definition.

Definition 3.4
Let $a = \left(a_0, a_1, \ldots, a_{m-1}\right) \in \mathbb{F}_q^m$, and C be a linear code of length $n = mr$ over $\mathbb{F}_q$. Let c be a codeword

$$c = (\overbrace{c_{1,0}, c_{1,1}, \ldots, c_{1,m-1}}^{c_1}, \overbrace{c_{2,0}, c_{2,1}, \ldots, c_{2,m-1}}^{c_2}, \ldots, \overbrace{c_{r,0}, c_{r,1}, \ldots, c_{r,m-1}}^{c_r}).$$

We say that C is

1. a right quasi-polycyclic code of length $n = mr$ and index r induced by the vector a, if r is the smallest integer such that

$$c_{\overrightarrow{a}} = \left(T_{\overrightarrow{a}}(c_1), T_{\overrightarrow{a}}(c_2), \ldots, T_{\overrightarrow{a}}(c_r)\right) \in C, \tag{3.4}$$

   where $T_{\overrightarrow{a}}$ is the right polyshift operator on $\mathbb{F}_q^m$ as in Definition 2.1.

2. a left quasi-polycyclic code of length $n = mr$ and index r induced by the vector a, if r is the smallest integer such that

$$c_{\overleftarrow{a}} = \left(T_{\overleftarrow{a}}(c_1), T_{\overleftarrow{a}}(c_2), \ldots, T_{\overleftarrow{a}}(c_r)\right) \in C, \tag{3.5}$$

   where $T_{\overleftarrow{a}}$ is the left polyshift operator on $\mathbb{F}_q^m$ as in Definition 2.1.

3. a quasi σ-cyclic code of length $n = mr$ and index r if r is the smallest integer such that

$$c_\sigma = (T_\sigma(c_1), T_\sigma(c_2), \ldots, T_\sigma(c_r)) \in C, \tag{3.6}$$

   where σ is an m-cycle and T_σ is the permutation operator on $\mathbb{F}_q^m$ as in Definition 2.1.

4. a quasi-monomial code of length $n = mr$ and index r induced by the vector a, if r is the smallest integer such that

$$c_{\overline{a}} = (T_{\overline{a}}(c_1), T_{\overline{a}}(c_2), \ldots, T_{\overline{a}}(c_r)) \in C \tag{3.7}$$

   where $T_{\overline{a}}$ is the monomial operator on $\mathbb{F}_q^m$ as in Definition 2.1.

5. a quasi σ-monomial code of length $n = mr$ and index r induced by the vector a if r is the smallest integer such that

$$c_{\sigma,\overline{a}} = \left(T_{\sigma,\overline{a}}(c_1), T_{\sigma,\overline{a}}(c_2), \ldots, T_{\sigma,\overline{a}}(c_r)\right) \in C \tag{3.8}$$

   where $T_{\sigma,\overline{a}}$ is the σ-monomial operator on $\mathbb{F}_q^m$ as in Definition 2.1.

Remark 3.1 (Quasi σ-cyclic codes)
With the same notations as in the above definition, if C is a quasi σ-monomial code of length $n = mr$ and index r induced by the vector $a = (a_0, a_1, \ldots, a_{m-1})$ such that $a_i = 1,\ 0 \leq i \leq m-1$, we say that C is a quasi σ-cyclic code.

In line with Example 2.1, we give the following Proposition.

Proposition 3.4
The following codes are quasi f-cyclic codes

1. Quasi-twisted codes of length $n = mr$ and index r are quasi f_λ-cyclic codes, where $f_\lambda(x) = x^m - \lambda$.

2. Quasi-cyclic codes of length $n = mr$ and index r are quasi f_1-cyclic codes, where $f_1(x) = x^m - 1$.

3. Quasi-negacyclic codes of length $n = mr$ and index r are quasi f_{-1}-cyclic codes, where $f_{-1}(x) = x^m + 1$.

4. Right quasi-polycyclic codes of length $n = mr$ and index r are induced by the vector $a = (a_0, a_1, \ldots, a_{m-1}) \in \mathbb{F}_q^m$ are $f_{\vec{a}}$-cyclic codes, where $f_{\vec{a}}(x) = x^m - \sum_{i=0}^{m-1} a_i x^i$.

On the other hand, the following codes are M-cyclic codes.

1. Left quasi-polycyclic codes of length $n = mr$ and index r induced by the vector $a = (a_0, a_1, \ldots, a_{m-1})$ are quasi $M_{\overleftarrow{a}}$-cyclic codes, where $M_{\overleftarrow{a}}$ is the cyclic matrix with minimal polynomial $f_{\overleftarrow{a}} = x^m - \sum_{i=0}^{m-1} a_i x^{n-1-i}$ as in Proposition 2.4.

2. Quasi-monomial codes of length $n = mr$ and index r induced by the vector $a = (a_0, a_1, \ldots, a_{m-1})$ are $M_{\overline{a}}$-cyclic codes, where $M_{\overline{a}}$ is the cyclic matrix with minimal polynomial $f_{\overline{a}} = x^m - \prod_{i=0}^{m-1} a_i$ as in Proposition 2.4.

3. Quasi σ-cyclic codes of length $n = mr$ and index r are quasi P_σ-cyclic codes, where σ is an m-cycle with associated permutation matrix P_σ.

4. Quasi σ-monomial codes of length $n = mr$ and index r induced by the vector $a = (a_0, a_1, \ldots, a_{m-1})$ are quasi $M_{\sigma,\overline{a}}$-cyclic codes, where σ is an m-cycle and

$$M_{\sigma,\overline{a}} = \mathrm{diag}(a_0, a_1, \ldots, a_{m-1}) P_\sigma$$

is a cyclic matrix with minimal polynomial $f_{\sigma,\overline{a}} = x^m - \prod_{i=0}^{m-1} a_i$.

Proposition 3.5
Let C be a linear code of length $n = mr$ over $\mathbb{F}_q$.

1. C is a **right** quasi-polycyclic code of index r induced by the vector $a = \left(a_0, a_1, \ldots, a_{m-1}\right)$ if and only if $\mathrm{diag}(\overbrace{P, P, \ldots, P}^{r \text{ times}})C$ is a **left** quasi-polycyclic code of index r induced by the vector $b = \left(a_{m-1}, a_{m-2}, \ldots, a_0\right)$, where P is the $m \times m$ square matrix given by

$$P^{-1} = P = \begin{pmatrix} 0 & \ldots & \ldots & 0 & 1 \\ 0 & \ldots & 0 & 1 & 0 \\ \vdots & \iddots & \iddots & \iddots & \vdots \\ 0 & \iddots & \iddots & & \vdots \\ 1 & 0 & \ldots & \ldots & 0 \end{pmatrix}$$

2. C is a quasi-monomial code of index r induced by the vector $a = (a_0, a_1, \ldots, a_{m-1})$ if and only if $\mathrm{diag}(\overbrace{Q^{-1}, Q^{-1}, \ldots, Q^{-1}}^{r \text{ times}})C$ is a λ-quasi-twisted code, where $\lambda = \prod_{i=0}^{m-1} a_i$ and Q is the

$m \times m$ square matrix given by

$$Q = \begin{pmatrix} 0 & 0 & \dots & 0 & \prod_{i=0}^{m-1} a_i \\ a_0 & 0 & \dots & 0 & 0 \\ 0 & a_0 a_1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & \prod_{i=0}^{m-2} a_i & 0 \end{pmatrix}, \; Q^{-1} = \begin{pmatrix} 0 & a_0^{-1} & 0 & \dots & 0 \\ 0 & 0 & (a_0 a_1)^{-1} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & \vdots & & \ddots & \prod_{i=0}^{m-2} a_i^{-1} \\ \prod_{i=0}^{m-1} a_i^{-1} & 0 & \dots & \dots & 0 \end{pmatrix}$$

3. C is a quasi σ-cyclic code of index r if and only if $\mathrm{diag}(P_\rho^{-1}, P_\rho^{-1}, \dots, P_\rho^{-1})C$ is a cyclic code, where P_ρ is the associated permutation matrix of $\rho \in S_m$ such that $\sigma = \rho\sigma^{'}\rho^{-1}$ and $\sigma^{'}$ of standard form.

4. C is a σ-quasi-monomial code of index r induced by the vector $a = (a_0, a_1, \dots, a_{m-1})$ if and only if $\mathrm{diag}(Q_\rho^{-1}, Q_\rho^{-1}, \dots, Q_\rho^{-1})C$ is a λ-quasi-twisted code, where $\lambda = \prod_{i=0}^{n-1} a_i$ and $Q_\rho = P_\rho Q$ such that $\rho, \sigma^{'} \in S_m$ with $\sigma = \rho\sigma^{'}\rho^{-1}$ and $\sigma^{'}$ is an m-cycle of standard form.

Proof
This result follows from the similarity between these matrices in play as in the proof of Proposition 2.5 for M-cyclic codes.

In line with the Remark 2.9, we make also the following remark.

Remark 3.2
Let $a = \left(a_0, a_1, \dots, a_{m-1}\right) \in \mathbb{F}_q^m$, and C a linear code of length $n = mr$ over $\mathbb{F}_q$.

1. C is a right quasi-polycyclic code of length n, if and only if $\varphi_{\vec{a},r}(C)$ is an $R_{\vec{a}}$-submodule of $R_{\vec{a}}^r$, where $R_{\vec{a}} := \mathbb{F}_q[x]/\langle f_{\vec{a}}(x)\rangle$, $f_{\vec{a}} = x^n - \sum_{i=0}^{n-1} a_i x^i$, $\varphi_{\vec{a},r} = (\overbrace{\varphi_{\vec{a}}, \varphi_{\vec{a}}, \dots, \varphi_{\vec{a}}}^{r \text{ times}})$ and $\varphi_{\vec{a}}$ is the right isomorphism induced by the vector $a \in \mathbb{F}_q^m$ as in Remark 2.9.

2. A similar remark remains valid, as in the above statement, for the left quasi-polycyclic codes via $\varphi_{\overleftarrow{a},r} = (\overbrace{\varphi_{\overleftarrow{a}}, \varphi_{\overleftarrow{a}}, \dots, \varphi_{\overleftarrow{a}}}^{r \text{ times}})$, and for quasi monomial codes via $\varphi_{\overline{a},r} = (\overbrace{\varphi_{\overline{a}}, \varphi_{\overline{a}}, \dots, \varphi_{\overline{a}}}^{r \text{ times}})$, where $\varphi_{\overleftarrow{a}}$ and $\varphi_{\overline{a}}$ are respectively the left and the monomial isomorphism induced by the vector $a \in \mathbb{F}_q^m$ as in Remark 2.9.

3. A similar remark also applies to quasi σ-cyclic codes and quasi σ-monomial codes.

3.4 Structure of 1-generator quasi f-cyclic codes

An important class of quasi f-cyclic codes is the class of 1-generator quasi f-cyclic codes. We can obtain properties of this class more easily by fixing a generator polynomial and a parity check polynomial, and then computing the dimension of each 1-generator quasi f-cyclic code. In this section we consider C to be a 1-generator quasi f-cyclic code. We write $C = R_f \cdot \underline{\mathbf{a}}(x)$ to denote that C is a 1-generator quasi f-cyclic code generated by $\underline{\mathbf{a}}(x) = (a_1(x), a_2(x), \dots, a_r(x))$.

Definition 3.5
Let C be a **1**-generator quasi f-cyclic code generated by $\underline{\mathbf{a}}(x) = (a_1(x), a_2(x), \dots, a_r(x))$. The polynomial

1. $g(x) = \gcd(\underline{\mathbf{a}}(x), f(x))$ is called the **generator polynomial** (g.p.) of C.

2. $h(x) = \dfrac{f(x)}{g(x)}$ is called the **parity check polynomial** (p.c.p.) of C.

From Theorem 2.3 and Proposition 2.3, we easily deduce the following characterization of **1**-generator quasi f-cyclic codes.

Proposition 3.6
Let C be a **1**-generator quasi f-cyclic code generated by $\underline{\mathbf{a}}(x) = (a_1(x), a_2(x), \ldots, a_r(x)) \in R_f^r$ with $g(x)$, $h(x)$ as g.p and p.c.p. respectively, $g_i(x) = \gcd(a_i(x), f(x))$, $i = 1, \ldots, r$, and $\underline{\mathbf{g}}(x) = (g_1(x), g_2(x), \ldots, g_r(x))$. Then,

1. C has a generator of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

where $\gcd(p_i(x), h(x)) = 1$.

2. $g(x) = \gcd(g_1(x), g_2(x), \ldots, g_r(x))$ and $h(x) = \operatorname{lcm}\{h_i(x) \ : \ 1 \leq i \leq r\}$.

3. $\dim_{\mathbb{F}_q}(C) = \deg(h(x))$, and

$$S = \left\{(x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ : \ 0 \leq i \leq \deg(h(x)) - 1\right\}$$

forms a basis of C.

4. a generator matrix of C is given by

$$G = \begin{pmatrix} \underline{\mathbf{g}}(x) \\ x\underline{\mathbf{g}}(x) \\ \vdots \\ x^{k-1}\underline{\mathbf{g}}(x) \end{pmatrix} = \left(\begin{array}{c|c|c|c} g_1(x) & g_2(x) & \ldots & g_r(x) \\ xg_1(x) & xg_2(x) & \ldots & xg_r(x) \\ \vdots & \vdots & \vdots & \vdots \\ x^{k-1}g_1(x) & x^{k-1}g_2(x) & \ldots & x^{k-1}g_r(x) \end{array}\right) \tag{3.9}$$

where $k = \deg(h(x))$.

5. $\underline{\mathbf{b}}(x) = (b_1(x), b_2, \ldots, b_r(x)) \in R_f^r$ is another generator of C if and only if there is $p(x) \in \mathbb{F}_q[x]$ such that $\gcd(p(x), h(x)) = 1$ and $\underline{\mathbf{b}}(x) = p(x)\underline{\mathbf{a}}(x)$.

Proof
Let $\underline{\mathbf{a}}(x) = (a_1(x), a_2(x), \ldots, a_r(x)) \in R_f^r$, and $C = R_f \cdot \underline{\mathbf{a}}(x)$ be a **1**-generator quasi f-cyclic code generated by $\underline{\mathbf{a}}(x)$.

1. From Theorem 3.4 we have that $C = C_1 \oplus C_2 \oplus \ldots \oplus C_r$, with each $C_i = R_f \cdot a_i(x)$ is an f-cyclic code with generator polynomial $g_i(x) = \gcd(a_i(x), f_i(x))$. As $\gcd(p_i(x), g_i(x)) = 1$, by Proposition 2.3 $p_i(x)g_i(x)$ generates C_i. Therefore $\underline{\mathbf{b}}(x)$ is a generator of C.

2. Set $h'(x) = \operatorname{lcm}\{h_i(x) \ : \ 1 \leq i \leq r\}$. As $h(x) \cdot a_i(x) = 0$, $i = 1, \ldots, r$, by the minimality of $h_i(x)$, we obtain that each $h_i(x)$ is a divisor of $h(x)$, and so $h'(x)$ divides $h(x)$. Finally, by the minimality of $h(x)$ we deduce that $h(x) = h'(x) = \operatorname{lcm}_{1 \leq i \leq r}\{h_i(x)\}$, with $f_i(x) = h_i(x)g_i(x)$.

3. Let us prove that the set

$$S = \left\{ x^i \underline{\mathbf{g}}(x) = (x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ : \ 0 \leq i \leq \deg(h(x)) - 1 \right\},$$

is a basis for C. Suppose that $v(x)\underline{\mathbf{g}}(x) = 0$ in R_f^r, then

$$\left(v(x)g_1(x) \text{ in } R_f, v(x)g_2(x) \text{ in } R_f, \ldots, v(x)g_r(x) \text{ in } R_f \right) = (0, 0, \ldots, 0).$$

It follows that for each $i = 1, \ldots, r$, we have $v(x)g_i(x) = 0$ in R_f. Therefore $h_i(x)$ divides $v(x)$, so $h(x) = \text{lcm}\{h_i(x),\ 1 \leq i \leq r\}$ as well, which provides that $\deg(v(x)) \geq \deg(h(x)) - 1$. This implies the linear independence of vectors in S.

To complete the proof we need to prove that for every $v(x) \in \mathbb{F}_q[x]$ with $\deg(v(x)) \geq \deg(h(x)) - 1$, there exists $r(x) \in \mathbb{F}_q[x]$ with $\deg(r(x)) < \deg(h(x)) - 1$ such that $v(x)\underline{\mathbf{g}}(x) = r(x)\underline{\mathbf{g}}(x)$. By the Euclidean division algorithm in $\mathbb{F}_q[x]$, we write $v(x) = q(x)h(x) + r(x)$, for some $q(x), r(x) \in \mathbb{F}_q[x]$ where $\deg(r(x)) < \deg(h(x))$. It follows that

$$\begin{aligned} v(x)\underline{\mathbf{g}}(x) &= v(x)(g_1(x), g_2(x), \ldots, g_r(x)) \\ &= (v(x)g_1(x) \bmod f(x), v(x)g_2(x) \bmod f(x), \ldots, v(x)g_r(x) \bmod f(x)) \\ &= (r(x)g_1(x) \bmod f(x), r(x)g_2(x) \bmod f(x), \ldots, r(x)g_r(x) \bmod f(x)) \\ &= r(x)(g_1(x), g_2(x), \ldots, g_r(x)) \\ &= r(x)\underline{\mathbf{g}}(x) \end{aligned}$$

4. Since $S = \left\{ x^i \underline{\mathbf{g}}(x) = (x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ : \ 0 \leq i \leq \deg(h(x)) - 1 \right\}$ forms a basis of C.

5. Suppose that $R_f \underline{\mathbf{a}}(x) = R_f \underline{\mathbf{b}}(x)$, then $\underline{\mathbf{a}}(x) = \alpha(x)\underline{\mathbf{b}}(x)$ and $\underline{\mathbf{b}}(x) = p(x)\underline{\mathbf{a}}(x)$ so that $\underline{\mathbf{a}}(x) = \alpha(x)p(x)\underline{\mathbf{a}}(x)$. Consequently, with $g(x) = f(x)/h(x)$, we have that

$$\begin{aligned} g(x) = \ & \gcd\left(\underline{\mathbf{a}}(x), f(x)\right) = \gcd\left(\alpha(x)\beta(x)\underline{\mathbf{a}}(x), f(x)\right) \\ & = g(x) \gcd\left(\alpha(x)p(x)a_1(x)/g(x), \ldots, \alpha(x)\beta(x)a_r(x)/g(x), h(x)\right) \end{aligned}$$

It follows that $\gcd\left(a_1(x)/g(x), \ldots, a_r(x)/g(x), h(x)\right) = 1$ and so $\gcd(\alpha(x)p(x), h(x)) = 1$. Finally, we obtain that $\gcd(p(x), h(x)) = 1$

Conversely, let $\gcd(p(x), h(x)) = 1$ and let $\underline{\mathbf{b}}(x) = p(x)\underline{\mathbf{a}}(x)$, then $R_f \underline{\mathbf{b}}(x) = R_f p(x)\underline{\mathbf{a}}(x) \subset R_f \underline{\mathbf{a}}(x)$. So as $\dim_{\mathbb{F}_q}(R_f \underline{\mathbf{b}}(x)) = \dim_{\mathbb{F}_q}(R_f \underline{\mathbf{a}}(x))$, the result holds.

Example 3.1 (Quasi f-cyclic codes)
Let $q = 7,\ m = 8,\ r = 2$ and $n = r \times m = 16$. Let $f(x) = x^8 + x^3 + 1$ whose factorization into irreducibles over $\mathbb{F}_7$ is given by

$$f(x) = (x^3 + x^2 + 3) \cdot (x^5 + 6x^4 + x^3 + 3x^2 + 5).$$

Let $g_1(x) = x^5 + 6x^4 + x^3 + 3x^2 + 5$ and consider the linear code $C \subseteq \mathbb{F}_7^{16}$ generated by $\mathbf{a}(x) = (a_1(x), a_2(x)) \in R_f^2$, where

$$\begin{aligned} a_1(x) &= x^9 + x^7 + x^6 + x^5 + 4x^4 + x^3 + 4x^2 + 4x + 3 \\ &= (x^4 + x^3 + x^2 + 5x + 2)g_1(x) \\ a_2(x) &= x^{11} + 4x^{10} + 5x^9 + 5x^8 + 5x^7 + 6x^6 + 3x^5 + 5x^4 + 5x^3 + 6x^2 + 6x + 4 \\ &= (x^6 + 5x^5 + 2x^4 + 6x^3 + x^2 + 4x + 5)g_1(x) \end{aligned}$$

This is a **1**-generator quasi f-cyclic code with parity check polynomial $h(x) = x^3 + x^2 + 3$. Its actual distance is $d = 12$, and according to [27] it is an optimal code with parameters $[16, 3, 12]_7$.

On the other hand, the **1**-generator quasi f-cyclic code generated by $\mathbf{a}(x) = (a_1(x), a_2(x)) \in R_f^2$, where

$$\begin{aligned} g_2(x) &= x^3 + x^2 + 3 \\ a_1(x) &= x^5 + x^4 + 4x^3 + 5 = (x^2 + 4)g_2(x) \\ a_2(x) &= x^{16} + x^{15} + 5x^{14} + x^{12} + 5x^{10} + x^9 + 3x^8 + 5x^7 + 6x^6 + 5x^5 + 3x^4 + x^3 + 4x^2 + 2x + 5 \\ &= (x^{13} + 5x^{11} + 6x^{10} + 2x^9 + 4x^8 + 4x^7 + 5x^6 + 5x^3 + 3x + 4)g_2(x) \end{aligned}$$

is a **1**-generator quasi f-cyclic code with parity check polynomial

$$h(x) = x^5 + 6x^4 + x^3 + 3x^2 + 5.$$

Hence its dimension is 5. Its actual minimum distance is $d = 9$, very close to the optimal code of parameters $[16, 5, 10]_7$.

Theorem 3.5
Let C be a 1-generator quasi f-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

where for all $1 \leq i \leq r$, $f(x) = h_i(x)g_i(x)$ and each $p_i(x)$ is coprime to $h_i(x)$. Also let d and d_i be, respectively, the minimum Hamming distance of C and $C_i = \langle p_i(x)g_i(x) \rangle$ for $i = 1, \ldots, r$. Then

1. $d \geq \sum_{i \notin K} d_i$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) = \operatorname{lclm}\{h_i(x) \ : \ 1 \leq i \leq r\} \neq \operatorname{lcm}\{h_i(x) : i \in K\}$.

2. If $h_1(x) = h_2(x) = \ldots = h_r(x)$, then $d \geq \sum_{i=1}^{r} d_i$.

Proof
Let C be a 1-generator quasi f-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

Then C can be decomposed as a direct sum of right polycyclic codes $C_i = \langle p_i(x)g_i(x) \rangle$ as $C = C_1 \oplus C_2 \oplus, \ldots, \oplus C_r$. Let $c(x) = (c_1(x), c_2(x), \ldots, c_r(x)) = u(x)\underline{\mathbf{a}}(x)$ be a codeword of C, then a component $c_i(x)$ is 0 if and only if $h_i(x)$ divides $u(x)p_i(x)$. As $\gcd(p_i(x), h_i(x)) = 1$, this happens only if $h_i(x)$ is a divisor of $u(x)$. Therefore, $c(x) = 0$ if and only if $h(x)$ is a divisor of $u(x)$. It follows that $c(x) \neq 0$ if and only if $h(x)$ does not divide $u(x)$. Thus, a non-zero codeword has the most number of zero components whenever $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$ and $\operatorname{lcm}\{h_i(x) : i \in K\}$ divides $u(x)$, where K is a maximal subset of $\{1, 2, \ldots, r\}$. Therefore, $d \geq \sum_{i \notin K} d_i$. On the other hand, we have $|K| = 0$ if and only if $h_1(x) = h_2(x) = \ldots = h_r(x)$. In this case, $d \geq \sum_{i=1}^{r} d_i$.

Let e be the order of $f(x)$ over $\mathbb{F}_q$ such that $\gcd(e, q) = 1$ and α be a primitive e^{th} root of unity in some extension of $\mathbb{F}_q$. In the following theorems, we show BCH-like and Hartmann-Tzeng-like bounds for 1-generator quasi f-cyclic codes.

Theorem 3.6 (BCH-like bound)
Let C be a **1**-generator quasi f-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

where for all $1 \leq i \leq r$, $f_i(x) = h_i(x)g_i(x)$ and each $p_i(x)$ is coprime to $h_i(x)$. For $1 \leq i \leq r$, let δ_i be an integer such that $\alpha^{a_i+jb_i}$, $j = 0, \ldots, \delta_i - 2$, are among the roots of $g_i(x)$, with a_i and b_i are some integers such that $\gcd(e, b_i) = 1$. Then,

1. $d \geq \sum_{i \notin K} \delta_i$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$.

2. If $h_1(x) = h_2(x) = \cdots = h_r(x)$, then $d \geq \sum_{i=1}^{r} \delta_i$.

Proof
By Theorem 3.5, we have that $d \geq \sum_{i \notin K}(d_i)$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$ and d_i is the minimum distance of $C_i = \langle p_i(x)g_i(x)\rangle$. As $\alpha^{a_i+jb_i}$, $j = 0, \ldots, \delta_i - 2$, are among the roots of $g_i(x)$, by Theorem 1.6, we obtain $d_i \geq \delta_i$, and so the results holds.

Theorem 3.7 (Hartmann-Tzeng-like bound)
Let C be a 1-generator quasi f-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

where for any $1 \leq i \leq r$, $f(x) = h_i(x)g_i(x)$ and $p_i(x)$ is coprime to $h_i(x)$. Let $\delta_i \geq 2$ and r_i be integers such that for $j = 0, \ldots, \delta - 2$, and $k = 0, \ldots, r_i$, the elements $\alpha^{a+jb_1+kc_1}$ are among the zeros of $g(x)$ where a_i, b_i and c_i are some integers with $\gcd(e, b_i) = 1$ and $\gcd(e, c_i) = 1$. Then,

1. $d \geq \sum_{i \notin K}(\delta_i + r_i)$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$.

2. If $h_1(x) = h_2(x) = \ldots = h_r(x)$, then $d \geq \sum_{i=1}^{r}(\delta_i + r_i)$.

Proof
By Theorem 3.5, we have that $d \geq \sum_{i \notin K}(d_i)$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$ and d_i is the minimum distance of $C_i = \langle p_i(x)g_i(x)\rangle$. As $\alpha^{a+jb_1+kc_1} \ : \ j = 0, \ldots, \delta - 2, k = 0, \ldots, r_i$, are among the zeros of $g(x)$, we have, by Theorem 2.9, $d_i \geq \delta_i + r_i$, and so the result holds.

Remark 3.3
In many works such as [8], the authors have been interested in quasi-twisted codes with generators of the form

$$(p_1(x)g(x), p_2(x)g(x), \ldots, p_r(x)g(x))$$

where for all $1 \leq i \leq r$, $x^n - \lambda = h(x)g(x)$ and each $p_i(x)$ is coprime to $h(x)$. This type of quasi-twisted codes have been very effective in finding new linear codes with better parameters. The authors developed a search algorithm called **ASR** that has produced many new linear codes from 1−generator QC and QT codes. A similar result in our case is given in the second statement of each of the three theorems above. In fact, if $h(x) = h_1(x) = h_2(x) = \ldots = h_r(x)$ then $g(x) = g_1(x) = g_2(x) = \ldots = g_r(x)$, so C has a generator of the form $(p_1(x)g(x), p_2(x)g(x), \ldots, p_r(x)g(x))$.

Example 3.2
In this Example, we consider quasi f-cyclic codes of length 21 over $\mathbb{F}_4 = \mathbb{F}_2(\alpha)$, where $\alpha^2+\alpha+1=0$ and $f(x) = x^7 + (\alpha+1)\,x + \alpha + 1$.
One such code C is generated by $\underline{\mathbf{a}}(x) = (a_1(x), a_2(x), a_3(x)) \in R_f^3$, where

$$\begin{aligned} a_1(x) &= x^4 + x^2 + (\alpha+1)\,x + 1 = g(x) \\ a_2(x) &= x^5 + x^4 + x^3 + \alpha x^2 + \alpha x + 1 = (x+1)g(x) \\ a_3(x) &= x^6 + (\alpha+1)\,x^3 + (\alpha+1)\,x + 1 = (x^2+1)g(x). \end{aligned}$$

This is a 1-generator quasi f-cyclic code with generator polynomial

$$g(x) = \gcd(a_1(x), a_2(x), a_3(x), f(x)) = x^4 + x^2 + (\alpha+1)\,x + 1,$$

and parity check polynomial $h(x) = x^3+x+\alpha+1$. It follows that $\dim_{\mathbb{F}_4}(C) = 3$. From Theorem 3.5 and the table in Example 1, the minimum distance of C is $d \geq 3\times 4 = 12$. Its actual distance is 14, close to the optimal code with parameters $[21, 3, 16]_4$.

Next, we give the following example as an illustration of the Theorem 3.6.

Example 3.3
In this Example, we consider f-cyclic codes of length 21 with $f(x) = x^{11}+x^5+x^4+2$. The order of $f(x)$ over $\mathbb{F}_9$ is $e = 364$, and the multiplicative order of 364 modulo 9 is 3. Let β be a primitive e^{th} root of unity. Consider the linear code $C \subseteq \mathbb{F}_4^{21}$ generated by $\underline{\mathbf{a}}(\mathbf{x}) = (a_1(x), a_2(x), a_3(x)) \in R_f^3$, where

$$\begin{aligned} a_1(x) &= x^4 + (\alpha+2)\,x^3 + (2\alpha+1)\,x^2 + (2\alpha+2)\,x + \alpha + 1 = g(x) \\ a_2(x) &= (x^4+1)g(x) \\ a_3(x) &= (x^5 + x^4 + x^3 + 1)g(x). \end{aligned}$$

This is a 1-generator quasi f-cyclic code with generator polynomial

$$g(x) = \gcd(a_1(x), a_2(x), a_3(x), f(x)) = x^4 + (\alpha+2)\,x^3 + (2\alpha+1)\,x^2 + (2\alpha+2)\,x + \alpha + 1,$$

and parity check polynomial $h(x) = x^7+(2\alpha+1)\,x^6+x^5+\alpha x^4+2\alpha x^3+2\alpha x^2+(\alpha+1)\,x+\alpha+1$. Since $\beta^{30}, \beta^{30+61}$ are among the roots of $g(x)$, $\delta = 3$. By Theorem 3.6, we obtain that the minimum distance of C is $d \geq 3\times 3 = 9$. Its actual minimum distance $d = 19$.

3.5 Structural properties and enumeration of 1-generator quasi f-cyclic codes.

Let $\mathbb{F}_{q^r}/\mathbb{F}_q$ be an extension field of $\mathbb{F}_q$ of degree r, $S_f := \mathbb{F}_{q^r}[x]/\langle f(x)\rangle$ and

$$f(x) = f_1^{\alpha_1}(x) f_2^{\alpha_2}(x), \ldots, f_s^{\alpha_s}(x) \tag{3.10}$$

be the irreducible factorization of $f(x)$ over $\mathbb{F}_q$. Set $\widehat{f_i^{\alpha_i}}(x) = \dfrac{f(x)}{f_i^{\alpha_i}(x)}$, $i = 1, \ldots, s$. Then as $\gcd(\widehat{f_i^{\alpha_i}}(x), f_i^{\alpha_i}(x)) = 1$, there is $u_i(x)$ and $v_i(x)$ in $\mathbb{F}_q[x]$ such that

$$u_i(x)\widehat{f_i^{\alpha_i}}(x) + v_i(x) f_i^{\alpha_i}(x) = 1. \tag{3.11}$$

Let us denote

$$\theta_i(x) = u_i(x)\widehat{f_i^{\alpha_i}}(x) = 1 - v_i(x) f_i^{\alpha_i}(x),\ i = 1, \ldots, s, \tag{3.12}$$

and

$$g(x) = f_{i_1}^{\alpha_{i_1}}(x) f_{i_2}^{\alpha_{i_2}}(x), \ldots, f_{i_t}^{\alpha_{i_t}}(x) \tag{3.13}$$

where $\{i_1, i_2, \ldots, i_t\} \subseteq \{1, 2, \ldots, s\}$.

Theorem 3.8
The polynomials $\theta_i(x),\ i = 1, \ldots, s$, satisfy the following assertions.

1. For $i \neq j \in \{1, \ldots, s\}$, $\theta_i(x)\theta_j(x) = 0 \mod f(x)$, and $\theta_i(x)^2 = \theta_i(x) \mod f(x)$.
2. $\theta_{_i}(x)$ is the identity element of $S_{_f}\widehat{f_{_i}^{\alpha_i}}(x)$, and $S_{_f}\widehat{f_{_i}^{\alpha_i}}(x) = S_{_f}\theta_{_i}(x)$.
3. $\theta_{i_1}(x), \theta_{i_2}(x), \ldots, \theta_{i_r}(x)$ are mutually orthogonal non-zero idempotent elements of $Sg(x)$, with $\{i_1, i_2, \ldots, i_r\} = \{1, 2, \ldots, s\} \setminus \{i_1, i_2, \ldots, i_t\}$.
4. $\theta(x) = \theta_{i_1}(x) + \theta_{i_2}(x) + \ldots + \theta_{i_r}(x)$ is the identity element of $S_{_f}g(x)$, and $S_{_f}g(x) = S_{_f}\theta(x)$.
5. $\theta_1(x) + \theta_2(x) + \ldots + \theta_s(x) = 1$ is the identity element of $S_{_f}$.
6. $Sg(x) = S_{_f}\widehat{f_{i_1}^{\alpha_{i_1}}}(x) \oplus S_{_f}\widehat{f_{i_2}^{\alpha_{i_2}}}(x) \oplus \ldots \oplus S_{_f}\widehat{f_{i_r}^{\alpha_{i_r}}}(x)$.
7. $S_{_f} = S_{_f}\widehat{f_{_1}^{\alpha_1}}(x) \oplus S_{_f}\widehat{f_{_2}^{\alpha_2}}(x) \oplus \ldots \oplus S_{_f}\widehat{f_{_s}^{\alpha_s}}(x)$.
8. $S_{_f}\widehat{f_{_i}^{\alpha_i}}(x) \cong \mathbb{F}_{q^{r\alpha_i m_i}}$, and if $\gcd(e, q) = 1$, $S_{_f}\widehat{f_{_i}^{\alpha_i}}(x) \cong \mathbb{F}_{q^{rm_i}}$, where $m_i = \deg f_i(x)$.

Proof

1. Let $i, j \in \{1, \ldots, s\}$, with $i \neq j$. Then

$$\begin{aligned} \theta_{_i}(x)\theta_{_j}(x) &= u_{_i}(x)\widehat{f_{_i}^{\alpha_i}}(x)u_{_j}(x)\widehat{f_{_j}^{\alpha_j}}(x) \\ &= u_{_i}(x)\widehat{f_{_i}^{\alpha_i}}(x)u_{_j}(x)\widehat{f_{_j}^{\alpha_j}}(x) \\ &= K_{i,j}(x)f(x) \\ &= 0 \mod f(x). \end{aligned} \tag{3.14}$$

   where $K_{i,j}(x) = \dfrac{u_{_i}(x)u_{_j}(x)\widehat{f_{_j}^{\alpha_j}}(x)}{f_{_i}^{\alpha_i}(x)} \in \mathbb{F}_{_q}[x]$.
   If $i = j$ we have

$$\begin{aligned} \theta_{_i}(x)^2 &= u_{_i}(x)\widehat{f_{_i}^{\alpha_i}}(x)[1 - v_{_i}(x)f_{_i}^{\alpha_i}(x)] \\ &= u_{_i}(x)\widehat{f_{_i}^{\alpha_i}}(x) \\ &= \theta_{_i}(x) \mod f(x). \end{aligned} \tag{3.15}$$

2. Let $c(x) \in S_{_f}\widehat{f_{_i}^{\alpha_i}}(x),\ i = 1, \ldots, s$ Then, $c(x) = \alpha(x)\widehat{f_{_i}^{\alpha_i}}(x)$, for some $\alpha(x) \in S$. Since $u_{_i}(x)\widehat{f_{_i}^{\alpha_i}}(x) + v_{_i}(x)f_{_i}^{\alpha_i}(x) = 1$, we have

$$\begin{aligned} c(x) &= u_{_i}(x)\widehat{f_{_i}^{\alpha_i}}(x)c(x) + v_{_i}(x)f_{_i}^{\alpha_i}(x)\alpha(x)\widehat{f_{_i}^{\alpha_i}}(x) \\ &= u_{_i}(x)\widehat{f_{_i}^{\alpha_i}}(x)c(x) + 0 \\ &= u_{_i}(x)\widehat{f_{_i}^{\alpha_i}}(x)c(x) \\ &= \theta_{_i}(x)c(x) \end{aligned} \tag{3.16}$$

   Then $\theta_i(x)$ is the identity element of $S_{_f}\widehat{f_{_i}^{\alpha_i}}(x)$, and $S_{_f}\theta_i(x) = S_{_f}\widehat{f_{_i}^{\alpha_i}}(x)$.
3. From the first statement the result can be easily deduced.
4. Let $c(x) \in S_{_f}g(x)$. By the assertions (2) and (3) we have for $k \in \{1, 2, \ldots, r\}$

$$\theta(x)c(x) = \theta_{i_k}(x)c(x) = c(x) \mod \widehat{f_{i_k}^{\alpha_{i_k}}}(x),$$

and

$$\theta(x)c(x) = c(x) \mod g(x).$$

It follows that

$$\theta(x)c(x) = c(x) \mod f(x),$$

and

$$\theta(x)c(x) = c(x) \text{ in } S_f g(x).$$

Thus, $\theta(x)$ is the identity element of $S_f g(x)$.

5. From the above statement $\theta_1(x) + \theta_2(x) + \ldots + \theta_s(x)$ is the identity element of $S_f = S_f.1$. Thus $\theta_1(x) + \theta_2(x) + \ldots + \theta_s(x) = 1$.

6. Let $c(x) \in S_f g(x)$. Then

$$\theta(x)c(x) = \theta_{i_1}(x)c(x) + \theta_{i_2}(x)c(x) + \ldots + \theta_{i_r}(x)c(x)$$

is an element of

$$S_f \widehat{f_{i_1}^{\alpha_{i_1}}}(x) + S_f \widehat{f_{i_2}^{\alpha_{i_2}}}(x) + \ldots + S_f \widehat{f_{i_r}^{\alpha_{i_r}}}(x).$$

As $S_f g(x)$ and $S_f \widehat{f_{i_1}^{\alpha_{i_1}}}(x) + S_f \widehat{f_{i_2}^{\alpha_{i_2}}}(x) + \ldots + S_f \widehat{f_{i_r}^{\alpha_{i_r}}}(x)$ are subrings of S_f with same identity element $\theta(x)$, we have

$$S_f g(x) = S_f \widehat{f_{i_1}^{\alpha_{i_1}}}(x) + S_f \widehat{f_{i_2}^{\alpha_{i_2}}}(x) + \ldots + S_f \widehat{f_{i_r}^{\alpha_{i_r}}}(x)$$

Now let $c(x) = c_{i_1}(x) + c_{i_2}(x) + \ldots + c_{i_r}(x) = 0$ in $S_f g(x)$. Multiplying by $\theta_{i_k}(x),\ k = 1, \ldots, r$, we obtain that each $0 = \theta_{i_k}(x)c_{i_k}(x) = c_{i_k}(x)$, and so

$$S_f g(x) = S_f \widehat{f_{i_1}^{\alpha_{i_1}}}(x) \oplus S_f \widehat{f_{i_2}^{\alpha_{i_2}}}(x) \oplus \ldots \oplus S_f \widehat{f_{i_r}^{\alpha_{i_r}}}(x).$$

7. Follows from the above statement.

8. Define for each $i = 1, \ldots, s$, the following homomorphism of rings.

$$\begin{array}{rccc} \varphi_i : & \mathbb{F}_{q^r}[x] & \longrightarrow & S_f \widehat{f_i^{\alpha_i}}(x) \\ & c(x) & \longmapsto & c(x)\theta_i(x) \end{array} \tag{3.17}$$

Clearly φ_i is surjective, and $\ker(\varphi_i) = \mathbb{F}_{q^r}[x]/\langle f_i^{\alpha_i}(x)\rangle$. Thus by the first Isomorphism Theorem $\mathbb{F}_{q^r}[x]/\langle f_i^{\alpha_i}(x)\rangle \cong S_f \widehat{f_i^{\alpha_i}}(x)$. Hence, as $f_i(x)$ is irreducible,

$$S_f \widehat{f_i^{\alpha_i}}(x) \cong \mathbb{F}_{q^{rm_i}}^{\alpha_i} \cong \mathbb{F}_{q^{r\alpha_i m_i}}.$$

However, $\gcd(e, q) = 1$ yields $\alpha_i = 1$, and we deduce that

$$S_f \widehat{f_i^{\alpha_i}}(x) \cong \mathbb{F}_{q^{rm_i}}.$$

As the polynomials $\theta_i(x),\ i = 1, \ldots, s$, lie in $\mathbb{F}_q[x]$, the result of the above theorem holds for $R_f := \mathbb{F}_q[x]/\langle f(x)\rangle$.

Theorem 3.9
The polynomials $\theta_i(x),\ i = 1, \ldots, s$, satisfy the following assertions.

1. $\theta_i(x)$ is the identity element of $R_f \widehat{f_i^{\alpha_i}}(x)$, and $R_f \widehat{f_i^{\alpha_i}}(x) = R_f \theta_i(x)$.

2. $\theta_{i_1}(x), \theta_{i_2}(x), \ldots, \theta_{i_r}(x)$ are mutually orthogonal non-zero idempotent elements of $R_f g(x)$, with $\{i_1, i_2, \ldots, i_r\} = \{1, 2, \ldots, s\} \setminus \{i_1, i_2, \ldots, i_t\}$.

   $\theta(x) = \theta_{i_1}(x) + \theta_{i_2}(x) + \ldots + \theta_{i_r}(x)$ is the identity element of $R_f g(x)$, and $R_f g(x) = R_f \theta(x)$.

3. $R_f g(x) = R_f \widehat{f_{i_1}^{\alpha_{i_1}}}(x) \oplus R_f \widehat{f_{i_2}^{\alpha_{i_2}}}(x) \oplus \ldots \oplus R_f \widehat{f_{i_r}^{\alpha_{i_r}}}(x)$.

4. $R_f \widehat{f_i^{\alpha_i}}(x) \cong \mathbb{F}_{q^{\alpha_{i_k} m_i}}$, and if $\gcd(e, q) = 1$, $R_f \widehat{f_i^{\alpha_i}}(x) \cong \mathbb{F}_{q^{m_i}}$, where $m_i = \deg f_i(x)$.

Remark 3.4
Note that if $\gcd(e, q) = 1$, where e is the polynomial order of $f(x)$, then all factors of $f(x)$ are simple. Hence the result presented in the above theorems can be proved for any divisor $g(x)$ of $f(x)$.

Now we consider the following map ψ, which is a homomorphism of R_f-modules.

$$\begin{array}{rcccc} \psi & & : R_f^r & \longrightarrow & S_f \\ & \underline{\mathbf{a}}(x) = (a_1(x), a_2(x), \ldots, a_r(x)) & & \longmapsto & A(x) = \sum_{i=1}^{r} \beta_i a_i(x) \end{array} \tag{3.18}$$

where $\{\beta_0, \beta_1, \ldots, \beta_{r-1}\}$ is a basis of $\mathbb{F}_{q^r}$ over $\mathbb{F}_q$. Moreover ψ is an R_f-module isomorphism. It follows that each 1-generator quasi f-cyclic code has an equivalent description in S_f.

Now, let us give the following Lemma which was already proved by G.E. Seguin in [79] for the quasi-cyclic case, and by L. Xiuli in [92] for the quasi-negacyclic case. So in our case, a similar result can be proved for quasi f-cyclic codes, by substituting $x^m - 1$ or $x^m + 1$ by $f(x)$.

Lemma 3.1 (Lemma 3 in [79], Lemma 3.4 in [92])
Suppose that $\gcd(e, q) = \gcd(r, \mathrm{ord}_e(q)) = 1$, where $\mathrm{ord}_e(q)$ is the multiplicative order of q modulo e. Let $\underline{\mathbf{a}}(x) \in R_f^r$ and $A(x) = \psi(\underline{\mathbf{a}}(x)) \in S_f$. Then $\gcd(A(x), f(x)) = \gcd(\underline{\mathbf{a}}(x), f(x))$.

Let us denote by $(R_f g(x))^\star$ (resp. $(S_f g(x))^\star$) the group of units of $R_f g(x)$ (respectively, of $S_f g(x)$), where $g(x)$ is a non-zero divisor of $f(x)$ over $\mathbb{F}_q$ as in (3.13). The following Theorem gives an important characterization of a 1-generator quasi f-cyclic code, where the proof is similar to the quasi-cyclic and quasi-negacyclic cases given in [79, Theorem 1, Theorem 2] and [92, Theorem 3.5] respectively.

Theorem 3.10
Suppose that $\gcd(e, q) = \gcd(r, \mathrm{ord}_e(q)) = 1$, and $g(x) = f_1(x) f_2(x), \ldots, f_t(x)$, where $\{1, 2, \ldots, t\} \subseteq \{1, 2, \ldots, s\}$, be a non-zero divisor of $f(x)$ as in (3.13).

1. There is a one-to-one correspondence between the 1-generator f-cyclic codes having $h(x) = \dfrac{f(x)}{g(x)}$ as a parity check polynomial and the elements of the group $\Big(S_f g(x)\Big)^\star \Big/ \Big(R_f g(x)\Big)^\star$.

3. The number of the 1-generator quasi f-cyclic codes having $h(x)$ as a parity check polynomial is

$$\prod_{i=1}^{r} L_i = \prod_{i=1}^{r} \frac{q^{r m_i} - 1}{q^{m_i} - 1}.$$

2. Let $f_i(x),\ i \in \{1,2,\ldots,r\} := \{1,2,\ldots,s\}\backslash\{1,2,\ldots,t\}$, be the primitive idempotent element of $S_f\widehat{f}_i(x)$, and $L_i = \dfrac{q^{rm_i}-1}{q^{m_i}-1}$. Then

$$f_1^{e_1}(x) + f_2^{e_2}(x) + \ldots + f_r^{e_r}(x)$$

with $0 \le e_i \le L_i - 1,\ 1 \le i \le r$, is a complete set of representatives of the $\left(R_f g(x)\right)^\star$ cosets in $\left(S_f g(x)\right)^\star$.

Proof

1. As $\theta(x)$ is the identity element of $R_f g(x)$ and $R_f g(x)$, $\left(R_f g(x)\right)^\star$ is a subgroup of $\left(S_f g(x)\right)^\star$.
Let $A(x) \in \left(S_f g(x)\right)^\star$, then $\gcd(A(x), f(x)) = g(x)$. The above Lemma implies that

$$\gcd(\underline{\mathbf{a}}(x), f(x)) = g(x).$$

Hence $R_f\underline{\mathbf{a}}(x)$ has $h(x) = \frac{f(x)}{g(x)}$ as a p.c.p.
Conversely, if $R_f\underline{\mathbf{a}}(x)$ has $h(x)$ as p.c.p. then

$$\gcd(A(x), f(x)) = \gcd(\underline{\mathbf{a}}(x), f(x)) = g(x),$$

where $f(x) = g(x)h(x)$, and so $A(x) \in S_f\underline{\mathbf{a}}(x)$.
Let now $M(x), B(x) \in \left(S_f g(x)\right)^\star$ generate $C = S_f g(x)$ then $B(x) = w(x)A(x)$, where $\gcd(w(x), h(x)) = 1$. It follows that

$$B(x)A(x)^{-1} = w(x)\theta(x) \in \left(S_f g(x)\right)^\star.$$

Thus $M(x)$ and $B(x)$ belong to the same coset of $\left(R_f g(x)\right)^\star$ in $\left(S_f g(x)\right)^\star$.
Conversely, if $M(x), B(x)$ belong to the same coset of $\left(R_f g(x)\right)^\star$ in $\left(S_f g(x)\right)^\star$, then

$$B(x)A(x)^{-1} = w(x) \in \left(R_f g(x)\right)^\star$$

and $B(x) = w(x)A(x) \in \left(R_f g(x)\right)^\star$. Let us write $w(x) = r(x)\theta(x) \in \left(R_f g(x)\right)^\star$, with $\gcd(r(x), h(x) = 1$. Then $B(x) = r(x)A(x)$ and so they generate the **1**-generator quasi f-cyclic code with a p.c.p $h(x)$.

2. As

$$\left(S_f g(x)\right)^\star \Big/ {}_{\left(R_f g(x)\right)^\star} = \bigoplus_{i=1}^{r} \left(\left(S_f \widehat{f_i^{\alpha_i}}(x)\right)^\star \Big/ {}_{\left(R_f \widehat{f_i^{\alpha_i}}(x)\right)^\star} \right)$$

and

$$\left| \left(S_f \widehat{f_i^{\alpha_i}}(x)\right)^\star \Big/ {}_{\left(R_f \widehat{f_i^{\alpha_i}}(x)\right)^\star} \right| = \frac{q^{rm_i}-1}{q^{m_i}-1} = L_i$$

the cardinality of the factor group $\left(S_f g(x)\right)^\star \Big/ {}_{\left(R_f g(x)\right)^\star}$ is $\prod_{i=1}^{r} L_i = \prod_{i=1}^{r} \dfrac{q^{rm_i}-1}{q^{m_i}-1}$.

3. It is sufficient to prove that

$$f_1^{e_1}(x) + f_2^{e_2}(x) + \ldots + f_r^{e_r}(x)$$

with $0 \leq e_i \leq L_i - 1,\ 1 \leq i \leq r$, belongs to distinct cosets of $\left(R_f g(x)\right)^\star$ in $\left(S_f g(x)\right)^\star$. If $M(x) = \sum_{i=0}^{r} f_i^{e_i}(x)$ and $B(x) = \sum_{i=0}^{r} f_i^{f_i}(x)$ belong to the same coset of $\left(R_f g(x)\right)^\star$ in $\left(S_f g(x)\right)^\star$, then

$$A(x)B(x)^{-1} = \sum_{i=0}^{r} f_i^{e_i - f_i}(x) \in \left(R_f g(x)\right)^\star.$$

It follows that $\left(f_i^{e_i - f_i}(x)\right)^{q^{m_i}-1} = \theta_i(x)$ and $q^{rm_i} - 1$ divides $e_i - f_i(q^{m_i} - 1)$. Hence L_i divides $e_i - f_i$, with $-L_i \leq e_i - f_i \leq L_i$, which means that $e_i = f_i,\ i = 1, \ldots, r$. Thus the result holds.

Example 3.4
Let $f(x) = x^7 + 2x^3 + x + 2 \in \mathbb{F}_3[x]$, whose irreducible factorization is given by:

$$\begin{array}{rl} f(x) = & (x+2) \cdot (x^2 + 2x + 2) \cdot (x^4 + 2x^3 + x^2 + x + 2) \\ = & f_1(x) \cdot f_2(x) \cdot f_3(x) \end{array} \tag{3.19}$$

The linear code generated by

$$\underline{\mathbf{a}}(x) = (x^3 + x^2 + 1\ ;\ x^5 + x^4 + 2x^3 + x + 1\ ;\ x^3 + x^2 + 1) \in R_f^3$$

is a **1**-generator quasi f-cyclic code of length $n = 21$ over $\mathbb{F}_3$. Its generator polynomial is

$$g(x) = \gcd(\underline{\mathbf{a}}(x), f(x)) = x + 2$$

Then the dimension of C is $7 - \deg(g(x)) = 6$. Now, let

$$h(x) = \frac{f(x)}{x+2} = f_2(x) \cdot f_3(x)$$

where $f_i(x),\ i = 2, 3$, are the irreducible factors of $f(x)$. Then as

$$\gcd(80, 7) = \gcd(3, \mathrm{ord}_{80}(7)) = 1,$$

where $\mathrm{ord}_{80}(7) = 4$, so according to Theorem 3.10, there are

$$L_2 \times L_3 = 91 \times 6643 = 604513$$

1-generator quasi f-cyclic codes of length $n = 21$ over $\mathbb{F}_3$ having $h(x)$ as a parity check polynomial.

Let

$$\widehat{f_2}(x) = \frac{f(x)}{f_2(x)} = \frac{f(x)}{x^2 + 2x + 2} = x^5 + x^4 + 2x^3 + x + 1,$$

$$\widehat{f_3}(x) = \frac{f(x)}{f_3(x)} = \frac{f(x)}{x^4 + 2x^3 + x^2 + x + 2} = x^3 + x^2 + 1.$$

We generate $\mathbb{F}_{27}$ with $\beta^3 + 2\beta + 1 = 0$, so that

$$\mathbb{F}_{27} = \left\{ x_1 + x_2\beta + x_3\beta^2 \quad : \quad x_1, x_2, x_3 \in \mathbb{F}_3 \right\}$$

with a primitive element β.

The idempotent and primitive elements for $S\widehat{f_2}(x)$ and $S\widehat{f_3}(x)$ may be found as

$$\begin{aligned}
\theta_2(x) &= x \cdot \widehat{f_2}(x) = x^6 + x^5 + 2x^4 + x^2 + x \\
\pi_2(x) &= (\beta x + \beta^2)\theta_2(x) \\
&= (\beta^2 + \beta)\, x^6 + \beta^2 x^5 + 2\beta x^4 + (2\beta^2 + 2\beta)\, x^3 \\
&\quad + (\beta^2 + 2\beta)\, x^2 + (2\beta^2 + 2\beta)\, x + \beta \\
\theta_3(x) &= (2x + 2) \cdot \widehat{f_3}(x) = 2x^4 + x^3 + 2x^2 + 2x + 2 \\
\pi_3(x) &= (\beta x^2 + 1)\theta_3(x) \\
&= 2\beta x^6 + \beta x^5 + (2\beta + 2)x^4 + (2\beta + 1)x^3 \\
&\quad + (2\beta + 2)x^2 + 2x + 2
\end{aligned} \tag{3.20}$$

It follows that

$$A(e_2, e_3, x) = \pi_2(x)^{e_2} + \pi_3(x)^{e_3},\ 0 \le e_2 \le 90,\ 0 \le e_3 \le 6642,$$

are all generators of **1**-generator quasi f-cyclic codes of length $n = 21$ over $\mathbb{F}_3$ having $h(x)$ as a parity check polynomial.

If we choose $e_2 = e_3 = 0$, we get

$$A(0, 0, x) = \theta_2(x) + \theta_3(x) = x^6 + x^5 + 2x^4 + x^3 + 2$$

which generates the **1**-generator quasi f-cyclic code $R_f\underline{\mathbf{a}}(x)$, where

$$\underline{\mathbf{a}}(x) = (x^6 + x^5 + 2x^4 + x^3 + 2, f(x), f(x)) \in R_f^3.$$

As a second example let $e_2 = 1,\ e_3 = 1$ then we get

$$\begin{aligned}
A(1, 1, x) &= \pi_2(x) + \pi_3(x) \\
&= 2 + 2x + 2x^2 + x^3 + 2x^4 + \beta(x^5 + x^4 + x^3 + x^2 + 2x + 1) + \beta^2(x^6 + x^5 + 2x^3 + x^2 + 2x)
\end{aligned} \tag{3.21}$$

which generates the **1**-generator quasi f-cyclic code $R_f\underline{\mathbf{a}}(x)$, where

$$\underline{\mathbf{a}}(x) = (2x^4 + x^3 + 2x^2 + 2x + 2, \ldots x^5 + x^4 + x^3 + x^2 + 2x + 1, x^6 + x^5 + 2x^3 + x^2 + 2x) \in R_f^3. \tag{3.22}$$

3.6 Additive structure of quasi f-cyclic codes

In this section we are interested in $\mathbb{F}_{q^r}$-linear images of quasi f-cyclic codes seen as additive f-cyclic codes over $\mathbb{F}_{q^r}$. Let us start with the CRT decomposition of a quasi f-cyclic code as in the case of quasi-cyclic codes.

Assume that $\gcd(e, q) = 1$, where e is the order of $f(x)$, and let the irreducible factorization of $f(x)$ be

$$f(x) = f_1(x)f_2(x)\ldots f_s(x)$$

By the Chinese Remainder Theorem (CRT) we have the following ring isomorphism:

$$R_f \cong \bigoplus_{i=1}^{s} \mathbb{F}_q[x]/\langle f_i(x)\rangle$$

Let ξ be a primitive $e^{\text{ th}}$ root of unity. As each $f_i(x), i = 0, \ldots, n-1$ is irreducible, $\mathbb{E}_i := \mathbb{F}_q(\xi^{u_i})$ is a field, where u_i is the smallest integer such that ξ^{u_i} is a root of $f_i(x)$. Hence R_f^r can be decomposed as follows

$$R_f^r \cong \mathbb{E}_1^r \oplus \mathbb{E}_2^r \oplus \ldots \oplus \mathbb{E}_s^r.$$

So, each quasi f-cyclic code $\varphi_{f,r}(C)$ can be viewed as an $(\mathbb{E}_1 \oplus \ldots \oplus \mathbb{E}_s)$-submodule of $\mathbb{E}_1^r \oplus \ldots \oplus \mathbb{E}_s^r$ and can be decomposed as

$$\varphi_{f,r}(C) = C_1 \oplus \ldots \oplus C_s \tag{3.23}$$

where C_i is a linear code of length r over $\mathbb{E}_i$ and is called **a constituent of** C**.** As in [33] one can establish that each C_j is of the form

$$C_i := \mathrm{Span}_{\mathbb{E}_i}\left\{\left(c_1\left(\xi^{u_j}\right), \ldots, c_r\left(\xi^{u_j}\right)\right)\right\}$$

for each $c(x) = (c_1(x), \ldots, c_r(x)) \in \varphi_{f,r}(C)$.
Let now $B = \{e_1, \ldots, e_r\}$ be a basis of the extension of $\mathbb{F}_{q^r}/\mathbb{F}_q$. As in [32], we consider the following R_f-module isomorphism between R_f^r and $S_f := \mathbb{F}_{q^r}/\langle f(x)\rangle$, given by:

$$\begin{array}{ccccc} \Phi_B & : & R_f^r & \longrightarrow & S_f \\ & & c(x) = (c_1(x), c_2(x), \ldots, c_r(x)) & \longmapsto & \Phi_B(c(x)) = \sum_{i=1}^{r} c_i(x) e_i. \end{array} \tag{3.24}$$

with $c_k(x) = \sum_{i=0}^{m-1} c_i x^i$, for each $k = 1, 2, \ldots, r$.

Thus Φ_B gives a correspondence between quasi f-cyclic codes of length $n = mr$ over $\mathbb{F}_q$, and f-cyclic codes of length m over $\mathbb{F}_{q^r}$ (additive f-cyclic code induced by the vector a). The proof of the following result is omitted, as is similar to the case of quasi-cyclic codes [32].

Proposition 3.7
A linear code $C \subseteq \mathbb{F}_q^{mr}$ is a quasi f-cyclic code if and only if $\Phi_B(C)$ is an additive f-cyclic code of length m over $\mathbb{F}_{q^r}$.

Now, we ask when the image $\Phi_B(C)$ is a linear f-cyclic code (not just additive f-cyclic) of length m over $\mathbb{F}_{q^r}$. To answer this question, let for each $1 \leq i, j \leq r$,

$$e_i e_j = \sum_{k=1}^{r} \lambda_k^{i,j} e_k \in \mathbb{F}_{q^r},$$

and define the $r \times r$ matrices $M_i, 1 \leq i \leq r$ by

$$M_i = \begin{pmatrix} \lambda_1^{i,1} & \lambda_1^{i,2} & \ldots & \lambda_1^{i,r} \\ \vdots & \vdots & \vdots & \vdots \\ \lambda_r^{i,2} & \lambda_r^{i,1} & \ldots & \lambda_r^{i,r} \end{pmatrix} \tag{3.25}$$

Following [32], we give the following result for quasi f-cyclic codes.

Theorem 3.11
Let C be a quasi f-cyclic code of length $n = mr$, and $\varphi_{f,r}(C) = C_1 \oplus C_2 \oplus \ldots \oplus C_s$ be its constituent decomposition as in (3.23). Then,

$$\begin{array}{lll} \Phi_{\mathcal{B}}(C) \text{ is } \mathbb{F}_{q^r}\text{-linear} & \Longleftrightarrow & \varphi_{f,r}(C) \text{ is invariant under the matrices } M_i,\ 1 \leq i \leq r. \\ & \Longleftrightarrow & C_i \subseteq \mathbb{E}_i^r,\ i = 1, \ldots, s \text{ is invariant under the matrices } M_i,\ 1 \leq i \leq r. \end{array}$$

Proof
Let $c = (c_1, c_2, \ldots, c_r) \in C$, and $c(x) := (c_1(x), c_2(x), \ldots, c_r(x)) \in \varphi_{f,r}(C)$, then

$$
\begin{aligned}
e_i\Phi_B(c(x)) = \; & e_i\sum_{j=1}^{r} c_j(x)e_j. \\
= \; & \sum_{j=1}^{r} c_j(x)e_ie_j. \\
= \; & \sum_{j=1}^{r} c_j(x)\sum_{k=1}^{r}\lambda_k^{i,j}e_k \\
= \; & c_1(x)\left(\lambda_1^{i,1}e_1+\ldots+\lambda_{r-1}^{i,1}e_r\right)+\ldots+c_r(x)\left(\lambda_1^{i,r}e_1+\ldots+\lambda_r^{i,r}e_r\right) \\
= \; & \left(\lambda_1^{i,1}c_1(x)+\ldots+\lambda_1^{i,r}c_r(x)\right)e_1+\ldots+\left(\lambda_r^{i,1}c_1(x)+\ldots+\lambda_r^{i,r}c_r(x)\right)e_r
\end{aligned}
$$

$$
\begin{aligned}
\Phi_B(C) \text{ is } \mathbb{F}_{q^r}\text{-linear} \quad & \Longleftrightarrow \quad \forall i=1,\ldots,r, \;\; e_i\Phi_B(C)\in\Phi_B(C) \\
& \Longleftrightarrow \quad \forall i=1,\ldots,r, \; \forall 1\le j\le r, \;\; \left(\lambda_j^{i,1}c_1(x),\ldots,\lambda_j^{i,r}c_r(x)\right)\in\varphi_{f,r}(C) \\
& \Longleftrightarrow \quad \forall i=1,\ldots,r, \;\; M_i\; c(x)^t\in\varphi_{f,r}(C)
\end{aligned}
$$

This means that $\varphi_{f,r}(C)$ is invariant under multiplication by the matrix $M_i,\ 1\le i\le r$.

Remark 3.5
Suppose that $B=\{1,\alpha,\alpha^2,\ldots,\alpha^{r-1}\}$ is a basis of the extension $\mathbb{F}_{q^r}=\mathbb{F}_q(\alpha)$, and $f_\alpha(x)=x^r+\alpha_{r-1}x^{r-1}+\ldots+\alpha_0\in\mathbb{F}_q[x]$ be the minimal polynomial of α. Then the following assertions hold.

1. The matrix M_1 of the above theorem is the identity matrix and M_i is the matrix corresponding to the multiplication by α^i, for each $i=1,\ldots,r-1$.
2. The matrix M_2 is the companion matrix C_{f_α} of the polynomial $f_\alpha(x)$.

From the above remark, we deduce the following characterization about the $\mathbb{F}_{q^r}$-linearity of $\Phi_B(C)$.

Corollary 3.3
Let C be a quasi f-cyclic code of length $n=mr$, and $\varphi_{f,r}(C)=C_1\oplus C_2\oplus\ldots\oplus C_s$, its constituent decomposition as in (3.23). Then the following assertions are equivalent.

1. $\Phi_B(C)$ is $\mathbb{F}_{q^r}$-linear
2. $\varphi_{f,r}(C)$ is invariant by the matrix $M_1=C_{f_\alpha}$.
3. Each $C_i\subseteq\mathbb{E}_i^r,\ i=1,\ldots,s$ is invariant under the matrix C_{f_α}.
4. $C_i\subseteq\mathbb{E}_i^r$, is a right polycyclic code induced by the vector $(-\alpha_0,-\alpha_1,\ldots,-\alpha_{r-1})\in\mathbb{E}_i^r$.
5. There is a monic divisor $g(x)$ of $f_\alpha(x)$ such that C_i is seen as an ideal of $\mathbb{E}_i[x]/\langle f_\alpha(x)\rangle$ and $\dim_{\mathbb{E}_i}=\deg(g(x))$.

Assume that $\gcd(m,q)=1$ and let $t_i=[\mathbb{E}_i:\mathbb{F}_q]$ be the degree of the extension $\mathbb{E}_i/\mathbb{F}_q$, $b_i=\gcd(r,t_i)$, and $d_i=\dfrac{r}{b_i}$. According to [52, Theorem 3.46], the irreducible factorization of $f_\alpha(x)$ over each $\mathbb{E}_i$ is

$$f_\alpha(x)=f_{i,1}(x)f_{i,2}(x)\cdots f_{i,b_i}(x),$$

where each $f_{i,j}(x)\in\mathbb{E}_i[x], j=1\ldots,b_i$, is of degree d_i.

As in section 2.5, Chapter 2, let for $i = 1, \ldots, s,\ j = 1, \ldots, b_i$ the null space $U_{i,j}$ of each $f_{i,j}(x)(C_{f_\alpha})$, where C_{f_α} is the companion matrix of f_α which is viewed as an $r \times r$ matrix in $\mathbb{M}_r(\mathbb{E}_i)$, given by

$$U_{i,j} := \ker\left(f_{i,j}(C_{f_\alpha})\right) = \left\{v \in \mathbb{E}_i^r \ :\ f_{i,j}(C_{f_\alpha})\, v = 0,\ j = 0, \ldots, b_i\right\}. \tag{3.26}$$

Then we have

- $\mathbb{E}_i^r = U_{i,1} \oplus U_{i,2} \oplus \cdots \oplus U_{i,b_i}$,
- Each $U_{i,j},\ j = 1, \ldots, b_i$ is a minimal invariant subspace of $\mathbb{E}_i^r$ under C_{f_α}.
- If W_i is an invariant subspace of $\mathbb{E}_i^r$ under C_{f_α} and $W_{i,j} = W_i \cap U_{i,j}$, for $j = 1, \ldots, b_i$, then $W_{i,j}$ is also invariant under C_{f_α} and $W_i = W_{i,1} \oplus W_{i,2} \oplus \cdots \oplus W_{i,b_i}$.
- $\dim_{\mathbb{E}_i}(U_{i,j}) = \deg(f_{i,j}(x))$. Hence we deduce the following result on the $\mathbb{F}_{q^r}$-linearity of $\Phi_B(C)$ when $\gcd(m, q) = 1$.

Theorem 3.12
Let C be a quasi f-cyclic code of length $n = mr$ over $\mathbb{F}_q$ such that $\Phi_B(C)$ is $\mathbb{F}_{q^r}$-linear, and

$$\varphi_{f,r}(C) = C_1 \oplus C_2 \oplus \ldots \oplus C_s \subseteq \mathbb{E}_1^r \oplus \mathbb{E}_2^r \oplus \ldots \oplus \mathbb{E}_s^r,$$

be the constituent decomposition of C. Then,

1. if $\gcd(r, t_i) = 1$, then C_i is either $\{0\}$ or the full space $\mathbb{E}_i^r$.
2. otherwise, $\dim_{\mathbb{E}_i}(C_i) = k_i d_i$ for some $0 \leq k_i \leq b_i$.

Proof
If $\Phi_B(C)$ is $\mathbb{F}_{q^r}$-linear, then each $C_i \subseteq \mathbb{E}_i^r$ is invariant under M_α. It follows that

$$C_i = C_i \cap \mathbb{E}_i^r = C_{i,1} \oplus C_{i,2} \oplus \ldots \oplus C_{i,b_i}, \quad \text{where each } \ C_{i,j} := C_i \cap U_{i,j} \text{ as in (3.26)}.$$

1. If $\gcd(r, t_i) = 1$, then $f_\alpha(x)$ is irreducible in $\mathbb{E}_i[x]$, and so there is no trivial subspaces of $\mathbb{E}_i^r$ which are invariant under M_α. Thus C_i is either $\{0\}$ or the full space $\mathbb{E}_i^r$.
2. Otherwise, since each $U_{i,j}$ are minimal invariant subspaces, then

$$\dim_{\mathbb{E}_i}(C_{i,j}) = \dim_{\mathbb{E}_i}(C_i \cap U_{i,j}) = \begin{cases} 0 & \text{if } C_i \cap U_{i,j} = \{0\} \\ \deg(f_{i,j}) = d_i & \text{if } C_i \cap U_{i,j} \neq \{0\} \end{cases}$$

It follows that $\dim_{\mathbb{E}_i}(C_i) = \sum_{j=1}^{b_i} \dim_{\mathbb{E}_i}(C_{i,j}) = k_i d_i$ for some $0 \leq k_i \leq b_i$.

Corollary 3.4
Let C be a **1**-generator quasi f-cyclic code generated by $\underline{\mathbf{a}}(x) = (a_0(x), a_1(x), \ldots, a_{\ell-1}(x)) \in R_f^r$, and $g(x)$ be its generator polynomial. If $\Phi_B(C)$ is $\mathbb{F}_{q^r}$-linear then for each $i = 1, \ldots, s$ such that $\gcd(r, t_i) = 1$, the corresponding constituent $C_i = \{0\}$. This is equivalent to saying that the irreducible factor $f_i(x)$ of $f(x)$ corresponding to $\mathbb{E}_i := \mathbb{F}_q(\xi^{u_i})$ must divide $g(x)$.

Example 3.5
In this example, we are interested in quasi f-cyclic codes of index 4 and of length 20 over $\mathbb{F}_3$ with associated vector $a = (2, 1, 1, 2, 2) \in \mathbb{F}_3^5$. Let α be a primitive element of $\mathbb{F}_{3^4}$ with minimal polynomial $C_{f_\alpha}(x) = x^4 + 2x^3 + x^2 + 2x + 1 = 0$. Hence $B = \{1, \alpha, \alpha^2, \alpha^3\}$ is a basis of the extension $\mathbb{F}_{3^4}/\mathbb{F}_3$. Let the irreducible factorization of $f(x)$ be

$$\begin{aligned} f(x) &= (x+1) \cdot (x^4 + 2x^3 + x^2 + 2x + 1) \\ &= f_1(x) f_2(x). \end{aligned}$$

Then $\mathbb{E}_1 := \mathbb{F}_3[x]/\langle f_1(x)\rangle = \mathbb{F}_3$ and $\mathbb{E}_2 := \mathbb{F}_3[x]/\langle f_2(x)\rangle = \mathbb{F}_{3^4}$. Let C be the quasi f-cyclic code generated by $a(x) = (a_1(x), a_2(x), a_3(x), a_4(x)) \in R_{3,5}^4$, where

$$\begin{aligned} a_1(x) &= x^3 + x^2 \\ a_2(x) &= 2x^3 + 2x^2 \\ a_3(x) &= x^4 + 2x^3 + x + 2 \\ a_4(x) &= x^3 + 1 \end{aligned}$$

It follows that the constituent components of C are

$$\begin{aligned} &C_1 = \mathrm{Span}_{\mathbb{E}_1}(a_1(-1), a_2(-1), a_3(-1), a_4(-1)) = \{0\} \subseteq \{0\}. \\ &C_2 = \mathrm{Span}_{\mathbb{E}_2}\{(a_1(\alpha), a_2(\alpha), a_3(\alpha), a_4(\alpha))\} = \mathrm{Span}_{\mathbb{E}_2}\{(1,\ 2,\ 2\alpha^3 + 2\alpha^2 + 2\alpha,\ 2\alpha^2 + \alpha)\} \subseteq \mathbb{E}_2^4. \end{aligned}$$

As $\gcd(4, 3) = 1$, the polynomials f_1 and f_2 are irreducible over $\mathbb{E}_1$ and $\mathbb{E}_2$ respectively. Recall that the companion matrix of $C_{f_\alpha}(x)$ is

$$C_{f_\alpha} = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

It follows that

$$C_{f_\alpha} C_2 = \mathrm{Span}_{\mathbb{E}_2}\left\{\left(2\alpha^3 + 2,\ 2\alpha^3 + \alpha^2 + 1,\ \alpha^3 + 2\alpha^2 + 2,\ \alpha^3 + 2\alpha^2 + 2\alpha + 2\right)\right\} \subseteq C_2$$

which means that $C = C_1 \oplus C_2 \subseteq \mathbb{E}_1^4 \oplus \mathbb{E}_2^4$ has an $\mathbb{F}_{3^4}$-linear image with respect to the basis B.

Chapter 4

Linear codes invariant under a linear transformation

In this chapter, we are interested in linear codes invariant under an operator T, that we call T-codes. Since each operator T can be represented by a matrix, we will, for simplicity, consider linear codes invariant under right multiplication by a square matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$ and we call them M-codes. In particular we distinguish three fundamental types: M-cyclic codes, quasi M-cyclic codes and $(M_1, M_2, \ldots, M_r)$-multi-cyclic codes. The approach developed here allows us to realize cyclic codes and many of their generalizations as special cases. It also helps us better understand the structure of linear codes invariant under some special matrices such as a permutation matrix or a monomial matrix. Next, we study the duality of M-codes, where we show that the b-dual of an M-code is an M^*-code, where M^* is the adjoint matrix of M with respect to a non-degenerate bilinear form b, and we explore some important results on the duality of these codes. In order to use polynomial rings in the study of M-codes, we show a one-to-one correspondence between M-codes and $\mathbb{F}_q[x]$-submodules of $\prod_{i=1}^{r} R_{f_i}$, where $R_{f_i} := \mathbb{F}_q[x]/\langle f_i(x)\rangle$ and $(f_1(x), f_2(x), \ldots, f_r(x))$ is the sequence of invariant factors of M. Finally, we investigate the structure of **1**-generator M-codes and we prove BCH-like and Hartmann-Tzeng-like bounds for them. As an application of the theory developed here, we construct some optimal linear codes.

4.1 $(M_1, M_2, \ldots, M_r)$-multi-cyclic codes

We start with the following definition.

Definition 4.1
Let C be a linear code of length n over $\mathbb{F}_q$. Then we say that C is a/an

1. **M-code** if it is invariant under T_M, where M is an $n \times n$ square matrix.
2. **$(M_1, M_2, \ldots, M_r)$-multi-cyclic code** if it is invariant under $\mathrm{diag}(M_1, M_2, \ldots, M_r)$ where each M_i is a square $n_i \times n_i$ cyclic matrix and $n = n_1 + n_2 + \ldots + n_r$.
3. **$(f_1, f_2, \ldots, f_r)$-multi-cyclic code** if it is invariant under $\mathrm{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$, where $f_i(x),\ i = 1, \ldots, r$ are non-zero polynomials of degree n_i.

Proposition 4.1
Let $C \subseteq \mathbb{F}_q^n$ be a linear code and M is a square matrix with the sequence of invariant factors $(f_1(x), f_2(x), \ldots, f_r(x))$. Then C is an M-code if and only if $P^{-1}C$ is an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code of length n, with $\mathrm{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r}) = P^{-1}MP$.

Proof
As M and $D = \mathrm{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$ are similar, then there is an invertible matrix P such $D = P^{-1}MP$, and so the result holds from Proposition 2.2.

In the next theorem, we prove that every $(f_1, f_2, \ldots, f_r)$-multi-cyclic (resp. $(M_1, M_2, \ldots, M_r)$-multi-cyclic) code can be decomposed as the direct sum of f_i-cyclic (resp. M_i-cyclic) codes.

Theorem 4.1
Let $C \subseteq \mathbb{F}_q^n$ be a linear code. Then

1. C is an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code if and only if $C = C_1 \oplus C_2 \oplus \cdots \oplus C_r$, where each C_i is an f_i-cyclic code of length n_i.

2. C is an $(M_1, M_2, \ldots, M_r)$-multi-cyclic code if and only if $C = C_1 \oplus C_2 \oplus \cdots \oplus C_r$, where each C_i is an M_i-cyclic code of length n_i.

Proof
1. Let C be an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code and $D = \mathrm{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$. Then $T_D(C) \subseteq C$. From Theorem 3.1, one can write $C = C_1 \oplus C_2 \oplus \cdots \oplus C_r \subseteq \mathbb{F}_q^n$, with $C_i = C \cap F_{w_i}$ for each i. It follows that

$$T_D(C) = T_{C_{f_1}}(C_1) \oplus T_{C_{f_2}}(C_2) \oplus \cdots \oplus T_{C_{f_r}}(C_r) \subseteq C_1 \oplus C_2 \oplus \cdots \oplus C_r.$$

   Hence $T_{C_{f_i}}(C_i) \subseteq C_i$ for $i = 1, \ldots, r$, and each C_i is an f_i-cyclic code.

2. Similar argument as in the proof of the first statement.

In order to make use of polynomial rings in the study of $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes. Let us consider the $\mathbb{F}_q[x]$-module isomorphisms φ_{f_i}, $i = 1, \ldots, r$ between $\mathbb{F}_q^{n_i}$ and $R_{f_i} := \mathbb{F}_q[x]/\langle f_i(x)\rangle$ as in (2.2). Then the $\mathbb{F}_q[x]$-module $\varphi := (\varphi_{f_1}, \varphi_{f_2}, \ldots, \varphi_{f_r})$, defined by

$$\begin{array}{rccc} \varphi : & \mathbb{F}_q^n & \longrightarrow & \prod_{i=1}^{r} R_{f_i}, \\ & v = (v_1, v_2, \ldots, v_r) & \longmapsto & \varphi(v) = (\varphi_{f_1}(v_1), \varphi_{f_2}(v_2), \ldots, \varphi_{f_r}(v_r)) \end{array} \tag{4.1}$$

is an $\mathbb{F}_q[x]$-module isomorphism between $\mathbb{F}_q^n$ and $\prod_{i=1}^{r} R_{f_i}$, with $v_i = (v_{i,0}, v_{i,1}, \ldots, v_{i,n_1}) \in \mathbb{F}_q^{n_i}$.

Theorem 4.2
Let C be a linear code of length n over $\mathbb{F}_q$, and let M be a square matrix with sequence of invariant factors $(f_1(x), f_2(x), \ldots, f_r(x))$. Then

1. C is an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code if and only if $\varphi(C)$ is an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{f_i}$.

2. C is an M-code if and only if $\psi(C)$ is an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{f_i}$ where $\psi := \varphi \circ P^{-1}$ and $\mathrm{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r}) = P^{-1}MP$.

Proof
1. Let C be an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code. By Theorem 4.1, C can be decomposed as

$$C = C_1 \oplus C_2 \oplus \cdots \oplus C_r,$$

where each C_i is an f_i-cyclic code. Then $\varphi(C) = (\varphi_{f_1}(C_1), \varphi_{f_2}(C_2), \ldots, \varphi_{f_r}(C_r)) \subseteq \prod_{i=1}^{r} R_{f_i}$. By Theorem 2.2, we have that $\varphi_{f_i}(C_i),\ i = 1, \ldots, r$, is an ideal of R_{f_i} and so it is an $\mathbb{F}_q[x]$-submodule of R_{f_i}. It follows that $\varphi(C)$ is an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{f_i}$.

2. If C is an M-code, then by Proposition 4.1, $P^{-1}C$ is an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code where M is similar to $\mathrm{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$ by P. So, by the above statement $\psi(C) = \varphi(P^{-1}C)$ is an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{f_i}$.

We now give the definition of an **s**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic and an $(M_1, M_2, \ldots, M_r)$-multi-cyclic code.

Definition 4.2

1. An $(f_1, f_2, \ldots, f_r)$-multi-cyclic code C is called **s**-generator if $\varphi(C)$ is generated by s elements as an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{f_i}$, i.e.,

$$\varphi(C) = \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_1}(x) + \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_2}(x) + \ldots + \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_s}(x)$$

where $\underline{\mathbf{a}_i}(x) = (a_{i,1}(x), a_{i,2}(x), \ldots, a_{i,r}(x)) \in \prod_{i=1}^{r} R_{f_i}$.

2. An $(M_1, M_2, \ldots, M_r)$-multi-cyclic code C is called **s**-generator if $\psi(C)$ is generated by s elements as an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{f_i}$, i.e.,

$$\psi(C) = \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_1}(x) + \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_2}(x) + \ldots + \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_s}(x).$$

Proposition 4.2

Let C be an s-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code generated by $\underline{\mathbf{a}_1}(x), \underline{\mathbf{a}_2}(x), \ldots, \underline{\mathbf{a}_s}(x)$ where $\underline{\mathbf{a}_i}(x) = (a_{i,1}(x), a_{i,2}(x), \ldots, a_{i,r}(x)) \in \prod_{j=1}^{r} R_{f_j}$ for every $1 \leq i \leq s$. Then, there exist polynomials $b_{i,j}(x)$ in $\mathbb{F}_q[x]$, such that $a_{i,j}(x) = b_{i,j}(x) g_{i,j}(x),\ 1 \leq i \leq s,\ 1 \leq j \leq r$, where $g_{i,j}(x)$ is a divisor of $f_i(x)$.

Proof

Let $\varphi(C) = \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_1}(x) + \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_2}(x) + \ldots + \mathbb{F}_q[x] \cdot \underline{\mathbf{a}_s}(x)$ be an s-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code. Then for each $i = 1, \ldots, s$, $\mathbb{F}_q[x] \cdot \underline{\mathbf{a}_i}(x)$ is a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code. According to Theorem 4.1, we can decompose $\mathbb{F}_q[x] \cdot \underline{\mathbf{a}_i}(x),\ i = 1, \ldots, s$, as

$$\mathbb{F}_q[x] \cdot \underline{\mathbf{a}_i}(x) = \mathbb{F}_q[x] \cdot a_{i,1}(x) \oplus \mathbb{F}_q[x] \cdot a_{i,2}(x) \oplus \cdots \oplus \mathbb{F}_q[x] \cdot a_{i,r}(x),$$

where each $\mathbb{F}_q[x] \cdot a_{i,j}(x),\ j = 1, \ldots, r$ is an f_j-cyclic code of length $n_j = \deg(f_j(x)),\ j = 1, \ldots, r$. Let $g_{i,j}(x)$ be the standard generator polynomial of $\mathbb{F}_q[x] \cdot a_{i,j}(x)$, i.e., $f_i(x) = h_{i,j}(x) g_{i,j}(x)$, for some $h_{i,j}(x) \in \mathbb{F}_q[x]$. Finally, as $a_{i,j}(x) \in \mathbb{F}_q[x] \cdot a_{i,j}(x)$, there exists $b_{i,j}(x) \in \mathbb{F}_q[x]$ such that $a_{i,j}(x) = b_{i,j} g_{i,j}(x)$.

4.2 Connection between $(M_1, M_2, \ldots, M_r)$-multi-cyclic codes and other types of codes

This section is devoted to the connections between M-codes and some well-known types of codes in the literature. We start by defining more classes of codes in the literature as special cases.

Definition 4.3
Let $\Lambda = (a_1, a_2, \ldots, a_r)$ where $a_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$ for every i, and let C be a linear code of length $n = n_1 + n_2 + \cdots + n_r$ over $\mathbb{F}_q$. Let

$$c = \left(\overbrace{c_{1,0}, c_{1,1}, \ldots, c_{1,n_1-1}}^{c_1}, \overbrace{c_{2,0}, c_{2,1}, \ldots, c_{2,n_2-1}}^{c_2}, \ldots, \overbrace{c_{r,0}, c_{r,1}, \ldots, c_{r,n_r-1}}^{c_r} \right) \in C.$$

be an arbitrary codeword. We say that C is

1. a generalized right quasi-polycyclic code induced by Λ, if

$$(T_{\overrightarrow{a_1}}(c_1), T_{\overrightarrow{a_2}}(c_2), \ldots, T_{\overrightarrow{a_r}}(c_r)) \in C \tag{4.2}$$

   where $T_{\overrightarrow{a_i}}(c_i) = (0, c_{i,0}, c_{i,1}, \ldots, c_{i,n_i-2}) + c_{i,n_i-1}a_i$, for $1 \leq i \leq r$, as in Example 2.1.

2. a generalized left quasi-polycyclic code induced by Λ, if

$$(T_{\overleftarrow{a_1}}(c_1), T_{\overleftarrow{a_2}}(c_2), \ldots, T_{\overleftarrow{a_r}}(c_r)) \in C \tag{4.3}$$

   where $T_{\overleftarrow{a_i}}(c_i) = (c_{i,1}, c_{i,2}, \ldots, c_{i,n_i-1}, 0) + c_{i,0}a_i$ for $1 \leq i \leq r$, as in Example 2.1.

3. a generalized quasi-monomial code induced by Λ, if

$$\left(T_{\overline{a}_1}(c_1), T_{\overline{a}_2}(c_2), \ldots, T_{\overline{a}_r}(c_r)\right) \in C \tag{4.4}$$

   where $T_{\overline{a}_i}(c_i) = (a_{i,n_i-1}c_{i,n_i-1}, a_{i,0}c_{i,0}, \ldots, a_{i,n_i-2}c_{i,n_i-2})$, for $1 \leq i \leq r$, as in Example 2.1.

4. a $(\sigma_1, \sigma_2, \ldots, \sigma_r)$-generalized quasi-cyclic code, if

$$\left(T_{\sigma_1}(c_1), T_{\sigma_2}(c_2), \ldots, T_{\sigma_r}(c_r)\right) \in C, \tag{4.5}$$

   where σ_i is an n_i-cycle and $T_{\sigma_i}(c_i) = (c_{i,\sigma_i(0)}, c_{i,\sigma_i(1)}, \ldots, c_{i,\sigma_i(n_i-1)})$, for $1 \leq i \leq r$, as in Example 2.1.

5. a $(\sigma_1, \sigma_2, \ldots, \sigma_r)$-generalized quasi-monomial code induced by Λ, if

$$\left(T_{\sigma,\overline{a}_1}(c_1), T_{\sigma,\overline{a}_2}(c_2), \ldots, T_{\sigma,\overline{a}_r}(c_r)\right) \in C \tag{4.6}$$

   where σ_i is an n_i-cycle and $T_{\sigma_i,\overline{a}_i}(c_i) = (a_{i,\sigma_i(0)}c_{i,\sigma_i(0)}, a_{i,\sigma_i(1)}c_{i,\sigma_i(1)}, \ldots, a_{i,\sigma_i(n_i-1)}c_{i,\sigma_i(n_i-1)})$, for $1 \leq i \leq r$, as in Example 2.1.

In line with Definition 2.3, we give the following propositions.

Proposition 4.3 ($(f_1, f_2, \ldots, f_r)$-multi-cyclic codes)
Let $\Lambda = (a_1, a_2, \ldots, a_r)$ where $a_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$ for every i. Then

1. $(\lambda_1, \lambda_2, \ldots, \lambda_r)$-multi-twisted codes of length $n = n_1 + n_2 + \ldots + n_r$ are $(f_{\lambda_1}, f_{\lambda_2}, \ldots, f_{\lambda_r})$-multi-cyclic codes, where $f_{\lambda_i}(x) = x^{n_i} - \lambda_i$.

2. generalized quasi-cyclic codes of length $n = n_1 + n_2 + \ldots + n_r$ are $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes, where $f_i(x) = x^{n_i} - 1$.

3. generalized quasi-negacyclic codes of length $n = n_1 + n_2 + \ldots + n_r$ are $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes, where $f_i(x) = x^{n_i} + 1$.

4. generalized right quasi-polycyclic codes induced by Λ, of length $n = n_1 + n_2 + \ldots + n_r$, are $(f_{\overrightarrow{a}_1}, f_{\overrightarrow{a}_2}, \ldots, f_{\overrightarrow{a}_r})$-multi-cyclic codes, where $f_{\overrightarrow{a_i}}(x) = x^{n_i} - \sum_{j=0}^{n_i-1} a_{i,j}x^j$.

Proposition 4.4 ($(M_1, M_2, \ldots, M_r)$-multi-cyclic codes)
Let $\Lambda = (a_1, a_2, \ldots, a_r)$ where $a_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$ for every i. Then

1. generalized left quasi-polycyclic codes induced by Λ, of length $n = n_1 + n_2 + \ldots + n_r$, are $(M_{\overleftarrow{a}_1}, M_{\overleftarrow{a}_2}, \ldots, M_{\overleftarrow{a}_r})$-multi-cyclic codes, where each $M_{\overleftarrow{a}_i}$ is the representation matrix of $T_{\overleftarrow{a}_i}$ defined in Definition 4.3.

2. generalized quasi-monomial codes induced by Λ, of length $n = n_1 + n_2 + \ldots + n_r$, are $(M_{\overline{a}_1}, M_{\overline{a}_2}, \ldots, M_{\overline{a}_r})$-multi-cyclic codes, where each $M_{\overline{a}_i}$ is the representation matrix of $T_{\overline{a}_i}$ defined in Definition 4.3.

3. $(\sigma_1, \sigma_2, \ldots, \sigma_r)$-generalized quasi-cyclic codes of length $n = n_1+n_2+\ldots+n_r$ are $(P_{\sigma_1}, P_{\sigma_2}, \ldots, P_{\sigma_r})$-multi-cyclic codes, where each P_{σ_i} is the associated permutation matrix of the n_i-cycle σ_i.

4. $(\sigma_1, \sigma_2, \ldots, \sigma_r)$-generalized quasi-monomial codes of length n, are $(M_{\sigma_1,\overline{a}_1}, M_{\sigma_2,\overline{a}_2}, \ldots, M_{\sigma_r,\overline{a}_r})$-multi-cyclic codes, where each $M_{\sigma_i,\overline{a}_i}$ is the representation matrix of $T_{\sigma_i,\overline{a}_i}$ defined in Definition 4.3.

We give the following result about σ-monomial codes when σ is not an n-cycle.

Proposition 4.5
Let $C \subseteq \mathbb{F}_q^n$ be a σ-monomial code induced by the vector $a = (a_0, a_1, \ldots, a_{n-1})$, where $\sigma = \sigma_1\sigma_2\ldots\sigma_r$, is the product of r disjoint n_i-cycles. Then C is a $(M_{\sigma_1,\overline{a}_1}, M_{\sigma_2,\overline{a}_2}, \ldots, M_{\sigma_r,\overline{a}_r})$-multi-cyclic code of length $n = n_1 + n_2 + \ldots + n_r$, where each $M_{\sigma_i,\overline{a}_i}$ is the cyclic monomial matrix of minimal polynomial $f_{\overline{a}_i}(x) = x^{n_i} - \prod_{j\in \mathrm{Supp}\,(\sigma_i)} a_j$. Moreover,

1. If each σ_i is of the standard form, then C is a $(f_{\overline{a}_1}, f_{\overline{a}_2}, \ldots, f_{\overline{a}_r})$-multi-cyclic code of length n.

2. If each $\sigma_i = \rho_i\sigma_i'\rho_i^{-1}$, with σ_i' an n_i-cycle of standard form, then $Q_\rho^{-1}C$ is an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code of length n, where $Q_\rho = \mathrm{diag}(Q_{\rho_1}, Q_{\rho_2}, \ldots, Q_{\rho_r})$ and Q_{ρ_i} are the $n_i \times n_i$ matrices as in Proposition 2.5.

We now present some connections between different types of codes of Definition 4.3 and other well known linear codes in the literature.

Proposition 4.6
Let $\Lambda = (a_1, a_2, \ldots, a_r)$, where each $a_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$ and C be a linear code of length $n = n_1 + n_2 + \ldots + n_r$ over $\mathbb{F}_q$. Then

1. C is generalized **right** quasi-polycyclic code induced by Λ, if and only if $P^{-1}C$ is a generalized **left** quasi-polycyclic code induced by $\Lambda^{'}$, where $\Lambda^{'} = (b_1, b_2, \ldots, b_r)$, such that $b_i = (a_{i,n_i-1}, a_{i,n_i-2}, \ldots, a_{i,0}) \in \mathbb{F}_q^{n_i}$, $P = \mathrm{diag}(P_1, P_2, \ldots, P_r)$ and each P_i is the $n_i \times n_i$ square matrix given by
$$P_i^{-1} = P_i = \begin{pmatrix} 0 & \ldots & \ldots & 0 & 1 \\ 0 & \ldots & 0 & 1 & 0 \\ \vdots & \iddots & \iddots & \iddots & \vdots \\ 0 & \iddots & \iddots & & \vdots \\ 1 & 0 & \ldots & \ldots & 0 \end{pmatrix}$$

2. C is a generalized quasi-monomial code induced by Λ, if and only if $Q^{-1}C$ is a $(\lambda_1, \lambda_2, \ldots, \lambda_r)$-multi-twisted code, where for each i, $\lambda_i = \prod_{j=0}^{n_i-1} a_{i,j}$,
$Q = \mathrm{diag}(Q_1, Q_2, \ldots, Q_r)$ and Q_i is the $n_i \times n_i$ square matrix given by
$$Q_i = \begin{pmatrix} 0 & 0 & \ldots & 0 & \prod_{j=0}^{n_i-1} a_{i,j} \\ a_0 & 0 & \ldots & 0 & 0 \\ 0 & a_0 a_1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \ldots & 0 & \prod_{j=0}^{n_i-2} a_{i,j} & 0 \end{pmatrix}, \quad Q_i^{-1} = \begin{pmatrix} 0 & a_0^{-1} & 0 & \ldots & 0 \\ 0 & 0 & (a_0 a_1)^{-1} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & \vdots & & \ddots & \prod_{j=0}^{n_i-2} a_{i,j}^{-1} \\ \prod_{j=0}^{n_i-1} a_{i,j}^{-1} & 0 & \ldots & \ldots & 0 \end{pmatrix}$$

3. C is a $(\sigma_1, \sigma_2, \ldots, \sigma_r)$-generalized quasi-cyclic code if and only if $P_\rho^{-1}C$ is a generalized quasi-cyclic code , where $P_\rho = \mathrm{diag}(P_{\rho_1}, P_{\rho_2}, \ldots, P_{\rho_r})$ and each P_{ρ_i} is the associated permutation matrix of $\rho_i \in S_{n_i}$ such that $\sigma_i = \rho_i \sigma_i^{'} \rho_i^{-1}$ and $\sigma_i^{'}$ is of standard form.

4. C is a $(\sigma_1, \sigma_2, \ldots, \sigma_r)$-generalized quasi-monomial code induced by Λ if and only if $Q_\rho^{-1}C$ is a $(\lambda_1, \lambda_2, \ldots, \lambda_r)$-multi-twisted code, where for each i, $\lambda_i = \prod_{j=0}^{n_i-1} a_{i,j}$, $Q = \mathrm{diag}(Q_{\rho_1}, Q_{\rho_2}, \ldots, Q_{\rho_r})$ and $Q_{\rho_i} = P_{\rho_i} Q_i$ such that $\rho_i, \sigma_i^{'} \in S_{n_i}$ with $\sigma_i = \rho_i \sigma_i^{'} \rho_i^{-1}$ and $\sigma_i^{'}$ is an n_i-cycle of standard form.

Proof

1. Since $\mathrm{diag}(M_{\overrightarrow{b}_1}, M_{\overrightarrow{b}_2}, \ldots, M_{\overrightarrow{b}_r}) = P^{-1}\,\mathrm{diag}(M_{\overleftarrow{a}_1}, M_{\overleftarrow{a}_2}, \ldots, M_{\overleftarrow{a}_r})P$.

2. Since $\mathrm{diag}(M_{\lambda_1}, M_{\lambda_2}, \ldots, M_{\lambda_r}) = Q^{-1}\,\mathrm{diag}(M_{\overline{a}_1}, M_{\overline{a}_2}, \ldots, M_{\overline{a}_r})Q$.

3. Since $\mathrm{diag}(M_{\sigma_1^{'}}, M_{\sigma_2^{'}}, \ldots, M_{\sigma_r^{'}}) = P_\rho^{-1}\,\mathrm{diag}(P_{\sigma_1}, P_{\sigma_2}, \ldots, P_{\sigma_r})P_\rho$, where each $M_{\sigma_i^{'}}$ is the companion matrix of $x^{n_i} - 1$.

4. Since $\mathrm{diag}(M_{\lambda_1}, M_{\lambda_2}, \ldots, M_{\lambda_r}) = Q_\rho^{-1}\,\mathrm{diag}(M_{\sigma_1,\overline{a}_1}, M_{\sigma_2,\overline{a}_2}, \ldots, M_{\sigma_r,\overline{a}_r})Q_\rho$.

Remark 4.1 (Codes invariant under a permutation)
With same notations as in the above propositions, if each $a_i = 1, i = 0, \ldots, n-1$, then we speak of σ-codes and so we obtain cyclic codes, and their different generalizations such as quasi-cyclic codes and generalized quasi-cyclic codes.

In line with the Remark 2.9 and Definition 4.3, we also make the following remark.

Remark 4.2
Let $\Lambda = (a_1, a_2, \ldots, a_r)$, where each $a_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$ and C be a linear code of length $n = n_1 + n_2 + \ldots + n_r$ over $\mathbb{F}_q$.

1. C is a Λ-generalized right quasi-polycyclic code of length n, if and only if $\varphi_{\overrightarrow{\Lambda}}(C)$ is an $\mathbb{F}_q[x]$-submodule of $\prod_{k=1}^{r} R_{\overrightarrow{a}_i}$, where $f_{\overrightarrow{a}_i} = x^{n_i} - \sum_{j=0}^{n_i-1} a_{i,j}x^j$ and $\varphi_{\overrightarrow{\Lambda}} = (\varphi_{\overrightarrow{a_1}}, \varphi_{\overrightarrow{a_2}}, \ldots, \varphi_{\overrightarrow{a_r}})$ and each $\varphi_{\overrightarrow{a_i}}$ is the right isomorphism induced by the vector $\overrightarrow{a}_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$.

2. A similar remark remains valid, as in the above statement, for Λ-generalized left quasi-polycyclic codes via $\varphi_{\overleftarrow{\Lambda}} = (\varphi_{\overleftarrow{a_1}}, \varphi_{\overleftarrow{a_2}}, \ldots, \varphi_{\overleftarrow{a_r}})$, and for Λ-generalized monomial codes via $\varphi_{\overline{\Lambda}} = (\varphi_{\overline{a_1}}, \varphi_{\overline{a_2}}, \ldots, \varphi_{\overline{a_r}})$, where $\varphi_{\overleftarrow{a_i}}$ and $\varphi_{\overline{a_i}}$ are respectively the left and the monomial isomorphism as in Remark 2.9.

3. A similar remark remains valid for Λ-generalized quasi σ-cyclic codes and Λ-generalized quasi σ-monomial codes.

We close this section by giving some results about the dual of $(M_1, M_2, \ldots, M_r)$-multi-cyclic codes and M-codes with respect to a non-degenerate bilinear form b on $\mathbb{F}_q^n$. From Theorem 2.6, we easily deduce the following result for M-codes

Theorem 4.3
Let $C \subseteq \mathbb{F}_q^n$ be a linear code and let M be a square matrix with sequence of invariant factors $(f_1(x), f_2(x), \ldots, f_r(x))$. The following assertions are equivalent.

1. C is an M-code.

2. $C^{\perp b}$ is an M^*-code.

3. $P^{-1}C^{\perp b}$ is an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code where $\mathrm{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r}) = P^{-1}M^*P$.

In order to give a connection between duals of M-codes and duals of some well-known types of linear codes in the literature such as quasi-cyclic, quasi-twisted, quasi-monomial codes and others, let us consider $b = (b_1, b_2, \ldots, b_r)$, a non-degenerate bilinear form on $\mathbb{F}_q^n = \mathbb{F}_q^{n_1} \times \mathbb{F}_q^{n_2} \times \ldots \times \mathbb{F}_q^{n_r}$ where b_i is the restriction of b on $\mathbb{F}_q^{n_i} \times \mathbb{F}_q^{n_i}$. In this case the b-dual code of C is the linear code $C^{\perp b}$ given by

$$C^{\perp b} := C_1^{\perp b_1} \oplus C_2^{\perp b_2} \oplus \cdots \oplus C_r^{\perp b_r}$$

where $C_i^{\perp b_i} = \left\{ x_i \in \mathbb{F}_q^{n_i} \,:\, b_i(x_i, y_i) = 0 \text{ for all } y_i \in C_i \right\}$.

Theorem 4.4
Let C be a linear code of length n over $\mathbb{F}_q$, and let $b = (b_1, b_2, \ldots, b_r)$ be a non-degenerate bilinear form on $\mathbb{F}_q^n = \mathbb{F}_q^{n_1} \times \mathbb{F}_q^{n_2} \times \ldots \times \mathbb{F}_q^{n_r}$. Then,

1. the b-dual $C^{\perp b}$ of an $(M_1, M_2, \ldots, M_r)$-multi-cyclic code is an $(M_1^*, M_2^*, \ldots, M_r^*)$-multi-cyclic code where M_i^* is the adjoint matrix of M_i with respect to b_i.

2. the b-dual $C^{\perp b}$ of an $(f_1, f_2, \ldots, f_r)$-multi-cyclic code is a $(C_{f_1}^*, C_{f_2}^*, \ldots, C_{f_r}^*)$-multi-cyclic code where $C_{f_i}^*$ is the adjoint matrix of C_{f_i} with respect to b_i.

Proof
Follows from Theorem 2.6.

In line with Definition 4.3 and Proposition 4.6, we give the following result.

Corollary 4.1
1. The Euclidean dual of a $(\lambda_1, \lambda_2, \ldots, \lambda_r)$-multi-twisted code is a $(\lambda_1^{-1}, \lambda_2^{-1}, \ldots, \lambda_r^{-1})$-multi-twisted code.

2. The Euclidean dual of a generalized quasi-cyclic (resp. generalized quasi-negacyclic) code is also a generalized quasi-cyclic (resp. generalized quasi-negacyclic) code.

3. For $\Lambda = (a_1, a_2, \ldots, a_r)$, with $a_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$. The Euclidean dual of a Λ-generalized quasi-monomial code is a Λ^{-1}-generalized quasi-monomial code, where $\Lambda^{-1} = \{b_1, b_2, \ldots, b_r\}$, and each $b_i = (a_{i,0}^{-1}, a_{i,1}^{-1}, \ldots, a_{i,n_i-1}^{-1}) \in \mathbb{F}_q^{n_i}$ for all i.

Proof

1. Follows from the Theorem 2.7 and the fact $\mathrm{diag}(M_{\lambda_1}^t, M_{\lambda_2}^t, \ldots, M_{\lambda_r}^t)$ commutes with $\mathrm{diag}(M_{\lambda_1^{-1}}, M_{\lambda_2^{-1}}, \ldots, M_{\lambda_r^{-1}})$, where $M_{\lambda_i}^t$ is the transpose matrix of M_{λ_i}.

2. Follows from the above statement for $\lambda_i = \pm 1$

3. The Euclidean adjoint matrix of $M_{\overline{a}_i}$ is $M_{\overline{b}_i}$, with $b_i = (a_{i,0}^{-1}, a_{i,1}^{-1}, \ldots, a_{i,n_i-1}^{-1})$ for each i. We verify by direct computation that $\mathrm{diag}(M_{\overline{b}_1}, M_{\overline{b}_2}, \ldots, M_{\overline{b}_r})$ commutes with $\mathrm{diag}(M_{\overline{a}_1}^t, M_{\overline{a}_2}^t, \ldots, M_{\overline{a}_r}^t)$. Therefore, by Theorem 2.7, the result holds.

4.3 1-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes

In this section we are interested in the class of 1-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes generated by one element $\underline{\mathbf{a}}(x) = (a_0(x), a_1(x), \ldots, a_{r-1}(x)) \in \prod_{i=1}^{r} R_{f_i}$. We write $C = \mathbb{F}_q[x].\underline{\mathbf{a}}(x)$ to denote that C is a 1-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code generated by $\underline{\mathbf{a}}(x)$.

Definition 4.4
Let C be a **1**-generator quasi $(f_1, f_2, \ldots, f_r)$-multi-cyclic code. The **parity check polynomial** of C is defined to be the non-zero polynomial $h(x)$ of least degree in $\mathbb{F}_q[x]$ such that $h(x)\underline{\mathbf{a}}(x) = 0$ in $\prod_{i=1}^{r} R_{f_i}$.

Proposition 4.7
Let $C = \mathbb{F}_q[x] \cdot \underline{\mathbf{a}}(x)$ be a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code with parity check polynomial $h(x)$, and let $g_i(x) = \gcd(a_i(x), f_i(x))$, $i = 1, \ldots, r$, $\underline{\mathbf{g}}(x) = (g_1(x), g_2(x), \ldots, g_r(x))$. Then,

1. $h(x) = \mathrm{lcm}\,\{h_i(x),\ 1 \le i \le r\}$, where $f_i(x) = h_i(x)g_i(x)$.

2. Any element of the form $\underline{\mathbf{b}}(x) = (p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$, where $\mathrm{gcrd}(p_i(x), h_i(x)) = 1$, generates C as an $\mathbb{F}_q[x]$-submodule of $\prod_{i=1}^{r} R_{f_i}$.

3. $\dim_{\mathbb{F}_q}(C) = \deg(h(x))$, and

$$S = \left\{ x^i.\underline{\mathbf{g}}(x) = (x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ :\ 0 \le i \le \deg(h(x)) - 1 \right\}$$

forms a basis of C.

4. A generator matrix of C is given by

$$G = \begin{pmatrix} \underline{\mathbf{g}}(x) \\ x\underline{\mathbf{g}}(x) \\ \vdots \\ x^{k-1}\underline{\mathbf{g}}(x) \end{pmatrix} = \left(\begin{array}{c|c|c|c} g_1(x) & g_2(x) & \ldots & g_r(x) \\ xg_1(x) & xg_2(x) & \ldots & xg_r(x) \\ \vdots & \vdots & \vdots & \vdots \\ x^{k-1}g_1(x) & x^{k-1}g_2(x) & \ldots & x^{k-1}g_r(x) \end{array}\right) \tag{4.7}$$

where $k = \deg(h(x))$.

Proof
Let $\underline{\mathbf{a}}(x) = (a_1(x), \ldots, a_r(x)) \in \prod_{i=1}^{r} R_{f_i}$, and $C = \mathbb{F}_q[x] \cdot \underline{\mathbf{a}}(x)$ be a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code.

1. Set $h^{'}(x) = \operatorname{lcm}\{h_i(x) \ : \ 1 \leq i \leq r\}$. As $C = C_1 \oplus C_2 \oplus \cdots \oplus C_r$, where each $C_i = \mathbb{F}_q[x]a_i(x)$ is an f_i-cyclic code with generator polynomial $g_i(x)$ and parity check polynomial $h_i(x)$, we have $h(x) \cdot a_i(x) = 0, \ i = 1, \ldots, r$ by the minimality of the degree of $h_i(x)$. Hence each $h_i(x)$ is a divisor of $h(x)$, and so $h^{'}(x)$ divides $h(x)$. Finally, by the minimality of $h(x)$ we deduce that $h(x) = h^{'}(x) = \operatorname{lcm}_{1 \leq i \leq r}\{h_i(x)\}$, with $f_i(x) = h_i(x)g_i(x)$.

2. From Theorem 4.1 we have that $C = C_1 \oplus C_2 \oplus \cdots \oplus C_r$, where each $C_i = \mathbb{F}_q[x] \cdot a_i(x)$ is an f_i-cyclic code with generator polynomial $g_i(x) = \gcd(a_i(x), f_i(x))$. Since $\gcd(p_i(x), g_i(x)) = 1$, $p_i(x)g_i(x)$ generates C_i. Therefore $\underline{\mathbf{b}}(x)$ is a generator of C.

3. We prove that the set

$$S = \left\{ x^i \underline{\mathbf{g}}(x) = (x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ : \ 0 \leq i < k \right\},$$

where $k = \deg(h(x))$ is a basis for C. Suppose, for any $i \in \{0, 1, .., k-1\}$, that

$$a_0 x^i g_1(x) + a_1 x^i g_2(x) + \cdots + a_k x^i g_r(x) = 0,$$

where $a_i \in \mathbb{F}_q$. This equation can be rewritten as $A(x)\underline{\mathbf{g}}(x) = 0$ where $\deg(A(x)) < k$. Equivalently,

$$(A(x)g_1(x) \text{ in } R_{f_1}, A(x)g_2(x) \text{ in } R_{f_2}, \ldots, A(x)g_r(x) \text{ in } R_{f_r}) = (0, 0, \ldots, 0).$$

It follows that for each $i = 1, \ldots, r$, we have $A(x)g_i(x) = 0$ in R_{f_i}. Therefore $h_i(x)$ divides $A(x)$ and so does $h(x) = \operatorname{lcm}\{h_i(x), \ 1 \leq i \leq r\}$. This would contradict the minimality of the degree of $h(x)$ unless $A(x)$ is the zero polynomial. Hence, we conclude that $a_i = 0$ for all $i = 0, 1, .., k-1$, and S is linearly independent.

To complete the proof we need to show that S spans C, i.e., for every $v(x) \in C$ there exists $r(x) \in \mathbb{F}_q[x]$ with $\deg(r(x)) < \deg(h(x)) - 1$ such that $v(x)\underline{\mathbf{g}}(x) = r(x)\underline{\mathbf{g}}(x)$. By the Euclidean algorithm in $\mathbb{F}_q[x]$, we write $v(x) = q(x)h(x) + r(x)$, for some $q(x), r(x) \in \mathbb{F}_q[x]$ where $\deg(r(x)) < \deg(h(x))$. It follows that

$$\begin{aligned}
v(x)\underline{\mathbf{g}}(x) &= v(x)(g_1(x), g_2(x), \ldots, g_r(x)) \\
&= (v(x)g_1(x) \bmod f_1(x), v(x)g_2(x) \bmod f_2(x), \ldots, v(x)g_r(x) \bmod f_r(x)) \\
&= (r(x)g_1(x) \bmod f_1(x), r(x)g_2(x) \bmod f_2(x), \ldots, r(x)g_r(x) \bmod f_r(x)) \\
&= r(x)(g_1(x), g_2(x), \ldots, g_r(x)) \\
&= r(x)\underline{\mathbf{g}}(x)
\end{aligned}$$

4. Since $S = \left\{ x^i \underline{\mathbf{g}}(x) = (x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ : \ 0 \leq i \leq \deg(h(x)) - 1 \right\}$ forms a basis of C.

Remark 4.3
If C is a **1**-generator $(M_1, M_2, \ldots, M_r)$-multi-cyclic code such that $\psi(C) = \mathbb{F}_q[x] \cdot \underline{\mathbf{a}}(x)$, then all of the statements of the above theorem are still valid for C, except for the statement on the generator matrix where we need to compute $\psi^{-1}(x^i \underline{\mathbf{g}}(x)), 1 \leq i \leq \deg(h(x)) - 1$.

4.4 A Bound on the Hamming distance

In this section, following [28], we give a lower bound on the minimum Hamming distance of a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code.

Theorem 4.5
Let C be a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

where for all $1 \leq i \leq r$, $f_i(x) = h_i(x)g_i(x)$ and each $p_i(x)$ is coprime to $h_i(x)$. Also let d and d_i be, respectively, the minimum Hamming distance of C and $C_i = \langle p_i(x)g_i(x)\rangle$ for $i = 1, ..., r$. Then

1. $d \geq \sum_{i \notin K} d_i$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) = \operatorname{lcm}\{h_i(x) \ : \ 1 \leq i \leq r\} \neq \operatorname{lcm}\{h_i(x) : i \in K\}$.

2. If $h_1(x) = h_2(x) = \cdots = h_r(x)$, then $d \geq \sum_{i=1}^{r} d_i$.

Proof
Let C be a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x)).$$

Then C can be decomposed as a direct sum of f_i-cyclic codes $C_i = \langle p_i(x)g_i(x)\rangle$ as $C = C_1 \oplus C_2 \oplus, \cdots, \oplus C_r$. Let $c(x) = (c_1(x), c_2(x), \ldots, c_r(x)) = u(x)\underline{\mathbf{a}}(x)$ be a codeword of C, then a component $c_i(x)$ is 0 if and only if $h_i(x)$ divides $u(x)p_i(x)$. As $\gcd(p_i(x), h_i(x)) = 1$, this happens only if $h_i(x)$ is a divisor of $u(x)$. Therefore, $c(x) = 0$ if and only if $h(x)$ is a divisor of $u(x)$. It follows that $c(x) \neq 0$ if and only if $h(x)$ does not divide $u(x)$. Thus, a non-zero codeword has the most number of zero components whenever $h(x) \neq \operatorname{lclm}\{h_i(x) : i \in K\}$ and $\operatorname{lcm}\{h_i(x) : i \in K\}$ divides $u(x)$, where K is a maximal subset of $\{1, 2, \ldots, r\}$. Therefore, $d \geq \sum_{i \notin K} d_i$. On the other hand, we have $|K| = 0$ if and only if $h_1(x) = h_2(x) = \cdots = h_r(x)$. In this case, $d \geq \sum_{i=1}^{r} d_i$.

For $i = 1, \ldots, r$, let e_i be the order of $f_i(x)$ and $e = \operatorname{lcm}(e_1, e_2, \ldots, e_r)$ such that $\gcd(e, q) = 1$. Then the roots of every $f_i(x)$, $i = 1, \ldots, r$ are simple and can be expressed as powers of a primitive e^{th} root of unity α in an extension $\mathbb{F}_{q^m}$ where m is the multiplicative order of q modulo e. Then we obtain the following bound on the Hamming distance of a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code.

Theorem 4.6 (BCH-like bound for $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes)
Let C be a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

where for all $1 \leq i \leq r$, $f_i(x) = h_i(x)g_i(x)$ and each $p_i(x)$ is coprime to $h_i(x)$. For $1 \leq i \leq r$, let δ_i be an integer such that $\alpha^{a_i + jb_i}$, $j = 0, \ldots, \delta_i - 2$, are among the roots of $g_i(x)$, with a_i and b_i are some integers such that $\gcd(e, b_i) = 1$. Then,

1. $d \geq \sum_{i \notin K} \delta_i$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$.

2. If $h_1(x) = h_2(x) = \cdots = h_r(x)$, then $d \geq \sum_{i=1}^{r} \delta_i$.

Proof
By Theorem 4.5, we have that $d \geq \sum_{i \notin K}(d_i)$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$ and d_i is the minimum distance of $C_i = \langle p_i(x)g_i(x)\rangle$. As $\alpha^{a_i+jb_i},\ j = 0, \ldots, \delta_i - 2$, are among the roots of $g_i(x)$, by Theorem 2.8, we obtain $d_i \geq \delta_i$, and so the results hold.

Theorem 4.7 (Hartmann-Tzeng-like bound for $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes)
Let C be a **1**-generator $(f_1, f_2, \ldots, f_r)$-multi-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

where for all $1 \leq i \leq r$, $f_i(x) = h_i(x)g_i(x)$ and each $p_i(x)$ is coprime to $h_i(x)$. Let $\delta_i \geq 2$ and $s_i,\ i = 1, \ldots, r$ be integers such that for $j = 0, \ldots, \delta_i - 2$, and $k = 0, \ldots, s_i$, the elements $\alpha^{a_i+jb_i+kc_i}$ are among the zeros of $g_i(x)$ where $a_i,\ b_i$ and c_i are some integers with $\gcd(e, b_i) = 1$ and $\gcd(e, c_i) = 1$. Then,

1. $d \geq \sum_{i \notin K}(\delta_i + s_i)$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$.
2. If $h_1(x) = h_2(x) = \cdots = h_r(x)$, then $d \geq \sum_{i=1}^{r}(\delta_i + s_i)$.

Proof
By Theorem 4.5, we have $d \geq \sum_{i \notin K}(d_i)$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) \neq \operatorname{lcm}\{h_i(x) : i \in K\}$ and d_i is the minimum distance of $C_i = \langle p_i(x)g_i(x)\rangle$. As $\alpha^{a_i+jb_i+kc_i} : j = 0, \ldots, \delta_i - 2, k = 0, \ldots, s_i$, are among the zeros of $g_i(x)$, we have, by Theorem 2.9, $d_i \geq \delta_i + s_i$, and so the result holds.

4.5 Examples

In this section, we present a few examples of $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes with good parameters. These examples of codes are optimal and most of them are presented in the online database of best-known linear codes [27]. Moreover, some of them cannot be obtained as a constacyclic code, or as a quasi-twisted code, or a multi-twisted code. Considering the class of $(f_1, f_2, \ldots, f_r)$-multi-cyclic codes and their generalizations in a computer search to discover new linear codes is both more promising and more challenging than only considering one of the special cases, because the search space is larger.

Example 4.1 ((f_1, f_2)-multi-cyclic codes)
Let $q = 7,\ n_1 = 16,\ n_2 = 7$ and $n = n_1 + n_2 = 23$. For $f_1(x) = x^{16} + 4x^{14} + 6x^{13} + x^{12} + 3x^{11} + 3x^{10} + 5x^8 + x^7 + x^6 + 6x^5 + 4x^4 + 5x^3 + 4x^2 + 4x + 1$, $f_2(x) = 3x^7 + x^6 + x^5 + 3x^4 + 3x^3 + 3x^2 + 6x + 5$. Consider the linear code $C \subseteq \mathbb{F}_7^{23}$ generated by $\mathbf{a}(x) = (a_1(x), a_2(x)) \in R_{f_1} \times R_{f_2}$, such that $g_1(x) = x^{13} + 5x^{12} + 3x^{11} + 4x^{10} + 4x^9 + 6x^8 + 3x^7 + 3x^6 + 4x^5 + 2x^4 + x^3 + 5x^2 + 5x + 6$, $g_2(x) = 1$ and

$$\begin{aligned} a_1(x) &= x^{17} + x^{16} + x^{15} + 3x^{14} + 3x^{13} + 2x^{12} + 5x^{11} + 3x^{10} + 2x^9 + 5x^6 + 2x^5 + 2x^4 + 6x^3 + 2x^2 + 4x + 1 \\ &= (x^{13} + 5x^{12} + 3x^{11} + 4x^{10} + 4x^9 + 6x^8 + 3x^7 + 3x^6 + 4x^5 + 2x^4 + x^3 + 5x^2 + 5x + 6)g_1(x) \\ a_2(x) &= x^9 + 6x^8 + 5x^7 + 2x^5 + 2x^4 + 4x^3 + 5x^2 + 2 \\ &= (x^9 + 6x^8 + 5x^7 + 2x^5 + 2x^4 + 4x^3 + 5x^2 + 2)g_2(x) \end{aligned}$$

This is a **1**-generator (f_1, f_2)-multi-cyclic code with parity check polynomial $h(x) = x^3 + 2x^2 + 5x + 6$. Its actual distance is $d = 18$, and according to [27], it is an optimal code with parameters $[23, 3, 18]_7$. Moreover we have also constructed the following (f_1, f_2)-multi-cyclic codes with good parameters:
$[23, 2, 19]_7$, $[23, 4, 16]_7$, $[23, 5, 14]_7$, $[23, 6, 13]_7$, $[23, 7, 12]_7$, $[23, 8, 11]_7$, $[23, 9, 10]_7$
$[23, 10, 9]_7$, $[23, 11, 8]_7$, $[23, 12, 8]_7$, $[23, 13, 7]_7$, $[23, 14, 6]_7$, $[23, 15, 5]_7$.

Example 4.2 (M-codes)
Let M be a 10×10 square matrix over $\mathbb{F}_5$ given by

$$M = \begin{pmatrix} 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

According to the cyclic decomposition theorem we prove that M is similar to $D = \mathrm{diag}(C_{f_1}, C_{f_2})$ where $f_1(x) = x^3 - x - 1$, and $f_2(x) = x^7 - 4x^6 - 3x^5 - 4x^4 - x^2 - 2x - 4$, are the invariant factors of M, i.e.,

$$D = \left(\begin{array}{ccc|ccccccc} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 4 \end{array}\right).$$

From Theorem 4.2, we obtain the M-codes of length 10 over $\mathbb{F}_5$ corresponding to the $\mathbb{F}_5[x]$-submodules of $R_{f_1} \times R_{f_2}$, via the isomorphism $\psi := \varphi \circ P^{-1}$ where $\varphi = (\varphi_{f_1}, \varphi_{f_2})$ as in (4.1). Let the irreducible factorizations of $f_1(x)$ and $f_2(x)$ be

$$\begin{aligned} f_1(x) &= (x+3) \cdot (x^2 + 2x + 3) \\ f_2(x) &= (x+3) \cdot (x^2 + 2x + 3) \cdot (x^2 + 2x + 4) \cdot (x^2 + 4x + 1) \end{aligned}$$

For example the code $C \subseteq \mathbb{F}_5^{10}$ such that $\varphi(C) = \mathbb{F}_5[x]\underline{\mathbf{a}}(x)$ with $\underline{\mathbf{a}}(x) = (x+3, x+3)$, is a $\mathbf{1}$-generator (f_1, f_2)-multi-cyclic code with parity check polynomial

$$h(x) = \mathrm{lcm}(x^2+2x+3, x^6+3x^5+3x^4+2x^3+4x^2+2x+2) = x^6+3x^5+3x^4+2x^3+4x^2+2x+2.$$

Hence its dimension is 6. Its actual distance is $d = 3$, and so it is of parameters $[10, 6, 3]_5$ close to the optimal code $[10, 6, 4]_5$.

It follows that if P is the invertible matrix such that $D = P^{-1}MP$, then PC is a $\mathbf{1}$-generator M-code of dimension 6, and of minimum distance $d = 3$ if P is a monomial matrix. Moreover we can construct the following (f_1, f_2)-multi-cyclic codes: $[10, 2, 6]_5$, $[10, 3, 6]_5$, $[10, 7, 2]_5$.

Example 4.3
In the following table we collect some optimal codes with additional desirable properties from the class of $f-$cyclic codes, quasi $f-$cyclic codes and multi $(f_1, f_2, \ldots, f_r)-$cyclic codes.

Table 4.1: Examples of best known linear codes, generated by $\langle g_1(x), g_2(x), ...\rangle$ with some additional desired properties

$[n,k,d]$	$g_i(x)$	
$[36,18,8]_2$	$f_1 = x^{13}+x^{11}+x^{10}+x^9+x^2+x+1$, $f_2 = x^{15}+x^{11}+x^8+x^5+x^3$	reversible
$[52,26,10]_2^L$	$f_1 = x^{22}+x^{19}+x^{18}+x^{17}+x^{15}+x^{13}+x^9+x^7+x^6+x^5+x^4+x^3+x^2+x+1$, $f_2 = x^{24}+x^{18}+x^{17}+x^{15}+x^{14}+x^{13}+x^{12}+x^{11}+x^7+x^6+x^4+x^2+x$	LCD
$[12,6,6]_3^*$	$f_1 = x^5+x^4+x^3+x^2+x$, $f_2 = 2x^5+2x^3+2x^2+2x$	self-dual
$[18,6,9]_3^\circ$	$f_1 = 2x^5+x^4+x^3+x$ $f_2 = 2x^3+2x^2$, $f_3 = 2x^5+x^3+2x^2$, $f_4 = x^4+2x^3+x+2$, $f_5 = 2x^4+2x+2$, $f_6 = 2x^5+2x^4+x^2+x+1$	self-orthogonal
$[20,5,12]_3^\circ$	$f_1 = 2x^4+2$ $f_2 = 2x^3+x^2+1$, $f_3 = 2x^4+x^3+x^2+2x+1$, $f_4 = x^3+x$, $f_5 = 2x^2+x+2$, $f_6 = x^2$	self-orthogonal
$[14,7,6]_5$	$f_1 = 3x^{10}+4x^9+x^7+2x^5+3x^4+2x^2+x+3$ $f_2 = 4x^{10}+x^8+2x^7+x^5+2x^4+x^2+4x+2$	LCD
$[22,11,8]_5$	$f_1 = 2x^5+2x^4+4x^3+4x^2+4x+3$,$f_2 = 4x^6+3x^5+3x^4+x^3+3x^2+3x+1$	self-orthogonal

Table 4.2: Examples of best known linear codes, generated by $\langle g(x)\rangle$ that are LCD codes

$[n,k,d]$	$f(x)$	$g(x)$
$[32,26,4]_5$	$2x^{32}+2x^{31}+3x^{30}+3x^{29}+2x^{27}+4x^{26}+2x^{25}+4x^{24}+2x^{23}+x^{22}+3x^{21}+4x^{20}+4x^{19}+4x^{18}+x^{17}+2x^{15}+2x^{12}+3x^{11}+3x^{10}+2x^8+4x^7+x^6+4x^5+2x^4+4x^3+x^2+1$	$x^6+x^5+4x^3+4x^2+4$
$[33,27,4]_5$	$x^{33}+x^{32}+2x^{31}+2x^{30}+2x^{29}+4x^{28}+x^{27}+2x^{26}+2x^{25}+4x^{24}+x^{23}+x^{22}+4x^{21}+2x^{20}+2x^{18}+2x^{17}+2x^{16}+3x^{15}+3x^{13}+4x^{11}+2x^{10}+4x^9+3x^8+x^6+4x^5+2x^4+x^3+2x^2+3x+3$	$x^6+2x^4+2x^2+x+4$
$[33,29,3]_5$	$x^{33}+x^{32}+2x^{31}+2x^{30}+2x^{29}+4x^{28}+x^{27}+2x^{26}+2x^{25}+4x^{24}+x^{23}+x^{22}+4x^{21}+2x^{20}+2x^{18}+2x^{17}+2x^{16}+3x^{15}+3x^{13}+4x^{11}+2x^{10}+4x^9+3x^8+x^6+4x^5+2x^4+x^3+2x^2+3x+3$	$x^4+3x^3+4x^2+x+2$
$[36,30,4]_5$	$4x^{36}+x^{34}+2x^{33}+x^{32}+4x^{31}+x^{30}+4x^{29}+2x^{28}+3x^{27}+2x^{26}+2x^{25}+x^{23}+4x^{21}+x^{20}+x^{19}+2x^{18}+x^{16}+x^{15}+3x^{14}+4x^{13}+x^{12}+3x^{11}+3x^{10}+3x^9+3x^8+4x^6+3x^5+x^4+2x^3+x+3$	x^6+x^2+x+4
$[38,34,3]_5$	$2x^{38}+3x^{37}+x^{36}+3x^{35}+3x^{34}+3x^{33}+3x^{32}+2x^{30}+2x^{29}+x^{28}+x^{27}+4x^{26}+3x^{24}+4x^{23}+x^{22}+x^{20}+3x^{18}+4x^{17}+4x^{16}+4x^{13}+2x^{12}+4x^9+x^7+3x^6+2x^5+3x^4+4x+4$	$x^4+x^3+4x^2+x+2$
$[38,32,4]_5^L$	$2x^{38}+3x^{37}+x^{36}+3x^{35}+3x^{34}+3x^{33}+3x^{32}+2x^{30}+2x^{29}+x^{28}+x^{27}+4x^{26}+3x^{24}+4x^{23}+x^{22}+x^{20}+3x^{18}+4x^{17}+4x^{16}+4x^{13}+2x^{12}+4x^9+x^7+3x^6+2x^5+3x^4+4x+4$	$x^6+3x^5+x^4+3x^3+4x^2+3$
$[38,26,7]_5$	$2x^{38}+3x^{37}+x^{36}+3x^{35}+3x^{34}+3x^{33}+3x^{32}+2x^{30}+2x^{29}+x^{28}+x^{27}+4x^{26}+3x^{24}+4x^{23}+x^{22}+x^{20}+3x^{18}+4x^{17}+4x^{16}+4x^{13}+2x^{12}+4x^9+x^7+3x^6+2x^5+3x^4+4x+4$	$x^{12}+2x^{10}+x^9+4*x^8+x^7+2x^6+x^5+4x^4+3x^3+2x+4$

Chapter 5

Linear codes invariant under a (σ, δ)-pseudo-linear transformation

In this chapter, we are interested in linear codes invariant under a (σ, δ)-pseudo-linear transformation T, called (T, σ, δ)-codes, where σ is an automorphism of $\mathbb{F}_q$ and δ is a σ-derivation of $\mathbb{F}_q$. As in the previous chapter we restrict ourselves to linear codes invariant under right multiplication by the (σ, δ)-action of a given matrix M, that we call (M, σ, δ)-skew codes. Various subclasses of these codes, depending on the (σ, δ)-cyclical property of M, are derived. In particular, we distinguish between three fundamental types: (M, σ, δ)-cyclic, quasi (M, σ, δ)-cyclic and (M, σ, δ)-multi-cyclic. These different subclasses allow us to obtain some of the well-known types of linear codes as special cases depending on the choice of M. We observe that the (σ, δ)-similarity in $\mathbb{M}_n(\mathbb{F}_q)$ gives rise to various relations between these codes. As common in the study of cyclic codes and their generalizations, we use the skew polynomial ring $\mathbb{F}_q[x, \sigma, \delta]$ to study (M, σ, δ)-skew codes and we investigate their algebraic structure. As an application of the theory developed here, we construct some optimal linear codes and MDS codes over different alphabets with additional desirable properties such as self-orthogonality and linear complementary duality (LCD).

5.1 Skew polynomial ring

Let $\mathbb{F}_q$ be the finite field with q elements, where $q = p^s$ for some prime number p and $s \in \mathbb{Z}_{\geq 1}$. Let $\sigma : \mathbb{F}_q \to \mathbb{F}_q$ be an automorphism of $\mathbb{F}_q$, and let $\delta : \mathbb{F}_q \to \mathbb{F}_q$ be a $\sigma-$derivation, that is, for all $a, b \in \mathbb{F}_q$

$$\delta(a + b) = \delta(a) + \delta(b) \qquad \text{and} \qquad \delta(ab) = \sigma(a)\delta(b) + \delta(a)b.$$

It is shown in [21, Chapter 8, Theorem 3.1] that every $\sigma-$derivation δ on $\mathbb{F}_q$ has the form $\delta(a) = \beta(\sigma(a) - a)$, for some $\beta \in \mathbb{F}_q^*$. Let us recall that the *Frobenius automorphism of* $\mathbb{F}_q$ is the automorphism $\theta : \mathbb{F}_q \to \mathbb{F}_q$ defined by $\theta(a) = a^p$. Note that θ fixes $\mathbb{F}_p$, and generates the cyclic group of automorphisms of $\mathbb{F}_q$. A (σ, δ)-skew polynomial ring is the set

$$R := \mathbb{F}_q[x, \sigma, \delta] = \left\{a_0 + a_1 x + \ldots + a_{n-1}x^{n-1} : a_i \in \mathbb{F}_q, n \in \mathbb{N}\right\}$$

endowed with the usual polynomial addition and the multiplication is defined by

$$xa = \sigma(a)x + \delta(a).$$

We recall the result of [11], where the authors give an explicit way to calculate $ax^i.bx^j$. For a scalar $a \in \mathbb{F}_q$, let $\mathcal{C}_{d,s}(a)$ be the sum of all possible compositions (as functions) of d copies of δ

and s copies of σ evaluated at a for $d, s \in \mathbb{Z}_{>0}$, $\mathcal{C}_{0,0}(a) = a$ and $\mathcal{C}_{d,s}(a) = 0$ otherwise. Then by [11, Lemma 2.10.], we have

$$ax^i \cdot bx^j = \sum_{k=0} a \cdot \mathcal{C}_{k,i-k}(b)x^{i+j-k}$$

Recall that an element $f \in R$ is called central if $f(x)g(x) = g(x)f(x)$ for all $f(x) \in R$, i.e., f commutes with every element of R. The set of all elements of R that commute with every element of R is called the center of R which is defined and denoted by

$$\mathrm{Z}(R) = \{f(x) \in R \ : \ f(x)g(x) = g(x)f(x), \quad \text{for all} \quad f(x) \in R\}\,.$$

In the following Proposition we collect some properties of the skew polynomial ring R.

Proposition 5.1 ([70], p. 483-486)
Let s be the order of the automorphism σ, and denote by $\mathrm{Fix}_\sigma(\mathbb{F}_q)$ the fixed subfield of $\mathbb{F}_q$ by σ.

1. The ring R is, in general, a non-commutative ring unless σ is the identity automorphism of $\mathbb{F}_q$.
2. An element $f \in R$ is central if and only if f belongs to $\mathrm{Fix}_\sigma(\mathbb{F}_q)[x^s, \sigma, \delta]$ i.e., $\mathrm{Z}(R) = \mathrm{Fix}_\sigma(\mathbb{F}_q)[x^s, \sigma, \delta]$.
3. Two-sided ideals of R are generated by elements of the form $h(x^s)x^m$, where m is an integer and $h(x) \in R$.
4. The ring R is a right (resp. left) Euclidean domain.

Since R is a right Euclidean domain, we have the notion of the greatest common right divisor of polynomials in R.

Definition 5.1
Let $f(x), g(x) \in R$. A polynomial $d(x)$ is called a **greatest common right divisor** (gcrd) of $f(x)$ and $g(x)$ if

- $d(x)$ is a right divisor of $f(x)$ and $g(x)$, and
- Any right divisor $d^{'}(x)$ of $f(x)$ and $g(x)$ is also a right divisor of $d(x)$.

In particular, $f(x)$ and $g(x)$ are said to be right coprime if $d(x) = 1$. In this case, there exist polynomials $a(x), b(x) \in R$ such that $a(x)f(x) + b(x)g(x) = 1$. Similarly, the **least common left multiple** of $f(x)$ and $g(x)$, denoted by $(\mathrm{lclm}(f(x), g(x)))$, is defined as the polynomial $\ell(x) \in R$ of least degree such that $f(x)$ and $g(x)$ are both **right** divisors of $\ell(x)$.

It is important to note that in the polynomial ring R, the evaluation of skew polynomials is different from the evaluation of polynomials in the usual polynomial rings over commutative rings.

Definition 5.2
For $a \in \mathbb{F}_q$, and $f(x) \in R$, the right evaluation of $f(x)$ at a, is the element $f(a) := r$ such that $f(x) = q(x)(x - a) + r$.

In [45, Proposition 2.9], Lam and Leroy proved that, for any $f(x) = \sum_{i=0}^{n-1} a_i x^i$ and $a \in \mathbb{F}_q$ we have

$$f(a) = \sum_{i=0}^{n} a_i N_i^{(\sigma,\delta)}(a).$$

where $N_i^{(\sigma,\delta)} : \mathbb{F}_q \longrightarrow \mathbb{F}_q$ are functions that play the same role as the powers in the commutative case, defined recursively by:

$$N_0^{(\sigma,\delta)}(a) = 1 \quad \text{and} \quad N_i^{(\sigma,\delta)}(a) = \sigma(N_{i-1}^{(\sigma,\delta)}(a))a + \delta(N_{i-1}^{(\sigma,\delta)}(a)).$$

In the rest of this section, we recall some of the fundamental properties of (σ, δ)-pseudo-linear transformations ($(\sigma,\delta)-$PLT). For more details we refer the reader to [49] and [40].

Definition 5.3 (Definition 1 in [40])
A map $T : \mathbb{F}_q^n \to \mathbb{F}_q^n$ is called (σ, δ)-PLT if for any $\alpha \in \mathbb{F}_q$, $u, v \in \mathbb{F}_q^n$, we have

1. $T(u + v) = T(u) + T(v)$
2. $T(\alpha u) = \sigma(\alpha)T(u) + \delta(\alpha)u$.

For a $v = (v_0, v_1, \ldots, v_{n-1}) \in \mathbb{F}_q^n$, we naturally extend σ and δ from $\mathbb{F}_q$ to $\mathbb{F}_q^n$ by $\sigma(v) = (\sigma(v_0), \sigma(v_1), \ldots, \sigma(v_{n-1}))$, and $\delta(v) = (\delta(v_0), \delta(v_1), \ldots, \delta(v_{n-1})) \in \mathbb{F}_q^n$.

Let $B = \{e_1, \ldots, e_n\}$ be the standard basis of $\mathbb{F}_q^n$ and $M = (a_{ij})_{1 \le i,j \le n}$ be the representation matrix of T with respect to B. Then

$$T(v) = M\sigma(v^t) + \delta(v^t).$$

Conversely, any $n \times n$ square matrix M permits to define a $(\sigma,\delta)-$PLT, T_M, on $\mathbb{F}_q^n$, by

$$T_M(v) = M\sigma(v^t) + \delta(v^t).$$

We call T_M **the associated pseudo-linear transformation** to the matrix M.

Remark 5.1 (Notation)
From now to the end of this chapter we avoid the use of transpose, i.e., the use of $M\sigma(v)$ will be understood as $M\sigma(v^t)$, and $\delta(v^t)$ as $\delta(v)$. More generally, we identify the vector $(M\sigma(v^t))^t$ by $M\sigma(v^t)$.

To evaluate a skew polynomial $f(x) = \sum_{i=0}^{n-1} a_i x^i$ at a matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$, we extend σ and δ from $\mathbb{F}_q$ to $\mathbb{M}_n(\mathbb{F}_q)$ component-wise, and obtain

$$f(M) = \sum_{i=0}^{m} a_i N_i^{(\sigma,\delta)}(M),$$

where $N_i^{(\sigma,\delta)} : \mathbb{M}_n(\mathbb{F}_q) \longrightarrow \mathbb{M}_n(\mathbb{F}_q)$, are functions, defined recursively as above on $\mathbb{F}_q$ by:

$$N_0^{(\sigma,\delta)}(M) = 1 \quad \text{and} \quad N_i^{(\sigma,\delta)}(M) = \sigma(N_{i-1}^{(\sigma,\delta)}(M))M + \delta(N_{i-1}^{(\sigma,\delta)}(M)).$$

For a matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$, we say that M is $(\sigma,\delta)-$**algebraic** if there exists a monic polynomial of degree m, $f(x) = \sum_{i=0}^{m} a_i x^i \in R$ such that

$$f(M) = \sum_{i=0}^{m} a_i N_i^{(\sigma,\delta)}(M) = 0,$$

and a (σ,δ)-PLT T_M is **algebraic** if there is a non-zero polynomial $f(x) = \sum_{i=0}^{n} a_i x^i \in R$ such that

$$f(T_M) = \sum_{i=0}^{n} a_i T_M^i = 0.$$

Example 5.1
Let σ be an automorphism of $\mathbb{F}_q$, and δ be a σ-derivation. We give some examples of nontrivial pseudo-linear transformations.

1. Let λ be a non-zero element of $\mathbb{F}_q$. The map $T_{\sigma,\delta,\lambda}$ defined on $\mathbb{F}_q^n$ by

$$T_{\sigma,\delta,\lambda}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(\lambda\sigma(v_{n-1}), \sigma(v_0), \ldots, \sigma(v_{n-2})\right) + \delta(v),$$

   is a (σ,δ)-pseudo-linear transformation that we call **the (σ,δ,λ)-constacyclic shift.** For $\lambda = \pm 1$ we call it **the (σ,δ)-cyclic shift** and **the (σ,δ)-negacyclic shift** .

2. Let $a = (a_0, a_1, \ldots, a_{n-1}) \in \mathbb{F}_q^n$. The map $T_{\sigma,\delta,\overrightarrow{a}}$ defined by

$$T_{\sigma,\delta,\overrightarrow{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = (\sigma(v_{n-1})a_0, \sigma(v_0) + \sigma(v_{n-1})a_1, \ldots, \sigma(v_{n-2}) + \sigma(v_{n-1})a_{n-1}) + \delta(v),$$

   is a $(\sigma,\delta)-$pseudo-linear transformation that we call **the right (σ,δ)-polyshift** induced by the vector a.

3. Let $a = (a_0, a_1, \ldots, a_{n-1}) \in \mathbb{F}_q^n$. The map $T_{\sigma,\delta,\overleftarrow{a}}$ defined by

$$T_{\sigma,\delta,\overleftarrow{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(\sigma(v_1) + \sigma(v_0)a_0, \ldots, \sigma(v_{n-1}) + \sigma(v_0)a_{n-2}, \sigma(v_0)a_{n-1}\right) + \delta(v),$$

   is a (σ,δ)-pseudo-linear transformation that we call **the left (σ,δ)-polyshift** induced by the vector a.

4. Let $\pi \in S_n$ be a permutation. Then $T_{\sigma,\delta,\pi}$ defined by

$$T_{\sigma,\delta,\pi}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(\sigma(v_{\pi(0)}), \sigma(v_{\pi(1)}), \ldots, \sigma(v_{\pi(n-1)})\right) + \delta(v),$$

   is a (σ,δ)-pseudo-linear transformation that we call **the (π,σ,δ)-permutation shift.**

5. Let $a = (a_0, a_1, \ldots, a_{n-1}) \in \left(\mathbb{F}_q^*\right)^n$. The map $T_{\overline{a},\sigma,\delta}$ defined by

$$T_{\overline{a},\sigma,\delta}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(a_{n-1}\sigma(v_{n-1}), a_0\sigma(v_0), \ldots, a_{n-2}\sigma(v_{n-2})\right) + \delta(v),$$

   is a (σ,δ)-pseudo-linear transformation that we call **the (σ,δ)-monomial shift** induced by the vector a.

6. Let $a = (a_0, a_1, \ldots, a_{n-1}) \in \mathbb{F}_q^{*n}$ and $M_{\overline{a},\pi} = \text{diag}(a_0, a_1, \ldots, a_{n-1})P_\pi$ be a monomial matrix, where $\pi \in S_n$ is a permutation and P_π its associated permutation matrix. The map $T_{\sigma,\delta,\pi,\overline{a}}$ associated to $M_{\sigma,\overline{a}}$ defined by

$$T_{\sigma,\delta,\pi,\overline{a}}\left(v_0, v_1, \ldots, v_{n-1}\right) = \left(a_{\pi(0)}\sigma(v_{\pi(0)}), a_{\pi(1)}\sigma(v_{\pi(1)}), \ldots, a_{\pi(n-1)}\sigma(v_{\pi(n-1)})\right) + \delta(v),$$

is a $(\sigma, \delta)-$pseudo-linear transformation that we call **the (σ, δ, π)-monomial shift** induced by the vector a.

Remark 5.2
It is shown in [49, Remark 4], that each square matrix M is $(\sigma, \delta)-$algebraic, and so M has a minimal polynomial, but T_M is not necessarily algebraic in general. In fact, it is also proved in [49, Lemma 3.17, Theorem 3.18] that if T_M is algebraic with minimal polynomial $f(x)$ then all its representation matrices with respect to different bases are also $(\sigma, \delta)-$algebraic with the same minimal polynomial $f(x)$.

Based on the above remark, we can demonstrate that pseudo-linear transformations over finite fields are always algebraic.

Lemma 5.1
Each pseudo-linear transformation of $\mathbb{F}_q^n$ is algebraic.

Proof
Since $\mathbb{F}_q$ is of finite size then each pseudo-linear transformation of $\mathbb{F}_q^n$ can be represented by a finite number of matrices. By the above remark, there is a polynomial f of least degree which vanishes at each representation matrix of T, that means f is the minimal polynomial of T.

Following [49, Definition 3, Proposition 2.4], we define on $\mathbb{M}_n(\mathbb{F}_q)$ an equivalence relation, called (σ, δ)-similarity relation, where we say that M is (σ, δ)**-similar to** M' if there is exists $P \in \text{GL}_n(\mathbb{F}_q)$ such that

$$M' = P^{-1}M\sigma(P) + P^{-1}\delta(P).$$

For $n = 1$, and $a, b \in \mathbb{F}_q$ we say that a is $(\sigma, \delta)-$**conjugate to** b, if there exists a non-zero element $c \in \mathbb{F}_q$, such that $b = a^c := c^{-1}a\sigma(c) + c^{-1}\delta(c)$, and we write $a \sim_{(\sigma,\delta)} b$.

Note that a given matrix M allows us to define a left $R-$module structure on $\mathbb{F}_q^n$ where the multiplication is given by:

$$P(x).v := (P(T_M))(v) = \sum_{i=0}^{n} a_i T_M^i(v), \quad \text{for each } v \in \mathbb{F}_q^n \text{ and } P(x) = \sum_{i=0}^{n} a_i x^i \in R.$$

In [21, Chapter 8, Section 4] the authors show that $\mathbb{F}_q^n$ has a unique decomposition into cyclic left $R-$submodules. Indeed, there is a unique sequence $f_1(x), f_2(x), \ldots, f_r(x)$ of polynomials, called **the invariant factors of** M, such that $f_i(x)$ right divides $f_{i+1}(x)$ and

$$\mathbb{F}_q^n \equiv R/R \cdot f_1(x) \oplus R/R \cdot f_2(x) \oplus \cdots \oplus R/R \cdot f_r(x).$$

In particular, when we have just one invariant factor $f_1(x)$, we say that M is a (σ, δ)-cyclic matrix.

For each vector $v \in \mathbb{F}_q^n$, let F_v be the $\mathbb{F}_q$-vector subspace of $\mathbb{F}_q^n$, generated by $\{v, T_M(v), T_M^2(v), \ldots\}$. We call F_v the $(M, \sigma, \delta)-$**cyclic subspace** of $\mathbb{F}_q^n$ generated by v. In particular the case when $F_v = \mathbb{F}_q^n$, leads to the following definition of a (σ, δ)-cyclic matrix.

Definition 5.4 ($(\sigma, \delta)-$cyclic matrix)
A square matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$ is called (σ, δ)-cyclic if there is a vector $v \in \mathbb{F}_q^n$ such that the set $B_v := \left\{v, T_M(v), \ldots, T_M^{n-1}(v)\right\}$ forms a basis of $\mathbb{F}_q^n$. In this case v is called an (M, σ, δ)-cyclic vector of $\mathbb{F}_q^n$ and B_v an (M, σ, δ)-cyclic basis of $\mathbb{F}_q^n$.

Definition 5.5
A skew polynomial $f(x) \in R$ is called an invariant polynomial if $f(x) \cdot R = R \cdot f(x)$, i.e., there exist $g(x)$ and $g^{'}(x)$ such that $f(x)g(x) = g^{'}(x)f(x)$. In particular, each polynomial in $\mathrm{Z}(R) = \mathrm{Fix}_\sigma(\mathbb{F}_q)[x^s, \sigma, \delta]$ is invariant, where s is the order of σ.

Theorem 5.1
Let M be a $(\sigma, \delta)-$cyclic matrix with minimal polynomial $f(x)$, and let v be an $(M, \sigma, \delta)-$cyclic vector such that $T_M^n(v) = \sum_{i=0}^{n-1} a_i T_M^i(v)$. Then,

1. The degree of $f(x)$ is at least n.

2. If $P(x) = x^n - \sum_{i=0}^{n-1} a_i x^i$ is an invariant polynomial in R, then the minimal polynomial of T_M is $P(x)$, and M is (σ, δ)-similar to C_P, the companion matrix of the polynomial $P(x)$, by the transition matrix $P_v = [v, (T_M(v)), \ldots, (T_M^{n-1}(v))]$ from the canonical (standard) basis of $\mathbb{F}_q^n$ to the cyclic basis B_v, which means that
$$C_P = P_v^{-1} M \sigma(P_v) + P_v^{-1} \delta(P_v).$$

Proof

1. Let $f(x)$ be the minimal polynomial of M. Then $f(M) = f(T_M) = 0$, where T_M is the associated pseudo-linear transformation of M. Such a polynomial $f(x)$ exists since T_M is algebraic (Lemma 5.1). Let $g(x) = \sum_{i=0}^{m} g_i x^i \in R$, be a polynomial such that $\deg(g(x)) < n$ and $g(T_M) = 0$. Since v is a (σ, δ)-cyclic vector, $g(T_M)(v) = \sum_{i=0}^{m} g_i T_M^i(v) = 0$. This contradicts the fact that B_v is a basis of $\mathbb{F}_q^n$. Therefore, the degree of the polynomial $f(x)$ is at least n.

2. To prove that $P(x) = x^n - \sum_{i=0}^{n-1} a_i x^i$ is the minimal polynomial of M, we need to prove that $P(M) = 0$ and it is a polynomial of minimal degree with this property. First, we start by showing that $P(T_M^i(v)) = 0$, for $i = 0, \ldots, n-1$. Since v is an $(M, \sigma, \delta)-$cyclic vector, we have $T_M^n(v) - \sum_{i=0}^{n-1} a_i T_M^i(v) = 0$, and so $P(T_M)(v) = 0$.

   For any $i = 1, \ldots, n-1$, there is a polynomial $a_i(x) \in R$ such that $P(x)x^i = a_i(x)P(x)$, since $P(x)$ is an invariant polynomial in R. Therefore, for any $i = 1, \ldots, n-1$, we have that
   $$(P(T_M))(T_M^i(v)) = a_i(T_M)(P(T_M)(v)) = 0, \text{ since } P(T_M)(v) = 0.$$
   Next, we set $w = \alpha_1 v + \alpha_2 T_M(v) + \ldots + \alpha_{n-1} T_M^{n-1}(v) \in \mathbb{F}_q^n$. Then
   $$\begin{aligned} P(T_M)(w) &= P(T_M)(\alpha_1 v + \alpha_2 T_M(v) + \ldots + \alpha_{n-1} T_M^{n-1}(v)) \\ &= P(T_M)(\alpha_1 v)) + P(T_M)(\alpha_2 T_M(v)) + \ldots + P(T_M)(\alpha_{n-1} T_M^{n-1}(v)) \\ &= \sum_{i=0}^{n-1} K_i(T_M)(P(T_M)(T_M^i(v)), \text{ where } P(x)\alpha_i = K_i(x)P(x) \text{ for any } i. \\ &= 0, \text{ since } P(T_M)(T_M^i(v) = 0 \text{ for any } i = 0, \ldots, n-1. \end{aligned}$$

This shows that $P(x)$ is a polynomial of least degree such that $P(T_M) = P(M) = 0$. Therefore, according to [49, Theorm 3.18] $P(x)$ is the minimal polynomial of M.

For the second part of this assertion. We verify that C_f is the representation matrix of T_M with respect to the cyclic basis B_v, and M is its representation matrix with respect to the standard basis of $\mathbb{F}_q^n$. Then according to [49, Proposition 2.4], C_f and M are (σ, δ)-similar by the transition matrix $P_v = [v, T_M(v), \ldots, T^{n-1}(v)]$, which means that

$$C_f = P_v^{-1} M \sigma(P_v) + P_v^{-1} \delta(P_v).$$

In [40] and [21] the authors show that we can decompose $\mathbb{F}_q^n$ as a direct sum of $(M, \sigma, \delta)-$cyclic subspaces as follows.

Theorem 5.2 ([40], Theorem 5)
Let M be a square matrix with sequence of left invariant factors $f_1(x), f_2(x), \ldots, f_r(x)$. Then there exist vectors $w_1, w_2, \ldots, w_r$ in $\mathbb{F}_q^n$, such that

1. $\mathbb{F}_q^n = F_{w_1} \oplus F_{w_2} \oplus \cdots \oplus F_{w_r}$, where $\oplus$ is the direct sum of subspaces in $\mathbb{F}_q^n$.
2. M is $(\sigma, \delta)-$similar to $\text{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$, where each C_{f_i} is the companion matrix of $f_i(x)$.
3. T_M is $(\sigma, \delta)-$similar to the operator $T' := T_1 \oplus T_2 \oplus \ldots \oplus T_r$, where each T_i, $i = 1, \ldots, r$, is the (σ, δ)-PLT associated with C_{f_i}.

Now we give the definition of the skew rational form of a matrix M.

Definition 5.6 ([40], Skew rational form of a matrix)
Let M be a matrix with sequence of invariant factors $f_1, f_2, \ldots, f_r$. The matrix D given by

$$D = \begin{pmatrix} C_{f_1} & & & \\ & C_{f_2} & & \\ & & \ddots & \\ & & & C_{f_r} \end{pmatrix}$$

where C_{f_i} is the companion matrix of $f_i(x) = x^{n_i} - \sum_{i=0}^{n_i-1} a_{k,i} x^i$ is called the skew rational form (or the Smith normal form) of a matrix.

Remark 5.3
In [21, Chapter 8, Section 4], it is shown that if $f_1(x)$ is an invariant polynomial in R then $f_2(x), \ldots, f_r(x)$ are also invariant. From Theorem 5.1, we can conclude that if $x^{n_1} - \sum_{k=0}^{n_1-1} a_{1,k} x^k$ is invariant in R, where $T_M^{n_1}(v_1) = \sum_{k=0}^{n_1-1} a_{1,k} T_M^k$, then each $f_i(x)$ is invariant in R and has the form $f_i(x) = x^{n_i} - \sum_{k=0}^{n_i-1} a_{i,k} x^k$, $1 \leq i \leq r$.

5.2 Structure of skew $(M, \sigma, \delta)-$cyclic codes

This section is devoted to investigating the structure of linear codes invariant under the action of a (σ, δ)-cyclic matrix M. Let us start by giving the definitions of (M, σ, δ)-skew codes and (M, σ, δ)-skew cyclic codes.

Definition 5.7 ([15] ,[87], [73])
Let M be an $n \times n$ square matrix M and $C \subseteq \mathbb{F}_q^n$ a linear code of length n over $\mathbb{F}_q$. Then C is called

1. an $(M, \sigma, \delta)-$skew code if it is invariant under T_M, i.e., for each $c = (c_0, c_1, \ldots, c_{n-1}) \in C$, we have
$$T_M(c) = M\sigma(c) + \delta(c) \in C.$$
2. an $(M, \sigma, \delta)-$skew cyclic code if it is invariant under T_M, where M is a (σ, δ)-cyclic matrix.
3. an (f, σ, δ)-cyclic code if it is invariant under T_{C_f}, where C_f is the companion matrix of $f(x)$.

Note that if $\Delta(M) \subseteq \mathbb{M}_n(\mathbb{F}_q)$ is the $(\sigma, \delta)-$similarity class of M in $\mathbb{M}_n(\mathbb{F}_q)$, then $M' \in \Delta(M)$ if there exists $P \in \mathrm{GL_n}(\mathbb{F}_q)$ such that $M' = P^{-1}M\sigma(P) + P^{-1}\delta(P)$. This leads to the following Theorem.

Theorem 5.3
Let M be an $n \times n$ square matrix with $(\sigma, \delta)-$similarity class $\Delta(M)$ and let $C \subseteq \mathbb{F}_q^n$ be a linear code. Then the following assertions are equivalent.

1. C is an $(M, \sigma, \delta)-$skew code.
2. $P^{-1}C = \{P^{-1}c : c \in C\}$ is an $(M', \sigma, \delta)-$skew code for each $M' \in \Delta(M)$ such that $M' = P^{-1}M\sigma(P) + P^{-1}\delta(P)$.

In particular, C is (M, σ, δ)-skew cyclic if and only if $P^{-1}C$ is $(M', \sigma, \delta)-$skew cyclic.

Proof

$(1 \Longrightarrow 2)$, Let C be an (M, σ, δ)-skew code, then $T_M(C) = M\sigma(C) + \delta(C) \subseteq C$. It follows that

$$\begin{aligned}
T_{M'}(P^{-1}C) &= M'\sigma(P^{-1}C) + \delta(P^{-1}C) \\
&= P^{-1}M\sigma(P)\sigma(P^{-1}C) + P^{-1}\delta(P)\sigma(P^{-1}C) + \delta(P^{-1}C) \\
&= P^{-1}M\sigma(P^{-1}PC) + P^{-1}\left[\beta(\sigma(P) - P)\right]\sigma(P^{-1}C) + \beta(\sigma(P^{-1}C) - P^{-1}C) \\
&\text{as } \delta \text{ is a } \sigma-\text{derivation, we have } \delta = \beta(\sigma - \mathrm{I}) \text{ for some } \beta \in \mathbb{F}_q^*. \\
&= P^{-1}M\sigma(C) + \beta P^{-1}\sigma(P)\sigma(P^{-1}C) - \beta P^{-1}P\sigma(P^{-1}C) + \beta\sigma(P^{-1}C) - \beta P^{-1}C) \\
&= P^{-1}M\sigma(C) + \beta P^{-1}\sigma(C) - \beta\sigma(P^{-1}C) + \beta\sigma(P^{-1}C) - \beta P^{-1}C \\
&= P^{-1}M\sigma(C) + \beta P^{-1}\sigma(C) - \beta P^{-1}C \\
&= P^{-1}\left[M\sigma(C) + \beta\sigma(C) - \beta C\right] \\
&= P^{-1}\left[M\sigma(C) + \delta(C)\right] \\
&= \sigma(C)T_M(C) \subseteq P^{-1}C, \text{ since } T_M(C) \subseteq C.
\end{aligned}$$

Hence $P^{-1}C$ is an (M', σ, δ)-skew code.

$(2 \Longrightarrow 1)$, Conversely, suppose that $P^{-1}C$ is an $(M^{'}, \sigma, \delta)$-skew code. Since $M^{'} \in \Delta(M)$ then there is an invertible matrix $P \in \mathrm{GL_n}(\mathbb{F}_q)$ such that M is (σ, δ)-similar to $M^{'}$ by P, i.e., $M^{'} = P^{-1}M\sigma(P) + P^{-1}\delta(P)$. It follows that

$$\begin{aligned} M = PM^{'}\sigma(P^{-1}) - \delta(P)\sigma(P^{-1}) &= PM^{'}\sigma(P^{-1}) - \beta\sigma(P)\sigma(P^{-1}) + \beta P\sigma(P^{-1}) \\ &= PM^{'}\sigma(P^{-1}) - \beta\,\mathrm{I}_n + \beta P\sigma(P^{-1}) \\ &= PM^{'}\sigma(P^{-1}) + P[\beta\sigma(P^{-1} - \beta P^{-1})] \\ &= PM^{'}\sigma(P^{-1}) + P\delta(P^{-1}) \\ &= Q^{-1}M^{'}\sigma(Q) + Q^{-1}\delta(Q) \quad \text{where } Q = P^{-1} \end{aligned}$$

Hence $M^{'} = Q^{-1}M^{'}\sigma(Q) + Q^{-1}\delta(Q)$, that means $M^{'}$ is (σ, δ)-similar to M by Q. Then by the above implication (the forward direction of the theorem) we deduce that $Q^{-1}(P^{-1}C) = P(P^{-1}C) = C$ is an (M, σ, δ)-skew code.

Remark 5.4
The result proved in the above theorem deals with the equivalence of C and $P^{-1}C$, where P represents the transition matrix between two bases of $\mathbb{F}_q^n$. In fact C and $P^{-1}C$ are equivalent if P is a monomial matrix, but in general, C and $P^{-1}C$ are not equivalent.

From the above theorem, we easily deduce the following result.

Proposition 5.2
Let M be a (σ, δ)-cyclic matrix with minimal polynomial $f(x)$ and v an (M, σ, δ)-cyclic vector of $\mathbb{F}_q^n$. Then $C \subseteq \mathbb{F}_q^n$ is a $(M, \sigma, \delta)-$cyclic code if and only if P_vC is an (f, σ, δ)-cyclic code, with $P_v = \left[v, T_M(v), \ldots, T_M^{n-1}(v)\right] \in \mathrm{GL_n}\left(\mathbb{F}_q\right)$.

Proof
Since M is (σ, δ)-similar to C_f, the result follows from Theorem 5.3 and Definition 5.7.

In relation to some well known types of codes in the literature, we make the following remark.

Remark 5.5
Note that (f, σ, δ)-cyclic codes generalize the well known codes in the literature with cyclical and skew cyclical property. For example:

1. $f(x) = x^n - \lambda, C$ is a skew $(\sigma, \delta)-$constacyclic code and for $\lambda = \pm 1$ we obtain a skew $(\sigma, \delta)-$cyclic code and a skew $(\sigma, \delta)-$negacyclic code.
2. $f(x) = x^n - \overrightarrow{v}(x)$, where $\overrightarrow{v}(x) = v_0 + v_1x + \ldots + v_{n-1}x^{n-1}$, C is a right $(\sigma, \delta)-$polycyclic code induced by the vector $v = (v_0, v_1, \ldots, v_{n-1})$.

In [85, 30, 80, 81, 58], the authors are interested in different types of codes with cyclical property. These codes are considered as invariant subspaces under a given linear transformation. Similar to these works, in the following definition, we define various types of codes with the skew (σ, δ)-cyclical property, including skew cyclic codes, skew constacyclic codes, and right (left) skew polycyclic codes and others. Note also that some of these codes were introduced by different authors, such as M. Boulougouz and A. Leroy in [15], D. Boucher and F. Ulmer in [13].

Definition 5.8
Let $a = (a_0, a_1, \ldots, a_{n-1}) \in \mathbb{F}_q^n$ be a non-zero vector. A linear code $C \subseteq \mathbb{F}_q^n$ is called a

1. (σ,δ,λ)**-constacyclic** code if for each $\mathbf{c}=\left(c_0,c_1,\ldots,c_{n-1}\right)\in C$ we have
$$T_{\lambda,\sigma,\delta}(\mathbf{c})=\left(\lambda\sigma(c_{n-1}),\sigma(c_0),\ldots,\sigma(c_{n-2})\right)+\delta(c)\in C$$
where λ is a non-zero element of $\mathbb{F}_q$. For $\lambda=\pm1$ we call them (σ,δ)**-cyclic codes** and (σ,δ)**-negacyclic codes.**

2. **right** $(\sigma,\delta)-$**polycyclic** code induced by $a=(a_0,a_1,\ldots,a_{n-1})$ if it is invariant under the (σ,δ)-PLT $T_{\overrightarrow{a},\sigma,\delta}$, the associated $(\sigma,\delta)-$PLT with the matrix $M_{\overrightarrow{a}}$. This means, for each $c=(c_0,c_1,\ldots,c_{n-1})\in C$ we have:
$$T_{\overrightarrow{a},\sigma,\delta}(\mathbf{c})=(\sigma(c_{n-1})a_0,\sigma(c_0)+\sigma(c_{n-1})a_1,\sigma(c_1)+\sigma(c_{n-1})a_2,\ldots,\sigma(c_{n-2})+\sigma(c_{n-1})a_{n-1})+\delta(c)\in C.$$

3. **left** (σ,δ)**-polycyclic** code induced by $a=(a_0,a_1,\ldots,a_{n-1})$ if it is invariant under the $(\sigma,\delta)-$PLT $T_{\overleftarrow{a},\sigma,\delta}$, the associated $(\sigma,\delta)-$PLT with the matrix $M_{\overleftarrow{a}}$. This means, for each $c=(c_0,c_1,\ldots,c_{n-1})\in C$ we have:
$$T_{\overleftarrow{a},\sigma,\delta}(c)=(\sigma(c_1)+\sigma(c_0)a_0,\sigma(c_2)+\sigma(c_0)a_1,\ldots,\sigma(c_{n-1})+\sigma(c_0)a_{n-2},\sigma(c_0)a_{n-1})+\delta(c)\in C.$$

4. **skew** $(\pi,\sigma,\delta)-$**cyclic** code, where π is an n-cycle, if for each $\mathbf{c}=\left(c_0,c_1,\ldots,c_{n-1}\right)\in C$ we have
$$T_{\pi,\sigma,\delta}(\mathbf{c})=\left(\sigma(c_{\pi(0)}),\sigma(c_{\pi(1)}),\ldots,\sigma(c_{\pi(n-1)})\right)+\delta(c)\in C.$$

From the above definition we make the following remarks.

Remark 5.6

1. Right $(\sigma,\delta)-$polycyclic codes induced by the vector $a=(a_0,a_1,\ldots,a_{n-1})$ are exactly $(\sigma,\delta,f)-$cyclic codes where $f(x)=x^n-\sum_{i=0}^{n-1}a_ix^i$.

2. (σ,δ,λ)-constacyclic codes are right $(\sigma,\delta)-$polycyclic codes induced by the vector $(0,\ldots,0,\lambda)$, and left $(\sigma,\delta)-$polycyclic codes induced by the vector $(\lambda,0,\ldots,0)$. A similar remark is valid for skew $(\sigma,\delta)-$cyclic (resp. $(\sigma,\delta)-$negacyclic) codes with $\lambda=\pm1$.

3. Right $(\sigma,\delta)-$polycyclic codes induced by the vector $a=(a_0,a_1,\ldots,a_{n-1})$ are invariant by the $(\sigma,\delta)-$pseudo-linear transformation $T_{M_{\overrightarrow{a}}}$, and left $(\sigma,\delta)-$polycyclic codes by $T_{M_{\overleftarrow{a}}}$, where
$$A_{\overrightarrow{a}}=\begin{pmatrix}0&0&\ldots&0&a_0\\1&0&\ldots&0&a_1\\0&1&\ddots&\vdots&\vdots\\\vdots&\ddots&\ddots&0&\vdots\\0&0&\ldots&1&a_{n-1}\end{pmatrix}\qquad A_{\overleftarrow{a}}=\begin{pmatrix}a_0&1&0&\ldots&0\\a_1&0&1&\ldots&0\\\vdots&0&\ddots&\ddots&\vdots\\\vdots&\ldots&\ddots&\ddots&1\\a_{n-1}&0&0&\ldots&0\end{pmatrix}.$$

The next proposition gives a basic property of (σ,δ)-polycyclic codes.

Proposition 5.3

A linear code C is a left $(\sigma,\delta)-$polycyclic code induced by the vector $a=(a_0,a_1,\ldots,a_{n-1})$ if and only if $P^{-1}C$ is a right $(\sigma,\delta)-$polycyclic code induced by the vector $b=(a_{n-1},a_{n-2},\ldots,a_0)$, where P is the $n\times n$ square matrix given by
$$P=P^{-1}=\begin{pmatrix}0&\ldots&\ldots&0&1\\0&\ldots&0&1&0\\\vdots&\iddots&\iddots&\iddots&\vdots\\0&\iddots&\iddots&&\vdots\\1&0&\ldots&\ldots&0\end{pmatrix}$$

Proof
Since $\delta(1) = 0$ and $\sigma(1) = 1$, $\delta(P) = 0$ and $\sigma(P) = P$. We can directly verify that

$$M_{\overrightarrow{a}} = P^{-1} M_{\overleftarrow{b}} \sigma(P) + P^{-1}\delta(P).$$

It follows that $M_{\overrightarrow{a}}$ is $(\sigma, \delta)-$similar to $M_{\overleftarrow{b}}$ by P. By Theorem 5.3 the result holds.

Remark 5.7
As P is a monomial matrix in Proposition 5.3, we conclude that C and $P^{-1}C$ are equivalent.

As is common in coding theory, we consider the following isomorphism φ_f of $\mathbb{F}_q$-vector spaces between $\mathbb{F}_q^n$ and $R_f := R/R \cdot f(x)$, given by:

$$\begin{array}{rccc} \varphi_f : & \mathbb{F}_q^n & \longrightarrow & R_f := R/R \cdot f(x) \\ & v := (v_0, v_1, ..., v_{n-1}) & \longmapsto & v(x) = \sum_{i=0}^{n-1} v_i x^i \end{array} \tag{5.1}$$

which leads to the following characterization of $(M, \sigma, \delta)-$cyclic codes.

Theorem 5.4
Let $C \subseteq \mathbb{F}_q^n$ be a linear code, and let M be a $(\sigma, \delta)-$cyclic matrix with minimal polynomial $f(x) = x^n - \sum_{i=0}^{n-1} a_i x^i$. Then,

1. C is an $(f, \sigma, \delta)-$cyclic code if and only if $\varphi_f(C)$ is a left $R-$submodule of $R_f := R/R \cdot f(x)$.
2. C is an $(M, \sigma, \delta)-$cyclic code if and only if $\psi_f(C)$ is a left $R-$submodule of R_f, where $\psi_f : \mathbb{F}_q^n \longrightarrow R_f$ is the $\mathbb{F}_q-$vector space isomorphism defined by $\psi_f(v_0, v_1, \ldots, v_{n-1}) := \varphi_f(P^{-1}(v_0, v_1, \ldots, v_{n-1}))$, and $P \in \mathrm{GL}_n(\mathbb{F}_q)$ is such that $M = P^{-1}C_f\sigma(P) + P^{-1}\delta(P)$.

Proof
1. Let $C \subseteq \mathbb{F}_q^n$ be an $(f, \sigma, \delta)-$cyclic code of length n over $\mathbb{F}_q$. By linearity of φ_f, $\big(\varphi_f(C), +\big)$ is a group. Let $c = (c_0, c_1, ..., c_{n-1}) \in C$. Then

$$\begin{aligned} x\varphi_f(c) &= x(c_0 + c_1 x + \ldots + c_{n-1}x^{n-1}) \\ &= xc_0 + xc_1 x + \ldots + xc_{n-1}x^{n-1} \\ &= \sigma(c_0)x + \delta(c_0) + \sigma(c_1)x^2 + \delta(c_1)x + \ldots + \sigma(c_{n-1})x^n + \delta(c_{n-1})x^{n-1} \\ &= \sigma(c_0)x + \sigma(c_1)x^2 + \ldots + \sigma(c_{n-1})x^n + \delta(c_0) + \delta(c_1)x + \ldots + \delta(c_{n-1})x^{n-1} \\ &= \sigma(c_0)x + \ldots + \sigma(c_{n-2})x^{n-1} + \sum_{i=0}^{n-1}\sigma(c_{n-1})a_i x^i + \sum_{i=0}^{n-1}\delta(c_i)x^i, \text{ since } x^n = \sum_{i=0}^{n-1} a_i x^i (\mathrm{mod}\ f(x)) \\ &= \sum_{i=1}^{n-1}\sigma(c_{i-1})x^i + \sum_{i=0}^{n-1}\sigma(c_{n-1})a_i x^i + \sum_{i=0}^{n-1}\delta(c_i)x^i \ (\mathrm{mod}\ f(x)) \\ &= \sigma(c_{n-1})a_0 + \delta(c_0) + \sum_{i=1}^{n-1}\left(\sigma(c_{i-1}) + \sigma(c_{n-1})a_i + \delta(c_i)\right)x^i \ (\mathrm{mod}\ f(x)) \\ &= \varphi_f\left(\sigma(c_{n-1})a_0 + \delta(a_0), \sigma(c_0) + \sigma(c_{n-1})a_1 + \delta(a_1), \ldots, \sigma(c_{n-2}) + \sigma(c_{n-1})a_{n-1} + \delta(a_{n-1})\right) \\ &= \varphi_f\left(C_f\sigma(c) + \delta(c)\right) \\ &= \varphi_f\left(T_{C_f}(c)\right) \end{aligned} \tag{5.2}$$

By induction we show that for $k \in \{0, ..., n-1\}$

$$x^k\varphi_f(c) = \varphi_f(T_{C_f}^k(c)).$$

It follows that $\varphi_f(C)$ is a left $R-$submodule of R_f.

2. Let C be an $(M,\sigma,\delta)-$cyclic code. Since $M = P^{-1}C_{_f}\sigma(P) + P^{-1}\delta(P)$, then by Theorem 5.3, $P^{-1}C$ is an $(f,\sigma,\delta)-$cyclic code, and so $\psi_f(C) = \varphi_{_f}(P^{-1}C)$ is a left $R-$submodule of R_f.

From the above theorem, we make the following remarks.

Remark 5.8

1. The $\mathbb{F}_{_q}-$vector space isomorphism $\varphi_{_f}$ preserves the Hamming weight, and so C and its corresponding left $R-$module $\varphi_{_f}(C)$ have the same minimum Hamming distance.

2. The $\mathbb{F}_{_q}-$vector space isomorphism $\psi_{_f}$ does not, in general, preserve the Hamming weight, unless P is a monomial matrix. So, C and its corresponding left $R-$module $\psi_f(C)$ may have different minimum Hamming distances.

3. If the polynomial $f(x)$ generates a two-sided ideal in R, then $\varphi_{_f}(C)$ is an ideal of R_f, and we call them **central codes**, see [13].

Example 5.2
Let $q = 8$ and $\mathbb{F}_{_8} = \mathbb{F}_2(\alpha)$, where $\alpha^3 + \alpha + 1 = 0$. In $\mathbb{F}_{_8}$, let $\sigma : \mathbb{F}_8 \longrightarrow \mathbb{F}_8$ be the automorphism defined by $\sigma(a) = a^4$ and let the σ-derivation δ given by $\delta(a) = \alpha^2(\sigma(a) - a)$ for all $a \in \mathbb{F}_{_8}$. Let C be a $[5,3,2]$-linear code with generator matrix

$$G = \begin{pmatrix} \alpha^6 & 1 & \alpha^6 & 1 & 1 \\ \alpha^4 & 1 & \alpha^4 & \alpha^3 & 0 \\ 1 & 0 & 0 & 0 & \alpha^4 \end{pmatrix}$$

We verify by using Magma software that C is invariant under the $(\sigma,\delta)-$action of M, and $M = P^{-1}C_{_f}\sigma(P) + P^{-1}\delta(P)$, where $f(x) = x^5 + \alpha^3x^3 + x^2 + x + 1$ and M, P, P^{-1} are given bellow. Then C is an (M,σ,δ)-cyclic code. According to Theorem 5.4, $\psi_{_f}(C)$. is a left $R-$submodule of R_f, and $\psi_{_f}(C) := \varphi(P^{-1}C)$ is an $[5,3,3]$-linear code. So we observe here that $\psi_{_f}$ does not preserve the Hamming weight, where

$$\begin{aligned} &\psi_{_f}(c_0, c_1, \ldots, c_4) = \varphi_{_f}((c_0, c_1, \ldots, c_4)P^{-1}) \\ &= (\alpha^6c_0 + \alpha c_2 + c_3 + c_4) + (\alpha^6c_0 + c_3)x + (\alpha^6c_0 + c_1 + \alpha c_2 + c_3 + c_4)x^2 + c_3x^3 + c_4x^4. \end{aligned}$$

$$M = \begin{pmatrix} \alpha^3 & 0 & \alpha^4 & 0 & \alpha^4 \\ \alpha^6 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \alpha^6 & 1 & \alpha^3 & 1 & 0 \\ 0 & 1 & \alpha^2 & \alpha^4 & 1 \end{pmatrix} \quad ; \quad P = \begin{pmatrix} 0 & \alpha & 0 & 0 & \alpha \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & \alpha & 0 \end{pmatrix} \quad ; \quad P^{-1} = \begin{pmatrix} \alpha^6 & 0 & \alpha & 1 & 1 \\ \alpha^6 & 0 & 0 & 1 & 0 \\ \alpha^6 & 1 & \alpha & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

In line with the Proposition 5.3 and Remark 5.5 we give the following example.

Example 5.3
The goal of this example is to construct a map $\psi_{\overleftarrow{a}}$ that allows us to connect left (σ,δ)-polycyclic codes with left R-submodules of a given left module $R_{\overleftarrow{a}}$. Let C be a **left** skew (σ,δ)-polycyclic code induced by the vector $a = (a_0, a_1, \ldots, a_{n-1})$. According to Proposition 5.3, $P^{-1}C$ is a **right** skew polycyclic code induced by the vector $a = (a_{n-1}, a_{n-2}, \ldots, a_0)$. By Remark 5.6, $P^{-1}C$ is the $(f_{\overleftarrow{a}},\sigma,\delta)-$cyclic code, where $f_{\overleftarrow{a}}(x) = x^n - \sum_{i=0}^{n-1} a_i x^{n-1-i}$. It follows by Theorem 5.4 that $\varphi_{_f}(P^{-1}C)$ is a left $R_{_f}$-submodule of $R/R\cdot f_{\overleftarrow{a}}(x)$. Setting $\psi_{\overleftarrow{a}} := \varphi \circ P^{-1}$ and $R_{\overleftarrow{a}} := R/R\cdot f_{\overleftarrow{a}}(x)$, we obtain that

$$\begin{aligned} \psi_{\overleftarrow{a}} \quad : \quad \mathbb{F}_q^n \quad &\longrightarrow \quad R/R\cdot f_{\overleftarrow{a}}(x) \\ c := (c_0, c_1, \ldots, c_{n-1}) \quad &\longmapsto \quad \psi_{\overleftarrow{a}}(c) = c_{n-1} + c_{n-2}x + \ldots + c_0x^{n-1} \end{aligned}$$

gives a one-to-one correspondence between **left** skew $(\sigma, \delta)-$polycyclic codes and left $R-$submodules of the left module $R_{\overleftarrow{a}} := R/R \cdot f_{\overleftarrow{a}}(x)$.

Remark 5.9
Note that the $\mathbb{F}_q-$vector space isomorphism $\psi_{\overleftarrow{a}}$ in the above example is used in [80] to investigate the structure of left polycyclic codes induced by a vector $a = (a_0, a_1, \ldots, a_{n-1})$ in the commutative case i.e., when $\sigma = \mathrm{Id}$.

In next result we collect some polynomial characterizations of $(M, \sigma, \delta)-$cyclic codes. These characterizations are already given in [15, 87, 73] for $(f, \sigma, \delta)-$cyclic codes. In our case we deduce them for $(M, \sigma, \delta)-$cyclic codes.

Theorem 5.5
Let M be a (σ, δ)-cyclic matrix with minimal polynomial $f(x)$, and let P be an invertible matrix such that M is $(\sigma, \delta)-$similar to C_f by an invertible matrix P. If $C \subseteq \mathbb{F}_q^n$ is an $(M, \sigma, \delta)-$cyclic code of length n over $\mathbb{F}_q$, then

1. There is a monic polynomial of least degree $g(x) \in R$ such that $g(x)$ is a right divisor of $f(x)$ and $\psi_f(C) = R \cdot g(x)$, where $\psi_f = \varphi_f \circ P^{-1}$ as in Theorem 5.4.
2. The set $\{g(x), xg(x), \ldots, x^{k-1}g(x)\}$ forms a basis for $\psi_f(C)$ and the dimension of C is $k = n - \deg(g(x))$.
3. A generator matrix G of C is given by:

$$G = \begin{pmatrix} \psi_f^{-1}(g(x)) \\ \psi_f^{-1}(xg(x)) \\ \vdots \\ \vdots \\ \psi_f^{-1}\left(x^{k-1}g(x)\right) \end{pmatrix} \tag{5.3}$$

where $k = n - \deg(g(x))$.

We observe that a generator matrix of an (M, σ, δ)-cyclic code does not have a specific form. Consider the following two examples.

Example 5.4
Assume that $\delta = 0$. Then we give the following two cases of generator matrices of (M, σ, δ)-cyclic codes

1. If C is an $(f, \sigma, \delta)-$cyclic code generated by $g(x)$ then a generator matrix of C has the following form.

$$G = \begin{pmatrix} g_0 & g_1 & \cdots & g_{n-k} & 0 & \cdots & \cdots & 0 \\ 0 & \sigma(g_0) & \sigma(g_1) & \cdots & \sigma(g_{n-k}) & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \ddots & \vdots \\ 0 & \ldots & & 0 & \sigma^{k-1}(g_0) & \sigma^{k-1}(g_1) & \ldots & \sigma^{k-1}(g_{n-k}) \end{pmatrix} \tag{5.4}$$

where $k = n - \deg(g(x))$ and $g(x) = \sum_{i=0}^{n-k} g_i x^i$ is a right divisor of $f(x)$.

2. Let C be a left skew (σ,δ)-polycyclic code induced by the vector $a=(a_0,a_1,\ldots,a_{n-1})$. By 5.3, $\psi_{\overleftarrow{a}}$, defined by

$$\psi_{\overleftarrow{a}}((c_0,c_1,\ldots,c_{n-1}))=c_{n-1}+c_{n-2}x+\ldots+c_0x^{n-1},$$

gives a one-to-one correspondence between left skew polycyclic codes and left R-submodule of $R_{\overleftarrow{a}}$. Let $g(x)=\sum_{i=0}^{n-k}g_ix^i$ be a generator polynomial of C. This means $g(x)$ is a right divisor of $f_{\overleftarrow{a}}(x)=x^n-\sum_{i=0}^{n-1}a_ix^{n-i-1}$ such that
$\psi_{\overleftarrow{a}}(C)=R\cdot g(x)$. To determine a generator polynomial of C we need to compute $\psi_{\overleftarrow{a}}^{-1}(x^ig(x))$ for $i=0,\ldots,k-1$, where $k=n-\deg(g(x))$. We have

$$\begin{aligned}\psi_{\overleftarrow{a}}^{-1}(g(x)) &= \psi_{\overleftarrow{a}}^{-1}\left(g_0+g_1x+\ldots+g_{n-k}x^{n-k}+0x^{n-k-1}+\ldots+0x^{n-1}\right)\\ &=(0,\ldots,0,g_{n-k},\ldots,g_1,g_0),\ \text{and}\end{aligned}$$

$$\begin{aligned}\psi_{\overleftarrow{a}}^{-1}(xg(x)) &= \psi_{\overleftarrow{a}}^{-1}\left(0+\sigma(g_0)x+\sigma(g_1)x^2+\ldots+\sigma(g_{n-k})x^{n-k+1}+0x^{n-k-1}+\ldots+0x^{n-1}\right)\\ &=(0,\ldots,0,\sigma(g_{n-k}),\ldots,\sigma(g_1),\sigma(g_0),0).\end{aligned}$$

It follows that for any $i\leq k-1$, we have

$$\begin{aligned}\psi_{\overleftarrow{a}}^{-1}(x^ig(x)) &= \psi_{\overleftarrow{a}}^{-1}\left(0+\ldots+0x^{i-1}+\sigma^i(g_0)x^i+\sigma^i(g_1)x^{i+1}+\ldots+\sigma^i(g_{n-k})x^{n-k+i}+0x^{n-k+i+1}+\ldots+0x^{n-1}\right)\\ &=(0,\ldots,0,\sigma^i(g_{n-k}),\ldots,\sigma^i(g_1),\sigma^i(g_0),0,\ldots,0).\end{aligned}$$

Hence we obtain that its generator matrix has the form

$$G=\begin{pmatrix}0 & \ldots & \ldots & 0 & g_{n-k} & \ldots & g_1 & g_0\\ 0 & \ldots & 0 & \sigma(g_{n-k}) & \ldots & \sigma(g_1) & \sigma(g_0) & 0\\ \vdots & \iddots & \iddots & & \iddots & \iddots & \iddots & \vdots\\ 0 & \iddots & & \iddots & \iddots & \iddots & \iddots & \vdots\\ \sigma^{k-1}(g_{n-k}) & \ldots & \sigma^{k-1}(g_1) & \sigma^{k-1}(g_0) & 0 & \ldots & \iddots & 0\end{pmatrix}$$

where $k=n-\deg(g(x))$ and $g(x)=\sum_{i=0}^{n-k}g_ix^i$ is a right divisor of $f_{\overleftarrow{a}}=x^n-\sum_{i=0}^{n-1}a_ix^{n-i-1}$.

We now give the definition of generator polynomials for (f,σ,δ)-cyclic codes. Similar definitions can be given for $(M,\sigma,\delta)-$cyclic codes as well using ψ_f. Like an ordinary cyclic code, an (f,σ,δ)-cyclic code C has many generators but among all of its generators, there is a special unique one, called the standard generator of C. We sometimes refer to the standard generator of C as "the generator".

Definition 5.9
Let $C\subseteq\mathbb{F}_q^n$ be an (f,σ,δ)-cyclic code of length n over $\mathbb{F}_q$. Then the **standard generator polynomial** of C is the monic polynomial $g(x)$ of least degree in R such that $\varphi_f(C)=R\cdot g(x)$.

Proposition 5.4 (Theorem 2 in [15], and Proposition 3.6 [87])
Let C be an $(f,\sigma,\delta)-$cyclic code with generator polynomial $g(x)$, where $f(x)$ is a polynomial of degree n. Let $h(x)\in R$, such that $f(x)=h(x)g(x)=g(x)h^{'}(x)$. Then a set of $\deg(g(x))$

independent columns of the following $n \times n$ matrix H forms a basis of $C^{\perp}$, the Euclidean dual of C.

$$H = \begin{pmatrix} \varphi_f^{-1}(h'(x)) \\ \varphi_f^{-1}\left(xh'(x)\right) \\ \vdots \\ \vdots \\ \varphi_f^{-1}\left(x^{n-1}h'(x)\right) \end{pmatrix} \tag{5.5}$$

Proposition 5.5
Let $C \subseteq \mathbb{F}_q^n$ be an (f, σ, δ)-cyclic code of length n over $\mathbb{F}_q$ with the standard generator polynomial $g(x)$, and $f(x) = h(x)g(x)$.

1. If $\varphi_f(C) = R \cdot a(x)$ for some $a(x) \in R$ then $g(x) = \text{gcrd}(a(x), f(x))$.
2. If $\text{gcrd}(p(x), h(x)) = 1$, then $p(x)g(x)$ generates C, i.e $\varphi_f(C) = R \cdot (p(x)g(x))$.

Proof

1. Let $\alpha(x) = \text{gcrd}(a(x), f(x))$. As $\varphi_f(C) = R \cdot a(x)$, $a(x) \in \varphi_f(C)$ and so there is $q(x) \in R$ such that $a(x) = q(x)g(x)$, and by the fact that $g(x)$ right divides $f(x)$, we obtain that $g(x)$ is a common right divisor of $a(x)$ and $f(x)$. It follows $g(x)$ is a right divisor of $\alpha(x) = \text{gcrd}(a(x), f(x))$.

   On the other hand, let $g(x) = q_1(x)\alpha(x) + r(x)$, with $\deg(r(x)) < \deg(\alpha(x)) \leq \deg(g(x))$ or $r(x) = 0$. As $g(x)$ right divides $\alpha(x)$, there is $q_2(x) \in R$ such that $\alpha(x) = q_2(x)g(x)$, and so one can write $g(x) = q_1(x)\alpha(x) + r(x) = q_1(x)q_2(x)g(x) + r(x) = q'(x)g(x) + r(x)$, with $q'(x) = q_1(x)q_2(x)$. It follows that $r(x) = g(x) - q'(x)g(x) = (1 - q'(x))g(x) \in \varphi_f(C)$. Hence by minimality of the degree of $g(x)$ we deduce that $r(x) = 0$, which means that $\alpha(x)$ is a right divisor of $g(x)$. Therefore, $g(x) = \alpha(x) = \text{gcrd}(a(x), h(x))$.

2. Let C be an (f, σ, δ)-cyclic code with the generator polynomial $g(x)$. Since $p(x)$ and $h(x)$ are right coprime, there exist polynomials $a(x), b(x) \in R$ such that $a(x)p(x) + b(x)h(x) = 1$. The multiplication of both sides on the right by $g(x)$ leads to $a(x)p(x)g(x) + b(x)h(x)g(x) = g(x)$, i.e., $a(x)p(x)g(x) + b(x)f(x) = g(x)$. It follows that $a(x)p(x)g(x) = g(x) \mod f(x)$, which implies that $g(x) \in R \cdot (p(x)g(x))$, and so $p(x)g(x)$ generates C.

We now give some examples of (f, σ, δ)-cyclic codes and their generalizations. We used the Magma software [60] for these computations.

Example 5.5
Let $q = 8$ and $\mathbb{F}_8 = \mathbb{F}_2(\alpha)$, where $\alpha^3 + \alpha + 1 = 0$. In $\mathbb{F}_8$, let $\sigma : \mathbb{F}_8 \longrightarrow \mathbb{F}_8$ be the automorphism defined by $\sigma(a) = a^4$ and let the σ-derivation δ given by $\delta(a) = \alpha(\sigma(a) - a) = \alpha a^4 - \alpha a$ for all $a \in \mathbb{F}_8$. Let $f(x) = x^{11} + x^{10} + \alpha^3 x^9 + x^8 + \alpha^4 x^7 + \alpha^3 x^6 + x^2 + x + 1 \in \mathbb{F}_8[x, \sigma, \delta]$.

Let $g(x) = x^5 + x^4 + \alpha x^3 + x + 1$ be a right divisor of $f(x)$, with

$$f(x) = (x^6 + x^4 + x^2 + 1)g(x).$$

The linear code C generated by $g(x) = x^5 + x^4 + \alpha x^3 + x + 1$ is an $(f, \sigma, \delta)-$cyclic code of length $n = 11$ and of dimension $k = n - \deg(g(x)) = 11 - 5 = 6$. A generator matrix of C, as in Theorem 5.5, is given by

$$G = \begin{pmatrix} 1 & 1 & 0 & \alpha & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & \alpha^3 & \alpha^4 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \alpha & \alpha^3 & \alpha^2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \alpha & \alpha^6 & 0 & \alpha & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & \alpha^3 & 1 & \alpha^3 & \alpha^3 & \alpha^4 & 1 & 1 & 0 \\ 0 & 0 & 0 & \alpha^3 & \alpha^5 & \alpha & \alpha^2 & \alpha^3 & \alpha^2 & 1 & 1 \end{pmatrix}$$

and it is an $[11, 6, 3]_8$-linear code. Note that one can construct other (f, σ, δ)-cyclic codes of length $n = 11$ over $\mathbb{F}_8$ using other right divisors of $f(x)$. For example consider $f(x) = (x^4 + 1)g^{'}(x)$, where $g^{'}(x) = x^7 + x^6 + \alpha^6 x^5 + \alpha x^4 + x^2 + x + 1$. The polynomial $g^{'}(x)$ generates an (f, σ, δ)-cyclic code of length $n = 11$ and dimension 4 over $\mathbb{F}_8$. Let

$$M = \begin{pmatrix} 0 & 0 & \alpha^5 & 1 & 0 & 1 & \alpha^3 & \alpha^4 & \alpha^3 & 1 & 0 \\ \alpha^2 & \alpha^2 & \alpha^5 & 1 & 0 & 1 & \alpha^3 & 0 & \alpha^3 & 1 & 0 \\ 0 & 1 & \alpha^4 & 1 & 0 & \alpha^2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ \alpha^6 & \alpha^6 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \alpha^2 & \alpha^2 & 1 & 1 & 0 & \alpha^2 & \alpha^3 & 1 & \alpha & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \alpha^2 & \alpha & \alpha^5 & \alpha^5 & 0 & 0 & \alpha^3 & 0 & \alpha^5 & \alpha^3 & 0 \\ 1 & 1 & 0 & 1 & 0 & \alpha^6 & \alpha & 1 & \alpha & 1 & 0 \\ \alpha^6 & \alpha^6 & \alpha^4 & 0 & 0 & \alpha^2 & 0 & 0 & 1 & 0 & 0 \end{pmatrix},$$

and

$$P = \begin{pmatrix} 0 & 1 & \alpha^4 & 0 & 0 & \alpha^2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & \alpha^4 & 1 & 0 & \alpha^2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ \alpha^6 & \alpha^6 & \alpha^4 & 0 & 0 & \alpha^2 & 1 & 1 & 1 & 0 & 1 \\ \alpha^6 & \alpha^6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \alpha^4 & 0 & 0 & \alpha^2 & 0 & 0 & 1 & 1 & 0 \\ \alpha^6 & \alpha^6 & \alpha^4 & 0 & 0 & \alpha^2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & \alpha^4 & 0 & 0 & \alpha^2 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}, \; P^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \alpha & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & \alpha & 0 & 0 & 0 & 0 & 0 & \alpha^2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Then as $C_f = \sigma(P)MP^{-1} + \delta(P)P^{-1}$, M is a (σ, δ)-cyclic matrix and by Theorem 5.3, every (M, σ, δ)-skew code is an (f, σ, δ)-cyclic code. It follows that $P^{-1}C$ (by Theorem 5.4) is an (M, σ, δ)-skew code generated by $g(x)$, where C is the (f, σ, δ)-skew cyclic code generated by $g(x)$.

Remark 5.10
Note that $(f, \sigma, \delta)-$cyclic codes considered in the above example are exactly right (σ, δ)-polycyclic codes associated with the vector $a = (1, 1, 1, 0, 0, 0, \alpha^3, \alpha^4, 1, \alpha^3, 1)$. It follows that left $(\sigma, \delta)-$polycyclic codes associated with a are $(M_{\overleftarrow{a}}, \sigma, \delta)-$cyclic codes.

5.3 Structure of $(M, \sigma, \delta)-$multi-cyclic and $(M, \sigma, \delta)-$quasi-cyclic codes

This section is devoted to investigating the structure of linear codes invariant under the action of a matrix M, which is not necessarily $(\sigma, \delta)-$cyclic. So, let us start with the following definitions.

Definition 5.10
A linear code $C \subseteq \mathbb{F}_q^n$ is called

1. $(M, \sigma, \delta)-$quasi-cyclic of index r if it is invariant under T_D, where $D = \text{diag}(\overbrace{M, M, \ldots, M}^{r \text{ times}})$, and M is $(\sigma, \delta)-$cyclic.

2. $(f, \sigma, \delta)-$quasi-cyclic of index r if it is invariant under T_D, where $D = \text{diag}(\overbrace{C_f, C_f, \ldots, C_f}^{r \text{ times}})$.

3. $(M_1, M_2, \ldots, M_r, \sigma, \delta)-$multi-cyclic if it is invariant under T_D, where $D = \text{diag}(M_1, M_2, \ldots, M_r)$ and each M_i is a $(\sigma, \delta)-$cyclic matrix.

4. $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic if it is invariant under T_D, where $D = \text{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$.

In relation with Definition 5.8, we define more classes of codes in the literature as special cases of $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic codes. Note that we can define other special cases by taking some special $(\sigma, \delta)-$pseudo-linear transformations.

Definition 5.11
Let $\Lambda = (\mathbf{a}_1,\ \mathbf{a}_2, \ldots, \mathbf{a}_r)$ where $\mathbf{a}_i = (a_{i,0}, a_{i,1}, \ldots, a_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$ for every i, and let C be a linear code of length $n = n_1 + n_2 + \cdots + n_r$ over $\mathbb{F}_q$. Let

$$\mathbf{c} = \left(\overbrace{c_{1,0}, c_{1,1}, \ldots, c_{1,n_1-1}}^{\mathbf{c}_1}, \overbrace{c_{2,0}, c_{2,1}, \ldots, c_{2,n_2-1}}^{\mathbf{c}_2}, \ldots, \overbrace{c_{r,0}, c_{r,1}, \ldots, c_{r,n_r-1}}^{\mathbf{c}_r} \right) \in C.$$

be an arbitrary codeword. We say that C is

1. (σ, δ)**-skew** $(\lambda_1, \lambda_2, ..., \lambda_r)$**-multi-twisted** code if

$$(T_{\lambda_1,\sigma,\delta}(\mathbf{c}_1), T_{\lambda_2,\sigma,\delta}(\mathbf{c}_2), \ldots, T_{\lambda_r,\sigma,\delta}(\mathbf{c}_r)) \in C \tag{5.6}$$

   where each λ_i is a non-zero scalar of $\mathbb{F}_q$, and each $T_{\lambda_i,\sigma,\delta},\ 1 \leq i \leq r$ is the $(\sigma, \delta, \lambda_i)$-constacyclic shift as in Example 5.1. For $\lambda_i = \pm 1$ we call them (σ, δ)**-generalized quasi-cyclic codes** and (σ, δ)**-generalized quasi-negacyclic codes.**

2. a **right** (σ, δ)**-multi polycyclic** code induced by Λ, if

$$\left(T_{\overrightarrow{a_1},\sigma,\delta}(\mathbf{c}_1), T_{\overrightarrow{a_2},\sigma,\delta}(\mathbf{c}_2), \ldots, T_{\overrightarrow{a_r},\sigma,\delta}(\mathbf{c}_r)\right) \in C \tag{5.7}$$

   where each $T_{\overrightarrow{a_i},\sigma,\delta},\ 1 \leq i \leq r$, is the right $(\sigma, \delta)-$polyshift induced by $\mathbf{a}_i$ as in Example 5.1.

3. a **left** (σ, δ)**-multi polycyclic** code induced by Λ, if

$$\left(T_{\overleftarrow{a_1},\sigma,\delta}(\mathbf{c}_1), T_{\overleftarrow{a_2},\sigma,\delta}(\mathbf{c}_2), \ldots, T_{\overleftarrow{a_r},\sigma,\delta}(\mathbf{c}_r)\right) \in C \tag{5.8}$$

   where each $T_{\overleftarrow{a_i},\sigma,\delta},\ 1 \leq i \leq r$, is the left $(\sigma, \delta)-$polyshift induced by $\mathbf{a}_i$ as in Example 5.1.

4. a $(\pi_1, \pi_2, \ldots, \pi_i)$-generalized (σ, δ)-quasi-cyclic code, if

$$(T_{\pi_1,\sigma,\delta}(\mathbf{c}_1), T_{\pi_2,\sigma,\delta}(\mathbf{c}_2), \ldots, T_{\pi_r,\sigma,\delta}(\mathbf{c}_r)) \in C, \tag{5.9}$$

   where π_i is an n_i-cycle and each $T_{\pi_i,\sigma,\delta},\ 1 \leq i \leq r$, is the (π_i, σ, δ)-permutation shift as in Example 5.1.

In relation to some well known types of codes in the literature and the above definition we make the following remark.

Remark 5.11
Let $C \subseteq \mathbb{F}_q^n$ be a linear code.

1. If C is an $(f, \sigma, \delta)-$quasi-cyclic code of index r, then
   - For $f(x) = x^m - \lambda,\ m = \dfrac{n}{r}$, we call C a skew $(\sigma, \delta)-$quasi-constacyclic, or simply (σ, δ)-quasi-twisted (QT), code of index r, as introduced in the commutative case.
   - For $f(x) = x^m - \overrightarrow{v}(x)$, where $\overrightarrow{v}(x) = v_0 + v_1 x + \ldots + v_{m-1}x^{m-1},\ m = \dfrac{n}{r}$, we call C a **right** (σ, δ)-quasi polycyclic code of index r and associated vector $v = (v_0, v_1, \ldots, v_{m-1})$.
2. If C is a $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic code, then
   - For $f_i(x) = x^{n_i} - \lambda_i$, we call C a skew (σ, δ)-multi-twisted (MT) code.

- For $f_i(x) = x^{n_i} - \overrightarrow{v_i}(x)$, where $\overrightarrow{v_i}(x) = v_{i,0} + v_{i,1}x + \ldots + v_{i,n_i-1}x^{n_i-1}$, we call C a **right** (σ, δ)-multi polycyclic code induced by the vector $v = (v_0, v_1, \ldots, v_{r-1})$, where each $v_i = (v_{i,0}, v_{i,1}, \ldots, v_{i,n_i-1})$.

From Theorem 5.3, we deduce the following proposition.

Proposition 5.6

Let M be a square matrix with sequence of invariant factors $f_1(x), f_2(x), \ldots, f_r(x)$, and let $C \subseteq \mathbb{F}_q^n$ be a linear code of length n over $\mathbb{F}_q$.

1. If M is (σ, δ)-similar to $D = \text{diag}(\overbrace{C_{f_1}, C_{f_1}, \ldots, C_{f_1}}^{r \text{ times}})$ by the matrix P, then C is an $(M, \sigma, \delta)-$skew code if and only if $P^{-1}C$ is an (f_1, σ, δ)-QC code of index r.
2. If M is (σ, δ)-similar to $D = \text{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$ by the matrix P, then C is an (M, σ, δ)-skew code if and only if $P^{-1}C$ is an $(f_1, f_2, \ldots, f_r, \sigma, \delta)$-multi-cyclic code.

Proof

1. Suppose that M is (σ, δ)-similar to $D = \text{diag}(\overbrace{C_{f_1}, C_{f_1}, \ldots, C_{f_1}}^{r \text{ times}})$ by the matrix P, i.e., $M = P^{-1}D\sigma(P) + P^{-1}\delta(P)$. Let C be an $(M, \sigma, \delta)-$skew code, then according to Theorem 5.3, $P^{-1}C = \{P^{-1}c \ : \ c \in C\}$ is an $(D, \sigma, \delta)-$skew code. Hence it is invariant under the pseudo-linear transformation T_D, which means that $P^{-1}C$ is an (f_1, σ, δ)-QC code of index r (Definition 5.10 (2)). Conversely if $P^{-1}C$ is an (f_1, σ, δ)-QC code of index r, then $P^{-1}C$ is invariant under the $(\sigma, \delta)-$action of $D = \text{diag}(\overbrace{C_{f_1}, C_{f_1}, \ldots, C_{f_1}}^{r \text{ times}})$. By Theorem 5.3, we obtain that C is an $(M, \sigma, \delta)-$skew code.
2. The proof is similar to the first statement; we explore the $(\sigma, \delta)-$similarity between M and $D = \text{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$.

Let us recall that if C_1 and C_2 are two $\mathbb{F}_q-$linear codes of lengths n_1 and n_2 then the direct sum of C_1 and C_2

$$C_1 \oplus C_2 = \{(c_1, c_2) : c_1 \in C_1, \ c_2 \in C_2\}$$

is also an $\mathbb{F}_q-$linear code of length n_1+n_2. In the next theorem, we prove that every $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$mul cyclic code can be decomposed as the direct sum of $(f_i, \sigma, \delta)-$cyclic codes.

Theorem 5.6

Let $C \subseteq \mathbb{F}_q^n$ be a linear code. Then C is an $(f_1, f_2, \ldots, f_r, \sigma, \delta)$-multi-cyclic code if and only if $C = C_1 \oplus C_2 \oplus \cdots \oplus C_r$, where each C_i is an (f_i, σ, δ)-cyclic code.

Proof

Let C be an $(f_1, f_2, \ldots, f_r, \sigma, \delta)$-multi-cyclic code and $D = \text{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$. Then $T_D(C) \subseteq C$. From Theorem 5.2, one can write $C = C_1 \oplus C_2 \oplus \cdots \oplus C_r \subseteq \mathbb{F}_q^n$, where $C_i = C \cap F_{v_i}$ for $i = 1, \ldots, r$. It follows that

$$T_D(C) = T_{C_{f_1}}(C_1) \oplus T_{C_{f_2}}(C_2) \oplus \cdots \oplus T_{C_{f_r}}(C_r) \subseteq C_1 \oplus C_2 \oplus \cdots \oplus C_r.$$

Hence $T_{C_{f_i}}(C_i) \subseteq C_i$ for $i = 1, \ldots, r$, and each C_i is an $(f_i, \sigma, \delta)-$cyclic code.

Corollary 5.1

Let M be a matrix with sequence of invariant factors $f_1(x), f_2(x), \ldots, f_r(x)$, and let C be an $(M, \sigma, \delta)-$skew code of length n over $\mathbb{F}_q$. Then there is an invertible matrix $P \in \text{GL}_n(\mathbb{F}_q)$ such that M is $(\sigma, \delta)-$similar to $\text{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$ by P, and

$$P^{-1}C = C_1 \oplus C_2 \oplus \cdots \oplus C_r,$$

where each C_i is an $(f_i, \sigma, \delta)-$cyclic code.

Now set $R_{f_i} := R/R \cdot f_i(x)$ and define on $\mathbb{F}_q^n \equiv \mathbb{F}_q^{n_1} \times \mathbb{F}_q^{n_2} \times \ldots \times \mathbb{F}_q^{n_r}$ a left R-module structure where the multiplication is given by:

$$\begin{array}{ccc} R \times \mathbb{F}_q^n & \longrightarrow & \mathbb{F}_q^n \\ (P, (v_1, v_2, \ldots, v_r)) & \longmapsto & P.v := (P(C_{f_1})v_1, P(C_{f_2})v_2, \ldots, P(C_{f_r})v_r) \end{array} \tag{5.10}$$

for each $v_i = (v_{i,0}, v_{i,1}, \ldots, v_{i,n_i-1}) \in \mathbb{F}_q^{n_i}$. Then as $\prod_{i=1}^{r} R_{f_i}$ is a left $R-$module, we deduce that $\varphi_{f_1,\ldots,f_r} := (\varphi_{f_1}, \varphi_{f_2}, \ldots, \varphi_{f_r})$ is a left $R-$module isomorphism between $\mathbb{F}_q^n$ and $\prod_{i=1}^{r} R_{f_i}$, where $v_i(x) := \varphi_{f_i}((v_{i,0}, v_{i,1}, \ldots, v_{i,n_i-1}) = \sum_{j=0}^{n_i-1} v_{i,j} x^j$, as in (5.1).

Theorem 5.7
Let $C \subseteq \mathbb{F}_q^n$ be a linear code and M be a matrix with sequence of invariant factors $f_1, f_2, \ldots, f_r$.

1. C is an $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic code if and only if $\varphi_{f_1,\ldots,f_r}(C)$ is a left $R-$submodule of $\prod_{i=1}^{r} R_{f_i}$.

2. C is an $(M, \sigma, \delta)-$skew code if and only if $\psi_{f_1,\ldots,f_r}(C)$ is a left $R-$submodule of $\prod_{i=1}^{r} R_{f_i}$, where $\psi_{f_1,\ldots,f_r} := \varphi_{f_1,\ldots,f_r} \circ P^{-1}$, and $P \in \mathrm{GL_n}(\mathbb{F}_q)$ is such that $M = P^{-1}C_f\sigma(P) + P^{-1}\delta(P)$.

Proof

1. Let C be an $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic code. By Theorem 5.6, C can be decomposed as
$$C = C_1 \oplus C_2 \oplus \cdots \oplus C_r,$$
where each C_i is an $(f_i, \sigma, \delta)-$cyclic code.
Then $\varphi_{f_1,\ldots,f_r}(C) = (\varphi_{f_1}(C_1), \varphi_{f_2}(C_2), \ldots, \varphi_{f_r}(C_r)) \subseteq \prod_{i=1}^{r} R_{f_i}$, and by Theorem 5.4, we have that $\varphi_{f_i}(C_i)$, is a left $R-$submodule of R_{f_i}, $i = 1, \ldots, r$. It follows that $\varphi(C)$ is a left $R-$submodule of $\prod_{i=1}^{r} R_{f_i}$.

2. If C is an $(M, \sigma, \delta)-$skew code, then by Theorem 5.3, $P^{-1}C$ is an $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic code, where M is $(\sigma, \delta)-$similar to $\mathrm{diag}(C_{f_1}, C_{f_2}, \ldots, C_{f_r})$ by P. So, by the above statement $\psi_{f_1,\ldots,f_r}(C) = \varphi_{f_1,\ldots,f_r}(P^{-1}C)$ is a left $R-$submodule of $\prod_{i=1}^{r} R_{f_i}$.

We now give the definitions of s-generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)$-multi-cyclic codes and (M, σ, δ)-multi-cyclic codes.

Definition 5.12

1. An $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic code C is called $\mathbf{s}-$generator if and only if $\varphi_{f_1,\ldots,f_r}(C)$ is generated by s elements as a left $R-$submodule of $\prod_{i=1}^{r} R_{f_i}$, i.e.,
$$\varphi_{f_1,\ldots,f_r}(C) = R \cdot \underline{a_1}(x) + R \cdot \underline{a_2}(x) + \ldots + R \cdot \underline{a_s}(x),$$
where $\underline{a_i}(x) = (a_{i,1}(x), a_{i,2}(x), \ldots, a_{i,r}(x)) \in \prod_{i=1}^{r} R_{f_i}$ for $i = 1, ..., s$.

2. An $(M, \sigma, \delta)-$multi-cyclic code C is called $\mathbf{s}-$generator if and only if $\psi_{f_1,\ldots,f_r}(C)$ is generated by s elements as a left $R-$submodule of $\prod_{i=1}^{r} R_{f_i}$, i.e.,

$$\psi_{f_1,\ldots,f_r}(C) = R \cdot \underline{a_1}(x) + R \cdot \underline{a_2}(x) + \ldots + R \cdot \underline{a_s}(x).$$

Similar to the case of generalized quasi-cyclic codes [78, Lemma 1], we derive the following result on $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic codes.

Proposition 5.7
Let C be an $\mathbf{s}-$generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic code generated by $\underline{a_1}(x), \underline{a_2}(x), \ldots, \underline{a_s}(x)$ where $\underline{a_i}(x) = (a_{i,1}(x), a_{i,2}(x), \ldots, a_{i,r}(x)) \in \prod_{j=1}^{r} R_{f_j}$ for every $1 \leq i \leq s$. Then, there exist polynomials $b_{i,j}(x)$ in R, such that $a_{i,j}(x) = b_{i,j}(x)g_{i,j}(x),\ 1 \leq i \leq s,\ 1 \leq j \leq r$, where $g_{i,j}(x)$ is a right divisor of $f_i(x)$.

Proof
Let $\varphi_{f_1,\ldots,f_r}(C) = R\cdot\underline{a_1}(x)+R\cdot\underline{a_2}(x)+\ldots+R\cdot\underline{a_s}(x)$ be an $\mathbf{s}-$generator $(f_1, f_2, \ldots, f_r, \theta, \delta)-$multi-cyclic code. Then each $R \cdot \underline{a_i}(x),\ i = 1, \ldots, s$, is a $\mathbf{1}-$generator $(f_1, f_2, \ldots, f_r, \theta, \delta)-$multi-cyclic code. According to Theorem 5.6, we can decompose $R \cdot \underline{a_i}(x),\ i = 1, \ldots, s$, as

$$R \cdot \underline{a_i}(x) = R \cdot a_{i,1}(x) \oplus R \cdot a_{i,2}(x) \oplus \cdots \oplus R \cdot a_{i,r}(x),$$

where each $R \cdot a_{i,j}(x),\ j = 1, \ldots, r$ is an $(f_j, \sigma, \delta)-$cyclic code of length $n_j = \deg(f_j(x)),\ j = 1, \ldots, r$. Let $g_{i,j}(x)$ be the generator polynomial of $R\cdot a_{i,j}(x)$, i.e., $f_i(x) = h_{i,j}(x)g_{i,j}(x)$, for some $h_{i,j}(x) \in R$. Finally, as $a_{i,j}(x) \in R\cdot a_{i,j}(x)$, there exists $b_{i,j}(x) \in R$ such that $a_{i,j}(x) = b_{i,j}g_{i,j}(x)$.

Remark 5.12
All of the results presented in this section are also valid for $(f, \sigma, \delta)-$quasi-cyclic codes of index r and length $n = mr$, by considering the following left $R-$module isomorphism between $\mathbb{F}_q^n$ and R_f^r,

$$\begin{array}{ccccc} \varphi_{f,r}: & \mathbb{F}_q^n & \longrightarrow & R_f^r \\ & v := (v_1, v_1, ..., v_r) & \longmapsto & (v_1(x), v_2(x), \ldots, v_r(x)) \end{array} \tag{5.11}$$

Example 5.6
With the same notations as in the Example 5.5, we consider (f, σ, δ)-QC codes of index 3 and length $n = 3 \times 11 = 33$. For example, consider the code $C \subseteq \mathbb{F}_8^{33}$ with $\varphi_{f,3}(C) = R \cdot \underline{a}(x) = R \cdot (a_1(x), a_2(x), a_3)$, where

$$\begin{aligned} a_1(x) &= x^5 + x^4 + \alpha x^3 + x + 1 \\ a_2(x) &= x^6 + x^5 + \alpha^4 x^4 + \alpha^3 x^3 + x^2 + x = x a_1(x) \\ a_3(x) &= x^7 + x^6 + \alpha^6 x^5 + \alpha x^4 + x^2 + x + 1 = (x^2 + 1)a_1(x) \end{aligned}$$

This is a 1-generator $(f, \sigma, \delta)-$quasi-cyclic code of index $r = 3$ and length $n = 33$. A generator of C is the polynomial $g(x) = x^5 + x^4 + \alpha x^3 + x + 1$ with $f(x) = h(x)g(x)$ where $h(x) = x^6 + x^4 + x^2 + 1$. Hence, the dimension of C is $\dim_{\mathbb{F}_q}(C) = \deg(h(x)) = 6$. Equivalently, linear codes invariant under the matrix $D = \mathrm{diag}(M, M, M)$, where M is the (σ, δ)-cyclic matrix as in Example 5.5, are $(D, \sigma, \delta)-$quasi-cyclic codes of index 3 and length $n = 33$.

5.4 1−generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)$−multi-cyclic codes

In this section we are interested in the class of **1**-generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)$-multi-cyclic codes generated by one element $\underline{a}(x) = (a_1(x), a_2(x), \ldots, a_r(x)) \in \prod_{i=1}^{r} R_{f_i}$. We write $C = R \cdot \underline{a}(x)$ to denote that C is a **1**−generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)$−multi-cyclic code generated by $\underline{a}(x)$.

Theorem 5.8
Let $C = R \cdot \underline{a}(x)$ be a **1**−generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)$−multi-cyclic code. Put $g_i(x) = \text{gcrd}(a_i(x), f_i(x))$, $i = 1, \ldots, r$, $\underline{\mathbf{g}}(x) = (g_1(x), g_2(x), \ldots, g_r(x))$, and $h(x) = \text{lclm}\{h_i(x),\ 1 \leq i \leq r\}$, with $f_i(x) = h_i(x)g_i(x)$. Then,

1. each element of the form $\underline{\mathbf{b}}(x) = (p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$, where $\text{gcrd}(p_i(x), h_i(x)) = 1$, generates C as an R-submodule of $\prod_{i=1}^{r} R_{f_i}$.

2. $\dim_{\mathbb{F}_q}(C) = \deg(h(x))$, and

$$S = \left\{ x^i.\underline{\mathbf{g}}(x) = (x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ : \ 0 \leq i \leq \deg(h(x)) - 1 \right\}$$

   forms a basis of C.

3. a generator matrix of C is given by

$$G = \begin{pmatrix} \underline{\mathbf{g}}(x) \\ x\underline{\mathbf{g}}(x) \\ \vdots \\ x^{k-1}\underline{\mathbf{g}}(x) \end{pmatrix} = \left(\begin{array}{c|c|c|c} g_1(x) & g_2(x) & \ldots & g_r(x) \\ xg_1(x) & xg_2(x) & \ldots & xg_r(x) \\ \vdots & \vdots & \vdots & \vdots \\ x^{k-1}g_1(x) & x^{k-1}g_2(x) & \ldots & x^{k-1}g_r(x) \end{array}\right) \tag{5.12}$$

   where $k = \deg(h(x))$.

Proof
Let $\underline{a}(x) = (a_1(x), \ldots, a_r(x)) \in \prod_{i=1}^{r} R_{f_i}$, and $C = R \cdot \underline{a}(x)$ be a **1**−generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)$−multi-cyclic code.

1. From Theorem 5.6 and Proposition 5.5, we have that $C = C_1 \oplus C_2 \oplus \ldots \oplus C_r$, with each $C_i = R \cdot a_i(x)$ is an (f_i, σ, δ)−cyclic code with generator polynomial $g_i(x) = \text{gcrd}(a_i(x), f_i(x)$. As $\text{gcrd}(p_i(x), g_i(x)) = 1$, by the second statement of Proposition 5.5, $p_i(x)g_i(x)$ generates C_i. Therefore $\underline{\mathbf{b}}(x)$ is a generator of C.

2. Let us prove that the set

$$S = \left\{ x^i\underline{\mathbf{g}}(x) = (x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ : \ 0 \leq i \leq \deg(h(x)) - 1 \right\},$$

   is a basis for C. Suppose that $v(x)\underline{\mathbf{g}}(x) = 0$ in $\prod_{i=1}^{r} R_{f_i}$. Then

$$(v(x)g_1(x) \text{ in } R_{f_1}, v(x)g_2(x) \text{ in } R_{f_2}, \ldots, v(x)g_r(x) \text{ in } R_{f_r}) = (0, 0, \ldots, 0).$$

   It follows that for each $i = 1, \ldots, r$, we have $v(x)g_i(x) = 0$ in R_{f_i}. Therefore $h_i(x)$ is a right divisor of $v(x)$, so $h(x) = \text{lclm}\{h_i(x),\ 1 \leq i \leq r\}$ as well, which implies that $\deg(v(x)) \geq \deg(h(x)) - 1$. This shows the linear independence of the vectors in S.

To complete the proof we need to prove that for every $v(x) \in R$ with $\deg(v(x)) \geq \deg(h(x)) - 1$, there exists $r(x) \in R$ with $\deg(r(x)) < \deg(h(x)) - 1$ such that $v(x)\underline{\mathbf{g}}(x) = r(x)\underline{\mathbf{g}}(x)$. By the Right Euclidean division algorithm in R, we write $v(x) = q(x)h(x)+r(x)$, for some $q(x), r(x) \in R$ where $\deg(r(x)) < \deg(h(x))$. It follows that

$$\begin{aligned} v(x)\underline{\mathbf{g}}(x) &= v(x)(g_1(x), g_2(x), \ldots, g_r(x)) \\ &= (v(x)g_1(x) \bmod f_1(x), v(x)g_2(x) \bmod f_2(x), \ldots, v(x)g_r(x) \bmod f_r(x)) \\ &= (r(x)g_1(x) \bmod f_1(x), r(x)g_2(x) \bmod f_2(x), \ldots, r(x)g_r(x) \bmod f_r(x)) \\ &= r(x)(g_1(x), g_2(x), \ldots, g_r(x)) \\ &= r(x)\underline{\mathbf{g}}(x) \end{aligned}$$

3. Since $S = \left\{ x^i\underline{\mathbf{g}}(x) = (x^i g_1(x), x^i g_2(x), \ldots, x^i g_r(x)) \ : \ 0 \leq i \leq \deg(h(x)) - 1 \right\}$ forms a basis of C, the result follows.

Remark 5.13
If C is a $\mathbf{1}$-generator (M, σ, δ)-multi-cyclic code such that $\psi_{f_1,\ldots,f_r}(C) = R \cdot \underline{a}(x)$, then all statements of Theorem 5.8 are still valid for C, except for the statement about the generator matrix where we need to compute $\psi^{-1}_{f_1,\ldots,f_r}(x^i\underline{\mathbf{g}}(x)), 1 \leq i \leq \deg(h(x)) - 1$.

Like [28] and [67], we give a lower bound on the minimum Hamming distance of $\mathbf{1}$-generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)$-multi-cyclic codes.

Theorem 5.9
With the same notations as in Theorem 5.8, let C be a $\mathbf{1}-$generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x))$$

where for all $1 \leq i \leq r$, $f_i(x) = h_i(x)g_i(x)$ and each $p_i(x)$ is right coprime to $h_i(x)$. Also let d and d_i be, respectively, the minimum Hamming distance of C and $C_i = R \cdot (p_i(x)g_i(x))$ for $i = 1, ..., r$. Then

1. $d \geq \sum_{i \notin K} d_i$, where $K \subseteq \{1, 2, \ldots, r\}$ is a set of maximum size for which $h(x) = \operatorname{lclm}\{h_i(x) \ : \ 1 \leq i \leq r\} \neq \operatorname{lclm}\{h_i(x) : i \in K\}$.

2. If $h_1(x) = h_2(x) = \cdots = h_r(x)$, then $d \geq \sum_{i=1}^{r} d_i$.

Proof
Let C be a $\mathbf{1}-$generator $(f_1, f_2, \ldots, f_r, \sigma, \delta)-$multi-cyclic code generated by an element of the form

$$(p_1(x)g_1(x), p_2(x)g_2(x), \ldots, p_r(x)g_r(x)).$$

By Theorem 5.6,

$$C = C_1 \oplus C_2 \oplus \ldots \oplus C_r$$

where each $C_i = R \cdot (p_i(x)g_i(x))$ is an (f_i, σ, δ)-cyclic code generated by $g_i(x)$. Let $c(x) = (c_1(x), c_2(x), \ldots, c_r(x)) = u(x)\underline{a}(x)$ be a codeword of C. Then a component $c_i(x)$ is 0 if and only if $h_i(x)$ right divides $u(x)p_i(x)$. As $\operatorname{gcrd}(p_i(x), h_i(x)) = 1$, this happens only if $h_i(x)$ is a right divisor of $u(x)$. Therefore, $c(x) = 0$ if and only if $h(x)$ is a right divisor of $u(x)$. It follows that $c(x) \neq 0$ if and only if $h(x)$ does not divide $u(x)$ on the right. Thus, a non-zero codeword has the most number of zero components whenever $h(x) \neq \operatorname{lclm}\{h_i(x) : i \in K\}$ and $\operatorname{lclm}\{h_i(x) : i \in K\}$ right divides $u(x)$, where K is a maximal subset of $\{1, 2, \ldots, r\}$. Therefore, $d \geq \sum_{i \notin K} d_i$. On the other hand, we have $|K| = 0$ if and only if $h_1(x) = h_2(x) = \cdots = h_r(x)$. In this case, $d \geq \sum_{i=1}^{r} d_i$.

In the case of 1−generator (f,σ,δ)-QC codes of index r, all results presented in this section are still valid. Moreover, we can define the notion of a generator polynomial.

Remark 5.14
Let C be a 1-generator (f,σ,δ)-QC code of index r, and length $n=mr$, generated by $\underline{a}(x)=(a_1(x),a_2(x),\ldots,a_r(x))\in R_f^r$. Then

1. $g(x)=\gcd(\underline{a}(x),f(x))$ is called the **generator polynomial** (g.p.) of C.
2. Every $a_i(x)$ is of the form $a_i(x)=b_i(x)g(x)$ with $h(x)\underline{a}(x)=0$ in R_f^r.
3. $f(x)=h(x)g(x)$ and $\dim_{\mathbb{F}_q}(C)=n-\deg(g(x))=\deg(h(x))$.
4. $\underline{\mathbf{b}}(x)\in R_f^r$ is another generator of C if and only if there is $p(x)\in\mathbb{F}_q[x]$ such that $\gcd(p(x),h(x))=1$ and $\underline{\mathbf{b}}(x)=p(x)\underline{a}(x)$.

Example 5.7 (An $(f_1,f_2,f_3,\sigma,\delta)$-multi-cyclic code, case 1 of Theorem 9)
Let $q=27$ and $\mathbb{F}_{27}=\mathbb{F}_3(\alpha)$, where $\alpha^3+2\alpha+1=0$. In $\mathbb{F}_{27}$, consider the automorphism $\sigma:\mathbb{F}_{27}\longrightarrow\mathbb{F}_{27}$, defined by $\sigma(a)=a^9$ and the σ-derivation $\delta=0$. Let $f_1(x),f_2(x),f_3(x)$ be three skew polynomials in $\mathbb{F}_{27}[x,\sigma,\delta]$, with their factorizations into irreducibles given below. Note that factorization into irreducibles is not unique over a skew polynomial ring. For the polynomial f_1, there are six distinct factorizations of it into irreducibles. We list all of them below. For the polynomials f_2 and f_3, we give just one of the possible factorizations.

$$
\begin{aligned}
f_1(x) &= x^5+\alpha x^4+\alpha x^3+2\alpha x^2+x+\alpha+1\\
&= (x^3+2x+1)\cdot(x+2\alpha^2+2\alpha)\cdot(x+\alpha^2+2)\\
&= (x+\alpha^2+\alpha+2)\cdot(x^3+(\alpha^2+2\alpha+1)\,x^2+(2\alpha+1)\,x+\alpha^2+2)\cdot(x+\alpha^2+2)\\
&= (x^3+2x+1)\cdot(x+1)\cdot(x+\alpha+1)\\
&= (x+1)\cdot(x^3+2x+1)\cdot(x+\alpha+1)\\
&= (x+\alpha^2+\alpha+2)\cdot(x+\alpha^2+2)\cdot(x^3+\alpha^2x^2+(\alpha^2+\alpha+2)\,x+\alpha^2+2)\\
&= (x+1)\cdot(x+2\alpha^2+2\alpha)\cdot(x^3+\alpha^2x^2+(\alpha^2+\alpha+2)\,x+\alpha^2+2)\\
f_2(x) &= x^6+2x^5+2x^4+x^2+x+2\\
&= (x+2\alpha^2)\cdot(x+\alpha^2+\alpha+2)\cdot(x+\alpha+1)\cdot(x+\alpha+2)\cdot(x+\alpha)\cdot(x+1)\\
f_3(x) &= x^{10}+2x^9+x^6+x^5+2x^4+x^2+1\\
&= (x^2+x+2)\cdot(x^2+1)\cdot(x+\alpha^2)\cdot(x+2\alpha^2+2\alpha+1)\cdot(x+\alpha^2+2)^2\cdot(x+\alpha^2)\cdot(x+2).
\end{aligned}
$$

Consider the linear code $C\subseteq\mathbb{F}_{27}^{21}$ with

$$\varphi_{f_1,f_2,f_3}(C)=R\cdot\mathbf{g}(x)=R\cdot(g_1(x),g_2(x),g_3(x)),$$

where $g_1(x)=x^2+\alpha x+\alpha+1$, $g_2(x)=x^3+2$, $g_3(x)=x^5+2x^4+2x^3+1$. This is a 1-generator $(f_1,f_2,f_3,\sigma,\delta)$−multi-cyclic code of length $n=21$. Let

$$h(x)=\mathrm{lclm}(h_1(x),h_2(x),h_3(x))=x^{10}+x^9+x^8+2x^7+x^6+2x^5+x^4+x^3+x^2+1$$

with $h_1(x)=x^3+2x+1$, $h_2(x)=x^3+2x^2+2x+1$, and $h_3(x)=x^5+x^3+x^2+1$. Then the dimension of C is $\dim_{\mathbb{F}_{27}}(C)=10$, and a basis of C is given by

$$S=\left\{x^i\underline{\mathbf{g}}(x)\ :\ i=0,\ldots,9\right\}:=\left\{x^ig_1(x),x^ig_2(x),x^ig_3(x)\ :\ i=0,\ldots,9\right\}.$$

It follows that a generator matrix of C, as in Theorem 5.8, is given by

$$G = \left(\begin{array}{rrrrr|rrrrrr|rrrrrrrrrr}
\alpha+1 & \alpha & 1 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 & 0 & 0 & 0 & 0 \\
0 & \alpha+2 & \alpha+1 & 1 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 & 0 & 0 & 0 \\
0 & 0 & \alpha & \alpha+2 & 1 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 & 0 & 0 \\
2\alpha+2 & 2 & \alpha & 1 & 0 & 1 & 2 & 2 & 2 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 & 0 \\
0 & 2\alpha+1 & 2 & \alpha+1 & 1 & 1 & 0 & 1 & 2 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 \\
2\alpha+2 & 2 & 0 & 2\alpha+2 & 2 & 2 & 2 & 1 & 1 & 1 & 2 & 2 & 0 & 2 & 0 & 1 & 0 & 2 & 0 & 2 & 0 \\
\alpha+1 & 2\alpha+2 & 2\alpha+2 & \alpha & 1 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 2 & 0 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\
2\alpha+2 & \alpha+1 & 1 & \alpha+1 & 1 & 0 & 2 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 2 & 0 & 2 & 2 \\
2\alpha+2 & 2\alpha & 2\alpha+2 & 2\alpha+1 & 2 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 2 & 2 & 2 & 2 & 0 & 1 \\
\alpha+1 & 2\alpha+2 & \alpha+2 & 1 & 0 & 1 & 2 & 2 & 2 & 1 & 1 & 2 & 1 & 0 & 1 & 1 & 1 & 1 & 2 & 2 & 1
\end{array}\right)$$

Let us now compute the maximum size of the set K as in the first assertion of Theorem 5.9. Using SageMath we obtain

$$\begin{aligned}
\operatorname{lclm}(h_1, h_2) &= x^6 + 2x^5 + x^4 + x + 1 \neq h(x) \\
\operatorname{lclm}(h_2, h_3) &= x^7 + x^6 + 2x^5 + 2x^4 + 2x^3 + 2x^2 + x + 1 \neq h(x) \\
\operatorname{lclm}(h_3, h_1) &= x^8 + 2x^5 + 2x^4 + x^3 + x^2 + 2x + 1 \neq h(x)
\end{aligned}$$

It follows that the maximum size of the set K is 2. So, according to the first statement of Theorem 5.9, we get $d \geq d_1 = 3$ or $d \geq d_2 = 2$ or $d \geq d_3 = 4$, where each $d_i,\ i = 1, 2, 3$ is the minimum distance of the $(f_i, \sigma, \delta)-$cyclic code C_i generated by $g_i(x)$. Hence $d \geq 2$. Its actual minimum distance turns out to be 3.

Example 5.8 (An $(f_1, f_2, f_3, \sigma, \delta)$-multi-cyclic code, case 2 of Theorem 9)
Let $q = 27$ and $\mathbb{F}_{27} = \mathbb{F}_3(\alpha)$, where $\alpha^3 + 2\alpha + 1 = 0$. In $\mathbb{F}_{27}$, consider the automorphism $\sigma : \mathbb{F}_{27} \longrightarrow \mathbb{F}_{27}$, defined by $\sigma(a) = a^9$ and the $\sigma-$derivation $\delta = 0$. Let $f_1(x), f_2(x), f_3(x)$, be three skew polynomials in $\mathbb{F}_{27}[x, \sigma, \delta]$, with their factorizations into irreducibles. Note that factorization into irreducibles is not unique over a skew polynomial ring. For f_1 there are six distinct factorizations of it into irreducibles, for f_2 there are 4080, and for f_3 there are 240. We list below all factorizations of f_1. For the polynomials f_2 and f_3, we give just one of the possible factorizations.

$$\begin{aligned}
f_1(x) &= x^5 + \alpha x^4 + \alpha x^3 + 2\alpha x^2 + x + \alpha + 1 \\
&= (x^3 + 2x + 1) \cdot (x + 2\alpha^2 + 2\alpha) \cdot (x + \alpha^2 + 2) \\
&= (x + \alpha^2 + \alpha + 2) \cdot (x^3 + (\alpha^2 + 2\alpha + 1)\, x^2 + (2\alpha + 1)\, x + \alpha^2 + 2) \cdot (x + \alpha^2 + 2) \\
&= (x^3 + 2x + 1) \cdot (x + 1) \cdot (x + \alpha + 1) \\
&= (x + 1) \cdot (x^3 + 2x + 1) \cdot (x + \alpha + 1) \\
&= (x + \alpha^2 + \alpha + 2) \cdot (x + \alpha^2 + 2) \cdot (x^3 + \alpha^2 x^2 + (\alpha^2 + \alpha + 2)\, x + \alpha^2 + 2) \\
&= (x + 1) \cdot (x + 2\alpha^2 + 2\alpha) \cdot (x^3 + \alpha^2 x^2 + (\alpha^2 + \alpha + 2)\, x + \alpha^2 + 2) \\
f_2(x) &= x^8 + x^6 + x^4 + x^3 + 2x^2 + 2x + 1 \\
&= (x^3 + 2x + 1) \cdot (x + 2) \cdot (x + \alpha^2 + 1) \cdot (x + \alpha^2 + \alpha + 2) \cdot (x + \alpha^2 + 2\alpha + 2) \cdot (x + 1) \\
f_3(x) &= x^8 + 2x^7 + x^6 + 2x^5 + 2x + 1 \\
&= (x^3 + 2x + 1) \cdot (x^2 + x + 2) \cdot (x + 1)^2 \cdot (x + 2)
\end{aligned}$$

Consider the linear code $C \subseteq \mathbb{F}_{27}^{21}$ with

$$\varphi_{f_1, f_2, f_3}(C) = R \cdot \mathbf{g}(x) = R \cdot (g_1(x), g_2(x), g_3(x)),$$

where

$$g_1(x) = x^2 + \alpha x + \alpha + 1, \quad g_2(x) = x^5 + 2x^3 + 2x^2 + 1, \quad g_3(x) = x^5 + 2x^4 + 2x^3 + 1.$$

This is a 1-generator $(f_1, f_2, f_3, \sigma, \delta)-$multi-cyclic code of length $n = 21$. Let

$$h(x) = \operatorname{lclm}(h_1(x), h_2(x), h_3(x)) = x^3 + 2x + 1$$

where $h_1(x) = h_2(x) = h_3(x) = x^3 + 2x + 1$. Then the dimension of C is $\dim_{\mathbb{F}_{27}}(C) = 3$, and a basis of C is given by

$$S = \left\{ x^i \underline{\mathbf{g}}(x) \ : \ i = 0, \ldots, 2 \right\} := \left\{ x^i g_1(x), x^i g_2(x), x^i g_3(x) \ : \ i = 0, \ldots, 2 \right\}.$$

It follows that a generator matrix of C is given by

$$G = \left(\begin{array}{ccccc|cccccccc|cccccccc} \alpha+1 & \alpha & 1 & 0 & 0 & 1 & 0 & 2 & 2 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 & 0 & 0 \\ 0 & \alpha+2 & \alpha+1 & 1 & 0 & 0 & 1 & 0 & 2 & 2 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 & 0 \\ 0 & 0 & \alpha & \alpha+2 & 1 & 0 & 0 & 1 & 0 & 2 & 2 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 \end{array}\right)$$

According to the second assertion of Theorem 5.9 the Hamming distance d of C is at least 11. Its actual minimum distance turns out to be 11.

We also observe that if we take

$$\varphi(C) = R \cdot a(x) = R \cdot (a_1(x), a_2(x), a_3(x)),$$

where

$$\begin{aligned} a_1(x) &= (x^5+1)g_1 = x^7 + (\alpha+2)\,x^6 + \alpha x^5 + x^2 + \alpha x + \alpha + 1 \\ a_2(x) &= (x^3+1)g_2(x) = x^8 + 2x^6 + 2x^2 + 1 \\ a_3(x) &= (x^3+2)g_3(x) = x^8 + 2x^7 + 2x^6 + 2x^5 + x^4 + 2x^3 + 2 \end{aligned}$$

we obtain a $[21, 3, 13]$-code C, with a generator matrix

$$\left(\begin{array}{ccccc|cccccccc|cccccccc} 0 & \alpha+2 & 1 & 2\alpha+2 & 2 & 0 & 1 & 0 & 2 & 2 & 0 & 1 & 0 & 1 & 1 & 0 & 2 & 1 & 0 & 1 & 0 \\ \alpha+1 & 1 & 0 & \alpha+1 & 1 & 0 & 0 & 1 & 0 & 2 & 2 & 0 & 1 & 0 & 1 & 1 & 0 & 2 & 1 & 0 & 1 \\ 2\alpha+2 & \alpha+1 & \alpha+1 & 2\alpha & 2 & 2 & 1 & 1 & 0 & 2 & 2 & 1 & 0 & 2 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

This shows that it is possible to increase the minimum distance of such a code by changing the generator polynomials.

5.5 Duals of $(M, \sigma, \delta)-$skew codes

This section is devoted to the b-dual structure of (M, σ, δ)-skew codes.
Let $b : \mathbb{F}_q^n \times \mathbb{F}_q^n \longrightarrow \mathbb{F}_q$, be a non-degenerate bilinear form and C be a $(M, \sigma, \delta)-$skew code of length n. A b-dual code of C is the linear code $C^{\perp b}$ given by

$$C^{\perp b} = \left\{ x \in \mathbb{F}_q^n : b(x, y) = 0 \text{ for all } y \in C \right\}.$$

Lemma 5.2
Let b be a non-degenerate bilinear form on $\mathbb{F}_q^n$ and $C \subseteq \mathbb{F}_q^n$ be a linear code. Then for any vectors $x \in C^{\perp b}$ and $y \in C$, we have

$$b(\delta(x), y) = -b(\sigma(x), \delta(y)).$$

Proof
Let B be the representation matrix of the non-degenerate bilinear form b with respect to the standard basis of $\mathbb{F}_q^n$. For $x \in C^{\perp b}$ and $y \in C$, we have

$$\begin{aligned} 0 = \delta(b(x, y)) &= \delta(xBy^t) \\ &= \sigma(x)\delta(By^t) + \delta(x)By^t \\ &= \sigma(x)\delta(B)\sigma(y^t) + \sigma(x)B\delta(y^t) + b(\delta(x), y) \\ &= \sigma(x)\delta(B)\sigma(y^t) + b(\sigma(x), \delta(y^t)) + b(\delta(x), y) \\ &= \sigma(x)\delta(B)\sigma(y^t) + b(\sigma(x), \delta(y)) + b(\delta(x), y) \end{aligned} \tag{5.13}$$

Now since δ is a $\sigma-$derivation, there is a $\beta \in \mathbb{F}_q^*$ such that $\delta(a) = \beta(\sigma(a) - a)$, for any $a \in \mathbb{F}_q$. It follows that

$$\begin{aligned} \sigma(x)\delta(B)\sigma(y^t) &= \beta\sigma(b(x,y)) - \beta b(\sigma(x), \sigma(y)) \\ &= 0 - \beta b(\sigma(x), \sigma(y)) \\ &= -\beta b(\sigma(x), \sigma(y)) \end{aligned} \tag{5.14}$$

On the other hand, we have $b(\sigma(x), \sigma(y)) = 0$. In fact, as $x \in C^{\perp b}$, $\sigma(x) \in \sigma(C^{\perp b})$ which means that $b(\sigma(x), \sigma(z)) = 0$, for any $z \in C^{\perp b}$. In particular $b(\sigma(x), \sigma(y)) = 0$.
Finally from (5.13) we obtain that $b(\delta(x), y) = -b(\sigma(x), \delta(y))$.

Now we state the main result in this section, which gives a characterization of the b-dual of a given $(M, \sigma, \delta)-$skew code.

Theorem 5.10
Let $C \subseteq \mathbb{F}_q^n$ be an (M, σ, δ)-skew code. Then $C^{\perp b}$ is a $(\sigma^{-1}(M^*), \sigma, \delta')$-skew code, where M^* is the adjoint of M with respect to b, and $\delta' := \sigma^{-1} \circ \delta$ a σ-derivation.

Proof
Let $c \in C$ and $v \in C^{\perp b}$. Then we have

$$\begin{aligned} 0 &= b(T_M(c), v) \\ &= b(M\sigma(c) + \delta(c), v) \\ &= b(M\sigma(c), v) + b(\delta(c), v) \\ &= b(\sigma(c), M^*v) - b(\sigma(c), \delta(v)), \quad \text{by the above Lemma.} \\ &= b(\sigma(c), M^*v - \delta(v)) \\ &= \sigma(b(c, \sigma^{-1}(M^*v - \delta(v)))) \end{aligned} \tag{5.15}$$

Since σ is an automorphism, we have $b(c, \sigma^{-1}(M^*v - \delta(v))) = 0$. It follows that $T_{\sigma^{-1}(M^*)}(C^{\perp b}) \subseteq C^{\perp b}$, where $T_{\sigma^{-1}(M^*)}(v) = \sigma^{-1}(M)^*v - \sigma^{-1}(\delta)(v)$. Hence, $C^{\perp b}$ is a $(\sigma, \delta', M^*)-$skew code.

In particular when the non-degenerate bilinear form b is the Euclidean inner product on $\mathbb{F}_q^n$, the adjoint of the matrix M is its transpose M^t, and so we give the following result concerning the Euclidean dual of $(M, \sigma, \delta)-$skew codes.

Corollary 5.2 (Theorem 3.7 [87])
The Euclidean dual $C^{\perp}$ of any (M, σ, δ)-skew code is a $(\sigma^{-1}(M^t), \sigma, \delta')$-skew code.

5.6 Computational Results

In this section we collect some results obtained by computer searches for codes with good parameters and desirable properties. We searched for (f, σ, δ)-cyclic codes and quasi (f, σ, δ)-cyclic codes over $\mathbb{F}_q$ for $q = 4, 8, 9, 16, 25, 27$. We have found many codes with good parameters and properties. We list some of these codes in the tables below. All of the codes in these tables have the parameters of best known linear codes (BKLCs) according to the database [27] for $q = 4, 8, 9$, and they are also optimal. The ones in Table 4 are MDS codes over $\mathbb{F}_q$ for $q = 16, 25, 27$. In many cases, the BKLCs in the database ([27]) have indirect, multi-step constructions. In contrast, our codes are simple and direct to construct rather than a multiple step process that yields a code that does not necessarily have a good structure. For example, the $[11, 4, 6]_4$-code in the database has a multistep construction whereas we can obtain it as an (f, σ, δ) code (Table 1). Moreover, our codes have additional desirable properties such as being LCD or self-orthogonal. These properties are useful for cryptography and quantum error correcting codes.

The following tables list the parameters, properties, and generators of these new (f, σ, δ)-cyclic codes, quasi (f, σ, δ)-cyclic codes over $\mathbb{F}_q$, $q = 4, 8, 9, 16, 25, 27$. In these tables, g represents the (standard) generator polynomial of an (f, σ, δ)-cyclic code and β represents the scalar used to define a σ-derivation δ defined by $\delta = \beta(\sigma - \mathrm{Id})$. All polynomials are listed by their coefficients in ascending powers of x for a compact representation. For example, the polynomial

$$f = x^{10} + \alpha x^9 + \alpha x^8 + \alpha x^7 + \alpha^2 x^6 + x^5 + x^4 + \alpha^2 x + 1 \in \mathbb{F}_4[x, \sigma, \delta],$$

where $\alpha \in \mathbb{F}_4$ such that $\alpha^2 + \alpha + 1 = 0$, is represented by the sequence of its coefficients

$$1, \alpha^2, 0, 0, 1, 1, \alpha^2, \alpha, \alpha, \alpha, 1.$$

In Table 5.1, Table 5.2 and Table 5.3, we list some examples of (f, σ, δ)-cyclic codes over $\mathbb{F}_4$, $\mathbb{F}_8$ and $\mathbb{F}_9$ with best known parameters (most of these codes are in fact optimal) that are also LCD codes. In Table 5.4, we collect some examples of MDS $(f, \sigma, \delta)-$cyclic codes over $\mathbb{F}_{16}$, $\mathbb{F}_{25}$, $\mathbb{F}_{27}$ that are LCD codes. In table 5.5, we give some examples quasi (f, σ, δ)-cyclic codes over $\mathbb{F}_4$, generated by $(g(x), p_1 g(x))$, with best known (and often optimal) parameters that are also LCD or self-orthogonal.

Table 5.1: Examples of best known $(f, \sigma, \delta)-$cyclic codes over $\mathbb{F}_4$, generated by $g(x)$ that are LCD codes

q	$[n,k,d]$	$g(x)$	$f(x)$	σ	β	Properties
4	$[6,3,4]_4$	$1,\alpha,\alpha^2,1$	$\alpha^2,0,\alpha^2,\alpha^2,\alpha^2,0,1$	Frob	$\beta=0$	LCD
4	$[7,4,3]_4$	$1,\alpha,\alpha,1$	$1,1,1,\alpha,0,\alpha,0,1$	Frob	$\beta=\alpha^2$	LCD
4	$[7,3,4]_4$	$1,1,\alpha^2,\alpha^2,1$	$\alpha^2,1,0,0,1,0,0,1$	Frob	$\beta=1$	LCD
4	$[8,3,5]_4$	$1,\alpha,\alpha^2,\alpha^2,1,1$	$\alpha^2,0,\alpha,\alpha,0,0,\alpha,\alpha^2,1$	Frob	$\beta=\alpha^2$	LCD
4	$[8,4,4]_4$	$\alpha,\alpha^2,\alpha,0,1$	$0,\alpha^2,0,\alpha,1,\alpha,1,1,1$	Frob	$\beta=\alpha^2$	LCD
4	$[8,5,3]_4$	$\alpha,\alpha,\alpha,1$	$\alpha^2,\alpha,1,\alpha^2,0,0,0,\alpha,1$	Frob	$\beta=0$	LCD
4	$[9,3,6]_4$	$\alpha^2,0,\alpha^2,1,1,\alpha,1$	$1,\alpha,\alpha,\alpha,0,1,1,\alpha,\alpha,1$	Frob	$\beta=0$	LCD
4	$[9,4,5]_4$	$\alpha^2,\alpha^2,\alpha^2,1,\alpha,1$	$1,\alpha,1,1,\alpha,\alpha,\alpha^2,0,\alpha,1$	Frob	$\beta=1$	LCD
4	$[9,5,4]_4$	$\alpha^2,\alpha^2,\alpha^2,1,1$	$\alpha,1,1,\alpha^2,\alpha^2,\alpha,1,\alpha,\alpha^2,1$	Frob	$\beta=0$	LCD
4	$[9,6,3]_4$	$\alpha^2,0,1,1$	$\alpha^2,\alpha,\alpha^2,\alpha^2,\alpha,0,1,0,0,1$	Frob	$\beta=1$	LCD
4	$[10,3,6]_4$	$1,\alpha^2,\alpha^2,\alpha^2,\alpha,\alpha^2,0,1$	$1,0,0,\alpha,0,\alpha,0,\alpha^2,1,\alpha,1$	Frob	$\beta=0$	LCD
4	$[10,4,6]_4$	$1,1,1,\alpha^2,1,\alpha,1$	$\alpha,\alpha^2,0,1,0,\alpha^2,0,1,\alpha^2,0,1$	Frob	$\beta=0$	LCD
4	$[10,5,5]_4$	$\alpha,\alpha,1,1,0,1$	$\alpha,\alpha^2,\alpha^2,0,0,0,1,\alpha^2,1,0,1$	Frob	$\beta=0$	LCD
4	$[10,6,4]_4$	$\alpha,1,1,\alpha^2,1$	$1,\alpha^2,0,\alpha,\alpha^2,\alpha^2,1,\alpha^2,\alpha^2,0,1$	Frob	$\beta=0$	LCD
4	$[10,7,3]_4$	$\alpha^2,\alpha^2,\alpha^2,1$	$\alpha^2,\alpha^2,\alpha^2,\alpha^2,\alpha,1,1,\alpha,\alpha^2,\alpha^2,1$	Frob	$\beta=0$	LCD
4	$[11,3,7]_4$	$\alpha,0,0,1,1,\alpha^2,\alpha,\alpha^2,1$	$1,1,\alpha^2,1,0,0,\alpha,1,0,\alpha^2,\alpha^2,1$	Frob	$\beta=1$	LCD
4	$[11,4,6]_4$	$\alpha^2,\alpha,1,\alpha^2,1$	$1,\alpha,0,0,\alpha^2,0,0,\alpha^2,\alpha^2,0,\alpha,1$	Frob	$\beta=1$	LCD
4	$[11,7,4]_4$	$1,1,\alpha,\alpha^2,\alpha^2,\alpha,\alpha,1$	$\alpha^2,\alpha,\alpha,0,1,\alpha,0,1,0,0,0,1$	Frob	$\beta=0$	LCD
4	$[11,8,3]_4$	$1,\alpha^2,\alpha^2,1$	$\alpha,\alpha^2,\alpha^2,\alpha,\alpha^2,\alpha,\alpha^2,\alpha,0,\alpha,\alpha^2,1$	Frob	$\beta=1$	LCD
4	$[12,3,8]_4$	$1,0,1,1,1,\alpha^2,\alpha^2,0,\alpha^2,1$	$\alpha^2,\alpha^2,\alpha,\alpha^2,\alpha,\alpha^2,1$	Frob	$\beta=0$	LCD
4	$[12,4,7]_4$	$\alpha,0,1,\alpha^2,1,\alpha,0,\alpha^2,1$	$0,\alpha,\alpha,0,1,1,0,\alpha^2,\alpha^2,1,\alpha^2,1,1$	Frob	$\beta=\alpha^2$	LCD
4	$[12,5,6]_4$	$\alpha,\alpha^2,0,0,\alpha,\alpha,\alpha,1$	$\alpha^2,\alpha^2,\alpha^2,1,\alpha,\alpha,1,0,\alpha^2,0,0,\alpha^2,1$	Frob	$\beta=0$	LCD
4	$[12,7,4]_4$	$\alpha,1,\alpha^2,\alpha,\alpha,1$	$\alpha,1,1,0,1,\alpha,1,\alpha^2,1,0,\alpha,\alpha,1$	Frob	$\beta=1$	LCD
4	$[12,8,4]_4$	$1,1,\alpha,1,1$	$1,\alpha^2,\alpha,1,1,\alpha,0,\alpha^2,\alpha^2,1,\alpha^2,\alpha,1$	Id	$--$	LCD
4	$[12,9,3]_4$	$1,1,\alpha,1$	$\alpha^2,0,\alpha,\alpha^2,1,\alpha,\alpha,\alpha,\alpha^2,1,0,0,1$	Frob	$\beta=1$	LCD
4	$[13,6,6]_4$	$\alpha,0,1,\alpha^2,0,1,\alpha^2,1$	$\alpha,0,\alpha,\alpha,\alpha,1,\alpha,\alpha,0,\alpha,\alpha^2,\alpha^2,\alpha,1$	Frob	$\beta=0$	LCD
4	$[13,7,5]_4$	$1,0,\alpha,\alpha^2,\alpha^2,\alpha^2,1$	$\alpha,\alpha,0,\alpha^2,\alpha^2,1,\alpha,1,\alpha^2,\alpha^2,0,0,0,1$	Id	$--$	LCD
4	$[13,8,4]_4$	$1,0,\alpha^2,\alpha,\alpha^2,1$	$\alpha^2,\alpha^2,\alpha,\alpha^2,\alpha^2,\alpha,0,1,\alpha,1,\alpha^2,\alpha^2,0,1$	Id	$--$	LCD
4	$[13,10,3]_4$	$\alpha^2,\alpha^2,0,1$	$\alpha,1,1,1,\alpha,0,\alpha^2,\alpha,1,\alpha,1,1,\alpha^2,1$	Id	$--$	LCD
4	$[14,7,6]_4$	$\alpha^2,\alpha^2,\alpha,1,\alpha^2,\alpha,1,1$	$\alpha^2,0,\alpha,\alpha^2,\alpha,\alpha,0,\alpha^2,0,\alpha^2,1,0,0,0,1$	Frob	$\beta=0$	LCD
4	$[14,8,5]_4$	$\alpha,\alpha,1,\alpha^2,\alpha^2,1,1$	$1,\alpha,\alpha,1,0,\alpha,\alpha^2,\alpha,0,1,1,\alpha^2,\alpha,\alpha,1$	Id	$--$	LCD
4	$[14,9,4]_4$	$\alpha^2,0,\alpha,\alpha^2,0,1$	$\alpha^2,\alpha,1,\alpha^2,\alpha,\alpha,\alpha^2,\alpha,1,\alpha,\alpha,\alpha,\alpha,0,1$	Id	$--$	LCD
4	$[14,11,3]_4$	$\alpha^2,1,\alpha,1$	$\alpha,\alpha,1,1,\alpha,0,\alpha^2,\alpha,\alpha,1,0,\alpha^2,1,1,1$	Id	$--$	LCD
4	$[15,12,3]_4$	$\alpha^2,\alpha^2,\alpha^2,1$	$\alpha^2,1,0,0,\alpha,1,1,\alpha,\alpha^2,1,\alpha,0,0,1,\alpha^2,1$	Id	$--$	LCD
4	$[15,10,4]_4$	$1,0,0,\alpha^2,\alpha^2,1$	$1,\alpha,\alpha^2,1,\alpha^2,\alpha,1,1,0,0,\alpha^2,1,\alpha^2,\alpha,\alpha,1$	Frob	$\beta=0$	LCD
4	$[15,5,8]_4$	$\alpha^2,0,\alpha^2,0,\alpha,\alpha,\alpha,0,\alpha^2,\alpha^2,1$	$\alpha,0,1,1,\alpha,\alpha,\alpha^2,\alpha,\alpha,\alpha^2,\alpha^2,1,0,1,\alpha,1$	Frob	$\beta=0$	LCD
4	$[22,19,2]_4$	$1,\alpha^2,0,\alpha^2,\alpha^2,0,\alpha,0,\alpha,\alpha^2,$ $0,0,\alpha,\alpha^2,1,\alpha,1,\alpha^2,1,1$	$1,\alpha^2,0,\alpha^2,1,1,1,\alpha,\alpha^2,0,1,\alpha,0,\alpha^2,$ $0,0,\alpha^2,\alpha^2,\alpha,1,\alpha,1,1$ (From dual)	Frob	$\beta=1$	LCD
4	$[24,21,2]_4$	$\alpha^2,\alpha^2,\alpha,0,\alpha^2,0,\alpha,0,\alpha,\alpha^2,\alpha,$ $\alpha^2,\alpha^2,1,\alpha^2,\alpha^2,\alpha^2,\alpha,0,0,\alpha,1$	$1,0,1,\alpha,0,\alpha,0,0,1,0,0,\alpha,1,\alpha^2,\alpha,\alpha^2,\alpha^2,\alpha^2,$ $\alpha,1,1,1,0,\alpha,1$ (From dual)	Frob	$\beta=1$	LCD
4	$[24,21,2]_4$	$\alpha^2,\alpha^2,\alpha,0,\alpha^2,0,\alpha,0,\alpha,\alpha^2,\alpha,$ $\alpha^2,\alpha^2,1,\alpha^2,\alpha^2,\alpha^2,\alpha,0,0,\alpha,1$	$1,0,1,\alpha,0,\alpha,0,0,1,0,0,\alpha,1,\alpha^2,\alpha,\alpha^2,$ $\alpha^2,\alpha^2,\alpha,1,1,1,0,\alpha,1$(From dual)	Frob	$\beta=1$	LCD

Table 5.2: Examples of some best known $(f, \sigma, \delta)-$cyclic codes over $\mathbb{F}_8$, generated by $g(x)$ that are LCD codes

q	$[n, k, d]$	$g(x)$	$f(x)$	σ	β	Property
8	$[9, 3, 7]_8$	$\alpha^5, 1, \alpha^4, 1$	$1, \alpha^3, 1, \alpha^6, \alpha^5, \alpha^2, \alpha^5, \alpha^2, \alpha^4, 1$ (From dual)	Frob^2	$\beta = 0$	LCD
8	$[9, 4, 6]_8$	$\alpha^6, \alpha, \alpha^3, \alpha^4, 1$	$\alpha^6, \alpha^3, 1, \alpha^3, \alpha^2, \alpha, \alpha^6, \alpha^6, 1, 1$ (From dual)	Frob	$\beta = 0$	LCD
8	$[9, 5, 5]_8$	$\alpha^6, \alpha, \alpha^3, \alpha^4, 1$	$\alpha^6, \alpha^3, 1, \alpha^3, \alpha^2, \alpha, \alpha^6, \alpha^6, 1, 1$	Frob	$\beta = 0$	LCD
8	$[9, 6, 4]_8$	$\alpha^4, \alpha^3, \alpha^5, 1$	$\alpha^2, \alpha^3, \alpha^3, \alpha^6, \alpha^5, 1, \alpha^5, 1, \alpha^5, 1$	Frob	$\beta = 0$	LCD
8	$[9, 7, 3]_8$	$\alpha^4, 1, 1$	$\alpha^3, \alpha^4, \alpha^5, \alpha, 0, \alpha^5, \alpha, \alpha^5, \alpha^2, 1$	Frob	$\beta = 0$	LCD
8	$[10, 4, 6]_8$	$\alpha^6, \alpha^2, \alpha^5, 1, \alpha^2, 1, 1$	$\alpha^5, \alpha, \alpha^2, 0, \alpha^5, \alpha^6, \alpha^5, \alpha^3, \alpha^5, \alpha^3, 1$	Frob^2	$\beta = \alpha^6$	LCD
8	$[10, 5, 5]_8$	$\alpha^6, \alpha^6, \alpha^6, 0, \alpha, 1$	$\alpha^2, 1, \alpha^3, 1, 1, \alpha^2, \alpha^4, \alpha^4, \alpha^5, 1, 1$	Frob^2	$\beta = \alpha$	LCD
8	$[10, 6, 4]_8$	$1, \alpha^5, \alpha^3, \alpha^4, 1$	$\alpha^2, \alpha^6, 0, \alpha, \alpha^4, \alpha^6, \alpha^3, \alpha^5, \alpha^2, \alpha^3, 1$	Frob^2	$\beta = 0$	LCD
8	$[11, 3, 8]_8$	$\alpha^3, \alpha, 0, \alpha^6, \alpha^6, \alpha^2, \alpha^6, \alpha^3, 1$	$\alpha^6, \alpha, \alpha^3, \alpha, \alpha^3, \alpha^6, \alpha^4, \alpha, \alpha^3, \alpha^6, 0, 1$	Frob^2	$\beta = \alpha^3$	LCD
8	$[11, 4, 7]_8$	$\alpha, \alpha^6, \alpha, \alpha^4, 1, 1, \alpha^5, 1$	$\alpha^4, 1, \alpha^6, 1, \alpha^6, \alpha^2, 1, \alpha^4, \alpha^4, 0, \alpha^6, 1$	Frob^2	$\beta = 1$	LCD
8	$[11, 5, 6]_8$	$\alpha^6, \alpha^2, 1, \alpha^4, \alpha^3, \alpha^4, 1$	$\alpha, \alpha^2, \alpha^2, \alpha, \alpha^6, \alpha^3, 1, \alpha^2, \alpha^6, 1, \alpha^4, 1$	Frob^2	$\beta = 1$	LCD
8	$[11, 6, 5]_8$	$\alpha, \alpha^3, \alpha^6, \alpha^5, 0, 1$	$\alpha^4, \alpha^3, \alpha^2, \alpha^4, \alpha^4, \alpha^2, \alpha^5, \alpha^4, \alpha^4, 1, \alpha^3, 1$	Frob^2	$\beta = 1$	LCD
8	$[11, 7, 4]_8$	$1, 0, \alpha^2, 1, 1$	$\alpha, \alpha^2, 0, \alpha^2, 0, \alpha^2, \alpha^4, \alpha^2, \alpha^5, \alpha^2, \alpha^2, 1$	Frob^2	$\beta = 1$	LCD
8	$[11, 8, 3]_8$	$\alpha^4, \alpha^2, \alpha^3, 1$	$\alpha^4, \alpha^5, \alpha^6, 1, \alpha^3, \alpha^6, \alpha^2, \alpha^2, \alpha^6, \alpha, \alpha^6, 1$	Frob^2	$\beta = \alpha^3$	LCD
8	$[12, 3, 9]_8$	$\alpha^4, \alpha^2, \alpha, \alpha, \alpha^2, \alpha^5, 0, 1, \alpha^2, 1$	$\alpha^4, \alpha^6, \alpha^2, \alpha^6, \alpha, \alpha^4, 1, \alpha, \alpha^2, 0, \alpha^3, 0, 1$	Frob^2	$\beta = 0$	LCD
8	$[12, 4, 8]_8$	$\alpha^5, \alpha^4, 0, \alpha^3, \alpha^3, \alpha^3, \alpha^4, \alpha^2, 1$	$\alpha, \alpha, \alpha^5, \alpha^4, \alpha^4, \alpha^3, \alpha, 1, \alpha^4, \alpha^3, \alpha^6, \alpha^6, 1$	Id	$--$	LCD
8	$[12, 5, 7]_8$	$\alpha^4, \alpha, \alpha^6, 0, \alpha, 1, 1, 1$	$\alpha^4, \alpha^6, \alpha, \alpha^3, \alpha^2, \alpha^6, \alpha^6, 0, 0, 1, \alpha^5, \alpha^2, 1$	Frob^2	$\beta = 0$	LCD
8	$[12, 6, 6]_8$	$\alpha^5, \alpha^6, 0, \alpha^3, \alpha, \alpha^4, 1$	$\alpha^2, \alpha^4, \alpha^6, \alpha^4, \alpha^2, \alpha^4, \alpha^2, \alpha^4, \alpha^4, \alpha^2, 0, \alpha^5, 1$	Frob^2	$\beta = 0$	LCD
8	$[12, 7, 5]_8$	$\alpha, \alpha, \alpha^6, \alpha^2, \alpha^6, 1$	$\alpha, \alpha^4, \alpha^2, \alpha^2, 0, \alpha, \alpha^4, \alpha^3, \alpha^4, \alpha^3, \alpha^4, \alpha, 1$	Frob	$\beta = 0$	LCD
8	$[12, 8, 4]_8$	$\alpha, \alpha, \alpha^5, \alpha, 1$	$\alpha^6, 0, \alpha^2, \alpha^4, \alpha^4, \alpha^6, \alpha^5, \alpha^6, 1, \alpha^2, 1, 0, 1$	Frob	$\beta = 0$	LCD
8	$[12, 9, 3]_8$	$\alpha^6, \alpha^2, \alpha^4, 1$	$\alpha^2, \alpha^6, \alpha, \alpha^2, \alpha^6, 0, \alpha^3, \alpha^3, \alpha^5, \alpha^6, 1, 0, 1$	Frob^2	$\beta = \alpha^6$	LCD
8	$[13, 8, 5]_8$	$\alpha^3, 1, \alpha^6, \alpha^4, \alpha^5, 1$	$\alpha, \alpha^6, \alpha^3, 0, \alpha^6, 1, \alpha^6, \alpha^6, \alpha^3, \alpha^6, 0, \alpha^5, 0, 1$	Frob	$\beta = 0$	LCD
8	$[13, 9, 4]_8$	$\alpha, \alpha^5, \alpha, \alpha^5, 1$	$\alpha^6, \alpha^4, \alpha^2, \alpha^4, 1, \alpha^4, \alpha^4, \alpha^4, \alpha^3, \alpha^2, \alpha^5, \alpha^4, 1, 1$	Frob^2	$\beta = \alpha^2$	LCD
8	$[13, 10, 3]_8$	$\alpha^6, \alpha^4, \alpha^5, 1$	$\alpha^5, \alpha, \alpha, \alpha^5, \alpha^3, \alpha^2, 1, \alpha^6, \alpha^6, 0, \alpha^2, 1, 1, 1$	Frob^2	$\beta = \alpha$	LCD
8	$[14, 9, 5]_8$	$1, 0, \alpha^4, \alpha^2, \alpha^2, 1$	$\alpha, \alpha^2, 1, \alpha^4, \alpha^6, \alpha^6, \alpha^6, \alpha^5, \alpha^3, \alpha^5, 0, 1, \alpha^4, 0, 1$	Frob^2	$\beta = 0$	LCD
8	$[14, 10, 4]_8$	$1, 0, \alpha^2, \alpha^3, 1$	$\alpha^5, \alpha^3, \alpha, 1, \alpha, \alpha^4, 0, 1, \alpha^4, \alpha^6, \alpha^3, \alpha^6, 1, \alpha^3, 1$	Frob^2	$\beta = 0$	LCD
8	$[14, 11, 3]_8$	$1, \alpha^3, \alpha^5, 1$	$\alpha^5, 1, 0, \alpha^2, \alpha^4, 0, \alpha^2, \alpha^6, \alpha^4, \alpha, \alpha^4, \alpha^4, 1, \alpha^5, 1$	Frob^2	$\beta = \alpha$	LCD
8	$[15, 12, 3]_8$	$\alpha^3, \alpha^5, \alpha^4, 1$	$\alpha^4, \alpha^6, 1, \alpha^2, \alpha^6, \alpha^4, \alpha, \alpha, \alpha^3, 1, \alpha^5, 0, \alpha^2, 1, \alpha^4, 1$	Frob	$\beta = \alpha$	LCD
8	$[15, 12, 3]_8$	$\alpha^3, \alpha^5, \alpha^4, 1$	$\alpha^4, \alpha^6, 1, \alpha^2, \alpha^6, \alpha^4, \alpha, \alpha, \alpha^3, 1, \alpha^5, 0, \alpha^2, 1, \alpha^4, 1$	Frob	$\beta = \alpha$	LCD
8	$[20, 3, 16]_8$	$\alpha^3, \alpha^4, \alpha^6, \alpha, 0, \alpha^2, \alpha^5, 0, \alpha, \alpha^4, \alpha^4, \alpha^4, \alpha^2, \alpha^5, \alpha^2, \alpha^3, \alpha^6, 1$	$\alpha^5, 1, 0, \alpha^5, 0, 1, \alpha, \alpha, \alpha^2, \alpha^6, 1, \alpha^2, \alpha^3, \alpha^3, \alpha, 1, \alpha^4, \alpha^4, 1, \alpha^3, 1$	Frob	$\beta = 0$	LCD
8	$[20, 17, 3]_8$	$\alpha^3, \alpha^4, \alpha^6, \alpha, 0, \alpha^2, \alpha^5, 0, \alpha, \alpha^4, \alpha^4, \alpha^4, \alpha^2, \alpha^5, \alpha^2, \alpha^3, \alpha^6, 1$	$\alpha^5, 1, 0, \alpha^5, 0, 1, \alpha, \alpha, \alpha^2, \alpha^6, 1, \alpha^2, \alpha^3, \alpha^3, \alpha, 1, \alpha^4, \alpha^4, 1, \alpha^3, 1$ (From dual)	Frob	$\beta = 0$	LCD
8	$[22, 19, 3]_8$	$1, \alpha^6, 1, \alpha^2, \alpha^6, 1, \alpha^3, \alpha^6, \alpha^2, \alpha^5, \alpha^6, \alpha^6, \alpha^3, \alpha^4, \alpha^4, 1, \alpha^5, \alpha^6, \alpha^4, 1$	$1, \alpha^3, \alpha^4, \alpha^4, 1, \alpha, \alpha, \alpha^6, \alpha^3, \alpha^2, \alpha^3, \alpha^3, \alpha, \alpha^5, \alpha^5, \alpha^6, \alpha^2, \alpha^6, 0, \alpha, \alpha^4, \alpha^5, 1$ (From dual)	Frob	$\beta = \alpha^3$	LCD

Table 5.3: Examples of some best known $(f, \sigma, \delta)-$cyclic codes over $\mathbb{F}_9$, generated by $g(x)$ which are LCD codes

q	$[n,k,d]$	$g(x)$	$f(x)$	σ	β	Properties
9	$[11,3,8]_9$	$\alpha^2,\alpha^3,0,2,2,2,\alpha^3,\alpha^6,1$	$0,\alpha,\alpha^3,1,\alpha^6,\alpha,2,1,\alpha^5,\alpha^5,\alpha^6,1$	Frob	$\beta=\alpha^6$	LCD
9	$[11,4,7]_9$	$\alpha^5,\alpha^6,1,\alpha^7,\alpha^3,2,2,1$	$1,0,\alpha^2,\alpha^5,\alpha,0,2,\alpha^5,1,\alpha^2,1,1$	Frob	$\beta=\alpha^2$	LCD
9	$[11,5,6]_9$	$\alpha^6,0,\alpha^7,\alpha,2,\alpha,1$	$\alpha^2,\alpha^7,\alpha^7,1,\alpha^5,1,\alpha^2,\alpha^5,\alpha^2,\alpha^6,2,1$	Frob	$\beta=\alpha^7$	LCD
9	$[11,6,5]_9$	$\alpha,\alpha^3,\alpha^3,2,\alpha^2,1$	$\alpha^5,2,1,\alpha^5,\alpha^7,\alpha^6,\alpha^2,\alpha^7,\alpha,\alpha^3,\alpha^5,1$	Frob	$\beta=\alpha^2$	LCD
9	$[11,7,4]_9$	$1,\alpha,2,1,1$	$\alpha^2,\alpha^3,\alpha^3,\alpha^5,2,\alpha^5,\alpha^6,\alpha^7,1,\alpha^3,0,1$	Frob	$\beta=1$	LCD
9	$[11,8,3]_9$	$\alpha^6,\alpha^5,2,1$	$\alpha^5,1,0,1,\alpha^6,\alpha^6,\alpha,\alpha,\alpha^7,\alpha^3,\alpha^6,1$	Frob	$\beta=1$	LCD
9	$[12,3,9]_9$	$\alpha^5,\alpha,\alpha^2,\alpha,\alpha^2,0,\alpha^5,1,1,1$	$0,\alpha^3,\alpha^5,\alpha,\alpha^6,2,\alpha^2,\alpha^3,\alpha^2,\alpha^7,2,\alpha,1$	Frob	$\beta=\alpha^6$	LCD
9	$[12,4,8]_9$	$\alpha,1,\alpha,2,0,\alpha^7,\alpha^2,2,1$	$1,0,\alpha^2,0,\alpha^2,0,1,\alpha,\alpha^6,1,2,0,1$	Frob	$\beta=\alpha^5$	LCD
9	$[12,5,7]_9$	$2,2,\alpha,\alpha^2,\alpha^3,\alpha^7,0,1$	$2,\alpha,\alpha^2,\alpha,\alpha^5,1,2,\alpha,\alpha^7,\alpha^2,2,\alpha^7,1$	Frob	$\beta=\alpha^5$	LCD
9	$[12,6,6]_9$	$\alpha^5,2,\alpha,0,\alpha^6,2,1$	$1,0,\alpha^6,\alpha,\alpha,2,\alpha^2,0,1,0,1,2,1$	Frob	$\beta=\alpha^5$	LCD
9	$[12,7,5]_9$	$1,\alpha^5,\alpha^2,\alpha^3,\alpha^5,1$	$\alpha^7,\alpha^5,1,\alpha,1,\alpha,\alpha^6,\alpha^6,\alpha^7,1,\alpha,\alpha,1$	Frob	$\beta=2$	LCD
9	$[12,8,4]_9$	$\alpha^5,1,2,\alpha^3,1$	$0,0,\alpha,1,2,2,1,\alpha^3,1,\alpha^2,\alpha^7,1,1$	Frob	$\beta=2$	LCD
9	$[12,9,3]_9$	$\alpha^3,\alpha^6,\alpha,1$	$1,0,\alpha,1,\alpha^7,0,\alpha^6,\alpha^5,1,1,\alpha^6,1,1$	Frob	$\beta=\alpha^5$	LCD
9	$[13,3,10]_9$	$\alpha,\alpha^3,\alpha,1,2,0,\alpha^5,2,\alpha^6,2,1$	$0,0,2,\alpha^2,2,\alpha^7,\alpha^6,\alpha^5,\alpha^6,0,\alpha,1,\alpha^5,1$	Frob	$\beta=\alpha^5$	LCD
9	$[13,8,5]_9$	$\alpha^6,\alpha^7,0,\alpha^3,\alpha^7,1$	$\alpha^2,0,\alpha^5,\alpha^3,\alpha^3,\alpha^5,\alpha^6,\alpha,1,\alpha,\alpha^3,1,\alpha^6,1$	Frob	$\beta=\alpha^3$	LCD
9	$[13,9,4]_9$	$\alpha^5,2,\alpha^7,0,1$	$0,\alpha,\alpha^5,0,\alpha^6,0,\alpha^6,\alpha^5,0,0,\alpha,\alpha^5,\alpha^5,1$	Frob	$\beta=\alpha^6$	LCD
9	$[13,10,3]_9$	$\alpha^3,\alpha^3,2,1$	$0,\alpha,\alpha^6,\alpha,1,1,\alpha,2,\alpha^7,1,0,\alpha^2,\alpha,1$	Frob	$\beta=\alpha$	LCD
9	$[14,10,4]_9$	$\alpha^2,2,\alpha^2,\alpha^2,1$	$0,2,\alpha^2,\alpha^6,1,0,\alpha,1,\alpha^3,0,\alpha,1,0,\alpha^2,1$	Frob	$\beta=2$	LCD
9	$[14,11,3]_9$	$\alpha^2,\alpha^6,\alpha^6,1$	$\alpha,0,\alpha,\alpha^5,\alpha^2,\alpha^7,\alpha^3,\alpha^6,\alpha^6,\alpha^5,0,\ \alpha^5,\alpha^7,\alpha^2,1$	Frob	$\beta=\alpha$	LCD
9	$[16,13,3]_9$	$\alpha^5,2,\alpha^2,1$	$\alpha,\alpha^3,\alpha^5,\alpha^2,\alpha,\alpha^7,\alpha^7,0,\alpha,\alpha^3,0,1,\alpha^6,$ $\alpha^3,1,\alpha^2,1$	Frob	$\beta=\alpha^6$	LCD
9	$[16,12,4]_9$	$\alpha^3,\alpha^7,\alpha^3,\alpha^5,0,\alpha,\alpha^7,\ \alpha^6,1,1,2,\alpha^5,1$	$0,\alpha^3,1,\alpha^7,\alpha^5,0,2,\alpha^7,\alpha^2,\alpha^7,1,\ \alpha^6,1,0,\alpha^2,\alpha,1$ (From dual)	Frob	$\beta=\alpha^7$	LCD
9	$[18,3,14]_9$	$2,\alpha^6,1,2,\alpha^7,\alpha^6,\alpha^6,\alpha^2,\alpha^7,\alpha^2,$ $\alpha^3,\alpha^6,\alpha,0,\alpha^2,1$	$0,\alpha^3,\alpha^5,\alpha^3,\alpha^6,\alpha,\alpha^3,2,\alpha,0,\alpha^3,\alpha,$ $\alpha,\alpha^5,\alpha,\alpha^6,\alpha^5,2,1$	Frob	$\beta=1$	LCD
9	$[18,15,3]_9$	$2,\alpha^6,1,2,\alpha^7,\alpha^6,\alpha^6,\alpha^2,\alpha^7,\alpha^2,$ $\alpha^3,\alpha^6,\alpha,0,\alpha^2,1$	$0,\alpha^3,\alpha^5,\alpha^3,\alpha^6,\alpha,\alpha^3,2,\alpha,0,\alpha^3,\alpha,$ $\alpha,\alpha^5,\alpha,\alpha^6,\alpha^5,2,1$ (From dual)	Frob	$\beta=1$	LCD
9	$[20,13,17]_9$	$\alpha^5,2,\alpha^2,\alpha^6,\alpha^3,\alpha^5,\alpha^2,\alpha^6,\alpha^2,\alpha^5,$ $2,\alpha^7,\alpha,\alpha,\alpha^6,0,\alpha^5,1$	$0,0,\alpha^3,1,\alpha^3,\alpha^6,1,2,\alpha,\alpha^5,\alpha,\alpha^2,\alpha^5,1,$ $\alpha^2,\alpha^6,\alpha,1,\alpha^6,\alpha,1$ (From dual)	Frob	$\beta=1$	LCD

Table 5.4: Examples of some MDS $f-$cyclic codes over $\mathbb{F}_{16}$, $\mathbb{F}_{25}$, $\mathbb{F}_{27}$, generated by $g(x)$ that are LCD codes

q	$[n,k,d]$	$g(x)$	$f(x)$	σ	β	Properties
16	$[8,3,6]_{16}$	$\alpha^{14},\alpha^7,\alpha^9,\alpha^{12},\alpha^7,1$	$\alpha^6,\alpha^{10},\alpha,1,\alpha^{11},\alpha^6,\alpha,\alpha^{10},1$	Frob	$\beta=\alpha^{10}$	LCD, MDS
16	$[8,4,5]_{16}$	$\alpha^{10},\alpha^{11},\alpha^8,\alpha^{11},1$	$\alpha^9,\alpha^6,\alpha^2,\alpha^{14},\alpha^2,1,\alpha^{12},\alpha^7,1$	Frob^2	$\beta=\alpha^{10}$	LCD, MDS
16	$[8,5,4]_{16}$	$\alpha^7,\alpha^{14},\alpha^{11},1$	$\alpha^{22},\alpha^{15},\alpha^{21},\alpha^{12},\alpha^4,\alpha^{25},\alpha^8,\alpha,1$	Frob^2	$\beta=1$	LCD, MDS
16	$[11,8,4]_{16}$	$\alpha^6,\alpha^2,\alpha^5,1$	$\alpha^3,\alpha,\alpha,\alpha^9,\alpha^{12},\alpha^{13},\alpha^7,0,\alpha^{13},\alpha^2,\alpha^{12},1$	Frob^3	$\beta=0$	LCD, MDS
16	$[12,9,4]_{16}$	$\alpha^2,\alpha^4,1,1$	$\alpha^9,\alpha^8,0,\alpha^4,\alpha^{10},\alpha^{12},\alpha^6,\alpha^{14},1,\alpha^5,\alpha^3,\alpha^{14},1$	Frob^3	$\beta=0$	MDS
25	$[7,4,4]_{25}$	$1,\alpha^{19},1,1$	$\alpha^{16},\alpha^{16},\alpha^2,\alpha^{16},4,\alpha^3,\alpha^5,1$	Frob	$\beta=\alpha^{22}$	LCD, MDS
25	$[7,3,5]_{25}$	$\alpha^3,\alpha^{20},\alpha,\alpha^{19},1$	$\alpha^{20},\alpha,\alpha^5,\alpha^{10},\alpha^8,\alpha^{19},\alpha^{17},1$	Frob	$\beta=\alpha^9$	LCD, MDS
25	$[9,3,7]_{25}$	$\alpha^3,3,4,\alpha^{20},\alpha^9,\alpha^4,1$	$\alpha^{20},3,\alpha^8,\alpha^9,\alpha^{19},\alpha^{21},\alpha^9,\alpha^{10},\alpha^2,1$	Frob	$\beta=\alpha^{15}$	LCD, MDS
25	$[9,6,4]_{25}$	$\alpha^{20},3,\alpha^{23},1$	$2,\alpha^{10},\alpha^{16},\alpha^{23},\alpha^9,0,\alpha^{21},\alpha^{14},\alpha^{23},1$	Frob	$\beta=1$	LCD, MDS
25	$[10,3,8]_{25}$	$\alpha^{21},\alpha^2,\alpha^{10},\alpha^{15},\alpha^{10},4,\alpha^{21},1$	$\alpha^4,\alpha^{10},\alpha^{15},\alpha^5,\alpha^9,\alpha^{19},\alpha^4,\alpha^2,\alpha^{21},\alpha^{10},1$	Frob	$\beta=\alpha^5$	LCD, MDS
25	$[10,7,4]_{25}$	$4,2,4,1$	$3,\alpha^{22},\alpha^5,\alpha^{17},\alpha^{11},\alpha^{16},\alpha^8,\alpha^2,\alpha^3,\alpha^8,1$	Frob	$\beta=\alpha^{20}$	LCD, MDS
27	$[7,3,5]_{27}$	$\alpha^2,\alpha^6,2,\alpha^{18},1$	$\alpha^{17},\alpha^{21},\alpha^{23},1,2,2,\alpha^4,1$	Frob^2	$\beta=0$	LCD, MDS
27	$[7,4,4]_{27}$	$\alpha^{17},\alpha^{21},\alpha^{23},1$	$\alpha^{22},\alpha^{19},\alpha^{24},\alpha^{11},\alpha^{20},2,\alpha^8,1$	Frob	$\beta=0$	LCD, MDS
27	$[8,5,4]_{27}$	$\alpha^{20},\alpha^{18},\alpha^3,1$	$\alpha^{22},\alpha^{19},\alpha^{24},\alpha^{11},\alpha^{20},2,\alpha^8,1$	Frob^2	$\beta=1$	LCD, MDS
27	$[8,3,6]_{27}$	$\alpha^10,\alpha^8,\alpha^12,\alpha^15,\alpha^20,1$	$\alpha^4,\alpha^6,\alpha^14,\alpha^16,\alpha^23,\alpha^22,\alpha^10,1,1$	Frob	$\beta=\alpha^4$	LCD, MDS
27	$[8,4,5]_{27}$	$\alpha^{23},\alpha^{20},2,\alpha^4,1$	$\alpha^5,\alpha^2,\alpha^{16},\alpha^{23},\alpha^{25},\alpha^{12},\alpha^3,\alpha^8,1$	Frob	$\beta=1$	LCD, MDS
27	$[9,6,4]_{27}$	$\alpha^{25},\alpha^{10},\alpha,1$	$\alpha^{18},\alpha^7,\alpha^4,\alpha^{23},\alpha^2,\alpha^{15},\alpha^{25},\alpha^{25},\alpha^{24},1$	Frob^2	$\beta=\alpha^{14}$	LCD, MDS
27	$[9,5,5]_{27}$	$\alpha^{15},\alpha^{15},\alpha^{20},\alpha^3,1$	$\alpha^{19},\alpha^6,\alpha^{14},\alpha^{25},\alpha^9,\alpha^{12},\alpha^{25},2,2,1$	Frob	$\beta=\alpha^{23}$	LCD, MDS
27	$[9,4,6]_{27}$	$\alpha^7,\alpha^{11},\alpha^8,\alpha^{19},\alpha^{16},1$	$\alpha^{18},\alpha^{24},\alpha^{20},\alpha^{12},\alpha^2,\alpha^4,\alpha^{23},1,\alpha^{24},1$	Frob^2	$\beta=\alpha^{15}$	LCD, MDS
27	$[9,3,7]_{27}$	$\alpha^5,\alpha^{21},\alpha^8,\alpha^5,\alpha^{16},\alpha^{11},1$	$\alpha^7,\alpha^6,2,\alpha^{21},2,\alpha^{18},\alpha^5,\alpha^{25},\alpha^6,1$	Frob	$\beta=1$	LCD, MDS
27	$[10,7,4]_{27}$	$\alpha^{22},\alpha^{20},\alpha^6,1$	$\alpha^{15},\alpha^{20},\alpha^{16},\alpha^{21},\alpha^8,\alpha^{22},1,\alpha^{17},\alpha^{11},\alpha^3,1$	Frob^2	$\beta=0$	LCD, MDS
27	$[10,3,8]_{27}$	$\alpha^5,\alpha^2,\alpha^2,\alpha^{21},\alpha^{16},\alpha^5,\alpha,1$	$\alpha^6,\alpha^9,\alpha^9,\alpha^{12},\alpha^{24},\alpha^7,\alpha^9,\alpha^2,\alpha^4,\alpha^{25},1$	Frob	$\beta=\alpha^9$	LCD, MDS
27	$[11,3,9]_{27}$	$\alpha^{16},\alpha^{16},\alpha,\alpha^{14},\alpha^7,\alpha^{15},2,\alpha^{18},1$	$\alpha^{21},\alpha^{18},\alpha^{23},\alpha^{18},\alpha,\alpha^{12},\alpha^{15},\alpha^6,\alpha^4,\alpha,\alpha^{11},1$	Frob	$\beta=1$	LCD, MDS
27	$[11,8,4]_{27}$	$2,\alpha^{11},\alpha^{22},1$	$\alpha^{20},1,\alpha^2,1,\alpha^7,\alpha^3,\alpha^{12},\alpha^6,\alpha^9,\alpha^{10},\alpha^{12},1$	Frob^2	$\beta=\alpha^{14}$	LCD, MDS

Table 5.5: Examples of some best known quasi $(f,\sigma,\delta)-$cyclic codes over $\mathbb{F}_4$, generated by $(g(x),p_1g(x))$, with additional desirable properties.

$[n,k,d]$	g	$(p_1,p_2,..)$	f	β	Properties
$[10,3,6]_4$	$g=1,\alpha,1$	$p_1=\alpha,1,1$	$f=\alpha,\alpha^2,1,0,\alpha,1$	$\beta=\alpha$	LCD
$[10,3,6]_4$	$g=\alpha^2,1,1$	$p_1=1$	$f=0,1,1,0,0,1$	$\beta=\alpha$	self-orthogonal
$[10,2,8]_4$	$g=\alpha,\alpha,1,1$	$p_1=0,1$	$f=\alpha,\alpha,\alpha^2,\alpha^2,1,1$	$\beta=0$	self-orthogonal
$[12,3,8]_4$	$g=1,\alpha,0,1$	$p_1=\alpha,1$	$f=1,\alpha^2,1,\alpha^2,\alpha,\alpha,1$	$\beta=0$	LCD
$[14,3,10]_4$	$g=1,1,1,0,1$	$p_1=\alpha^2,\alpha^2,1$	$f=\alpha,1,\alpha^2,0,1,\alpha,\alpha,1$	$\beta=\alpha^2$	self-orthogonal

Discussion and Perspectives

In this dissertation, we studied the structure of linear codes that are invariant, respectively, under a linear and pseudo-linear transformation T, that we call T-codes and T-skew codes. Since each operator T can be represented by a matrix, for simplicity, we considered the invariance under a given matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$. In the first part, we studied the structure of codes invariant under right multiplication by a square matrix $M \in \mathbb{M}_n(\mathbb{F}_q)$ and we call them M-codes. In particular we distinguish three fundamental types: M-cyclic codes, quasi M-cyclic codes and $(M_1, M_2, \ldots, M_r)$-multi-cyclic codes. The approach developed here allows us to realize cyclic codes and many of their generalizations as special cases. It also helps us better understand the structure of linear codes invariant under some special matrices such as a permutation matrix or a monomial matrix. Next, we study the duality of M-codes, where we show that the b-dual of an M-code is an M^*-code, where M^* is the adjoint matrix of M with respect to a non-degenerate bilinear form b, and we explore some important results on the duality of these codes. In order to use polynomial rings in the study of M-codes, we show a one-to-one correspondence between M-codes and $\mathbb{F}_q[x]$-submodules of $\prod_{i=1}^{r} R_{f_i}$, where $R_{f_i} := \mathbb{F}_q[x]/\langle f_i(x)\rangle$ and $(f_1(x), f_2(x), \ldots, f_r(x))$ is the sequence of invariant factors of M. Finally, we investigate the structure of 1-generator M-codes and we prove BCH-like and Hartmann-Tzeng-like bounds for them. As an application of the theory developed here, we construct some optimal linear codes from various types of codes discussed.

In the second part, we studied the structure of linear codes invariant under the (σ, δ)-action of a given matrix M, that we call (M, σ, δ)-skew codes. Various subclasses of these codes, depending on the (σ, δ)-cyclical property of M, are derived. In particular, we distinguish between three fundamental types: (M, σ, δ)-cyclic, (M, σ, δ)-quasi-cyclic and (M, σ, δ)-multi-cyclic. These different subclasses allow us to obtain some of the well-known types of linear codes as special cases depending on the choice of M. We observe that the (σ, δ)-similarity in $\mathbb{M}_n(\mathbb{F}_q)$ gives rise to various relations between these codes. As common in the study of cyclic codes and their generalizations, we use the skew polynomial ring $\mathbb{F}_q[x, \sigma, \delta]$ to study (M, σ, δ)-skew codes. For future work, it would be interesting to investigate:

1. the duality of M-codes and M-skew codes with respect to various inner products.
2. the equivalence of codes using the theory of matrices
3. the generalization of this approach to codes over rings.
4. the question of the dependence of the Hamming distance on the basis of $\mathbb{F}_q^n$, and if there is a preferable cyclic matrix that allows us to maximize the Hamming distance of a given code.
5. the skew eigenvalues of M to potentially obtain some lower bounds on the minimum Hamming distance similar to the BCH bound, Hartmann-Tzeng bound, and Roos bound for skew f-cyclic codes and their generalizations.
6. the construction of quantum codes from this family of codes using the CSS method.

7. the exhaustive search for optimal codes in the class of f-cyclic codes and quasi f-cyclic codes, as in the case of cyclic and constacyclic codes.

8. the construction of new codes with additional desirable properties (like LCD, reversible, ...) from f-cyclic, skew f-cyclic and their generalizations.

9. the theory of cyclotomic cosets and their relation to factoring skew polynomials over a finite field and finite ring.

10. the theory of (σ, δ)-bilinear forms of $\mathbb{F}_q^n$.

11. the use of these codes in code-based cryptography.

12. the search for a new one-way function in a skew polynomial ring over a ring that could serve as the basis for a new cryptosystem that may be resistant to quantum attacks.

Appendix

In this chapter, we collect some SageMath programs that we used to construct various examples of this dissertation.

The following function is used to compute the order of polynomials over finite fields.

```
sage: F.<alpha>=GF(4);
....: FX.<x>=F[];
....: def order(f):
....:     k=f.valuation(x)
....:     P=f//x^k
....:     n=P.degree()
....:     Q=x^n-1
....:     R=Q.mod(P)
....:     while R!=0:
....:         n=n+1
....:         Q=x^n-1
....:         R=Q.mod(P)
....:     return (n,k)
....:
....: f=x^7 + (alpha + 1)*x + alpha + 1
....: order(f)
```

This function permits us to determine all possible divisors of $f(x)$, then all possible generators of an f-code.

```
sage: def f_codes(P):
....:     GPpoly=[];
....:     GPpoly1=[];
....:     n=P.degree();
....:     L=factor(P);
....:     factor1=[];
....:     factor2=[];
....:     u=1
....:     for i in range(len(L)):
....:         for j in range(L[i][1]):
....:                 factor1.append(L[i][0]**(j+1))
....:     for k in range(len(factor1)):
....:         SS=Subsets(factor1,k+1)
....:         factor2.append(SS.list())
....:     for i in range(len(factor2)):
....:         for j in range(len(factor2[i])):
```

```
....:             for k in range(len(factor2[i][j])):
....:                 u=u*factor2[i][j][k]
....:             if u.divides(P)and u!=P:
....:                 GPpoly.append(u)
....:                 u=1
....:     GPpoly1=list(set(GPpoly))
....:     return GPpoly1
....:
....: f_codes(f)
```

To compute the generator matrix of a given f-cyclic code generated by $g(x)$ we use the following two functions

```
sage: def generator_matrix(g,P):
....:     n=P.degree();
....:     d=g.degree() ;
....:     k=n-d;
....:     F=g.base_ring();
....:     M=matrix(F,k,n);
....:     coef=[];
....:     for i in range(k):
....:         coef=(((x^i*g)%P)).coefficients(sparse=False)
....:         for j in range(len(coef)):
....:             M[i,j]=coef[j]
....:     return M
```

or

```
def generator_matrix(g,P,k):
    n=P.degree();
    d=g.degree() ;
    F=g.base_ring();
    M=matrix(F,k,n);
    coef=[];
    for i in range(k):
        coef=(((x^i*g)%P)).coefficients(sparse=False)
        for j in range(len(coef)):
            M[i,j]=coef[j]
    return M
```

To determine a M-cyclic basis we use

```
sage: def cyclic_vector(A):
....:     F=A.base_ring()
....:     n=A.nrows()
....:     P=A.minimal_polynomial()
....:     C_A=companion_matrix(P, format='right')
....:     m=P.degree()
....:     base=[]
....:     if m==n:
....:         L=[K for K in F^n]
....:         for i in range(30):
```

```
....:             u= matrix(L[i]).transpose()
....:             P_v=matrix(F,n,n);
....:             for j in range(n):
....:                 P_v[:,j]=A^j*u
....:             if det(P_v)!=0:
....:                 CA=P_v^-1*A*P_v
....:                 base.append([u,CA,P_v, P_v^-1,A])
....:     return base
....:
....: L=cyclic_vector(A)
```

To determine all possible BCH-like f-cyclic codes with prescribed distance δ we use

```
....: def BchBounds(P,delta):
....:     gen=polycyclic_codes(P)
....:     F=P.base_ring();
....:     FX.<x>=F[]
....:     e=order(P)[0]
....:     n=P.degree();
....:     q = F.order();
....:     R = IntegerModRing(e);
....:     m = R(q).multiplicative_order();
....:     FF = GF(q**m,"beta");
....:     beta = FF.gen();
....:     ee = beta.multiplicative_order()//e;
....:     FFX = PolynomialRing(F,"x");
....:     x = FFX.gen()
....:     f= prime_factors(FFX(beta.minpoly()))[0];
....:     LF.<c> = F.extension(f);
....:     ci=c^ee
....:     Roots_P=[];
....:     BCH_delta=[];
....:     scosecutive=[]
....:     for i in srange(e):
....:         if P(ci**i)==0:
....:             Roots_P.append(i)
....:     S=Subsets(Roots_P,delta-1);
....:     Sdelta=S.list()
....:     N=[];
....:     VC=[];
....:     vc=[]
....:     if delta >2:
....:         for K in Sdelta:
....:             L1=list(K);L1.sort()
....:             L2=list(K); L2.sort()
....:             pas=L2[1]-L2[0]
....:             if gcd(e,pas)==1:
....:                 for i in range(1,delta-1):
....:                     L1[i]=L1[i-1]+pas
....:                 if L1==L2:
```

```
....:                     LL0 = [ci**kk for kk in L1]  ;
....:                     LL1 = [FFX(ai.minpoly()) for ai in LL0] ;
....:                     g = FFX(LCM(LL1))
....:                     if   g in gen:
....:                         VC.append([L1,pas,g]) ;
....:                         N.append(g)
....:     if delta==2:
....:         for jj in Roots_P:
....:             ai=ci**jj
....:             g = FFX(ai.minpoly())
....:             if   g.divides(P):
....:                 VC.append([jj,g]);
....:                 N.append(g)
....:     M=list(set(N))
....:     for c in M:
....:         G=generator_matrix(c,P,n-c.degree())
....:         code=codes.LinearCode(G)
....:         BCH_delta.append((c, code))
....:
....:     return BCH_delta,VC,Roots_P
```

To determine a Hartmann-Tzeng bound of an f-cyclic code.

```
sage: def Hratmannbound(P,delta,r):
....:     gen=polycyclic_codes(P)
....:     F=P.base_ring();
....:     e=order(P)[0]
....:     n=P.degree();
....:     q = F.order();
....:     R = IntegerModRing(e);
....:     m = R(q).multiplicative_order();
....:     FF = GF(q**m,"beta");
....:     beta = FF.gen();
....:     ee = beta.multiplicative_order()//e;
....:     FFX = PolynomialRing(F,"x");
....:     x = FFX.gen()
....:     Roots_P=[];
....:     BCH_delta=[]
....:     f= prime_factors(FFX(beta.minpoly()))[0];
....:     LF.<c> = F.extension(f);
....:     ci=c^ee
....:     for i in srange(e):
....:         if P(ci^i)==0:
....:             Roots_P.append(i)
....:
....:     S=Subsets(Roots_P,delta-1);
....:     R=Subsets(Roots_P,r+1);
....:     Sdelta=S.list() ;
```

```
....:     SR=R.list()
....:     N=[];
....:     liste=[]
....:     for K1 in Sdelta:
....:         L1=list(K1);
....:         L1.sort()
....:         L2=list(K1);
....:         L2.sort()
....:         if delta >2:
....:             pas1=L1[1]-L1[0]
....:             if gcd(e,pas1)==1:
....:                 for i in range(1,delta-1):
....:                     L1[i]=L1[i-1]+pas1
....:                 if L1==L2:
....:                     for K2 in SR:
....:                         R1=list(K2); R1.sort()
....:                         R2=list(K2);R2.sort()
....:                         pas2=R1[1]-R1[0]
....:                         for j in range(1,r+1):
....:                             R1[j]=R1[j-1]+pas2
....:                         if gcd(e,pas2)==1 and R1==R2 :
....:                             scosecutive=[]
....:                             for elm1 in L1:
....:                                 for elm2 in R2:
....:                                     scosecutive.append(elm1+elm2)
....:                             LL0 = [ci**j for j in scosecutive];
....:                             LL1 = [FX(ai.minpoly()) for ai in LL0]
....:                             g = FFX(LCM(LL1))
....:                             if g in gen:
....:                                 N.append(g);
....:                                 liste.append([g,scosecutive,pas1,pas2])
....:         if delta==2:
....:             Roots_P.sort()
....:             for i in Roots_P:
....:                 for K2 in SR:
....:                     R1=list(K2); R1.sort()
....:                     R2=list(K2);R2.sort()
....:                     pas2=R1[1]-R1[0]
....:                     for j in range(1,r+1):
....:                         R1[j]=R1[j-1]+pas2
....:                     scosecutive=[]
....:                     if gcd(e,pas2)==1 and R1==R2 :
....:                         for elm2 in R2:
....:                             scosecutive.append(i+elm2)
....:                         LL0 = [ci**jj for jj in scosecutive];
....:                         LL1 = [FFX(ai.minpoly()) for ai in LL0];
....:                         g = FFX(LCM(LL1))
....:                         if g in gen:
....:                             N.append(g);
....:                             liste.append([scosecutive,g,i,pas2])
```

```
....:     M=list(set(N))
....:     M=list(set(N))
....:     for c in M:
....:         #H=generator_matrix(c,P)
....:         #code=codes.LinearCode(H)
....:         #d=code.minimum_distance()
....:         BCH_delta.append((c))
....:
....:     return BCH_delta,liste,Roots_P
```

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